

A horizontal row of six small, solid-colored rectangular bars in blue, green, purple, orange, teal, and red.

Project Management Professional (PMP) Exam Prep

4-Day Instructor-Led Programme · 35 Contact Hours

Version: v2.0 Trainer: Dr. Alfred Ang

Course Code: C523

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Aligned to the PMI PMP Examination Content Outline — July 2026

Ground Rules

1 Phones on silent

Take calls outside the room so the session is not disrupted for others.

2 Participate actively

This course is built around labs and discussion — the more you contribute, the more you take away.

3 No question is too basic

If something is unclear, ask. Others in the room almost certainly share the same question.

4 Punctuality after breaks

Return on time so we can cover the full ECO syllabus within the four days.

5 Respect differing views

Project management is contextual — disagreement is useful when handled with professional respect.

6 Confidentiality

Workplace examples shared in class stay in the class; anonymise sensitive client or employer detail.

How You'll Learn



There is no assessment on this course. The mock practice exam in Lab 24 is a self-scored diagnostic that tells you where to study next.

About This Course

1 Course Code

C523 — quote this on your PMI application as evidence of formal project management education.

2 35 contact hours

This course satisfies the PMI education prerequisite of 35 contact hours for the PMP application.

3 ECO 2026 aligned

Built against the PMI Examination Content Outline effective July 2026, across all three domains.

4 Hybrid by design

The case study is deliberately hybrid, because roughly 60% of exam items sit in adaptive or hybrid contexts.

5 24 hands-on labs

One continuous project, worked end to end, producing a complete set of artifacts you keep.

6 No assessment

This is a commercial short course — there is no graded assessment. Your Lab 24 mock score is your readiness signal.

BY THE END OF THIS COURSE YOU WILL BE ABLE TO

Learning Outcomes

1 LO1

Scope medium-scale project requirements to drive timely completions.

2 LO2

Develop project plans based on realistic timelines and resource allocations.

3 LO3

Implement methodologies effectively to address project exigencies and derive solutions from identified root causes.

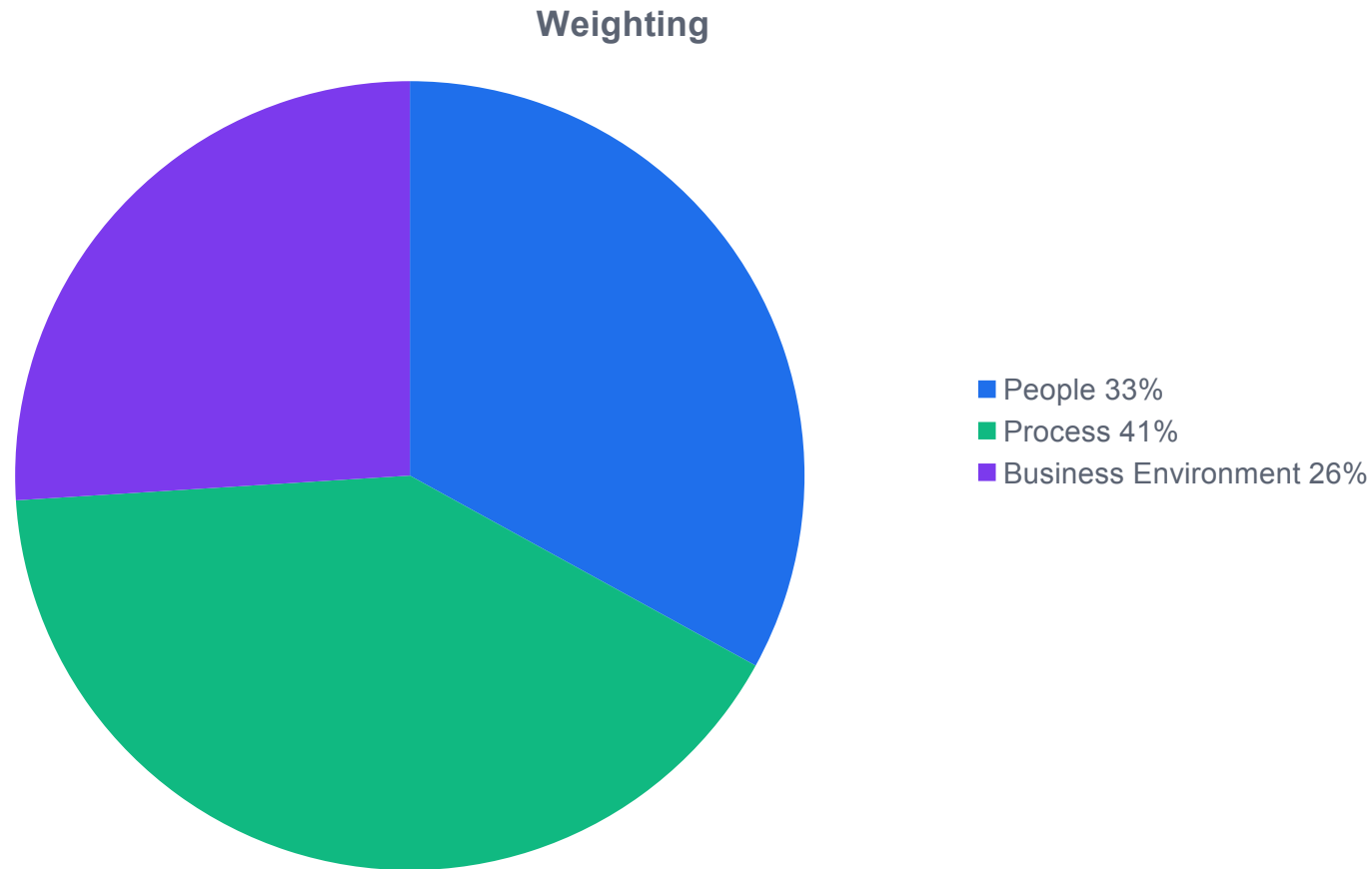
4 LO4

Analyze program risks and engage stakeholders through scheduled touchpoints to discuss potential issues.

5 LO5

Coordinate project deliverables against set objectives, costs, and timelines, and implement corrective actions.

The PMP Exam Content Outline — July 2026



WHAT IT TELLS YOU

- People 33% — leading, aligning and communicating with the team and stakeholders.
- Process 41% — planning, delivering, measuring and closing the work.
- Business Environment 26% — governance, compliance, risk and change. Up sharply from the old 8%.
- Approach mix: ~40% predictive, 60% adaptive/agile and hybrid.

What Changed in the July 2026 Exam Update

1 Business Environment tripled

Weighting rises from 8% to 26% — governance, compliance and external change are now central, not peripheral.

2 People eased slightly

From 42% to 33%, but the tasks are sharper: vision, conflict, expectations and knowledge transfer.

3 Process rebalanced

From 50% to 41%, with new emphasis on value-based delivery and financial management.

4 Sustainability is examinable

Named explicitly in compliance, quality (cost of quality) and risk response enablers.

5 Artificial intelligence enters scope

The JTA validated AI as a trend affecting practice — expect it in estimation, reporting and governance items.

6 Graphic-based questions are new

You will be asked to read charts, control charts and diagrams and answer from them — practise interpretation.

PMP Exam Format at a Glance

Element	Detail	What it means for you
Questions	180 total (170 scored + 10 pretest)	Unscored pretest items are indistinguishable — answer everything seriously.
Time	240 minutes (4 hours)	About 80 seconds per question — pace, do not linger.
Breaks	Two 10-minute breaks	First after the case-study section, second midway through the rest.
Section lock	No return after a break	Review your answers BEFORE you start a break.
Approach mix	About 40% predictive; the remaining 60% split between adaptive/agile and hybrid.	You cannot pass on waterfall knowledge alone.
Delivery	Test centre or online proctored	Case study questions are available on all modalities.

PMP Exam Question Types

1 Multiple choice

One best answer from four options — the bulk of the exam.

2 Multiple response

More than one correct answer; the item tells you how many.

3 Matching

Drag items from one column to match a second column.

4 Enhanced matching

Matching that includes an image or diagram for context.

5 Graphic-based (NEW)

Read a chart, graph, diagram or image, then answer from it.

6 Case study

A scenario with supporting visuals, then a series of linked questions.

Course Outline — Topics 1 to 3

1 Topic 1. Business Environment

Where projects come from, why they are funded, and the governance, compliance and change context they must survive in.

Foundation and project fundamentals ·
Strategic alignment and business value ·
Development approaches: predictive, agile, hybrid

2 Topic 2. Start the Project

Turning an idea into an authorised project: the business case, the charter, the stakeholders and the team.

Project integration and the charter · Business case, cost-benefit and benefits management · Identify, analyse and prioritise stakeholders

3 Topic 3. Plan the Project

Building the integrated plan — scope, schedule, cost, quality, resources, communications, risk and procurement.

The integrated project management plan ·
Scope: requirements, WBS and baselines ·
Schedule: activities, dependencies, critical path

Course Outline — Topics 4 to 6

1 Topic 4. Lead the Project Team

The People domain in practice — leadership style, motivation, conflict, coaching and communication.

Direct and manage project work · Craft your leadership skills · Build a collaborative team environment

2 Topic 5. Monitor and Control the Project

Measuring truth against plan: earned value, quality control, change control, issues and impediments.

Monitor and control project work · Evaluate project progress and earned value · Control quality with statistical tools

3 Topic 6. Close the Project

Landing the project: acceptance, transition, benefits realisation and the knowledge that outlives the team.

Project and phase closure · Benefits realisation and value confirmation · Knowledge transfer and lessons learned

The Four Days



Each day runs 9:30am–6:30pm with a one-hour lunch break and two tea breaks. Roughly half of each day is hands-on lab work.

DAY 1 SCHEDULE

Lesson Plan — Day 1: Business Environment & Starting the Project

Time	Session	Activity
9:30 – 10:00	Registration & Course Opening	Course introduction, trainer and learner introductions, ground rules, learning outcomes, ECO 2026 orientation
10:00 – 11:15	Topic 1. Business Environment	Concept delivery, worked examples and discussion
11:15 – 11:30	Tea Break	—
11:30 – 13:00	Topic 2. Start the Project	Concept delivery, worked examples and discussion
13:00 – 14:00	Lunch Break	—
14:00 – 15:45	Hands-on Labs	Lab 01, Lab 02, Lab 03, Lab 04 (LO1, LO3, LO4)
15:45 – 16:00	Tea Break	—
16:00 – 17:30	Hands-on Labs (continued)	Lab 05, Lab 06, Lab 07, Lab 08 — group share-back
17:30 – 18:30	Consolidation	Exam-style question drill on the day's material, recap and Q&A

Lesson Plan — Day 2: Planning the Project

Time	Session	Activity
9:30 – 10:00	Registration & Course Opening	Recap of Day 1, exam-style warm-up questions and Q&A
10:00 – 11:15	Topic 3. Plan the Project	Concept delivery, worked examples and discussion
11:15 – 11:30	Tea Break	—
11:30 – 13:00	Topic 3. Plan the Project	Concept delivery, worked examples and discussion
13:00 – 14:00	Lunch Break	—
14:00 – 15:45	Hands-on Labs	Lab 09, Lab 10, Lab 11 (LO1, LO2, LO4)
15:45 – 16:00	Tea Break	—
16:00 – 17:30	Hands-on Labs (continued)	Lab 12, Lab 13, Lab 14 — group share-back
17:30 – 18:30	Consolidation	Exam-style question drill on the day's material, recap and Q&A

DAY 3 SCHEDULE

Lesson Plan — Day 3: Leading the Team & Controlling Delivery

Time	Session	Activity
9:30 – 10:00	Registration & Course Opening	Recap of Day 2, exam-style warm-up questions and Q&A
10:00 – 11:15	Topic 4. Lead the Project Team	Concept delivery, worked examples and discussion
11:15 – 11:30	Tea Break	—
11:30 – 13:00	Topic 5. Monitor and Control the Project	Concept delivery, worked examples and discussion
13:00 – 14:00	Lunch Break	—
14:00 – 15:45	Hands-on Labs	Lab 15, Lab 16, Lab 17, Lab 18 (LO2, LO3, LO4, LO5)
15:45 – 16:00	Tea Break	—
16:00 – 17:30	Hands-on Labs (continued)	Lab 19, Lab 20, Lab 21, Lab 22 — group share-back
17:30 – 18:30	Consolidation	Exam-style question drill on the day's material, recap and Q&A

Lesson Plan — Day 4: Closing, Exam Strategy & Practice

Time	Session	Activity
9:30 – 9:45	Day 4 Opening	Recap of Day 3, exam-style warm-up questions and Q&A
9:45 – 11:00	Topic 6. Close the Project	Concept delivery: closure, benefits realisation, knowledge transfer
11:00 – 11:15	Tea Break	—
11:15 – 13:00	Hands-on Labs	Lab 23 (LO1, LO2, LO3, LO4, LO5)
13:00 – 14:00	Lunch Break	—
14:00 – 15:45	Capstone Lab	Lab 24 Part A — consolidate all 23 labs' artifacts into one integrated project management plan with a consistency audit
15:45 – 16:00	Tea Break	—
16:00 – 17:10	Mock Practice Exam	Lab 24 Part B — 30-question ECO 2026-format mock exam under timed conditions (self-scored, not an assessment)
17:10 – 18:00	Mock Exam Debrief	Score by domain against the 33/41/26 weights, work the rationale for every wrong answer, and build a dated remediation plan
18:00 – 18:30	Exam Strategy & Course Close	PMI application steps, study plan, exam-day tactics and Q&A

How to Get the Most From These Four Days

1 Work the arithmetic by hand

The exam gives you no spreadsheet. Do the PERT, EVM and EMV calculations manually until they are automatic.

2 Identify the approach first

Before answering anything, decide whether the scenario is predictive, adaptive or hybrid — it changes the correct answer.

3 Keep every artifact

Each lab feeds the next. By Lab 24 they assemble into one integrated plan you can reuse at work.

4 Ask about your own projects

The most valuable discussions come from real situations in the room — bring them.

5 Track your weak domains

Note which topics you find hardest; the Lab 24 score analysis will confirm them and set your study order.

6 Book the exam deliberately

Sit it while the material is fresh, but only once your mock scores are consistently above target.

01

TOPIC

Business Environment

Where projects come from, why they are funded, and the governance, compliance and change context every project must survive in. This topic sets the vocabulary used for the rest of the course.

- 1 What a project is, and what project management actually delivers
- 2 The 12 PMI principles and what project success really depends on
- 3 Life cycles and development approaches: predictive, agile and hybrid
- 4 Organisational culture, EEFs, OPAs and structured change management
- 5 Project governance, escalation thresholds and phase gates
- 6 Compliance, sustainability and the responsible use of AI in projects

DEFINITION OF A PROJECT

The foundation of everything that follows

A project is a temporary endeavour undertaken to create a unique product, service or result.

TEMPORARY

It has a definite start and a definite end — ending when objectives are met, or when it is clear they cannot be.

UNIQUE

The output has never been produced in exactly this form before, so some degree of uncertainty is always present.

DELIVERS CHANGE

A project moves the organisation from a current state to a future state that carries business value.

Projects versus operations — why the distinction matters

PROJECT WORK

Temporary, unique, change-driven

- Has a defined start and end date fixed by the charter
- Produces a unique deliverable that did not previously exist
- Staffed by a team that is assembled and later released
- Success is measured against scope, schedule, cost and benefits
- Progressively elaborated as more information becomes known
- Funded from a capital or change budget with a business case

OPERATIONAL WORK

Ongoing, repetitive, efficiency-driven

- Continues indefinitely as long as the business needs the output
- Produces the same product or service repeatedly to a standard
- Staffed by a permanent functional team reporting to a line manager
- Success is measured against throughput, cost per unit and quality
- Optimised for stability and predictable repetition, not novelty
- Funded from a recurring operating expense budget

Exam cue: if the scenario describes work that never ends or simply repeats, it is operations — the correct answer usually involves transitioning the deliverable to operations, not extending the project.

What counts as a project? Recognising project work in the wild

1 Developing a software product

Building a new mobile banking app from concept to launch — unique code, a fixed release date, and a team disbanded after go-live.

2 Constructing a new office

Design, permits, build and fit-out of a physical asset, with heavy regulatory compliance and irreversible, expensive change late on.

3 Filming a motion picture

Highly creative, schedule-driven work with named talent constraints and a single big-bang delivery to distribution.

4 An advertising campaign

Short, iterative, market-sensitive delivery where the external environment can invalidate the concept mid-flight.

5 Redesigning a business process

An internal change project whose real deliverable is adoption by people, not a document — heavy change management content.

6 Migrating to a cloud platform

Technical transformation with security, data-residency and compliance obligations plus a hard decommission deadline.

PROJECT MANAGEMENT DEFINED

What is project management?

The application of knowledge, skills, tools and techniques to project activities to meet the project requirements.

A DISCIPLINE

It spans initiating, planning, executing, monitoring and controlling, and closing — performed with intent, not by accident.

A VALUE ENGINE

Projects are how strategy actually happens; without them an organisation's strategic plan stays a slide deck.

A RISK REDUCER

Structured management makes uncertainty visible early, when responding to it is still cheap.

The role of the project manager — six jobs you are actually paid for

1 Integrator

Holds the whole picture together, making sure scope, schedule, cost, quality and risk decisions are consistent with one another.

2 Communicator

Spends roughly 90% of the working day communicating — translating between technical teams, sponsors and customers.

3 Decision maker

Makes calls within delegated authority and escalates cleanly and quickly when a threshold is breached.

4 Servant leader

Removes impediments, protects the team from noise, and creates the conditions in which skilled people can do good work.

5 Risk owner

Continuously scans internally and externally for threats and opportunities, and drives agreed responses to closure.

6 Business steward

Keeps the project honest about business value, stopping or reshaping work that no longer justifies its investment.

TRIPLE CONSTRAINT

The triple constraint — the physics of every project

Scope, time and cost are joined; move one and at least one other must move, or quality silently absorbs the difference.

SCOPE

The work required and the features delivered. Uncontrolled growth here is scope creep and is the single most common project failure.

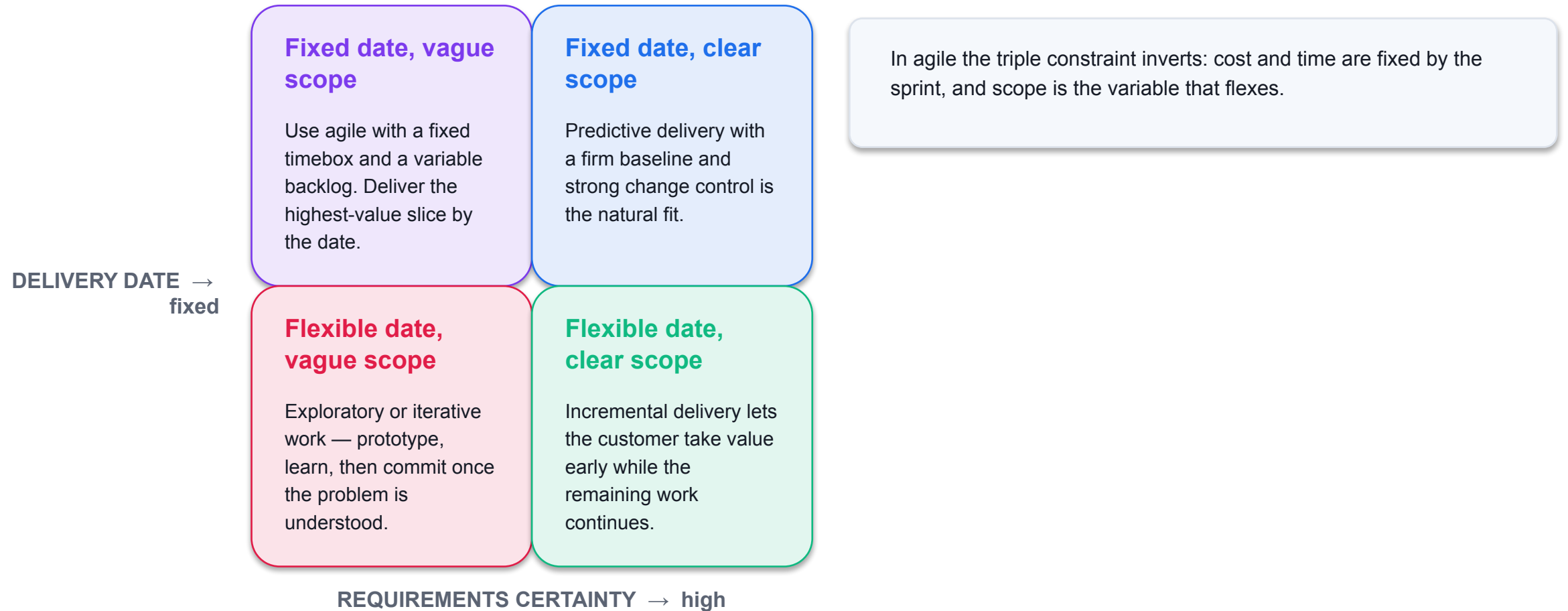
TIME

The schedule and its milestones. Compressing time without adding cost or cutting scope only borrows quality from the future.

COST

The approved budget including reserves. Cutting cost mid-flight usually means fewer people, which extends time.

Which constraint is fixed? Choosing your development approach from it



The PMI Talent Triangle — three competency areas the PMP tests

1 Ways of Working

Technical mastery of the many methods available — predictive, agile, hybrid — and the judgement to tailor the method to the environment.

2 Power Skills

Collaborative leadership, effective communication, empathy, conflict resolution and the influence needed without formal authority.

3 Business Acumen

Understanding priority, cost versus benefit, industry context and how to make decisions that hold up commercially, not just technically.

The 12 PMI project management principles (1 of 2)

1 Be a diligent, respectful, caring steward

Act with integrity and in the organisation's and society's interest, honouring both internal and external obligations.

2 Create a collaborative team environment

Teams that share agreements, structures and processes deliver more than individuals working in parallel silos.

3 Effectively engage with stakeholders

Engage proactively and to the degree needed to contribute to project success and customer satisfaction.

4 Focus on value

Continually evaluate and adjust project alignment to business objectives and the intended benefits and value.

5 Recognize and respond to system interactions

See the project as a system of interacting parts; a change in one area has consequences elsewhere in the organisation.

6 Demonstrate leadership behaviours

Leadership is a behaviour anyone on the team can show, not a position on an organisation chart.

The 12 PMI project management principles (2 of 2)

1 Tailor based on context

Design the delivery approach around the project's context, objectives, stakeholders, governance and environment.

2 Build quality into processes and deliverables

Quality is designed in and inspected continuously, not bolted on by testing at the end of the schedule.

3 Navigate complexity

Continually evaluate and address complexity arising from human behaviour, system interactions, ambiguity and uncertainty.

4 Optimize risk responses

Consistently evaluate exposure to both threats and opportunities, seeking responses proportional to the impact.

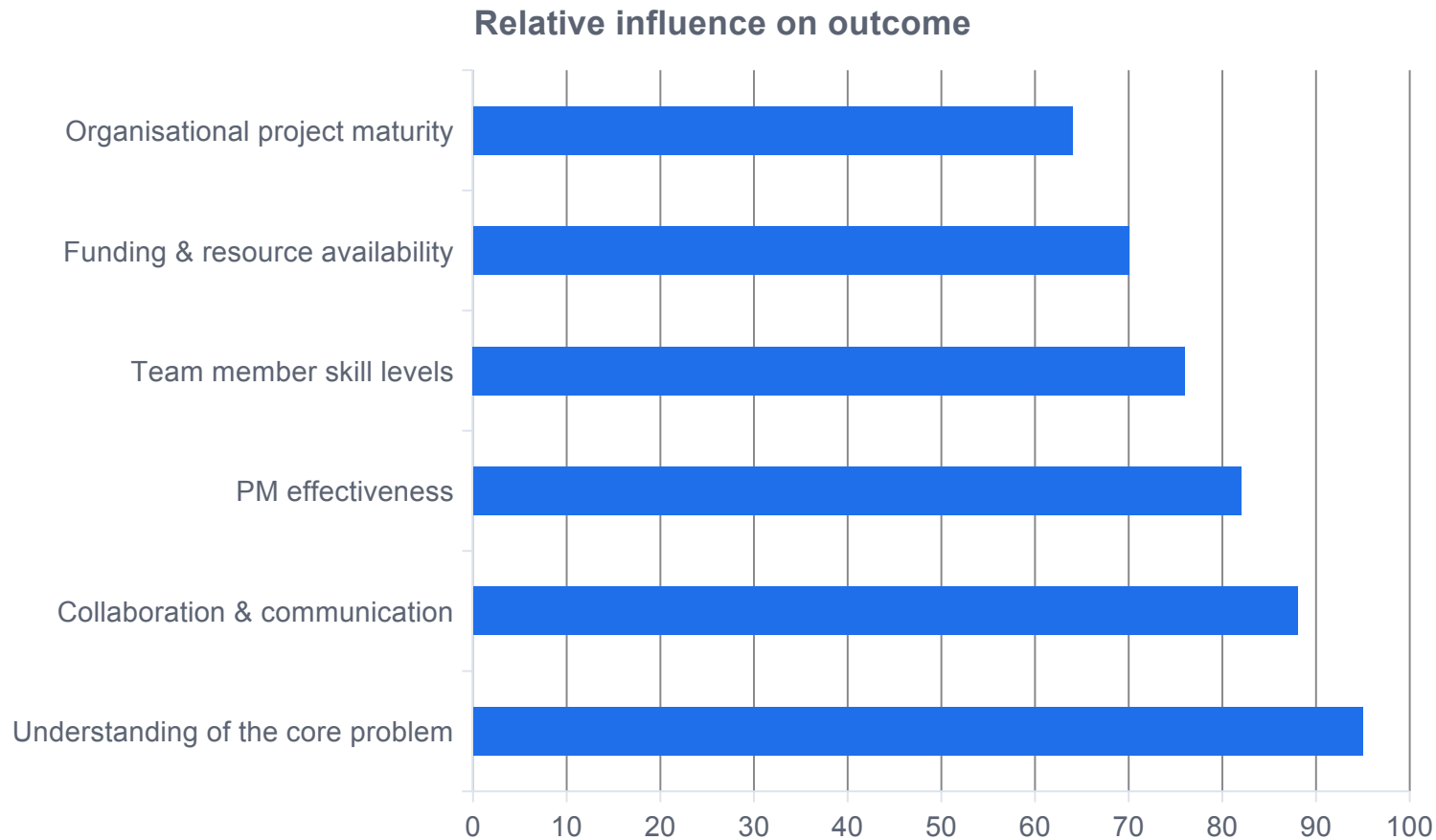
5 Embrace adaptability and resilience

Build capacity to absorb impacts and recover quickly, so setbacks do not become project failures.

6 Enable change to achieve the future state

Prepare those affected for adoption; a delivered output that nobody uses has produced zero value.

What project success actually depends on



WHAT IT TELLS YOU

- Most failures trace back to solving the wrong problem well, not to weak execution of the right one.
- Communication quality outranks tooling — no tool rescues a team that does not talk to its stakeholders.
- Organisational maturity sets the ceiling; a strong PM in an immature organisation still fights the system.
- Funding matters, but adequate funding on the wrong scope simply buys a faster failure.

Two different life cycles — do not confuse them on the exam

PROJECT LIFE CYCLE

The phases the project passes through

- The series of phases a project moves through from start to close
- Typical phases: feasibility, design, build, test, deploy, close
- Phases usually end in a phase gate or stage-gate review
- At each gate the sponsor decides go, go-with-change, hold or kill
- Defined by the organisation's methodology and governance
- Answers the question: where are we in the journey?

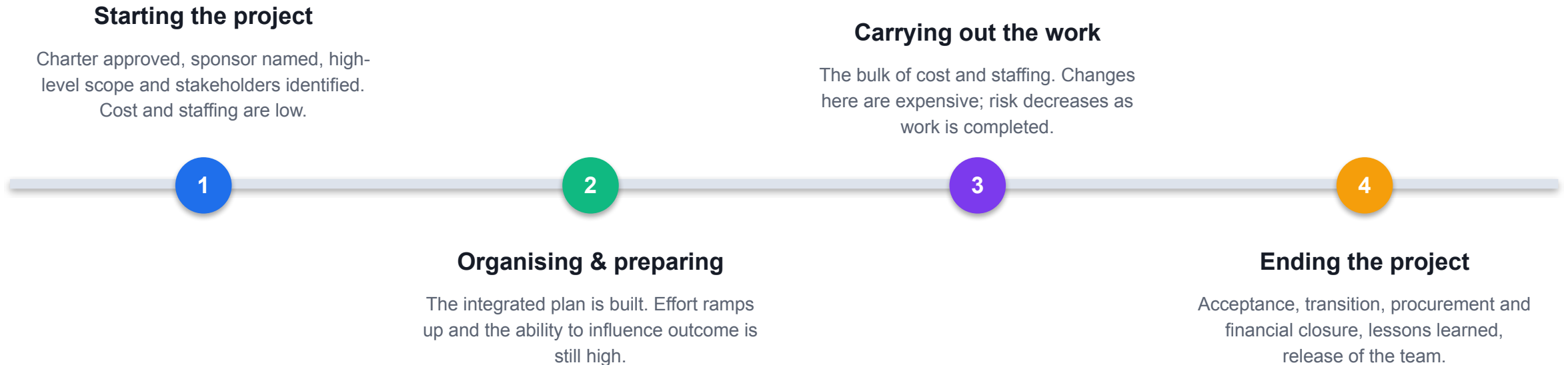
DEVELOPMENT LIFE CYCLE

How the deliverable is actually produced

- The approach used to create the product inside those phases
- Options: predictive, iterative, incremental, adaptive/agile, hybrid
- Chosen from requirements certainty, risk and delivery cadence
- Can differ per deliverable within one and the same project
- Defined by the team with the sponsor, and can be tailored
- Answers the question: how are we building the thing?

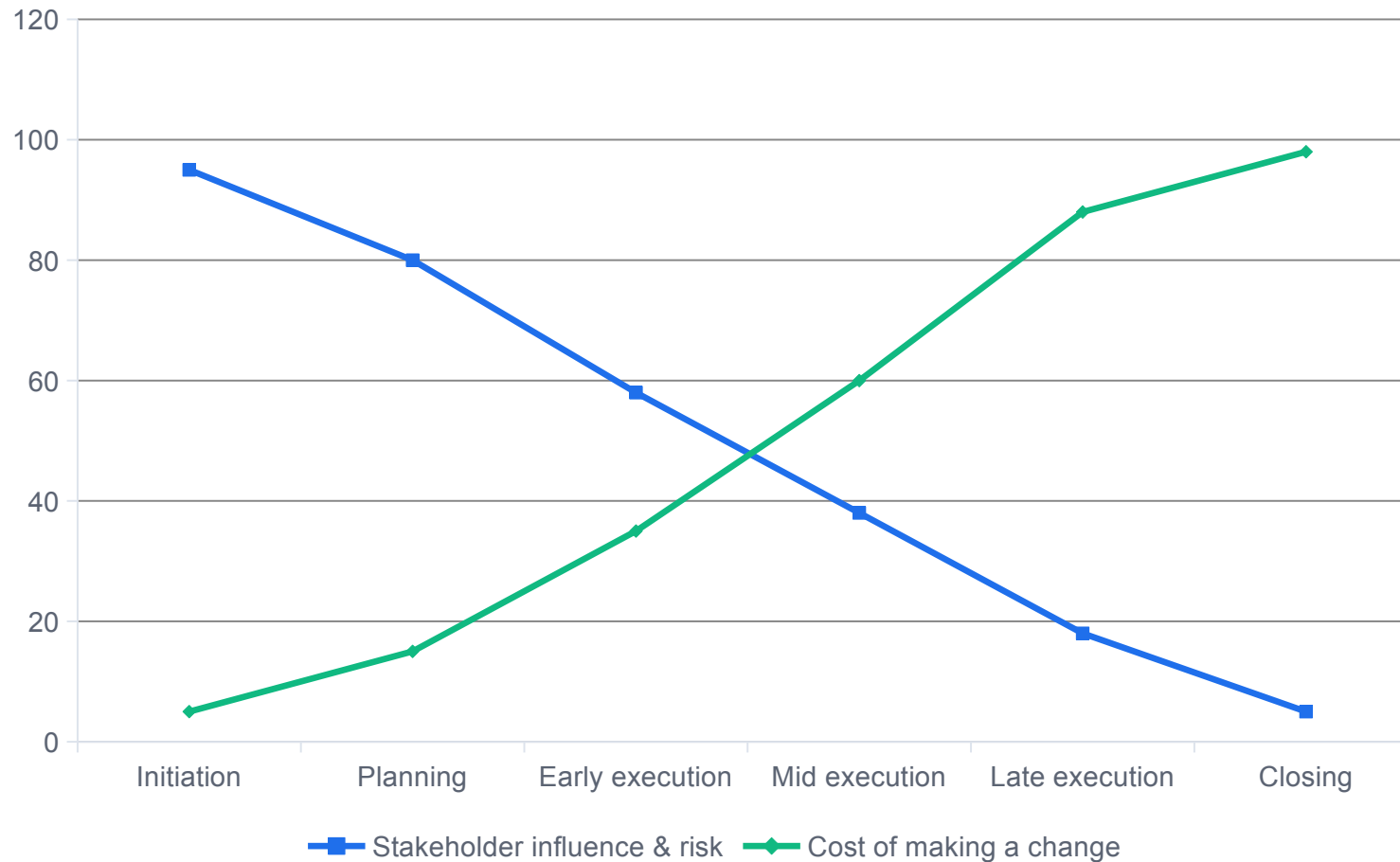
A single project can have a predictive project life cycle for governance while using an agile development life cycle inside the build phase — that combination is a hybrid.

The generic project life cycle and its effort profile



Risk and stakeholder influence are highest at the start; the cost of change is highest at the end. That single relationship explains most of PMI's guidance on planning and early engagement.

Why early decisions dominate: risk, influence and the cost of change



WHAT IT TELLS YOU

- The two curves cross early — after that point, changes cost more than the influence they buy.
- This is the mathematical argument for investing in requirements and stakeholder engagement up front.
- Agile flattens the cost-of-change curve by keeping increments small and reversible rather than by planning harder.
- Exam cue: 'the change was requested late in execution' signals a formal impact analysis, not immediate action.

Common life cycle definitions — the vocabulary PMI uses

Life cycle	Requirements	Delivery	Goal / when to use
Predictive	Fixed and baselined up front	One delivery at the end	Manage cost — best when the solution is well understood and change is expensive
Iterative	Dynamic; refined each cycle	One final delivery, repeated refinement	Correctness of the solution — prototype until the right thing is understood
Incremental	Dynamic; decomposed into slices	Frequent smaller usable deliveries	Speed to value — customer gets working capability before the whole is finished
Adaptive / agile	Dynamic, in a prioritised backlog	Frequent delivery via short timeboxes	Customer value through fast feedback — best under high uncertainty
Hybrid	Mixed by component or phase	Mixed cadence across the project	Fit the approach to each part — the most common shape in real organisations

Memory hook: iterative improves the same thing repeatedly; incremental adds new usable pieces. Agile deliberately does both at once.

The development approach continuum, from most to least predictive



The approach is a tailoring decision, not an identity. PMI expects you to justify it from requirements certainty, risk, regulatory constraint and how quickly value can be delivered.

Project cadence — the timing and frequency of delivery

1 Single delivery

One handover at the very end of the project.
Typical of construction, regulated pharmaceuticals and large infrastructure.

2 Multiple deliveries

Work is separated into parts released as they complete, not necessarily in a fixed sequence or on a fixed calendar.

3 Periodic delivery

Multiple deliveries released on a fixed drumbeat — monthly, bi-monthly or every sprint — which makes planning predictable.

4 Continuous delivery

Every accepted change flows to production as it is ready, supported by automated testing and deployment pipelines.

5 Cadence versus approach

Cadence is about when the customer receives value; the development approach is about how the work is produced.

6 Why it matters

Cadence drives the release plan, the funding profile, and how often the benefits measurement system must be read.

AGILE FUNDAMENTALS

The definition you must be able to state cold

Agile is both incremental — small, regular releases of value — and iterative — reflecting and improving regularly.

INCREMENTAL

Each sprint hands over a potentially releasable increment, so the customer accrues real value long before the end.

ITERATIVE

Each sprint also revisits and refines what exists, and the retrospective refines how the team works.

THE COMBINATION

Doing only one of the two is not agile: increments without reflection stagnate; reflection without increments stalls.

Predictive (waterfall) versus adaptive (agile) delivery

PREDICTIVE / WATERFALL

Plan the work, then work the plan

- Requirements are gathered once, then baselined and frozen
- Design, build and test are performed once, in sequence
- Success is conformance to the approved scope, schedule and budget
- Change is a formal event requiring a change request and CCB approval
- Detailed documentation is produced and is itself a deliverable
- Customer sees the working product late, usually at UAT

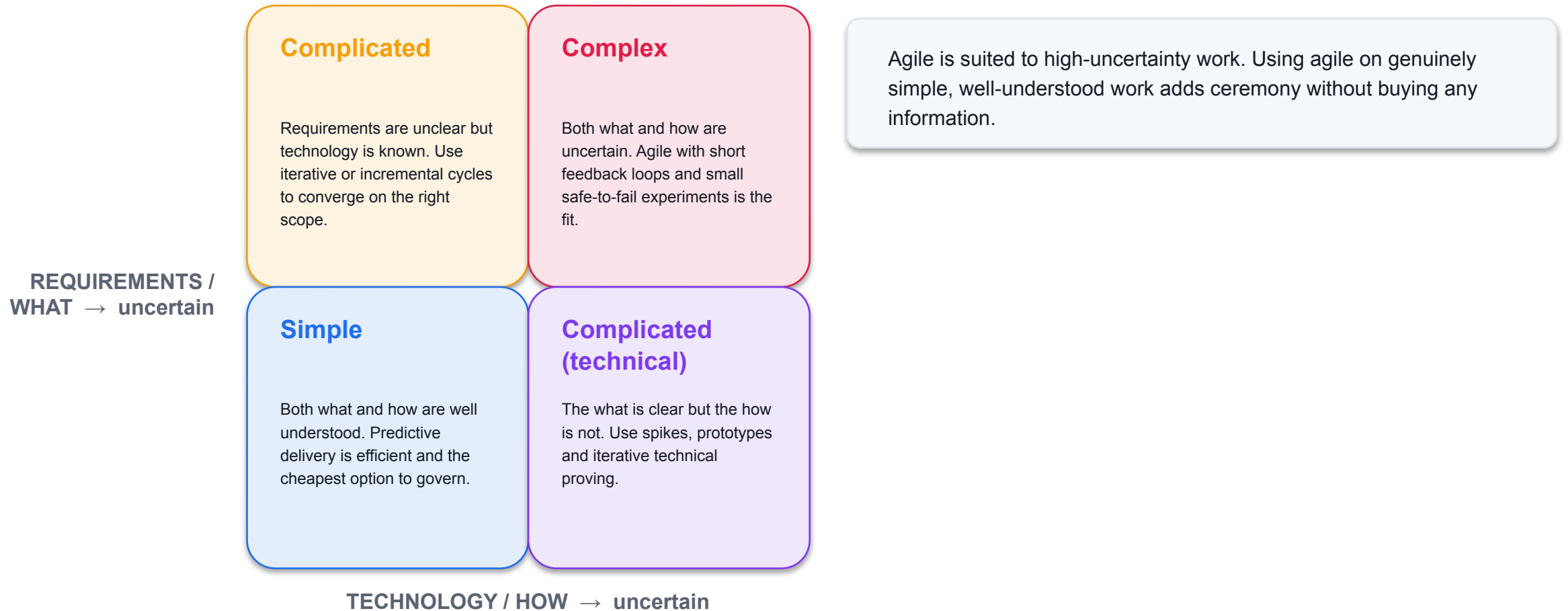
ADAPTIVE / AGILE

Inspect and adapt, every timebox

- Requirements live in a backlog and are refined continuously
- Design, build and test are performed once per increment
- Success is working product and realised customer value
- Change is expected and absorbed by reprioritising the backlog
- Working software is favoured over comprehensive documentation
- Customer sees working product at the end of every sprint

Predictive performs the full life cycle once; adaptive performs it once per increment. That single sentence answers a surprising number of exam questions.

The Stacey model — matching approach to complexity and uncertainty



The Agile Manifesto (2001) — four values, stated as preferences

WE VALUE MORE

The left-hand side of each pairing

- Individuals and interactions
- Working software (working product)
- Customer collaboration
- Responding to change
- Seventeen practitioners agreed these in Snowbird, Utah in 2001
- The values sit above the 12 principles and all frameworks

OVER

The right-hand side still has value

- ... over processes and tools
- ... over comprehensive documentation
- ... over contract negotiation
- ... over following a plan
- 'While there is value in the items on the right, we value the items on the left more'
- Agile never says documentation or plans are worthless

Exam trap: an option saying agile teams do not plan or do not document is always wrong. Agile teams plan more often, in smaller pieces, and document what is genuinely useful.

The 12 Agile Principles (1 of 2)

1 Satisfy the customer early

Highest priority is satisfying the customer through early and continuous delivery of valuable software.

2 Welcome changing requirements

Even late in development; agile processes harness change for the customer's competitive advantage.

3 Deliver frequently

Deliver working product from a couple of weeks to a couple of months, with a preference for the shorter timescale.

4 Business and developers together

Business people and developers must work together daily throughout the project, not meet at handover points.

5 Build around motivated individuals

Give them the environment and support they need, and trust them to get the job done.

6 Face-to-face conversation

The most efficient and effective method of conveying information within a development team.

The 12 Agile Principles (2 of 2)

1 Working product is the measure

Working software is the primary measure of progress — not percentage complete, not documents produced.

2 Sustainable pace

Sponsors, developers and users should be able to maintain a constant pace indefinitely; heroics are a warning sign.

3 Technical excellence

Continuous attention to technical excellence and good design enhances agility and keeps future change cheap.

4 Simplicity

The art of maximising the amount of work not done — building only what is needed is essential.

5 Self-organising teams

The best architectures, requirements and designs emerge from self-organising teams, not from imposed plans.

6 Reflect and adjust

At regular intervals the team reflects on how to become more effective, then tunes and adjusts its behaviour.

Scrum — the most widely used agile framework

1 Product Owner

Owns and prioritises the product backlog, represents the business, and is accountable for maximising the value the team delivers.

2 Scrum Master

Facilitates the events, coaches the team on Scrum, removes impediments, and shields the team from outside interference.

3 Development Team

Three to nine cross-functional people who self-organise to turn backlog items into a done increment each sprint.

4 Product Backlog

The single ordered list of everything that might be needed, continuously refined and re-ordered by the Product Owner.

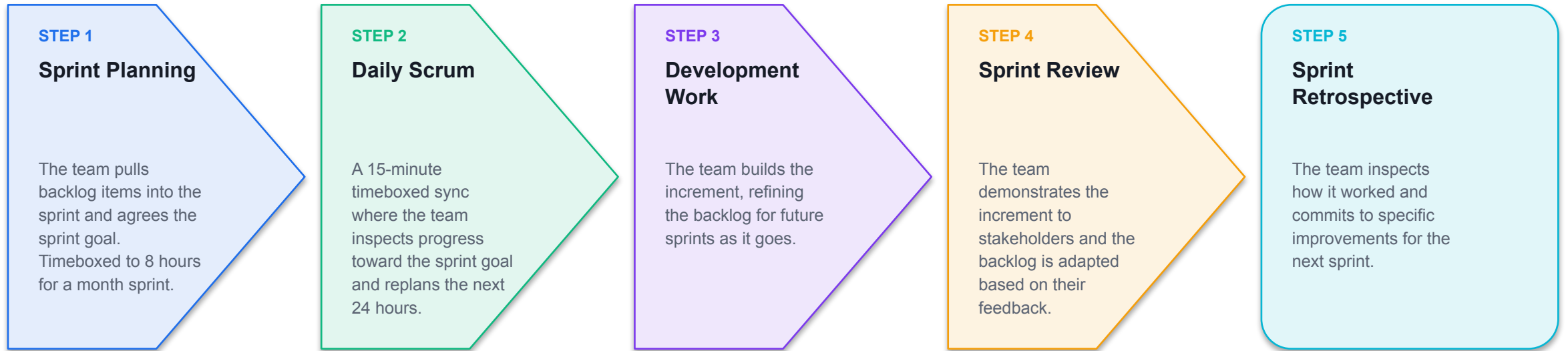
5 Sprint Backlog

The subset the team commits to this sprint, plus the plan for delivering it; owned entirely by the development team.

6 Increment

The sum of all completed backlog items this sprint plus all prior sprints, in a potentially releasable state.

The sprint — Scrum events in the order they occur



A sprint is a timebox of one month or less. Its length is fixed, its scope is not — the sprint end date never moves to accommodate unfinished work.

Agile ceremonies beyond the core Scrum events

1 Product strategy meeting

The Product Owner shares the product vision and roadmap so the team understands why the backlog is ordered the way it is.

2 Backlog refinement (grooming)

An ongoing session where items are clarified, split, estimated and re-ordered so the top of the backlog is always ready.

3 Daily standup

Five to fifteen minutes, timeboxed, and not necessarily daily on every team — the purpose is coordination, not status reporting.

4 Release planning

Maps multiple sprints to a release date so stakeholders outside the team can plan marketing, training and operations.

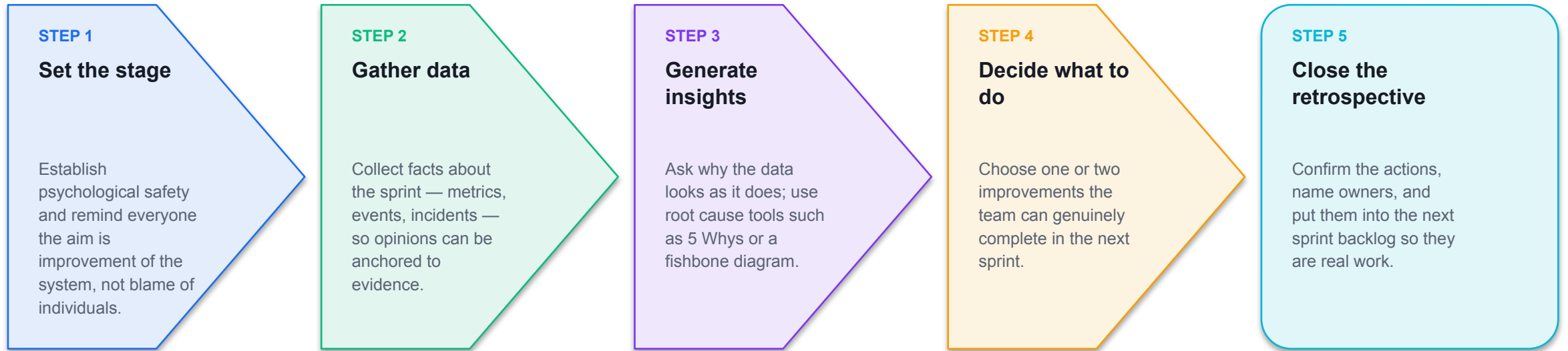
5 Project retrospective

Held at the end of the project to review both the work and the process — the agile equivalent of lessons learned.

6 Demo or showcase

An informal review with wider stakeholders, used to gather feedback before committing to the next block of work.

Running a retrospective that actually changes behaviour



A retrospective with no action items in the next sprint backlog is theatre. Track improvement actions with the same rigour as feature work.

The project team — makeup, sourcing and skills

1 Team composition

Refers to the team's makeup and how members are brought together; it varies with organisational culture, location and project scope.

2 Full-time and part-time

Members may be dedicated or shared. Shared members multitask, which reduces throughput and increases handover risk.

3 Generalists and specialists

Specialists bring depth for hard problems; generalists give the team the flexibility to redistribute work when plans change.

4 Avoid single points of failure

If only one person holds a required skill, their absence stops the project — cross-train deliberately.

5 Cross-functional teams

A team that contains every skill needed to take an item from idea to done removes waiting on external groups.

6 T-shaped people

Generalising specialists with one deep specialty plus broad general skills — the most valuable shape for agile teams.

Agile team characteristics and the agile team space

1 Small

Typically three to twelve members. Communication paths grow quadratically, so large teams spend their capacity coordinating.

2 Co-located or well-connected

Physical or virtual co-location creates an osmotic environment where useful information is absorbed without a meeting.

3 Dedicated

100% allocated to one team to avoid multitasking, which is the single largest source of hidden delay in knowledge work.

4 Information radiators

Burndown charts, Kanban boards and impediment lists on display, so status is visible at a glance without a report.

5 Pairing

Two people work on one item together, checking each other's thinking and spreading knowledge as a side effect.

6 Swarming

Multiple team members converge on one blocked or high-value item to finish it fast rather than starting new work.

Agile team roles and their traditional equivalents

Agile role	What they do	Closest predictive equivalent
Product Owner	Represents the customer and business, owns and prioritises the product backlog, accepts or rejects each increment	Business analyst plus sponsor's delegate
Team Facilitator	Also called Scrum Master, agile lead, team coach or servant leader; runs ceremonies, removes blockers, grows the team	Project manager (facilitation half of the role)
Cross-functional member	Generalising specialists — T-shaped people who self-organise to deliver the increment end to end	Team member from a functional department
Agile Coach	Works across several teams on practice maturity and organisational impediments beyond a single team	PMO methodology lead
Stakeholder / customer	Attends reviews, supplies the feedback that re-orders the backlog, and confirms value is real	Steering committee member or end user

Note what is missing: agile has no role that assigns tasks to individuals. Work is pulled by the team, not pushed by a manager.

Tailoring — the principle that makes everything else practical

We can, and should, tailor anything on our project so it meets the outcomes and delivers business value.

PROCESS

The development approach, the number of phases, the depth of planning and the frequency of governance reviews.

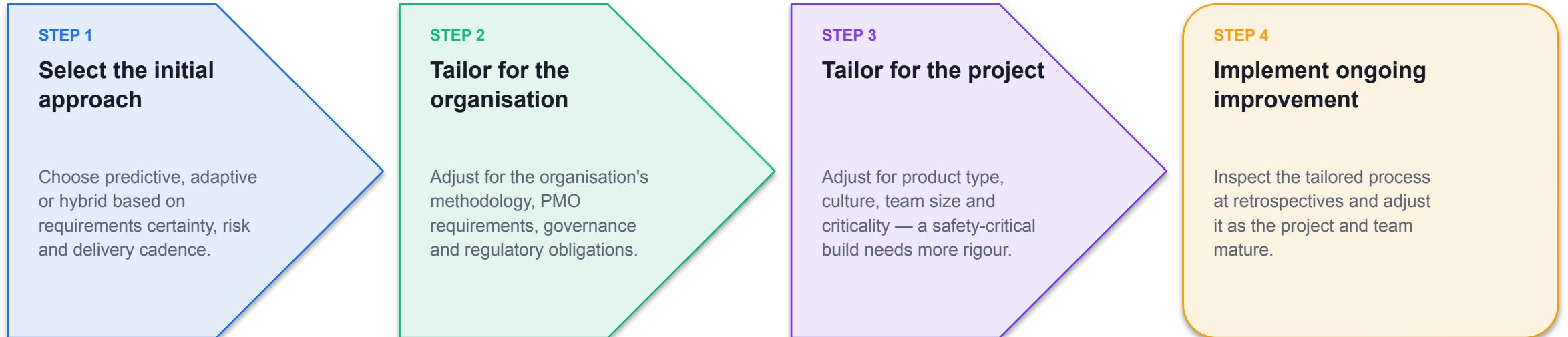
ARTIFACTS

Which plans, registers and reports genuinely help — and which exist only because a template said so.

METHODS & TOOLS

The estimating techniques, meeting formats, tooling and metrics that fit this team and this context.

The four-step tailoring approach



Tailoring is not permission to skip work you find inconvenient. Every removal should be a reasoned decision you could defend to the sponsor or an auditor.

Types of managers — who owns what in the organisation

1 Project Manager

Assigned by the performing organisation to lead the team responsible for achieving the project objectives and delivering business value.

2 Functional Manager

Manages a functional or business unit and provides the people, skills and physical resources the project consumes.

3 Operations Manager

Responsible for ensuring ongoing business operations run efficiently, and receives the project's output at transition.

4 Program Manager

Coordinates related projects to obtain benefits not available from managing them individually, and resolves cross-project conflicts.

5 Portfolio Manager

Selects and balances the mix of programs and projects so the investment aligns with organisational strategy.

6 Product Manager

Owens a product across its whole life — well before a project starts and long after it closes — driving the roadmap and value.

Product management versus project management

PRODUCT MANAGEMENT

Continuous, value-and-market driven

- Spans the entire product life cycle, potentially for years
- Owns the vision, roadmap and the ordered backlog of value
- Measures success by adoption, revenue, retention and outcomes
- Funding is often a standing capacity, not a one-off approval
- Ends only when the product is retired from the market
- Asks: are we building the right thing?

PROJECT MANAGEMENT

Temporary, delivery-and-constraint driven

- Spans one temporary endeavour with a defined start and end
- Owns the plan, the baselines and the delivery of agreed scope
- Measures success by scope, schedule, cost, quality and benefits
- Funding is approved against a specific business case
- Ends at acceptance, transition and administrative closure
- Asks: are we building the thing right?

Organisations increasingly fund persistent product teams rather than discrete projects — the PMP now expects you to operate comfortably in both models.

Organisational structure and project manager authority

Structure	PM authority	PM role	Resource control
Functional	Little or none	Part-time coordinator	Functional manager controls the budget and the people
Weak matrix	Low	Part-time expeditor or coordinator	Functional manager still controls the budget
Balanced matrix	Low to moderate	Full-time project manager	Budget is shared between PM and functional manager
Strong matrix	Moderate to high	Full-time project manager	Project manager controls the budget
Projectised	High to almost total	Full-time project manager	Project manager controls budget and staff entirely
Hybrid / composite	Varies by project	Varies	Mixed — the most common real-world arrangement

If an exam scenario says the PM has no authority over resources, you are in a functional or weak matrix — the correct action is almost always to negotiate with the functional manager.

Project Management Office types and their degree of control

PMO type	Authority and control	What it provides
Supportive	Low degree of control — consultative	Serves as a project repository; supplies templates, best practices, training and information from other projects
Controlling	Moderate degree of control — compliance	Provides support and ensures compliance through adoption of methods, frameworks, specific templates, forms and tools
Directive	High degree of control — direct management	Takes control by directly managing the projects; project managers are assigned by and report into the PMO
Agile / value delivery office	Enabling rather than controlling	Coaches teams, removes organisational impediments and reports on flow and benefits rather than on plan conformance

Memory hook: Supportive suggests, Controlling checks, Directive directs. The exam tests which one owns the project manager.

Reading the external business environment: PESTLE, TECOP and VUCA

1 PESTLE

Political, Economic, Socio-cultural, Technological, Legal and Environmental forces — the standard scan of the macro environment.

2 TECOP

Technical, Environmental, Commercial, Operational and Political — a risk-oriented scan used heavily in engineering and construction.

3 VUCA

Volatility, Uncertainty, Complexity and Ambiguity — describes the character of the environment rather than listing its factors.

4 Why the PM does this

External change can invalidate the business case mid-project; scanning turns an ambush into a managed risk with a response.

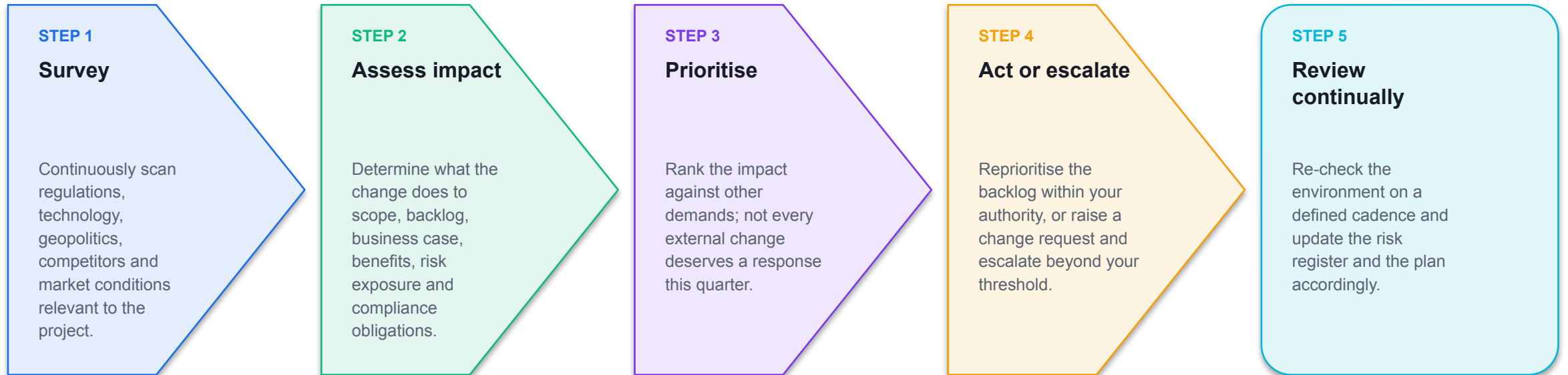
5 How often

Not once at initiation. ECO T8 requires the PM to continually review the external environment for impacts to scope and backlog.

6 What you do with it

Feed findings into the risk register, reprioritise the backlog, and escalate anything that threatens the business case.

Responding to a change in the external business environment



Classic exam scenario: a new regulation is announced mid-project. The best first action is to assess the impact on the project — not to stop work, and not to immediately change scope.

Enterprise Environmental Factors versus Organizational Process Assets

EEF — conditions you must live with

Usually inputs; largely outside your control

- Internal: resource capabilities, organisational culture, IT systems, distribution of facilities, employee capability
- External: marketplace conditions, laws, regulations and standards
- External: operating conditions, social and cultural influences
- Government or industry standards you cannot negotiate away
- Risk tolerances and the organisation's appetite for change
- You adapt the project to EEFs; you rarely change them

OPA — knowledge you can reuse

Inputs and outputs; you can and should update them

- Processes, policies and procedures: organisational charts, procurement rules, hiring and onboarding procedures
- Templates, checklists, standardised WBS structures and forms
- Organisational knowledge bases: engineering wikis, libraries, archives
- Lessons learned repositories and historical estimating databases
- Configuration management and financial control procedures
- ECO T6 requires you to update OPAs — they are an output too

Fastest test: can the project change it? If yes it is usually an OPA. If the project must simply conform to it, it is an EEF.

Exam drill: classify these as EEF or OPA

IF THE QUESTION SAYS

Economic demand for a new shopping area; local neighbourhood demand for a better town centre



THE BEST ANSWER IS

EEF — external marketplace and social conditions the project must respond to but cannot change.

IF THE QUESTION SAYS

Historical society conservation building regulations affecting the site



THE BEST ANSWER IS

EEF — a legal and regulatory constraint from outside the organisation; it becomes a compliance requirement.

IF THE QUESTION SAYS

An archive of past large infrastructure projects held by your company



THE BEST ANSWER IS

OPA — a knowledge base and historical information repository you can mine for estimates and lessons.

IF THE QUESTION SAYS

The approved vendor and contractor list, and the tenant selection process



THE BEST ANSWER IS

OPA — internal processes, policies and procedures the organisation maintains and the project must apply.

Assessing organisational culture — what to actually look at

1 Leadership, hierarchy and authority

How decisions get made and how far they must travel. This directly sets your realistic escalation lead time.

2 Shared vision, beliefs and expectations

What people believe the organisation is for. A project that contradicts this will meet quiet, persistent resistance.

3 Diversity, equity and inclusion practices

Whether varied perspectives are genuinely sought; this shapes team formation and the quality of risk identification.

4 Risk tolerance and appetite

How much uncertainty the organisation will accept determines your reserve levels and your response strategies.

5 Regulations, policies and code of conduct

The formal rules of the road, including the ethical standards the team is expected to uphold without supervision.

6 Motivation and reward systems

What behaviour actually gets rewarded. If individuals are rewarded, a team incentive scheme will not stick.

Why change management belongs to the project manager

Organisational change requires individual change — a delivered output that nobody adopts has produced exactly zero value.

YOU ARE A CHANGEMAKER

Whether or not your organisation has a PMO, the project manager is the person driving change into the business.

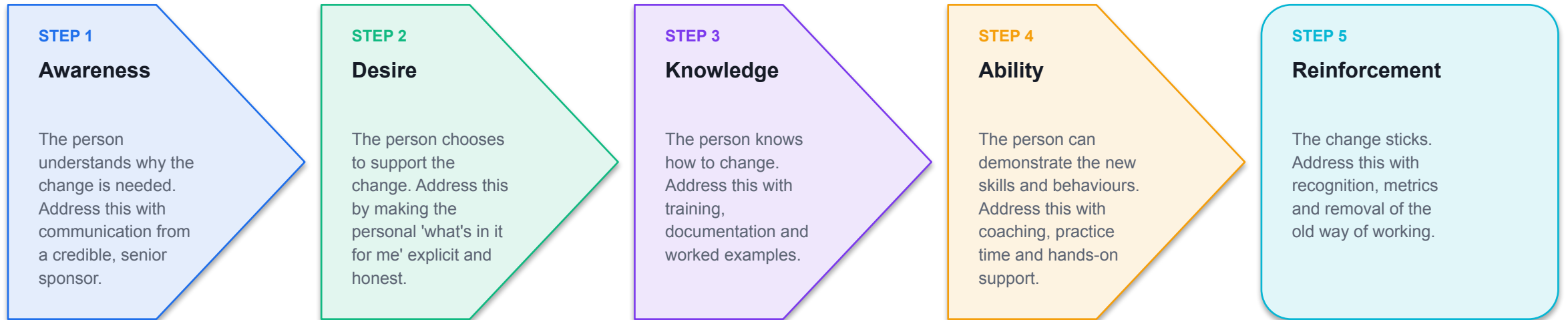
TAILOR THE STRATEGY

Change approach must fit the circumstances, the people affected and the timing — there is no universal script.

USE A ROBUST APPROACH

Structured models such as ADKAR give you a checklist you can measure adoption against, not just good intentions.

The ADKAR model — five milestones an individual must reach



Diagnostic power: if adoption is failing, find the first milestone the person has not reached. More training will never fix a Desire problem.

Plan for change — the practical activities to build into your plan

1 Define readiness activities

Specify the knowledge transfer, training and readiness work required to implement the change the project brings.

2 Run an attitudinal survey

Find out how affected people actually feel before go-live, so resistance is measured rather than assumed away.

3 Create an information campaign

Familiarise people with the coming change over time; a single announcement email is not a communication plan.

4 Be open about effects

Transparent about the potential negative effects too — hidden bad news destroys the trust the change depends on.

5 Create a rollout plan

Sequence adoption by group, region or business unit so support capacity matches the number of people changing at once.

6 Measure adoption

Define adoption metrics up front and track them after go-live; benefits realisation depends on them, not on delivery.

When the organisation changes around your project

1 Reorganisations

A new reporting structure can remove your sponsor or reassign your team mid-flight; re-confirm authority immediately.

2 Strategy or priority shifts

A change in strategic direction can reprioritise value, including removing deliverables you have already built.

3 New deliverables demanded

Internal business changes frequently create a need for deliverables that were never in the approved baseline.

4 Process changes

New internal processes may change how you must procure, hire, report or release — check them before you are blocked.

5 Stay informed

The PM, sponsor or product owner must be familiar with business plans, reorganisations and other internal activity.

6 Recommend options

Assess the impact, then recommend options for changes to the project rather than presenting a single fait accompli.

Project governance

The framework of structures, rules, procedures, reporting and ethics within which project decisions are legitimately made.

SINGLE ACCOUNTABILITY

Governance offers one clear point of accountability for the project, so decisions do not orbit unowned.

SPANS THE LIFE CYCLE

It encompasses the whole project life cycle, from authorisation through phase gates to closure.

ALREADY EXISTS

Governance is typically already in place — established by a PMO or aligned to organisational policy; you inherit it.

The governance board and what it is responsible for

1 Who sits on it

Typically the project sponsor, senior managers from affected business units, and PMO resources; sometimes customer representatives.

2 Provides oversight

Independent oversight of whether the project remains viable, compliant and aligned to the strategy that funded it.

3 Reviews key deliverables

Formally reviews and accepts major deliverables at defined checkpoints rather than only at the very end.

4 Guides project decisions

Provides direction on decisions that exceed the project manager's delegated authority or cross business unit boundaries.

5 Approves changes above threshold

Acts as, or feeds into, the change control board for changes larger than the PM can approve alone.

6 Governance varies

Its type and intensity depend on the project's strategic importance, constraints and external oversight requirements.

Governance defines escalation — the thresholds decision path



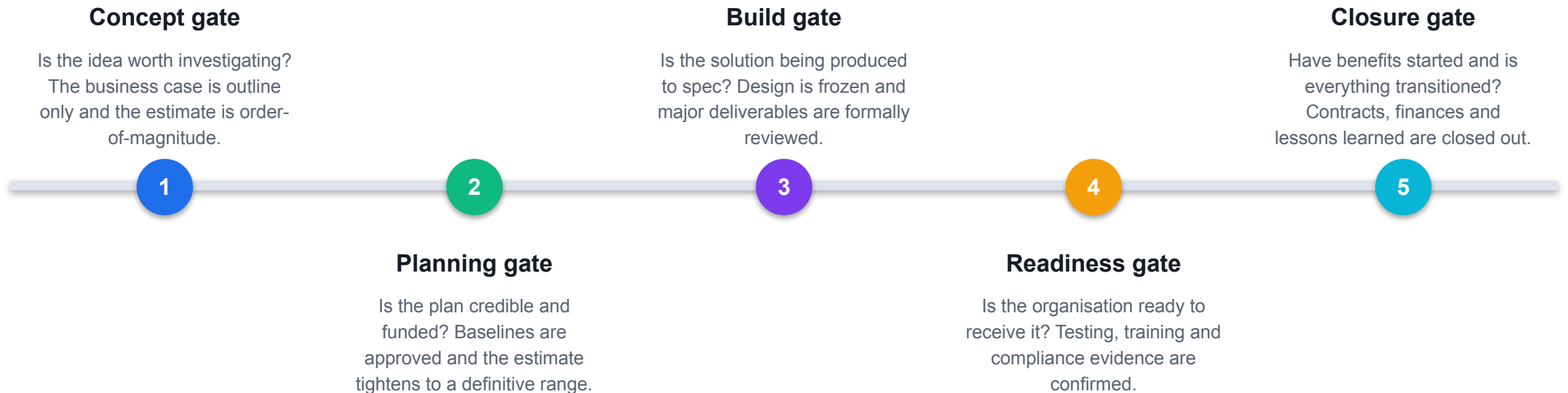
Escalating everything makes you look incapable; escalating nothing makes you look reckless. The written threshold is what makes the difference defensible.

Defining project success metrics that governance can act on

Schedule performance	$SPI = EV \div PV$	Below 1.00 means the project is earning value more slowly than planned; a common governance threshold is SPI below 0.90.
Cost performance	$CPI = EV \div AC$	Below 1.00 means each dollar spent is buying less value than planned; often escalated when CPI falls below 0.90.
Benefit realisation	$BRR = \text{Actual} \div \text{Planned benefit}$	Measured after transition to operations; it is the metric that proves the business case was real, not merely persuasive.
Return on investment	$ROI = (\text{Benefit} - \text{Cost}) \div \text{Cost}$	The financial justification governance used to fund the project; re-check it whenever scope or cost changes materially.
Compliance level	% requirements evidenced	The share of identified compliance requirements with current, auditable evidence — the measure ECO T2 asks for.

ECO Business Environment T1 explicitly requires you to define success metrics and escalation thresholds. Vague metrics make governance ceremonial.

Governance checkpoints: phase gates in a predictive project



At every gate the decision is one of four: continue, continue with modification, hold and re-plan, or terminate. 'Continue because we have already spent the money' is the sunk cost fallacy.

Project phase relationships — sequential versus overlapping

SEQUENTIAL PHASES

One phase completes before the next begins

- Each phase ends at a gate that must be passed to continue
- Lowest risk of rework because each stage is confirmed before build
- Longest overall duration — no phase runs in parallel
- Preferred where safety, regulation or irreversibility dominates
- Clearest governance because decisions are discrete events
- Change late in the sequence is the most expensive kind

OVERLAPPING PHASES

The next phase starts before the prior one ends

- Known as fast tracking; used to compress the total schedule
- Increases risk of rework because assumptions may prove wrong
- Requires far more communication between the overlapping teams
- Governance gates become partial or conditional approvals
- Common in construction and product development under time pressure
- The PM must explicitly accept and reserve for the rework risk

In adaptive delivery, the equivalent checkpoint is the sprint review — governance moves from a rare heavyweight gate to a frequent lightweight inspection of a working increment.

Applying governance in predictive versus adaptive projects

PREDICTIVE GOVERNANCE

Few, heavy, document-based checkpoints

- Phase gate reviews at defined milestones in the schedule
- Evidence is documentation: plans, designs, test reports, sign-offs
- Change control board approves deviations from the baseline
- Status reported against percentage complete and earned value
- Escalation follows a documented threshold and hierarchy
- Benefits are typically confirmed after the whole project closes

ADAPTIVE GOVERNANCE

Many, light, product-based checkpoints

- Sprint review at the end of every timebox, with stakeholders present
- Evidence is a working increment that can be inspected directly
- The Product Owner reprioritises the backlog instead of raising CRs
- Status reported through burndown, velocity and cumulative flow
- Impediments escalated immediately by the team facilitator
- Benefits can be measured from the first released increment

The governance obligation is identical in both; only the frequency, the artefacts and the evidence change. Regulated industries still require agile teams to produce audit trails.

Value-based delivery — the reason the project exists

Value is the worth, importance or usefulness of the output to the organisation and its stakeholders — not the volume of work completed.

IDENTIFY VALUE

Agree the value components with key stakeholders early, in measurable terms, so 'success' is not decided retrospectively.

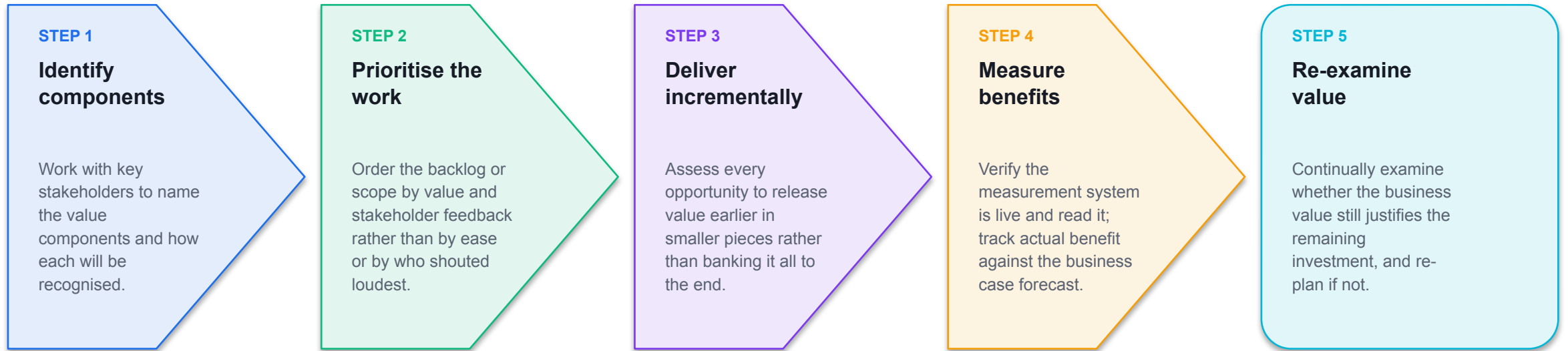
PRIORITISE BY VALUE

Order the work by value and stakeholder feedback so the most valuable capability is delivered first.

VERIFY MEASUREMENT

Confirm a measurement system is in place to track benefits — you cannot claim value you never measured.

Delivering and evaluating project value



A project that is on time and on budget but delivers no measured benefit has failed. PMI's value-delivery view treats the business case as a live document, not an entry ticket.

Project compliance — who is responsible for what

1 External standards

Government regulations, industry standards, privacy law and any licensing the product or the organisation requires.

2 Internal standards

Corporate policies, the quality management system, the code of conduct and the organisation's own methodology.

3 The PMO's role

Monitors compliance at organisational level, maintains the standards and audits projects against them.

4 The project team's role

Responsible for compliance in project activity — quality of processes and deliverables, and the work done by vendors.

5 Procurement compliance

Vendor work must meet the same obligations; contracts must pass the requirement down and give you audit rights.

6 The PM's stewardship

Compliance stewardship is your personal responsibility — it is an ethical obligation under the PMI Code of Conduct.

Classifying compliance categories

Category	Typical requirements	Consequence of failure
Environmental	Emissions limits, waste handling, environmental impact assessment, site remediation	Fines, forced remediation, loss of operating licence, reputational damage
Workplace health and safety	Risk assessments, permits to work, protective equipment, incident reporting	Injury, prosecution of individuals, stop-work orders, insurance loss
Ethical and anti-corruption	Anti-bribery policy, conflict of interest declarations, gift registers, procurement fairness	Criminal liability, contract voiding, blacklisting from public tenders
Data protection and security	Consent, data residency, encryption, breach notification, access control	Regulatory fines, breach notification cost, customer attrition
Social responsibility	Labour standards in the supply chain, accessibility, inclusion, community obligations	Reputational harm, activist and investor pressure, contract loss
Quality and process	Standards conformance, documented process rigour, traceability of test evidence	Rework, product recall, refused acceptance, warranty exposure

Categories vary with industry, solution scope and the organisation's unique legal and regulatory exposure — classify them for your project, not from a template.

Confirming compliance requirements for your project

1 Legal and regulatory constraints

Requirements for specific practices, mandatory standards, privacy laws and rules for handling sensitive information.

2 Where compliance is owned

Identify who in the organisation handles compliance — legal, quality, security, or an external regulator's inspector.

3 The quality policy

Understand the organisation's quality policy and how much process rigour and quality control it demands of this project.

4 Tailor the rigour

How much control is genuinely relevant here? A safety-critical system and an internal reporting tool do not need the same rigour.

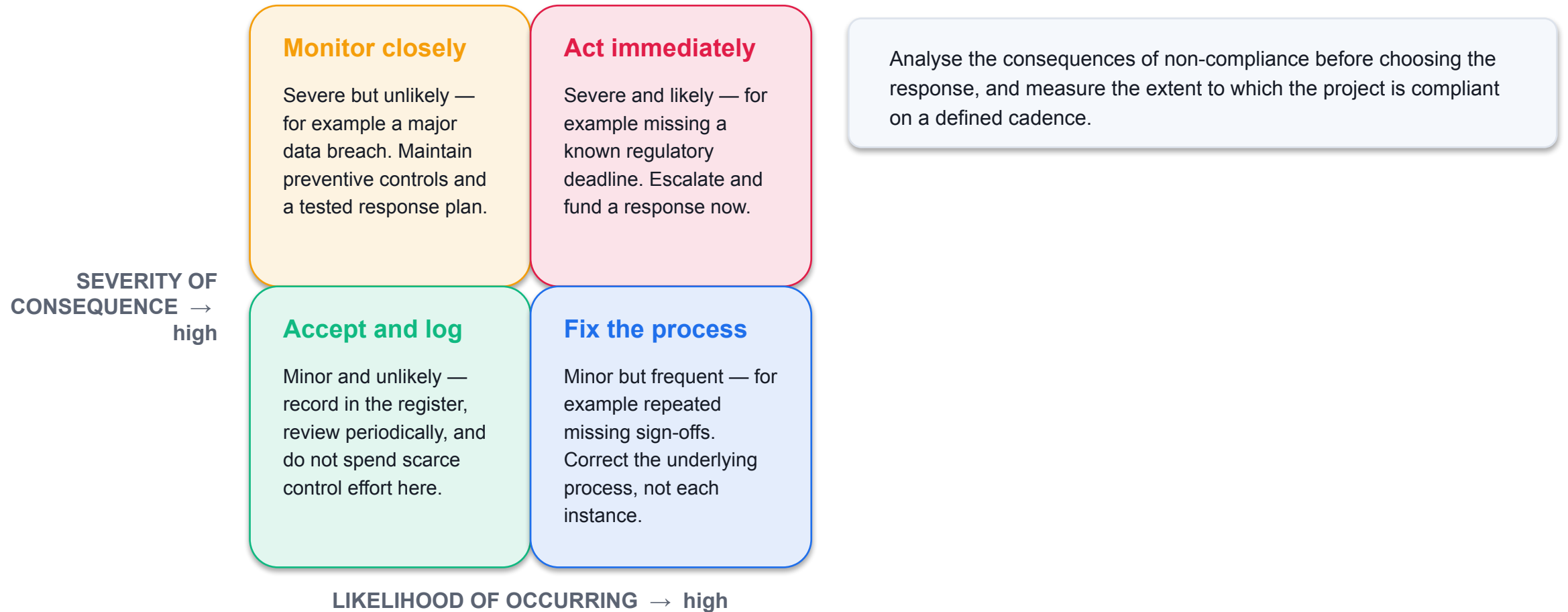
5 Awareness across the team

Confirm the team and stakeholders actually know the compliance matters that apply — assumed knowledge is a common failure.

6 Evidence, not intention

Every requirement needs a named owner, a control and auditable evidence that the control is actually operating.

Prioritising threats to compliance



Methods that support compliance through the project



Measure compliance the way you measure schedule: the percentage of identified requirements with current evidence, trended over time and reported to governance.

Sustainability is now a compliance and a value dimension

The 2026 ECO makes sustainability an explicit information requirement, a quality obligation and a category of project risk.

ENVIRONMENTAL

Carbon footprint of the solution, energy and water use, materials, waste and end-of-life disposal of what you build.

SOCIAL

Labour conditions across the supply chain, accessibility of the deliverable, and impact on the surrounding community.

GOVERNANCE

Transparent reporting of sustainability claims, ethical procurement, and avoidance of greenwashing in benefit claims.

Where sustainability appears in the 2026 ECO

ECO reference	What it requires	What you actually do
Process T1 — integrated plan	Determine critical information requirements, explicitly including sustainability	Add sustainability criteria to the plan and to the definition of done
Process T7 — quality	Manage cost of quality and sustainability together	Cost sustainable options in the business case rather than as an afterthought
Business Env T2 — compliance	Confirm sustainability among the compliance requirement categories	Register environmental and social obligations with owners and evidence
Business Env T5 — risk	Execute risk responses for security and sustainability risks	Log sustainability risks in the register with real, funded responses
Business Env T8 — external change	Survey regulatory change in the external environment	Track emerging disclosure and reporting regulation affecting the deliverable

Exam framing: sustainability is treated as a legitimate constraint and benefit, so 'ignore it because it is not in scope' is never the best answer.

AI in project management — where it genuinely helps

1 Estimation

Models trained on historical project data produce range estimates and flag where your estimate diverges from comparable past work.

2 Risk detection

Pattern analysis across schedule, issue logs and communications surfaces emerging risks earlier than a monthly review would.

3 Status reporting

Automated generation of status narratives from tool data, freeing PM time for stakeholder work rather than slide assembly.

4 Schedule optimisation

Scenario simulation across thousands of resource and sequence permutations to test compression options before committing.

5 Requirements analysis

Clustering and de-duplicating large volumes of stakeholder input, and flagging ambiguous or untestable requirements.

6 Meeting and knowledge capture

Transcription, action extraction and searchable project memory that makes lessons learned findable rather than archived.

Governing AI use on your project — the risks you own

1 Data confidentiality

Client data, contracts and personal information pasted into external tools may leave your control and breach your obligations.

2 Accuracy and hallucination

Generated estimates, text and analysis can be confidently wrong; a human must own and verify anything that reaches a stakeholder.

3 Bias in the training data

A model trained on past projects reproduces past bias — in vendor selection, in estimating, and in performance judgement.

4 Transparency and disclosure

Stakeholders should know when analysis or reporting they rely on was machine-generated; concealment is an ethics failure.

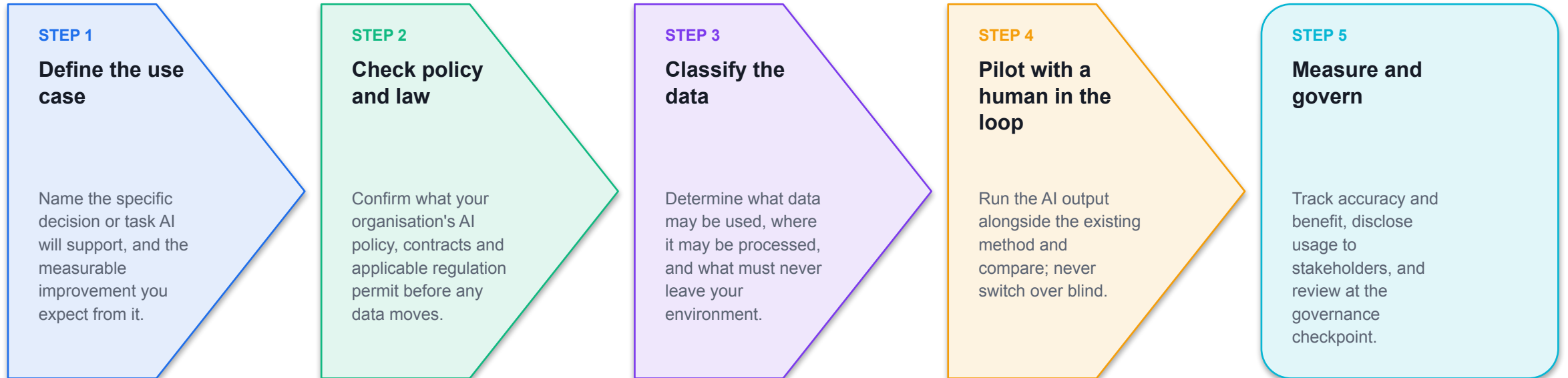
5 Accountability stays human

Delegating a task to AI never delegates the accountability; the project manager remains answerable for the decision.

6 Regulatory exposure

AI-specific regulation is emerging quickly — treat it as an external environment change to survey under ECO T8.

Introducing AI into your project practice responsibly



AI adoption is itself an organisational change project. Everything you learned about ADKAR applies to getting your own team to use it well.

Exam focus: Business Environment traps

IF THE QUESTION SAYS

A new law is announced that affects your project mid-execution



THE BEST ANSWER IS

Assess the impact on the project first, then raise a change request through the defined governance path.

IF THE QUESTION SAYS

An issue arises that exceeds your documented tolerance level



THE BEST ANSWER IS

Escalate to the stakeholder authorised to act, presenting options and a recommendation — do not decide alone.

IF THE QUESTION SAYS

A stakeholder asks you to skip a compliance step to hit the date



THE BEST ANSWER IS

Refuse and escalate. Compliance stewardship is non-negotiable and is an ethical obligation under the Code of Conduct.

IF THE QUESTION SAYS

The organisation reorganises and your sponsor is replaced



THE BEST ANSWER IS

Assess the impact on the project, then re-confirm the charter, authority and priorities with the new sponsor.

Exam focus: approach, governance and value traps

IF THE QUESTION SAYS

Requirements are unclear and are expected to keep evolving



THE BEST ANSWER IS

Recommend an adaptive or hybrid approach with short feedback loops — not a longer, more detailed requirements phase.

IF THE QUESTION SAYS

The sponsor asks why the project should continue after a market shift



THE BEST ANSWER IS

Re-examine the business value against the remaining investment and present the options, including terminating.

IF THE QUESTION SAYS

A team wants to drop retrospectives because they feel unproductive



THE BEST ANSWER IS

Fix the retrospective, do not remove it — check whether actions are actually reaching the next sprint backlog.

IF THE QUESTION SAYS

You are told 'we are agile so we do not need documentation'

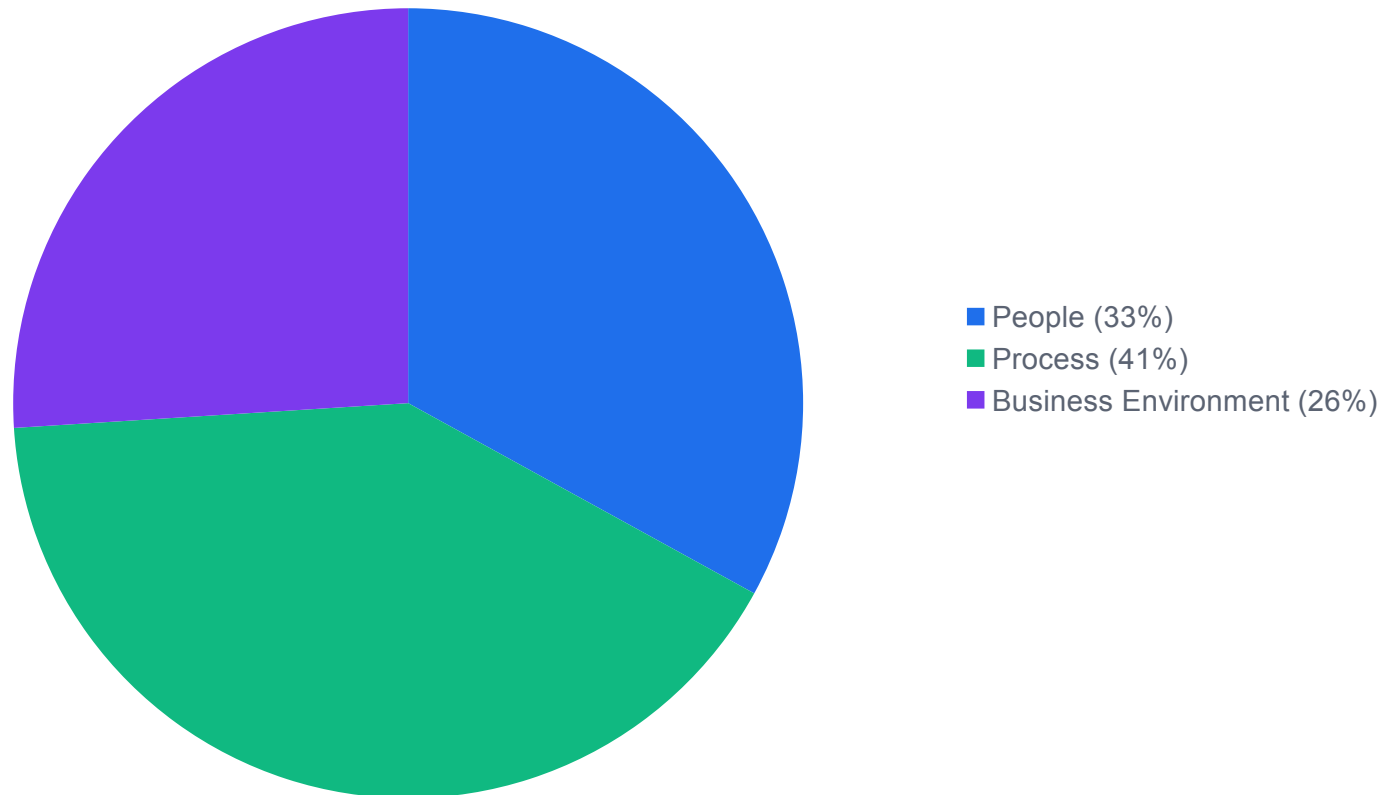


THE BEST ANSWER IS

Wrong. Agile values working product over comprehensive documentation; regulated projects still require compliance evidence.

Where Business Environment sits in the July 2026 exam

Share of scored questions



WHAT IT TELLS YOU

- Business Environment is roughly 44 of the 170 scored questions — too many to treat as a side topic.
- Its eight tasks cover governance, compliance, change control, impediments, risk, improvement, org change and external change.
- This topic delivers T1, T2, T7 and T8; the remaining tasks appear in Topics 5 and 6 of this course.
- About 40% of the exam is predictive and 60% adaptive or hybrid, so answer in the context each scenario states.

Topic 1 recap — Business Environment

1 A project is temporary and unique

It delivers change and business value, and ends when objectives are met or are shown to be unachievable.

3 Two life cycles, five approaches

The project life cycle is the phases; the development life cycle is predictive, iterative, incremental, agile or hybrid.

5 EEFs constrain, OPAs are reusable

You conform to EEFs; you apply and then update OPAs — updating them is an explicit ECO obligation.

7 Governance sets thresholds

Escalate outside tolerance with options; resolve inside tolerance with the team. Phase gates decide go, change, hold or kill.

9 Value, not activity, defines success

Prioritise by value, deliver incrementally, verify a benefits measurement system exists and read it.

2 Triple constraint and tailoring

Scope, time and cost are joined; tailor process, artifacts and methods to the context rather than applying a template.

4 Agile is incremental plus iterative

Four Manifesto values and twelve principles; Scrum supplies the roles, events and artifacts most teams actually use.

6 Change needs individuals to change

ADKAR gives five milestones — awareness, desire, knowledge, ability and reinforcement — to diagnose adoption failure.

8 Compliance and sustainability are owned by you

Classify requirements, analyse threats, evidence controls, and treat sustainability as both constraint and benefit.

10 AI helps but never takes accountability

Use it for estimation, risk detection and reporting under an explicit policy, with a human verifying every output.

Hands-on Labs — Topic 1

1 Lab 01. PMP Exam Orientation and 35-PDU Study Plan

All domains · LO1 · 20 min in class + 25 min self-study
Produces: Personal study plan, Domain self-assessment

2 Lab 02. PESTLE and TECOP External Environment Scan

BE T8 · LO1 · 20 min in class + 25 min self-study
Produces: PESTLE scan, TECOP risk themes

3 Lab 03. EEF/OPA Inventory with Governance and Escalation Thresholds

BE T1 · LO1 · 20 min in class + 25 min self-study
Produces: EEF/OPA inventory, Escalation threshold table

4 Lab 04. Compliance, Sustainability and AI Governance Register

BE T2 · LO4 · 20 min in class + 25 min self-study
Produces: Compliance register, AI governance controls

HANDS-ON ACTIVITY

Lab 01. PMP Exam Orientation and 35-PDU Study Plan

Case study: Contoso Training Portal Upgrade. All domains · LO1. Produces Personal study plan, Domain self-assessment.

20 min in class

1 Record your PMP eligibility route: degree plus 36 months leading projects, or diploma plus 60 months, and confirm the 35 contact hours this course supplies.

2 Score yourself 1-5 on each of the three ECO domains — People, Process and Business Environment — to expose where your knowledge is thinnest.

3 Weight your available study hours by the exam percentages: Process 41%, People 33%, Business Environment 26%.

4 Block an eight-week study calendar with named topics per week, not vague 'revision' entries.

5 Schedule three full 180-question mock exams into that calendar, spaced at weeks 4, 6 and 8.

6 Write down your target exam date and work backwards to confirm the plan is actually achievable.

DELIVERABLE

Personal study plan, Domain self-assessment

Lab 02. PESTLE and TECOP External Environment Scan

Case study: Contoso Training Portal Upgrade. BE T8 · LO1. Produces PESTLE scan, TECOP risk themes.

20 min in class

- 1 List the Contoso project's external drivers under each PESTLE heading: political, economic, socio-cultural, technological, legal, environmental.
- 2 Open the Mind Mapping tool and branch each PESTLE factor into the specific effect it has on the training portal project.
- 3 Repeat the scan using TECOP — technical, environmental, commercial, operational, political — and note which factors only TECOP surfaced.
- 4 Rate every factor for likelihood and impact on a 1-5 scale, then rank them.
- 5 Identify the three factors most likely to force a scope or schedule change.
- 6 Write a one-paragraph environmental summary suitable for the project charter's assumptions and constraints section.

DELIVERABLE

PESTLE scan, TECOP risk themes

TOOL

Mind Mapping

alfredang.github.io/mindmapping/

Lab 03. EEF/OPA Inventory with Governance and Escalation Thresholds

Case study: Contoso Training Portal Upgrade. BE T1 · LO1. Produces EEF/OPA inventory, Escalation threshold table.

20 min in class

1 Inventory the enterprise environmental factors constraining Contoso: existing systems, culture, staff capability, market and regulation.

2 Inventory the organisational process assets available to reuse: templates, historical data, procurement rules, lessons learned.

3 Classify each item explicitly as EEF or OPA and justify the classification in one line — this is a heavily examined distinction.

4 Define the governance structure: who sponsors, who sits on the steering committee, and what the project manager may decide alone.

5 Set numeric escalation thresholds for cost variance, schedule slip and risk severity, stating what triggers escalation to whom.

6 Define the phase gates and the go / no-go criteria applied at each.

DELIVERABLE

EEF/OPA inventory, Escalation threshold table

Lab 04. Compliance, Sustainability and AI Governance Register

Case study: Contoso Training Portal Upgrade. BE T2 · LO4. Produces Compliance register, AI governance controls.

20 min in class

1 Identify every compliance requirement touching the portal: data protection, accessibility, financial audit, and workplace health and safety.

2 Classify each requirement into a compliance category and name its owner inside the organisation.

3 Assess the consequence of non-compliance for each — financial penalty, reputational damage, or inability to launch.

4 Add sustainability requirements: energy use of the hosting platform, e-waste from decommissioned hardware, and supplier sustainability criteria.

5 Register the AI governance controls: where AI is used in the project, what data it consumes, and how its output is reviewed before acting on it.

6 Define how compliance will be MEASURED, not merely asserted — the 2026 outline expects a metric.

DELIVERABLE

Compliance register, AI governance controls

02

TOPIC

Start the Project

Turning an idea into an authorised project: the business case that justifies it, the charter that authorises it, the stakeholders who judge it, and the team that will build it.

- 1 Project integration and the Develop Project Charter process
- 2 Business value, business case, cost-benefit and benefits management
- 3 Identify, analyse and prioritise stakeholders
- 4 Align and manage stakeholder expectations (ECO 2026 People T5/T6)
- 5 Form the team, agree the team charter and kick off
- 6 Recommend the development approach for the integrated plan

What You Will Be Able To Do After Topic 2

1 Draft a project charter

Write a charter that names the sponsor and PM, states measurable objectives, and authorises the PM to apply organisational resources.

2 Justify a project financially

Build a cost-benefit analysis and read payback, NPV, IRR, ROI and BCR well enough to defend the selection decision to a steering committee.

3 Identify every stakeholder

Work an organisation breakdown structure and document analysis to find who is affected, then record them properly in a stakeholder register.

4 Analyse and prioritise

Apply power/interest and power/influence grids, the stakeholder cube and the salience model to decide who gets how much of your attention.

5 Align expectations

Facilitate discussions that surface conflicting expectations early and converge them onto one agreed set of project objectives.

6 Start the team well

Co-create a team charter, set ground rules and run a kick-off that leaves everyone sharing the same vision of success.

Why initiation deserves this much time

Projects are not lost at the end. They are lost at the start.

Wrong problem

A project with no validated business case delivers perfectly against an objective nobody needed.

Wrong people

A stakeholder discovered in month six brings requirements that should have shaped month one.

Wrong authority

Without a signed charter the PM has responsibility without the power to apply resources.

Project Integration Management: The Unifying Discipline

1 What integration is

The work of identifying, defining, combining, unifying and coordinating the other knowledge areas so they behave as one system, not ten silos.

2 Why it exists

Scope, schedule, cost, quality, resources, communications, risk and procurement all trade against each other; someone must own the trade-off.

3 Who owns it

Integration is the one area the project manager cannot delegate — it is the definition of the PM role in the PMBOK Guide.

4 The through-line

A researched business case, an approved charter and a well-planned project management plan tie every decision back to business value.

5 The test of integration

Change a date and you can immediately state the cost, scope, quality and risk consequence — that is an integrated project.

6 The failure mode

Silo optimisation: each function hits its own target while the project as a whole misses the business outcome entirely.

The Key Benefits of Project Integration

1 Strategic alignment

Every deliverable can be traced upward to an organisational objective, so the project earns its funding rather than merely consuming it.

2 A single decision point

Competing demands from stakeholders arrive at one integrating role that can weigh them against the whole rather than a part.

3 Coherent baselines

Scope, schedule and cost baselines are built to be consistent with each other, so variance analysis measures reality rather than mismatch.

4 Controlled change

Changes are assessed for their ripple effect across all knowledge areas before approval, not discovered downstream.

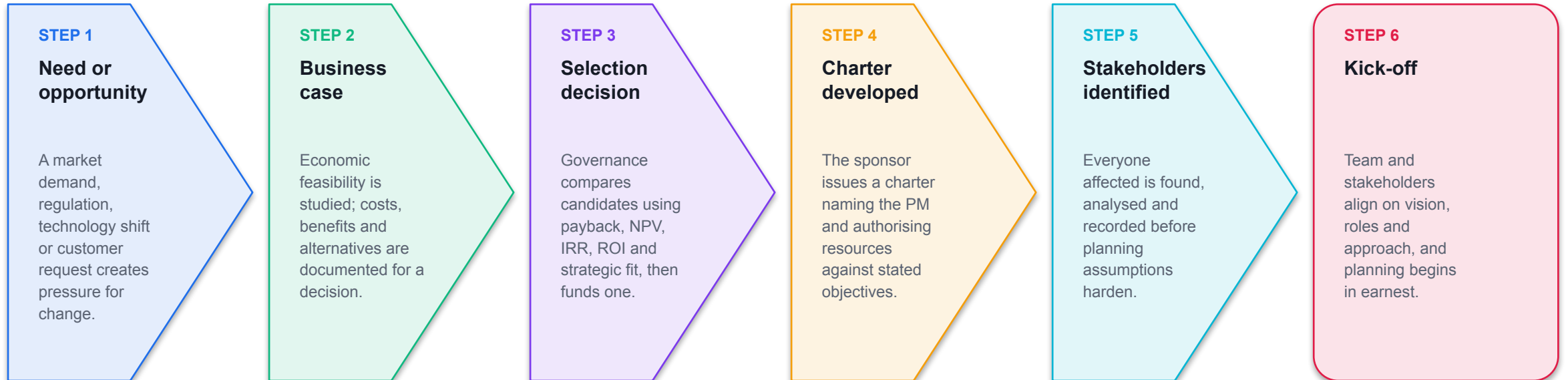
5 Value assurance

Business value is examined throughout the life cycle, so a project whose case has evaporated can be stopped honestly.

6 Clean handover

Integration carries the project through to transition, so benefits are actually realised by operations after the team disbands.

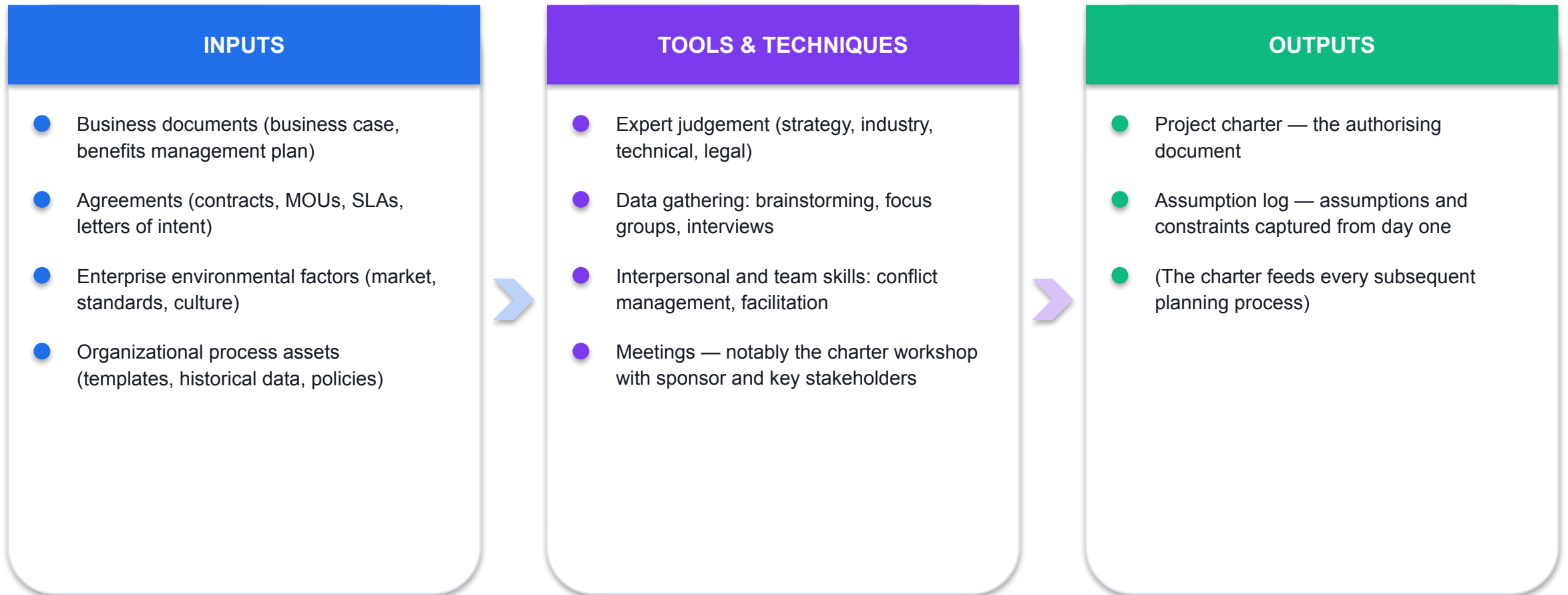
From Idea to Authorised Project: The Initiation Flow



Exam cue: the charter is an OUTPUT of initiating, not of planning. Nothing in the project management plan may be baselined before the charter authorises the project to exist.

Develop Project Charter — ITTO

Purpose: to produce a document that formally authorises the existence of the project and gives the project manager authority to apply organisational resources to project activities.



What the Project Charter Actually Does

1 Authorises the project

It is the formal statement that this project exists and may consume organisational resources — issued by a sponsor senior enough to fund it.

2 Empowers the project manager

It names the PM and states their authority level, converting a job title into the right to direct people, spend money and make decisions.

3 Defines rationale and need

It records why the organisation is doing this, so the reasoning survives sponsor turnover and mid-project pressure to change direction.

4 Verifies strategic alignment

It links the project explicitly to strategic goals, which is what makes the funding defensible at portfolio review.

5 Sets the shared vision

It gives every stakeholder one short reference for what success looks like, supporting ECO People T1 — develop a common vision.

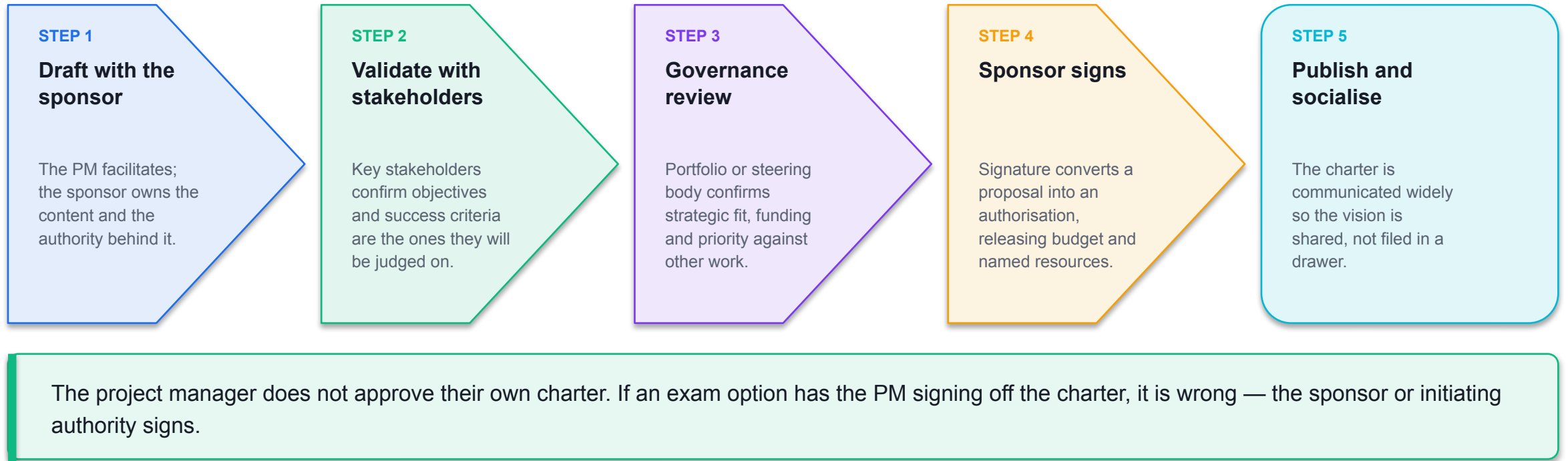
6 Anchors change control

Later change requests are assessed against charter objectives; a change that breaks the charter needs sponsor, not PM, approval.

What Is Included in the Project Charter

Charter element	What goes in it	Why it matters
Names and roles	Project sponsor, project manager, key stakeholders and their authority	Removes ambiguity about who decides and who must be consulted
Project description	High-level product description plus preliminary requirements	Bounds the solution space before detailed requirements work
Measurable objectives	Objectives with success criteria that can be objectively tested	Makes acceptance at closure a measurement, not an argument
Business need	The problem or opportunity, plus financial goals or milestones	Preserves the 'why' when the 'what' comes under pressure
Summary schedule	High-level milestones and any externally imposed dates	Signals feasibility and exposes immovable constraints early
Assumptions and constraints	Boundaries, overall project risk, approval requirements, approved budget	Seeds the assumption log and the initial risk register
Exit criteria	Success criteria and the conditions under which the project is stopped	Makes cancelling a failing project a planned option, not a failure

How a Charter Gets Approved



Business Value: The Six Reasons Organisations Fund Projects

1 Financial gain

Direct revenue increase or cost reduction, usually the easiest form of value to quantify and therefore the easiest to defend.

2 New customers

Access to a new segment, geography or channel, where the value is in the lifetime revenue of the relationship rather than the first sale.

3 Social benefit

Community, environmental or sustainability outcomes that build licence to operate — increasingly explicit in ECO 2026 sustainability language.

4 First to market

Speed itself is the value: capturing share, setting the standard or shaping customer expectations before competitors arrive.

5 Improvement

Technological or process uplift that raises capability, reduces cycle time or removes chronic defects in how work is done.

6 Regularization

Alignment or compliance with standards and regulations, where the value is the avoided penalty, avoided shutdown or preserved certification.

Tangible vs Intangible Business Value

TANGIBLE VALUE

Countable, auditable, defensible in a business case

- Monetary assets and direct revenue from new sales
- Stockholder equity and improved market capitalisation
- Utility, fixtures, tools and physical assets acquired
- Market share expressed as a measurable percentage
- Cost avoidance and hard headcount or licence savings
- Measured cycle-time or defect-rate improvement

INTANGIBLE VALUE

Real but harder to measure — still must be tracked

- Goodwill and brand recognition in the market
- Public benefit, sustainability and community standing
- Trademarks, patents and other intellectual property
- Strategic alignment and organisational reputation
- Employee capability, morale and retention
- Customer satisfaction and loyalty

Both count. ECO Process T3 asks you to identify value components WITH key stakeholders and to verify a measurement system exists — including for the intangibles.

The Three Business Documents

1 Business case

The documented economic feasibility study that justifies the investment and establishes the benefits the project work is expected to deliver.

2 Cost-benefit analysis

The quantitative core of the business case: all costs against all benefits over a defined horizon, used to compare candidate projects.

3 Benefits management plan

How benefits will be created, maximised and sustained after delivery, with owners, metrics and time frames for realisation.

Business Documents: Who Owns Them and When

OWNED BY THE SPONSOR / ORGANISATION

Created before the project — the PM is a consumer, not an author

- The business case is developed by the sponsor or a business analyst
- The benefits management plan is owned by a benefits owner
- Both predate the charter and are inputs to Develop Project Charter
- The PM may recommend updates but does not unilaterally change them
- They are maintained beyond project closure into benefits realisation

USED BY THE PROJECT MANAGER

Consumed continuously to keep the project honest about value

- Source the charter's rationale, objectives and success criteria
- Test every significant change request against continued value
- Feed the value discussion at each phase gate or stage review
- Trigger an escalation when the business case no longer holds
- Provide the acceptance and exit criteria used at closure

Exam cue: 'the business case no longer supports the project' means escalate to the sponsor for a stop/continue decision — not quietly re-plan.

How to Run a Cost-Benefit Analysis



Sunk costs are irrelevant to a go/no-go decision. Money already spent cannot be recovered by continuing; only future costs and future benefits belong in the analysis.

The Business Case as an Economic Feasibility Study

1 Documented, not verbal

It is a written study that can be reviewed, challenged and audited — an enthusiastic sponsor's confidence is not a business case.

2 Establishes the benefits

It states specifically what benefits the project work will produce, in units that can later be measured against reality.

3 Basis for authorisation

It provides the justification on which further project activities are authorised — no case, no charter, no funding.

4 Includes alternatives

Do nothing, buy, build and partial-scope options are compared, so the chosen route is a decision rather than a default.

5 States success criteria

Defines what 'it worked' will mean, which flows into charter objectives and eventually into acceptance at closure.

6 Is revisited, not filed

ECO Process T3 requires examining business value throughout the project, so the case is a live document across the life cycle.

Benefits Management Plan — Required Contents

Element	What it specifies	Typical example
Target benefits	The value expected, expressed in measurable terms	Reduce order-processing cost by 18% per transaction
Strategic alignment	Which organisational objective the benefit serves	Supports the 'digital-first operations' strategic pillar
Time frame	When short-term and long-term benefits will be realised	40% at go-live, full run-rate 12 months post-transition
Benefits owner	The accountable person who monitors and reports realisation	Head of Operations, named by the sponsor
Metrics	Direct and indirect measures used to demonstrate realisation	Cost per order, error rate, customer satisfaction score
Assumptions	Conditions that must hold for the benefits to appear	Volumes remain within +/-10% of forecast
Risks	Threats to benefit realisation, distinct from delivery risks	Operations does not adopt the new process after handover

The Benefits Owner: An Accountability Beyond Delivery

1 Works alongside the PM

Partners with the project manager or team lead during delivery to ensure planned benefits are actively managed as they are delivered.

2 Owns the transition

Assists in transitioning the requested benefits to the receiving organisation so value is not lost at handover.

3 Establishes measurement

Ensures measurement metrics and methods are established, baselined and monitored — you cannot claim a benefit you never measured.

4 Reports realised value

Reports to management on the realised results of delivered benefits, often long after the project team has been released.

5 Who holds the role

Typically a business analyst, the sponsor or an operations manager — a business-side role, not a project-side one.

6 In adaptive contexts

The product owner is responsible for making sure the work reaps benefits for the organisation, prioritising the backlog by value.

Benefit Measurement Formulas You Must Recognise

Return on investment

$$\text{ROI} = (\text{Benefit} - \text{Cost}) / \text{Cost}$$

Expressed as a percentage. Bigger is better. Ignores the timing of cash flows entirely.

Benefit-cost ratio

$$\text{BCR} = \text{Benefits} / \text{Costs}$$

Greater than 1.0 means benefits exceed costs. Bigger is better; used to rank competing proposals.

Net present value

$$\text{NPV} = \sum \text{CF}_t / (1+r)^t - \text{C}_0$$

Today's value of all future cash flows minus the investment. Positive NPV means proceed; choose the highest.

Payback period

$$\text{Payback} = \text{Cost} / \text{Annual net inflow}$$

How long until you recover the investment. Smaller is better; ignores everything after payback.

Rule of thumb for the exam: financial-return measures (NPV, IRR, ROI, BCR) are 'bigger is better'; time and cost measures (payback period, opportunity cost) are 'smaller is better'.

Benefit Measurement Methods Compared

Method	What it answers	Decision rule	Main weakness
Payback period	How fast do we get our money back?	Choose the shortest duration	Ignores all cash flow after payback and the time value of money
Opportunity cost	What is the value of the option we gave up?	Choose the option with the smaller opportunity cost	Depends entirely on which alternative you compare against
Net present value	What is this worth in today's dollars?	Choose the highest positive NPV	Highly sensitive to the discount rate chosen
Internal rate of return	What effective interest rate does it earn?	Choose the highest IRR above the hurdle rate	Misleads when comparing projects of very different sizes
Return on investment	What percentage return on money spent?	Choose the highest ROI	Ignores timing, duration and risk of the cash flows
Benefit-cost ratio	How many dollars of benefit per dollar spent?	Choose the highest ratio, must exceed 1.0	Sensitive to whether an item is classed as cost or negative benefit

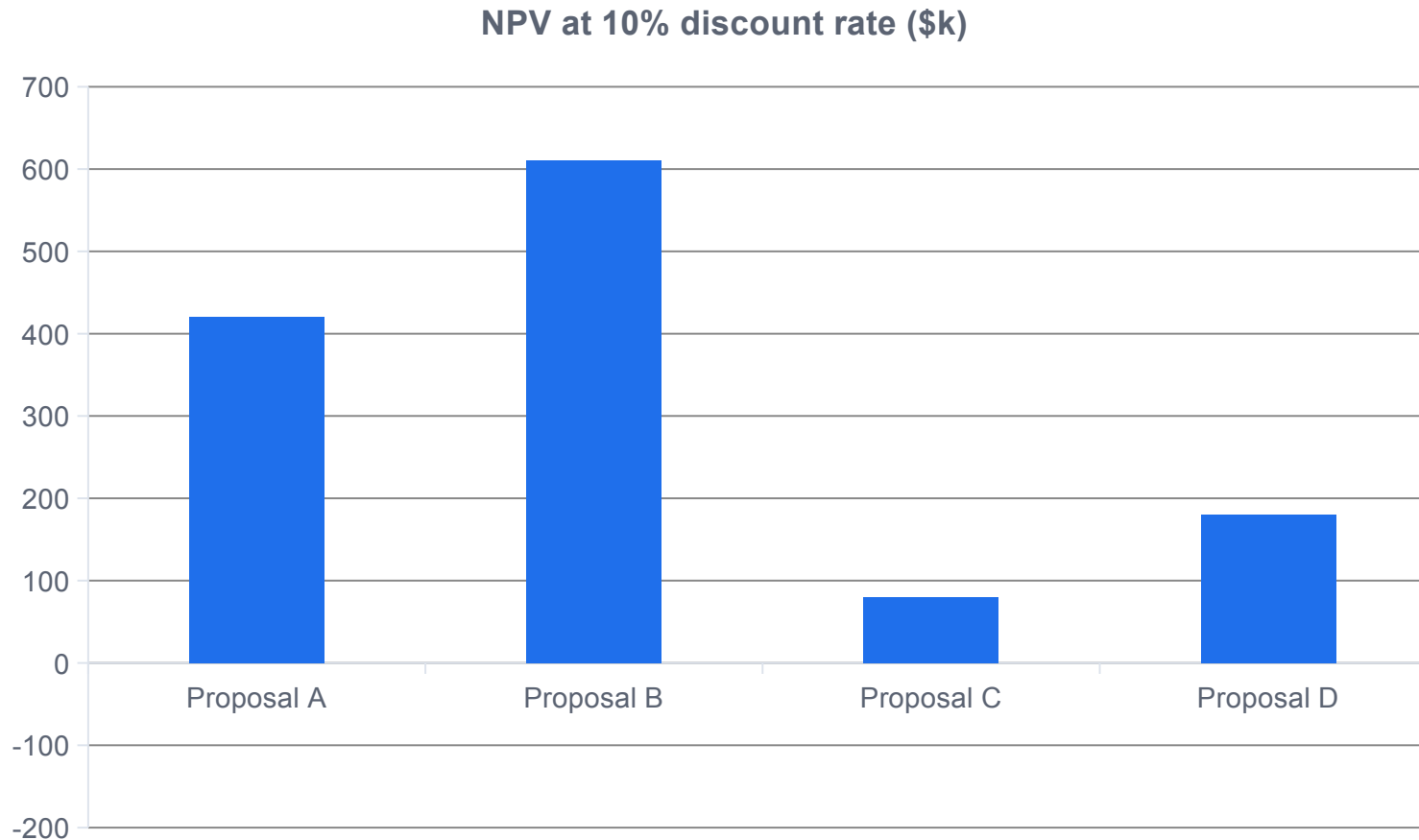
Payback Period: Cumulative Cash Flow of a \$500k Project



WHAT IT TELLS YOU

- Project A crosses zero during year 3 — payback is roughly 2.8 years, the shorter and therefore preferred payback.
- Project B pays back later but ends higher, so a payback-only rule would reject the project with more total value.
- This is exactly why payback is never used alone: pair it with NPV, which counts everything after the crossover point.
- Shorter payback also means less exposure to forecast error, which matters most in volatile business environments.

NPV of Four Competing Project Proposals



WHAT IT TELLS YOU

- Proposal B has the highest NPV and is the correct selection on financial grounds alone.
- Proposal C has a negative NPV — it destroys value and should not be funded regardless of how attractive it sounds.
- NPV is additive across projects, which is why portfolios are built on NPV rather than on IRR or ROI percentages.
- Financial ranking is only one input: strategic fit, risk and capacity can still override the highest NPV.

Exam Traps: Project Selection and Business Documents

IF THE QUESTION SAYS

Two projects, one has higher NPV and the other has shorter payback



THE BEST ANSWER IS

Choose the higher NPV — it accounts for the full value of every cash flow, not just the recovery point.

IF THE QUESTION SAYS

A large sum has already been spent and the project now looks unviable



THE BEST ANSWER IS

Ignore the sunk cost. Decide on future costs versus future benefits and escalate a stop recommendation to the sponsor.

IF THE QUESTION SAYS

You are asked who writes the business case



THE BEST ANSWER IS

The sponsor or business analyst — it exists before the project, and the PM is a consumer of it, not its author.

IF THE QUESTION SAYS

A stakeholder wants a change that breaks the charter objectives



THE BEST ANSWER IS

Escalate to the sponsor. The PM cannot approve a change that invalidates the authorising document.

The Project Vision Statement

1 Created by the sponsor

Written by the project sponsor or an executive, because vision requires the authority to commit the organisation to a direction.

2 Describes desired objectives

States clearly what the desired end state looks like, in language a non-specialist stakeholder can repeat accurately.

3 Shows strategic alignment

Makes explicit which organisational strategic goal this project advances, so the vision is anchored rather than aspirational.

4 Referred to throughout

Revisited at every major decision point to maintain alignment — a vision used once at kick-off is decoration.

5 Kept current

ECO People T1 requires keeping the vision current: when the business environment shifts, the vision is updated, not ignored.

6 Root-cause the drift

When people act inconsistently with the vision, break the situation down to find the root cause of the misunderstanding rather than blaming.

ECO People Task 1 — Develop a Common Vision

A vision only exists if the team can state it without reading it.

Help ensure it is shared

Co-create the vision with key stakeholders so they own it rather than receive it.

Promote it actively

Repeat the vision at stand-ups, reviews and steering meetings until it becomes the team's default language.

Keep it current

Revalidate the vision whenever strategy, market or sponsorship changes; a stale vision misdirects everyone.

Predictive Charter vs Agile Team Charter

PREDICTIVE PROJECT CHARTER

Authorisation document, issued top-down by the sponsor

- Purpose: authorise the project and empower the PM
- Author: sponsor or initiating authority signs it
- Contains measurable objectives and approved budget
- Contains summary milestones and approval requirements
- Changes require formal sponsor approval
- Written once, referenced throughout the life cycle

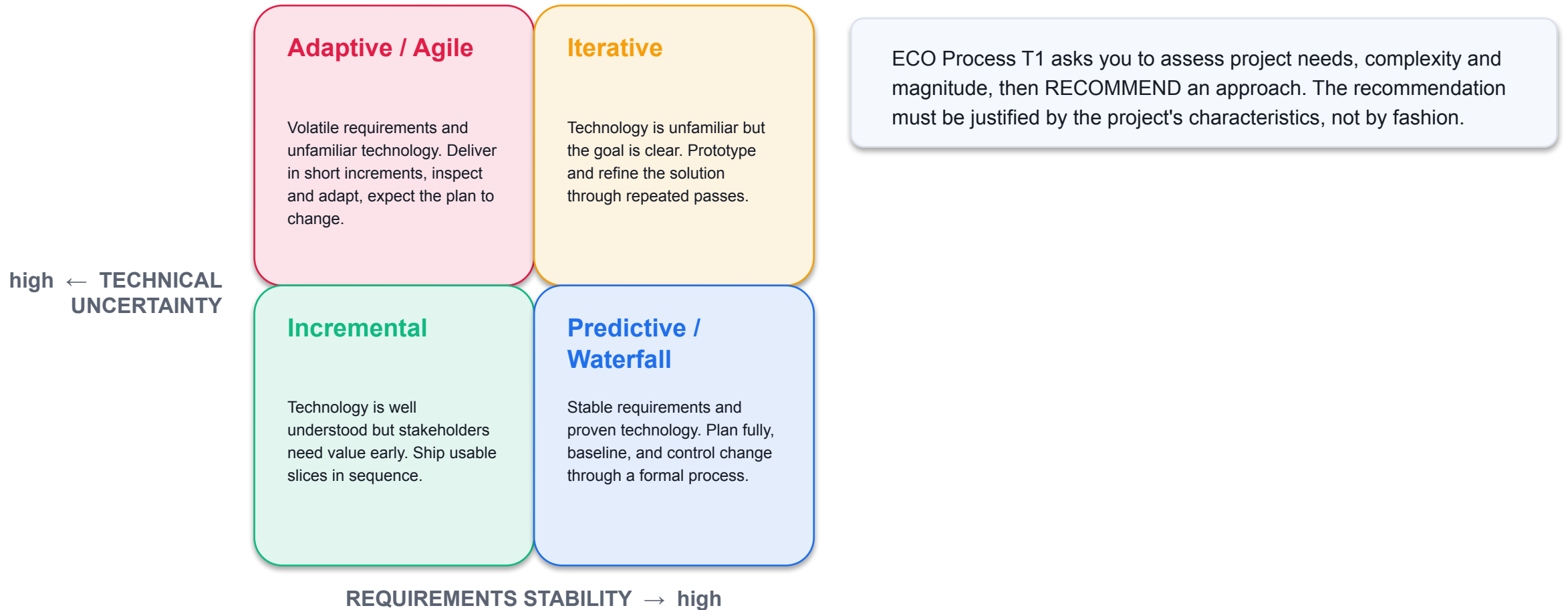
AGILE TEAM CHARTER

Working agreement, created bottom-up by the team itself

- Purpose: define how this team will work together
- Author: the whole team, facilitated by the servant leader
- Team vision or mission in the team's own words
- Team roles, responsibilities and the stakeholders served
- Team values and the ceremonies the team commits to
- Revisited at retrospectives and amended as the team learns

They are complements, not alternatives. A hybrid project commonly has both: a signed project charter for authority and a team charter for day-to-day working agreements.

Choosing the Development Approach



Stakeholder Definition

A stakeholder is anyone affected — or who believes they are affected — by your project.

Perception counts

Someone who merely believes they are affected can still block you, so they are a real stakeholder.

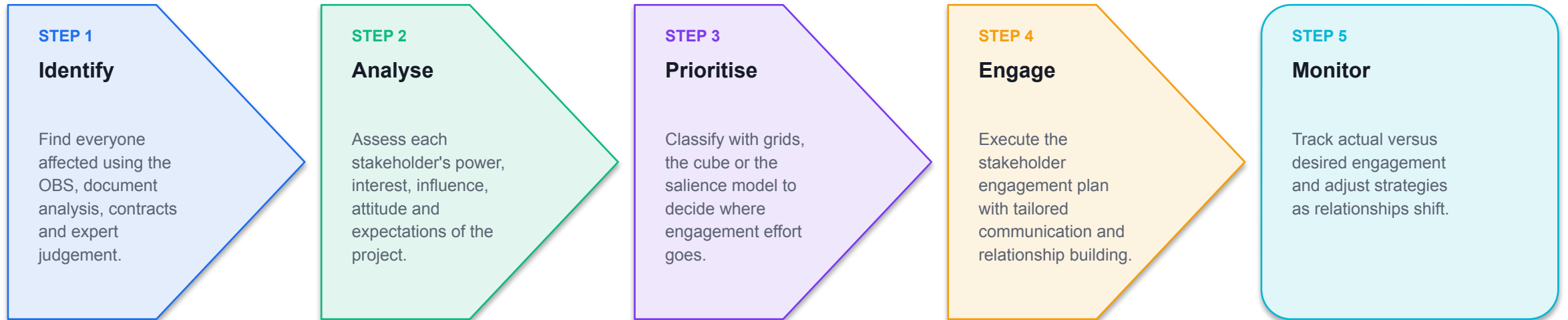
The outcome we want

Productive relationships in which stakeholders actively support project objectives rather than merely tolerate them.

The five-step cycle

Identify, analyse, prioritise and engage stakeholders — then monitor their engagement continuously.

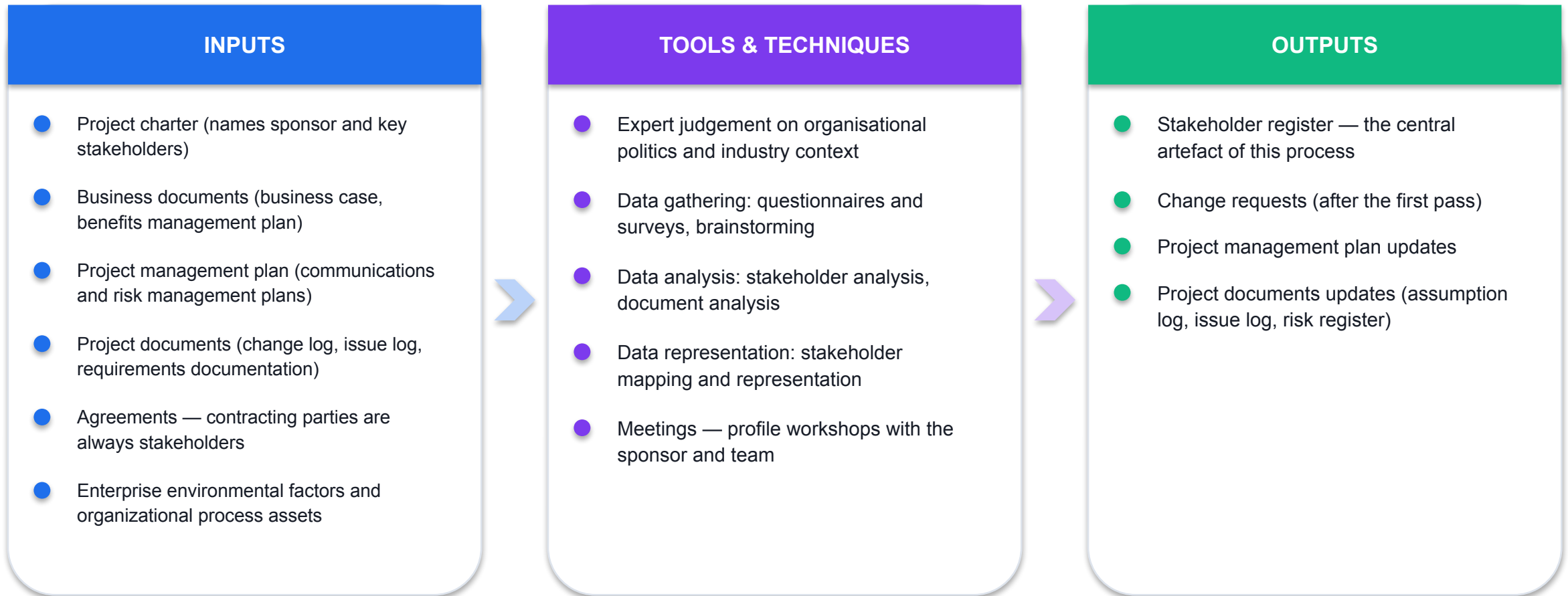
The Stakeholder Engagement Cycle



This cycle repeats. New stakeholders appear at every phase boundary, reorganisation and change of sponsor — identification is a continuous activity, not an initiating-phase task you complete once.

Identify Stakeholders — ITTO

Purpose: to identify project stakeholders and analyse and document relevant information about their interests, involvement, interdependencies, influence and potential impact on project success.



Stakeholder Identification in Practice

1 Start early, repeat often

Identify and analyse engagement early and throughout the project — the cost of a missed stakeholder rises steeply with every planning decision.

2 Use the OBS

Walk the organisational breakdown structure top to bottom to find the initial population of internal stakeholders systematically.

3 Mine the documents

Contracts, regulations, prior lessons learned and the business case each name parties who have a stake in the outcome.

4 Ask the stakeholders you have

Snowball sampling: every stakeholder you interview can name two more you had not considered, especially external ones.

5 Look outward deliberately

Regulators, unions, communities, suppliers and end users rarely appear on an internal org chart but can stop a project cold.

6 Record it, do not remember it

Everything found goes into the stakeholder register so the knowledge survives handover and team changes.

The Stakeholder Register — Fields and Purpose

Field	What you record	Why you need it
Name and organisation	Identity, department, employer and contact details	Basic addressability — you cannot engage whom you cannot reach
Role on the project	Sponsor, customer, end user, regulator, supplier, team member	Sets the default expectations and rights of that party
Power	Authority to grant, withhold or reverse decisions and resources	Determines how much of the PM's personal attention is warranted
Interest	How much the stakeholder cares about the project outcome	Predicts how actively they will seek involvement
Influence	Ability to affect the project through others, formally or informally	Reveals people with little authority but high real effect
Expectations and needs	What this stakeholder believes success looks like for them	The raw material for aligning expectations (ECO People T5)
Communication requirements	Preferred content, format, channel, frequency and language	Drives the communications management plan
Attitude / classification	Unaware, resistant, neutral, supportive or leading	Feeds the stakeholder engagement assessment matrix

Typical Project Stakeholders

1 End users

The people who will actually operate the product daily; their adoption decides whether benefits are realised or the system is worked around.

2 Customers

Those who acquire or pay for the product — sometimes but not always the same people as the end users, a distinction worth making explicit.

3 Employees and the team

Those doing the work and those whose jobs change because of it; both populations need engagement, not just the project team.

4 Sponsors and managers

The sponsor funds and champions; functional managers control the resources you must borrow and can quietly withdraw them.

5 Business partners and suppliers

Contractors, vendors and integrators whose commitments become your constraints and whose performance becomes your risk.

6 Government and community

Regulators, standards bodies and affected communities who cannot be negotiated with but must be satisfied.

Organisation Breakdown Structure as an Identification Tool

1 What an OBS is

A hierarchical decomposition of the organisation by department and reporting line, mirroring how the WBS decomposes the work.

2 Why it works

Walking it forces a systematic sweep of every function rather than listing the stakeholders who happen to be top of mind.

3 Map work to units

Cross the OBS against the WBS to see which organisational unit owns each work package — those owners are stakeholders by definition.

4 Find the silent affected

Departments that receive the output but do no project work — finance, legal, support — surface only through this sweep.

5 Its blind spot

An OBS only shows internal parties; regulators, customers and community groups must be found by other means.

6 Keep it alive

Reorganisations change the OBS mid-project, which is a trigger to re-run stakeholder identification.

Stakeholder and Communications Artefacts

Artefact	What it holds	When it is used
Stakeholder register	Identity, role, power, interest, influence, expectations, attitude	Created in Identify Stakeholders; updated continuously
Stakeholder engagement plan	Strategies and actions to move each stakeholder to the desired level	Part of the project management plan; drives daily engagement
Communications management plan	Who receives what information, in what format, how often, by whom	Executed throughout; the operational half of engagement
Stakeholder engagement assessment matrix (SEAM)	Current (C) versus desired (D) engagement per stakeholder	Reviewed in Monitor Stakeholder Engagement to target effort
Assessment grids and models	Power/interest, power/influence, impact/influence, cube, salience	Used during analysis to prioritise and classify
Issue log	Stakeholder-raised issues, owners and resolution status	Continuous — unresolved issues erode engagement fast

Assess Stakeholders: Data Gathering and Data Analysis

DATA GATHERING

Collecting raw information about stakeholders

- Questionnaires and surveys reach large, dispersed populations cheaply
- Surveys work well for end-user expectations across many sites
- Brainstorming with the team surfaces stakeholders nobody listed
- Brainwriting lets quieter contributors add names without pressure
- Interviews go deep with high-power individuals
- Focus groups expose disagreements between stakeholder groups

DATA ANALYSIS

Turning raw information into engagement decisions

- Stakeholder analysis produces a list of interests and their weight
- It classifies each party by power, interest, influence and attitude
- Document analysis mines charters, contracts and lessons learned
- Root-cause analysis explains WHY a stakeholder is resistant
- Assumption and constraint analysis tests what we believe about them
- Output feeds directly into the register and the engagement plan

Gathering tells you who they are and what they say they want; analysis tells you what to do about it. The exam frequently asks you to sort a technique into the correct bucket.

Data Representation: Five Ways to Map Stakeholders

1 2D grid: power/interest

Classifies by authority to act against level of concern — the default grid, giving manage-closely, keep-satisfied, keep-informed, monitor.

2 2D grid: power/influence

Classifies by formal authority against active involvement, useful when informal networks matter more than the org chart.

3 2D grid: impact/influence

Classifies by ability to cause change against active involvement, best when the risk is disruption rather than opposition.

4 3D grid: the stakeholder cube

Combines three dimensions at once so a stakeholder is a point in a volume, capturing combinations a flat grid flattens away.

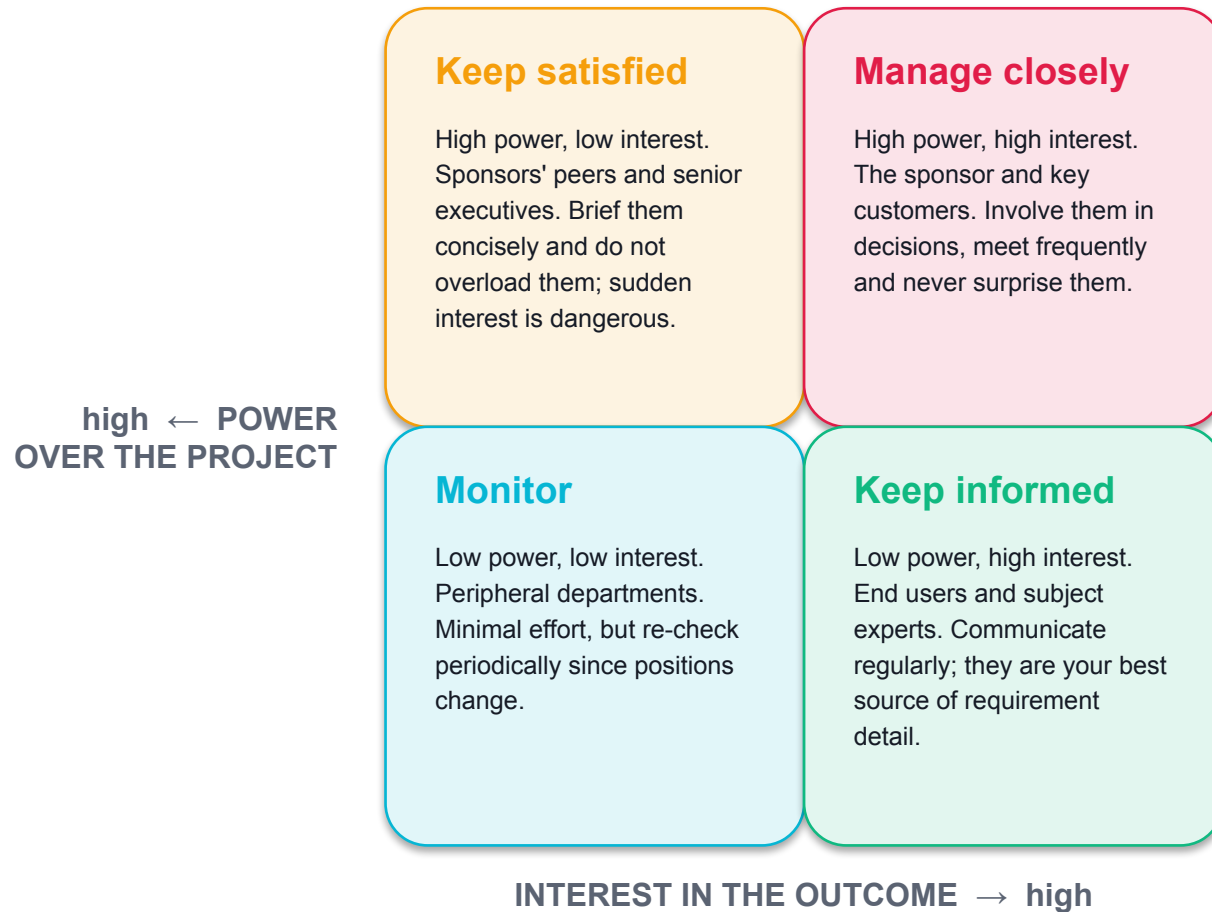
5 Salience model

Classifies by power, legitimacy and urgency, producing seven classes and identifying the definitive stakeholder who has all three.

6 Directions of influence

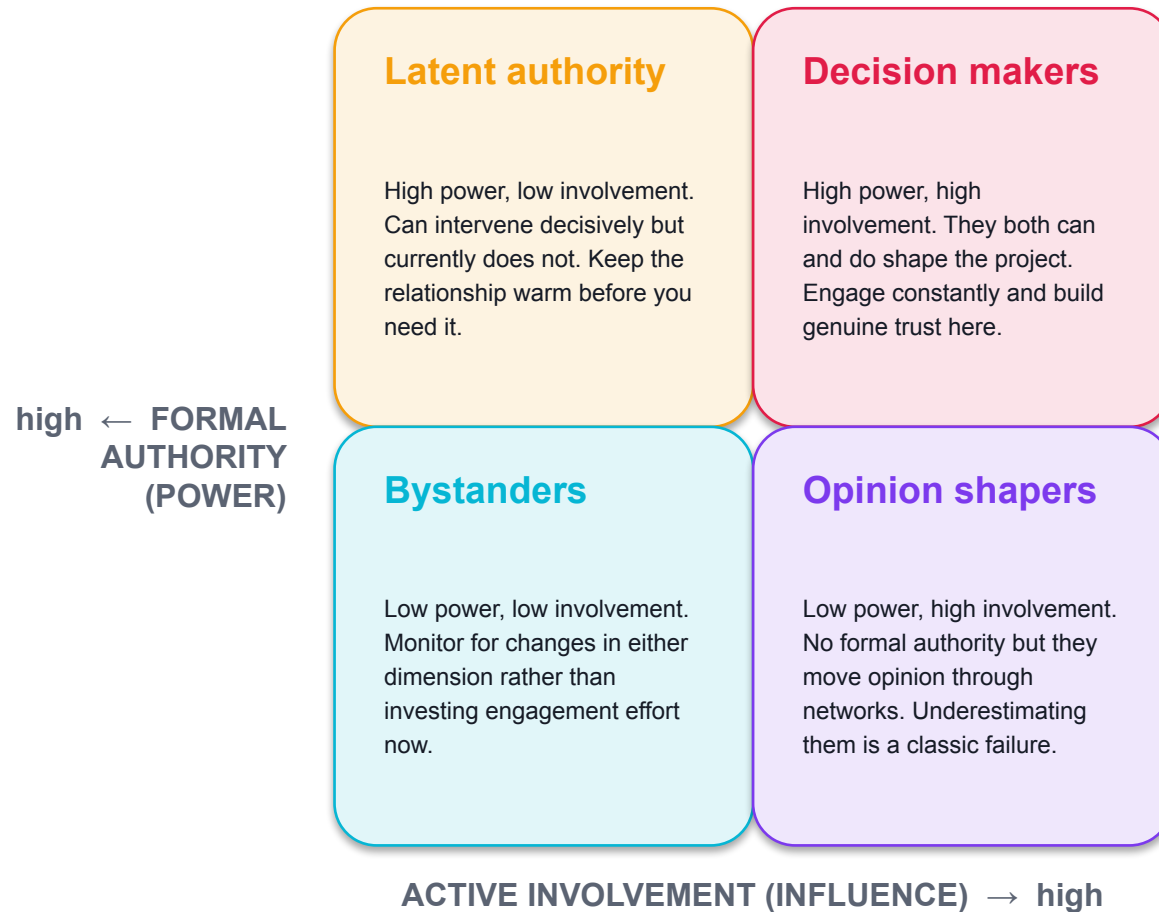
Classifies by where the stakeholder sits relative to the PM — upward, downward, sideward or outward — shaping the engagement style.

Power / Interest Grid



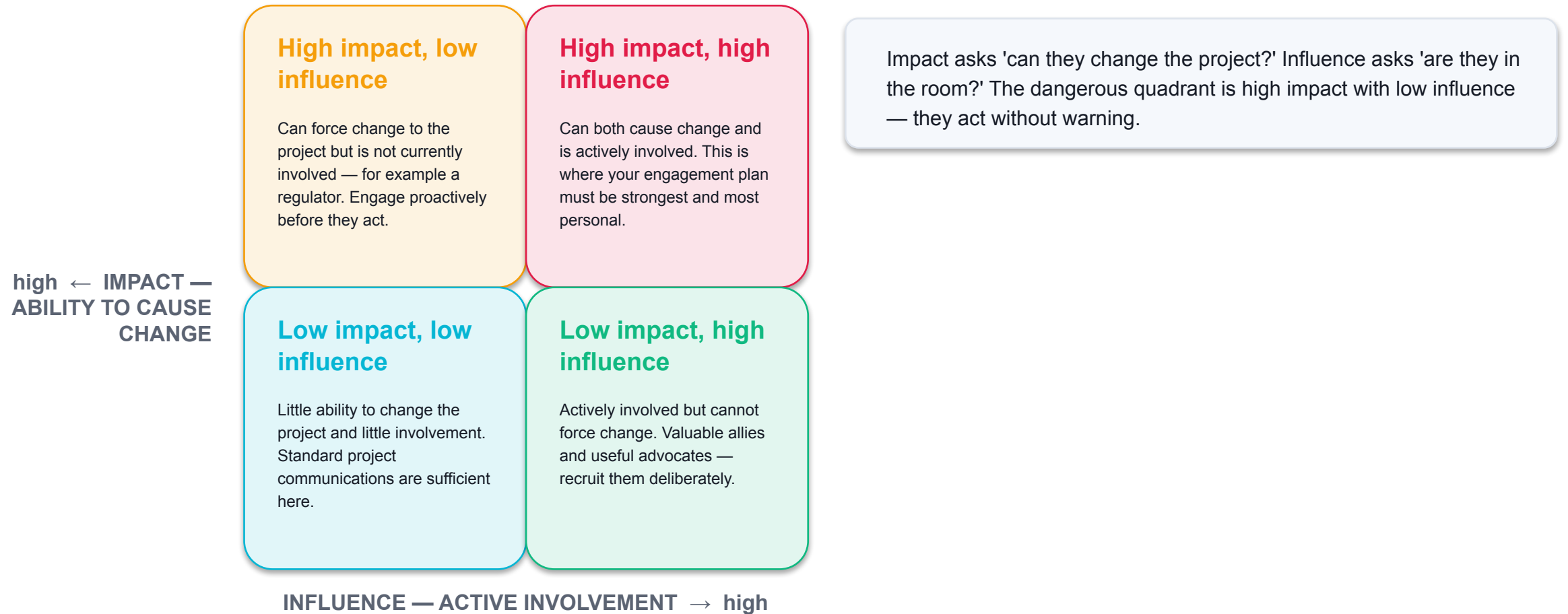
Power is the authority to impose your will. Interest is how much you care about the outcome. A stakeholder's position moves as the project progresses, so re-plot at every phase gate.

Power / Influence Grid



Power and influence are not the same thing. A respected long-serving engineer may have no authority and enormous influence — the power/influence grid exists precisely to expose that person.

Impact / Influence Grid



The Stakeholder Cube — Three Dimensions at Once

1 Why a cube

Two-dimensional grids lose information: two stakeholders can share a quadrant yet need completely different engagement approaches.

2 Dimension one: power

The stakeholder's level of authority to impose decisions on the project or withhold what the project needs.

3 Dimension two: interest

The degree to which the stakeholder cares about and attends to the project's outcome.

4 Dimension three: attitude

Whether the stakeholder is supportive, neutral or resistant — the dimension flat grids most often omit.

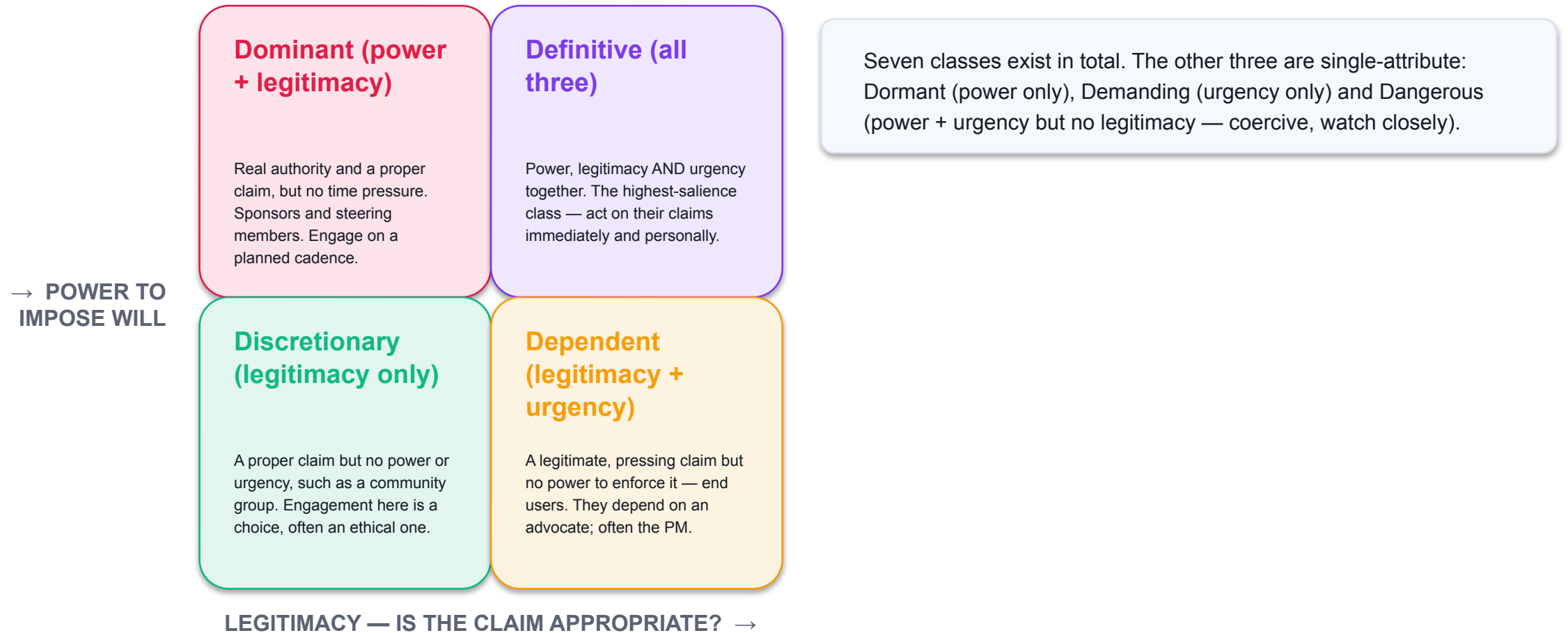
5 What it reveals

A high-power, high-interest, resistant stakeholder is a very different problem from a high-power, high-interest supporter.

6 How to use it

Treat each cell as a distinct engagement archetype and write a tailored strategy per archetype in the engagement plan.

Salience Model — Power, Legitimacy, Urgency



The Seven Salience Classes in Full

Class	Attributes held	Typical example	Engagement response
Dormant	Power only	A regulator not yet engaged	Monitor; the power can activate at any time
Discretionary	Legitimacy only	A community or charity group	Engage by choice; usually a reputational decision
Demanding	Urgency only	A vocal individual user	Acknowledge but do not let noise set priority
Dominant	Power + legitimacy	Sponsor, steering committee	Formal, planned, regular engagement
Dangerous	Power + urgency	A coercive group applying pressure	Manage carefully; escalate; involve governance
Dependent	Legitimacy + urgency	End users needing an advocate	Give them a voice through a powerful sponsor
Definitive	Power + legitimacy + urgency	Customer at a crisis point	Immediate, personal, highest-priority attention

Directions of Influence

1 Upward

Senior management, the sponsor and the steering committee. Engage with concise, decision-ready information and never surprise them in public.

2 Downward

The project team and specialists providing knowledge or skills. Engage by clarifying purpose, removing impediments and empowering decisions.

3 Sideward

Peers of the project manager and other middle managers who control the resources you borrow. Engage through negotiation and reciprocity.

4 Outward

Suppliers, users, government departments and the public. Engage through formal agreements, structured consultation and compliance evidence.

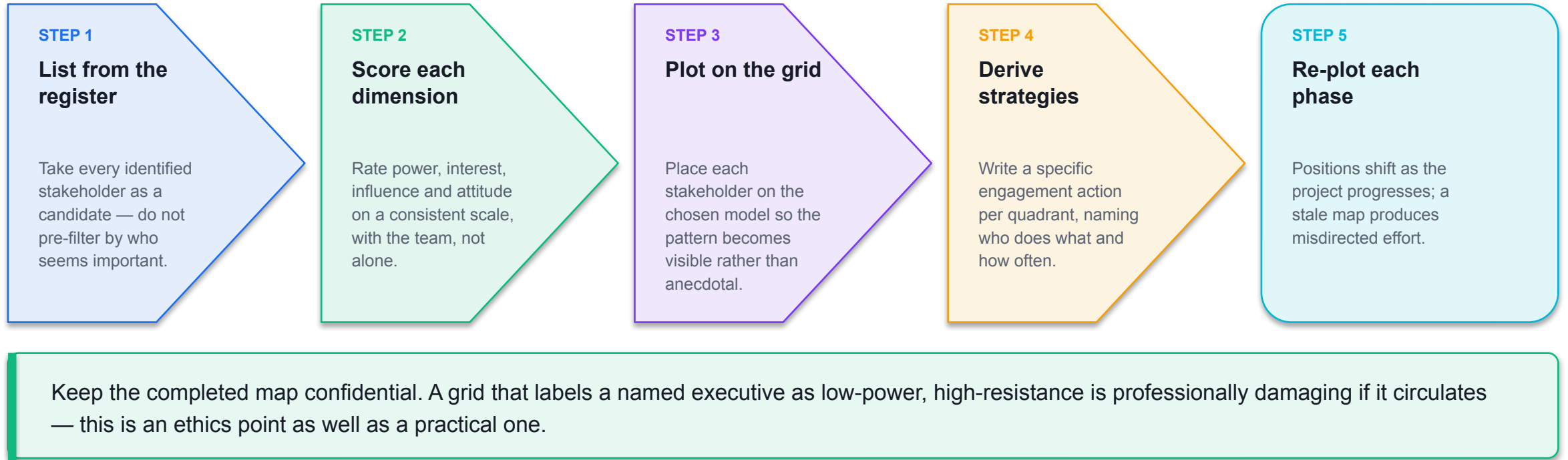
5 Why direction matters

The same message needs a different form in each direction; a status report that satisfies the sponsor will not motivate the team.

6 Where PMs under-invest

Sideward influence is the most commonly neglected direction, and it is where resource conflicts are actually resolved.

Stakeholder Mapping, Step by Step

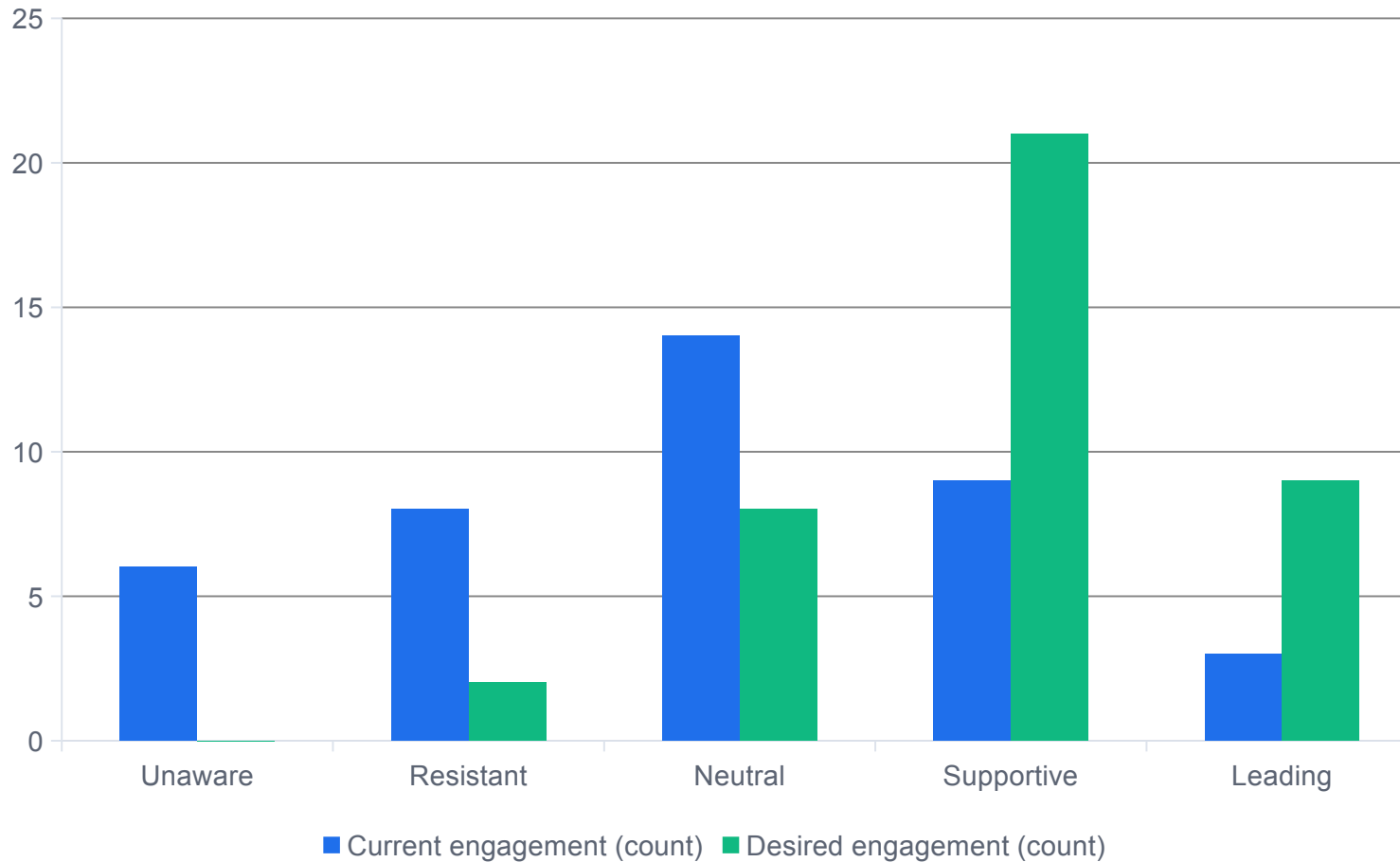


Stakeholder Engagement Assessment Matrix (SEAM)

Stakeholder	Unaware	Resistant	Neutral	Supportive	Leading
Operations Director		C		D	
Finance Manager			C	D	
End-user representative				C / D	
External regulator	C		D		
IT Infrastructure lead			C		D

C marks current engagement, D marks desired. Every gap between C and D is an action item for the stakeholder engagement plan. Where C equals D, protect the position rather than invest further.

Engagement Levels Across the Stakeholder Population



WHAT IT TELLS YOU

- Six unaware stakeholders is an identification and communication failure, not an attitude problem — inform them first.
- Eight resistant stakeholders need root-cause analysis: resistance usually signals an unaddressed expectation, not obstinacy.
- The desired profile deliberately keeps two resistant: total consensus is unrealistic and often means dissent is hidden.
- Moving fourteen neutrals to supportive is the highest-leverage engagement investment available on this project.

Monitor Stakeholder Engagement

1 What the process does

Monitors project stakeholder relationships and tailors strategies for engaging stakeholders through modification of plans and approaches.

2 Key input: the SEAM

The current-versus-desired matrix is re-scored regularly; movement in the wrong direction is an early warning signal.

3 Watch the issue log

A rising count of unresolved stakeholder issues predicts a drop in engagement long before anyone says so openly.

4 Watch participation

Declining attendance at reviews, slow sign-offs and delegated attendance are behavioural signals of falling engagement.

5 Monitor satisfaction explicitly

ECO People T6 requires monitoring internal and external customer satisfaction and expectations — and responding as needed.

6 Outputs

Work performance information, change requests, and updates to the plan, the register, the issue log and the risk register.

ECO 2026 People Task 5 — Align Stakeholder Expectations

Most project conflict is not disagreement. It is two people holding different definitions of done.

Categorize stakeholders

Group them so expectations can be surfaced and reconciled by group rather than one conversation at a time.

Identify expectations

Ask directly what each group believes success looks like — do not infer it from the requirements document.

Facilitate alignment

Bring conflicting parties into the same conversation and converge them on one agreed definition.

Facilitating an Expectation-Alignment Session



ECO People T5 also lists 'organize and act on mentoring opportunities'. Alignment sessions are where a PM can mentor stakeholders in how project trade-offs actually work.

ECO People Task 6 — Manage Stakeholder Expectations

1 Identify customer expectations

Distinguish internal customers (other departments receiving the output) from external customers (paying clients and end users) explicitly.

2 Align outcomes continuously

Maintain the fit between what is being built and what customers expect, rather than confirming alignment only at acceptance.

3 Monitor satisfaction

Use surveys, net promoter scores, review attendance and complaint rates as running indicators of customer satisfaction.

4 Respond as needed

The enabler explicitly requires response, not just measurement — a satisfaction survey nobody acts on is worse than none.

5 Manage expectation drift

Expectations inflate over a long project; periodic re-confirmation against the charter prevents a scope surprise at acceptance.

6 Under-promise deliberately

Committing to what you can reliably deliver protects trust far more than an optimistic commitment that slips.

Building Trust and Influence Without Authority

1 Trust is the engagement multiplier

ECO People T4 explicitly requires building trust and influencing stakeholders to accomplish project objectives — it is an assessed skill.

2 Competence

Demonstrate that your estimates, forecasts and status reports have been accurate before; credibility is built from a track record.

3 Consistency

Behave the same way under pressure as you do in calm periods; inconsistency is what stakeholders read as risk.

4 Transparency

Report bad news early and unvarnished. A PM who hides a slip once is never fully believed again.

5 Reciprocity

Help peers with their problems before you need help with yours — this is the currency of sideward influence.

6 Follow-through

Close every commitment visibly. Small kept promises compound into the influence you need when the stakes are large.

Exam Traps: Stakeholders and Engagement

IF THE QUESTION SAYS

A previously unknown stakeholder appears mid-project with requirements



THE BEST ANSWER IS

Add them to the stakeholder register and analyse them first, before assessing the impact of their requirements.

IF THE QUESTION SAYS

A key stakeholder is resistant and you must decide what to do



THE BEST ANSWER IS

Meet them to understand the root cause of the resistance. Do not escalate, exclude, or simply add more reports.

IF THE QUESTION SAYS

You are asked which model uses power, legitimacy and urgency



THE BEST ANSWER IS

The salience model. Power/interest, power/influence and impact/influence are all two-dimensional grids.

IF THE QUESTION SAYS

Stakeholders disagree about what the project should deliver



THE BEST ANSWER IS

Facilitate a discussion to align expectations against the charter — alignment first, change request only if truly needed.

Communication Channels and Why Team Size Matters

Communication channels

$$N \times (N - 1) / 2$$

N is the number of people. Channels grow quadratically, which is why small teams communicate so much more cheaply.

Team of 5

$$5 \times 4 / 2 = 10$$

Ten channels — everyone can plausibly stay in sync through informal conversation alone.

Team of 10

$$10 \times 9 / 2 = 45$$

Forty-five channels. Informal alignment starts failing; explicit communication planning becomes necessary.

Adding one person to 10

$$55 - 45 = 10 \text{ new channels}$$

The eleventh person adds ten channels, not one — the arithmetic behind 'adding people to a late project makes it later'.

Exam tip: if a question adds or removes people, compute channels before and after and report the DIFFERENCE — that is usually what is being asked, not the absolute number.

Agile Considerations for Starting Well

1 The whole-team approach

All skills needed to deliver sit inside one team, removing the hand-off delays and the finger-pointing that cross-functional silos create.

2 Small teams

Typically three to nine people, keeping communication channels low enough that alignment happens naturally rather than by process.

3 Co-location

Physical or deliberate virtual co-location maximises the osmotic communication that makes small teams fast.

4 Close customers

A customer representative available continuously replaces long requirement documents with short conversations and fast feedback.

5 Information radiators

Big visible charts — task boards, burndowns, impediment lists — in the team space so status is pulled, not reported.

6 Osmotic communication

Useful information overheard in a shared space; team members absorb context without anyone scheduling a meeting to transmit it.

Traditional Management vs Servant Leadership

TRADITIONAL COMMAND AND CONTROL

The manager holds the plan and issues the assignments

- The manager assigns tasks to named individuals
- Decisions escalate up and instructions come back down
- Status is reported to the manager who consolidates it
- The manager owns the estimates and commits on the team's behalf
- Problems are solved by the manager and communicated as decisions
- Performance is managed individually against assigned tasks

SERVANT LEADERSHIP

The leader removes impediments so the team can lead itself

- The team pulls work from a prioritised backlog
- Decisions are made at the level with the most information
- Status is visible to everyone on a radiator, always current
- The team estimates and commits to what it believes it can do
- The leader's first question is 'what is blocking you?'
- Performance is a team outcome, improved through retrospectives

Servant leadership: the practice of leading by understanding and addressing the needs and development of team members, in order to enable the highest possible team performance.

Self-Organising Teams

1 The definition

A cross-functional team in which people fluidly assume leadership as needed to achieve the team's objectives, rather than by fixed title.

2 Cross-functional by design

The team collectively holds every skill required to take an item from idea to done, so no external hand-off is needed.

3 Leadership is situational

Whoever has the most relevant expertise leads that decision, then hands leadership on — authority follows knowledge.

4 The leader still matters

Self-organising does not mean unled: the servant leader sets boundaries, protects focus and removes impediments from outside.

5 What it requires

Psychological safety, a clear goal, real decision authority and time — teams do not become self-organising by being told to be.

6 Why it is faster

Decisions are made where the information already is, eliminating the escalate-and-wait cycle that dominates traditional structures.

Project Team Roles You Must Distinguish

Role	Accountable for	Not accountable for
Project manager	Integration, the plan, stakeholder engagement and delivery of objectives	Doing the technical work or approving their own charter
Project sponsor	Funding, the business case, charter approval and removing senior blockers	Day-to-day management of the project team
Team members	Producing the deliverables and estimating their own work	Setting priority between competing business needs
Product owner	Backlog priority, maximising value and representing the customer voice	Telling the team how to build it or managing their performance
Scrum master / facilitator	Process health, removing impediments, coaching and shielding the team	Assigning work, estimating for the team or setting priority
Functional manager	Supplying skilled resources and developing their people's capability	Directing project scope or accepting project deliverables

T-Shaped People and Why Teams Want Them

1 What T-shaped means

Deep expertise in one discipline — the vertical stroke — plus working competence across several others, the horizontal stroke.

2 Individual value and versatility

T-shaped people contribute their specialism at full depth while still being able to pick up adjacent work when the queue demands it.

3 Flexibility for the organisation

Teams can rebalance around a bottleneck without waiting for a new hire or a formal resource request.

4 Avoids work stoppages

Reduces the key-person dependency that halts a whole team when one specialist takes leave or resigns.

5 Built deliberately

Train and coach team members to become T-shaped through pairing, rotation and shadowing — it does not happen by accident.

6 The contrast

I-shaped specialists create hand-off queues; generalists without depth create quality problems. T-shaped is the balance.

Team Charter and Ground Rules — What Goes In

Section	What the team agrees	Example agreement
Shared values	The principles the team holds itself to	We surface bad news early and without blame
Behaviour guidelines	How members treat each other day to day	One conversation at a time; no interrupting
Communication and tools	Which channel is used for what and expected response times	Chat for the day, email for decisions, board for status
Decision-making	How the team decides and what needs consensus	Fist-of-five; a two-thirds majority carries technical decisions
Performance expectations	What good contribution looks like on this team	Definition of done applies to every item, no exceptions
Conflict resolution	The agreed path when disagreement escalates	Direct conversation, then the team, then the servant leader
Meetings	Time, frequency and channel of each ceremony	Stand-up 09:15 daily, 15 minutes, cameras on
Other agreements	Shared core hours, improvement time and team norms	Core hours 10:00-16:00; Friday afternoon for improvement work

Critical point: the team charter is created WITH the team, not issued to it. A charter written by the PM alone produces compliance, not commitment — and ground rules nobody helped write get broken.

Making Ground Rules Actually Work

1 Make it visible

An electronic document or a poster on the team wall — ground rules filed in a shared drive stop influencing behaviour within a week.

2 Co-create it

Rules the team wrote itself are enforced by the team; rules imposed on the team are enforced only by the manager, and only when watching.

3 Keep it short

Six to ten rules the team can recall beats a thirty-item document nobody has finished reading.

4 Address violations early

ECO People T2 requires managing and rectifying ground-rule violations — an unaddressed breach becomes the new de facto rule.

5 Model it from the top

The project manager who breaks a ground rule cancels it for everyone, regardless of what the poster says.

6 Revisit at retrospectives

Rules that no longer serve the team are removed and new ones added as the team's context changes.

The Kick-Off Meeting: Two Distinct Audiences

ORGANISATIONAL / PUBLIC KICK-OFF

Announcing the project to the wider organisation

- Formally announce project initiation to the organisation
- Share a common understanding of the high-level vision and purpose
- State the business value the project will deliver
- Identify the sponsor, key stakeholders and the project manager
- Present high-level items drawn from the project charter
- Signal executive commitment so the project is taken seriously

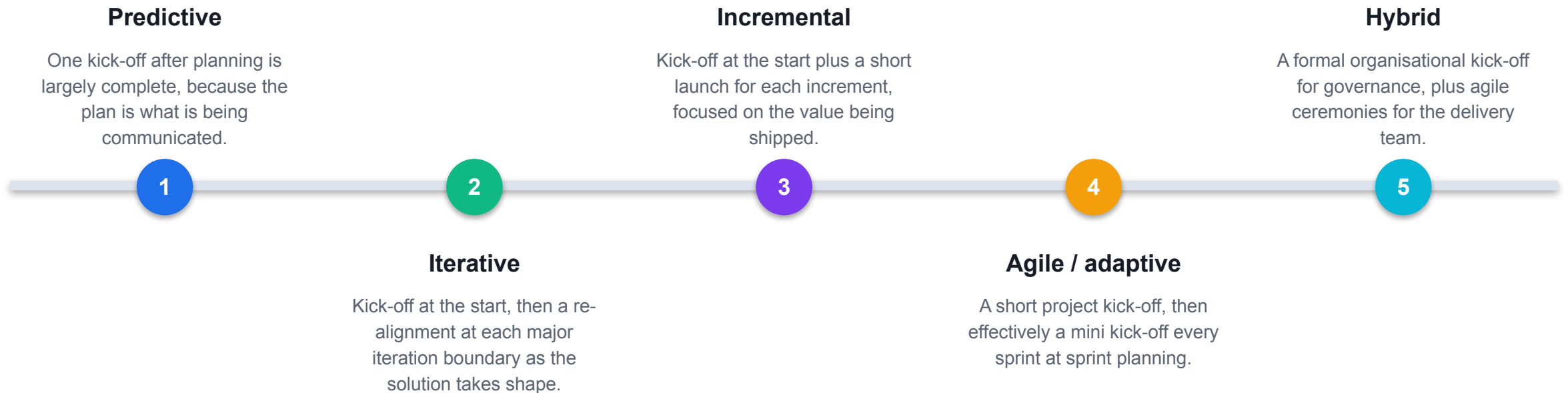
INTERNAL / TEAM KICK-OFF

Turning a group of assigned people into a project team

- Give an overview of the project charter and its objectives
- Clarify each team member's role and responsibilities
- Present the results of planning efforts completed so far
- Initiate the product backlog or the initial work breakdown
- Agree the team charter and ground rules together
- Establish the ceremony cadence and communication norms

Purpose of any kick-off: establish project context, assist in team formation, and align team and stakeholders with the project vision.

When Kick-Off Happens in Each Approach



Exam cue: in predictive projects the kick-off usually occurs at the END of planning; in adaptive projects it occurs at the start of execution and repeats each iteration.

Exam Traps: Charter, Team and Kick-Off

IF THE QUESTION SAYS

You are newly assigned as PM and asked what to do first



THE BEST ANSWER IS

Read the project charter and the business documents to understand objectives, then identify stakeholders.

IF THE QUESTION SAYS

A team member asks who assigns work on an agile team



THE BEST ANSWER IS

Nobody assigns it. The team self-organises and pulls work from the prioritised backlog.

IF THE QUESTION SAYS

The team is repeatedly missing agreed stand-up times



THE BEST ANSWER IS

Address the ground-rule violation with the team and revisit the team charter together — do not escalate first.

IF THE QUESTION SAYS

A predictive project asks when the kick-off should be held



THE BEST ANSWER IS

At the end of planning, once there is an integrated plan worth communicating to stakeholders.

Topic 2 Recap — Start the Project

1 Integration is the PM's job

Unifying scope, schedule, cost, quality, risk and stakeholders so they deliver one coherent business value.

3 Business documents justify

Business case, cost-benefit analysis and benefits management plan exist before the project and outlive it.

5 Financial metrics rank options

Bigger is better for NPV, IRR, ROI and BCR; smaller is better for payback period and opportunity cost.

7 Analyse before you engage

Power/interest, power/influence, impact/influence, the cube and the salience model each answer a different question.

9 Start the team deliberately

Co-create the team charter and ground rules; build T-shaped, self-organising capability through servant leadership.

2 The charter authorises

Signed by the sponsor, it empowers the PM, states measurable objectives and anchors all later change control.

4 Know the six value drivers

Financial gain, new customers, social benefit, first to market, improvement and regularization.

6 Identify stakeholders early

Use the OBS, document analysis and brainstorming, and record everything in the stakeholder register.

8 Align, then manage expectations

ECO People T5 and T6: surface expectations, converge them, then monitor customer satisfaction and respond.

10 Kick off with purpose

Establish context, form the team and align everyone with the project vision before execution begins.

Hands-on Labs — Topic 2

1 Lab 05. Business Case and Cost-Benefit Analysis

Process T3 · LO1 · 25 min in class + 25 min self-study

Produces: Business case, NPV/ROI/payback model

2 Lab 06. Project Charter

Process T1 · LO1 · 20 min in class + 25 min self-study

Produces: Approved project charter

3 Lab 07. Stakeholder Register, Power/Interest Grid and Salience Model

People T4, T5 · LO4 · 25 min in class + 25 min self-study

Produces: Stakeholder register, Power/interest grid, Salience classification

4 Lab 08. Team Charter, Ground Rules and Shared Vision

People T1, T2 · LO3 · 30 min in class + 15 min self-study

Produces: Team charter, Ground rules, Vision statement

HANDS-ON ACTIVITY

Lab 05. Business Case and Cost-Benefit Analysis

Case study: Contoso Training Portal Upgrade. Process T3 · LO1. Produces Business case, NPV/ROI/payback model.

25 min in class

1 State the business problem in one sentence and the opportunity in one more.

2 List three options including 'do nothing', with the cost and benefit of each.

3 Build the cash-flow table: initial investment, annual benefit, and running net position across five years.

4 Calculate payback period, NPV at a 6% discount rate, ROI and the benefit-cost ratio using the Statistics tool.

5 Identify the non-financial benefits and state how each will be measured.

6 Write the recommendation, naming the benefits owner accountable after the project closes.

DELIVERABLE

Business case, NPV/ROI/payback model

TOOL

Statistics

alfredang.github.io/novastats/

Lab 06. Project Charter

Case study: Contoso Training Portal Upgrade. Process T1 · LO1. Produces Approved project charter.

20 min in class

1 Draft the charter's purpose statement and its measurable success criteria — numbers, not adjectives.

2 Record high-level requirements and explicitly list what is OUT of scope.

3 Name the sponsor, project manager and key stakeholders, and state the authority delegated to the project manager.

4 Add the summary milestone schedule and the high-level budget with its confidence range.

5 Record assumptions, constraints and the top five high-level risks.

6 Define the approval criteria and who signs the charter into effect.

DELIVERABLE

Approved project charter

Lab 07. Stakeholder Register, Power/Interest Grid and Salience Model

Case study: Contoso Training Portal Upgrade. People T4, T5 · LO4. Produces Stakeholder register, Power/interest grid, Salience classification.

25 min in class

1 List every stakeholder affected by the portal upgrade, including those who merely believe they are affected.

2 For each, record interest, influence, expectations and their current attitude toward the project.

3 Plot all stakeholders on a power/interest grid and assign the four strategies: manage closely, keep satisfied, keep informed, monitor.

4 Re-classify the top eight using the salience model — power, legitimacy and urgency — and note where it disagrees with the grid.

5 Complete a SEAM recording Current (C) and Desired (D) engagement for each key stakeholder.

6 Write the engagement action that moves each stakeholder from C to D, with an owner and a date.

DELIVERABLE

Stakeholder register, Power/interest grid, Salience classification

HANDS-ON ACTIVITY

Lab 08. Team Charter, Ground Rules and Shared Vision

Case study: Contoso Training Portal Upgrade. People T1, T2 · LO3. Produces Team charter, Ground rules, Vision statement.

30 min in class

1 Draft the team's shared vision in a single sentence every member can repeat without reading it.

2 Define roles and responsibilities so no member is unclear what they own.

3 Agree working hours, core overlap for the distributed members, and expected response times.

4 Set the ground rules: meeting norms, decision rules, and how disagreement is raised and settled.

5 Define the Definition of Ready and Definition of Done the team will hold each other to.

6 Agree the conflict protocol and the escalation path before any conflict actually occurs.

DELIVERABLE

Team charter, Ground rules, Vision statement

03

TOPIC

Plan the Project

Building the integrated plan — scope, schedule, cost, quality, resources, communications, risk and procurement.

- 1 The integrated project management plan
- 2 Scope: requirements, WBS and baselines
- 3 Schedule: activities, dependencies, critical path
- 4 Cost: estimating, budget and reserves
- 5 Quality: planning and problem-solving tools
- 6 Resources, communications and procurement
- 7 Risk: identification, analysis and response

Why planning is the heart of the Process domain

A plan is a decision record, not a document

Reduces rework

Decisions made deliberately up front cost a fraction of the same decisions made reactively during delivery.

Creates alignment

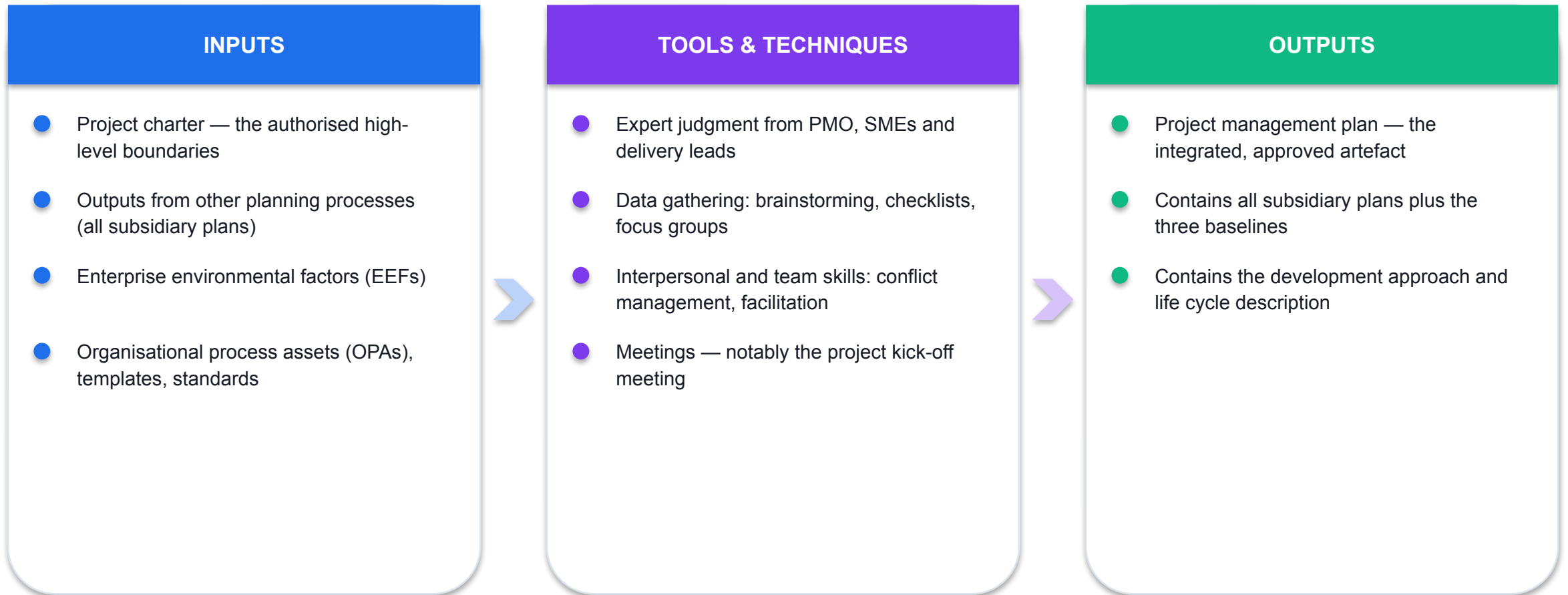
A shared plan lets thirty people make consistent choices without asking the project manager each time.

Enables control

You cannot detect variance without a baseline to measure the actual result against.

Develop Project Management Plan — ITTO

Integration is the only knowledge area whose job is to reconcile the other twelve — the PM owns the seams between the plans.



What the project management plan actually enables

1 Execute

Tells the team what work to do, in what order, with which resources and to which acceptance criteria.

2 Monitor

Defines the metrics, the reporting cadence and the thresholds that trigger a management conversation.

3 Control

Establishes the baselines against which variance is measured and the change control process that protects them.

4 Close

Defines the exit criteria, handover conditions and the evidence needed for formal acceptance of deliverables.

5 Guardrails

Sets the boundaries inside which the team may act autonomously without escalating every decision upward.

6 Tailoring

Records what was deliberately omitted or simplified for this project's size, risk and regulatory profile.

The subsidiary plans inside the integrated plan

1 Scope management plan

How scope is defined, validated and controlled, and how the WBS will be produced and maintained.

2 Schedule management plan

The scheduling method, tool, units of measure, accuracy thresholds and control rules.

3 Cost management plan

Estimating accuracy, units, control thresholds, earned value rules and reporting formats.

4 Quality management plan

Standards to be met, quality roles, review activities and the improvement approach.

5 Resource management plan

How people and physical resources are identified, acquired, developed and released.

6 Communications plan

Who needs what information, in what format, how often, and through which channel.

7 Risk management plan

Methodology, roles, budgeting, timing, categories, probability and impact definitions.

8 Procurement plan

What is bought versus built, contract types, and how sellers are selected and managed.

9 Stakeholder engagement plan

The strategies used to move each stakeholder from current to desired engagement level.

Plan-driven versus collaborative planning

PLAN-DRIVEN (PREDICTIVE)

The plan is authored, then approved, then followed

- The project manager drafts the plan and seeks formal sign-off
- Detail is pushed as far forward as the information allows
- Change after baseline requires a formal change request
- Strong on auditability, regulatory evidence and contract control
- Risk: the plan ages badly when the environment moves fast

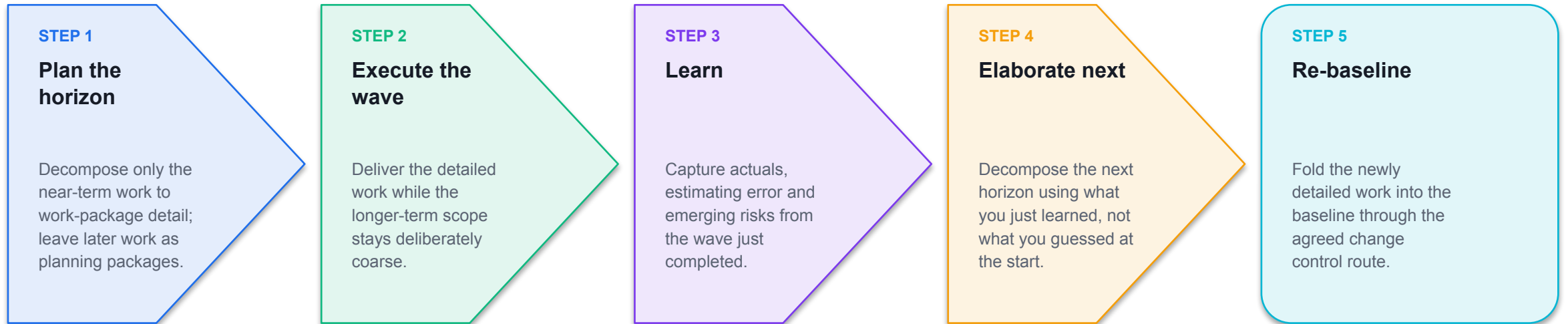
COLLABORATIVE (ADAPTIVE)

The plan is co-created continuously by those doing the work

- The product owner sets objectives from customer needs and value
- The team estimates and sequences its own work each iteration
- Team members are the local domain experts on how work gets done
- The PM or scrum master facilitates rather than dictates
- Risk: without discipline, planning degrades into improvisation

Most real projects are hybrid: governance-level milestones are plan-driven while the work inside each milestone is planned collaboratively by the delivery team.

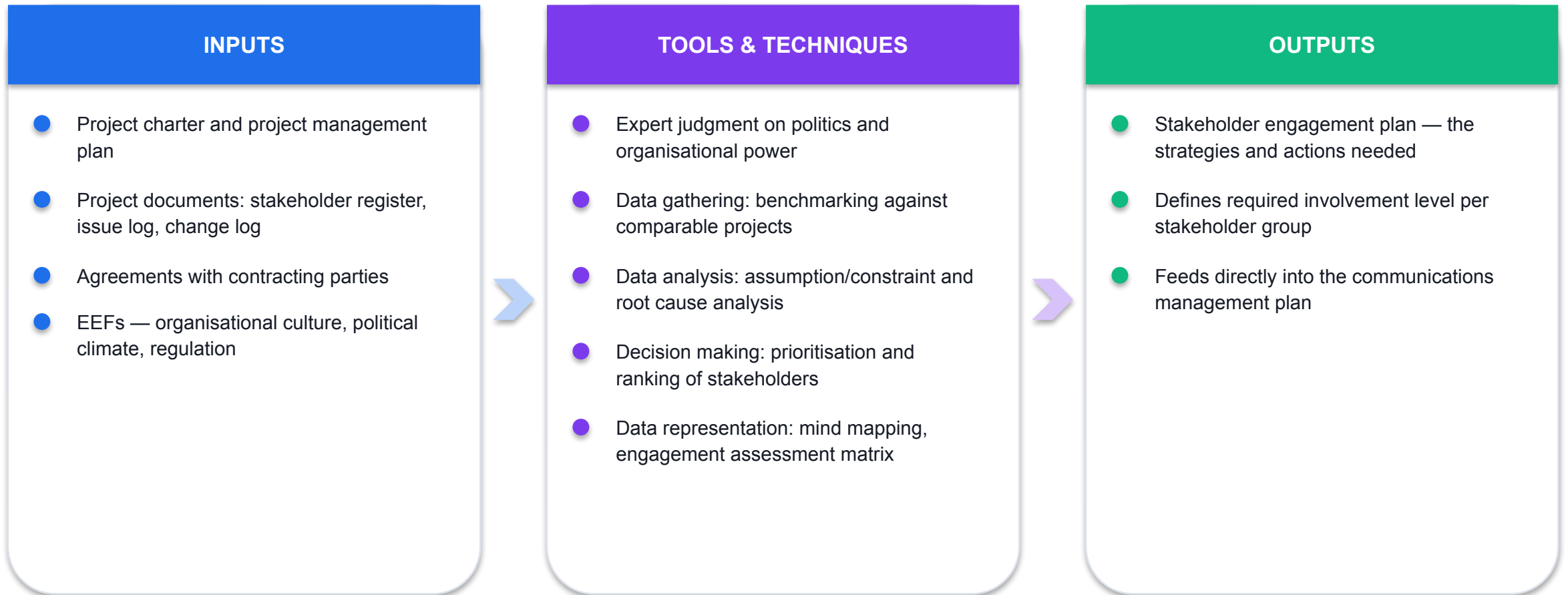
Rolling wave planning — progressive elaboration in practice



Rolling wave applies to BOTH predictive and adaptive approaches — it is applied to work packages, planning packages and release planning, and is the standard answer when a question says 'not enough information is available to plan the later phases'.

Plan Stakeholder Engagement — ITTO

Engagement is planned, not improvised: you decide deliberately who must be moved from resistant to supportive, and how.



Stakeholder Engagement Assessment Matrix (SEAM)

Level	What it looks like	Typical intervention
Unaware	Does not know the project exists or that it will affect them	Awareness communication; briefing pack; named contact point
Resistant	Aware of the project and actively opposed to the change	Surface the objection, negotiate, find the win, escalate if needed
Neutral	Aware but neither supportive nor opposed; will not help	Show the personal and unit-level benefit; invite to a demo
Supportive	Aware and in favour, will speak well of the project	Keep informed, use as a reference and as an internal advocate
Leading	Actively engaged in ensuring the project succeeds	Give real influence: steering seat, sponsorship of a workstream

Mark C for the CURRENT level and D for the DESIRED level in each row. Where C and D differ you have an engagement gap that needs a named action and an owner — not a hope.

Exam focus — planning and stakeholder engagement

IF THE QUESTION SAYS

A stakeholder is surprised by a project decision that affected them



THE BEST ANSWER IS

Review and update the stakeholder engagement plan and the communications plan — the omission is a planning defect

IF THE QUESTION SAYS

Later phases cannot be planned in detail yet



THE BEST ANSWER IS

Apply rolling wave planning; use planning packages now and elaborate them when information becomes available

IF THE QUESTION SAYS

The team asks who decides how the work will be done



THE BEST ANSWER IS

The team members do — they are the local domain experts; the PM facilitates and removes impediments

IF THE QUESTION SAYS

Sponsor asks 'what is the project management plan for?'



THE BEST ANSWER IS

It is the approved basis for executing, monitoring, controlling and closing the work — not a status report

Knowledge area 1 of 8 in the planning block

Scope: agree what is in, and prove what is out

Product scope

The features and functions that characterise the product, service or result being delivered.

Project scope

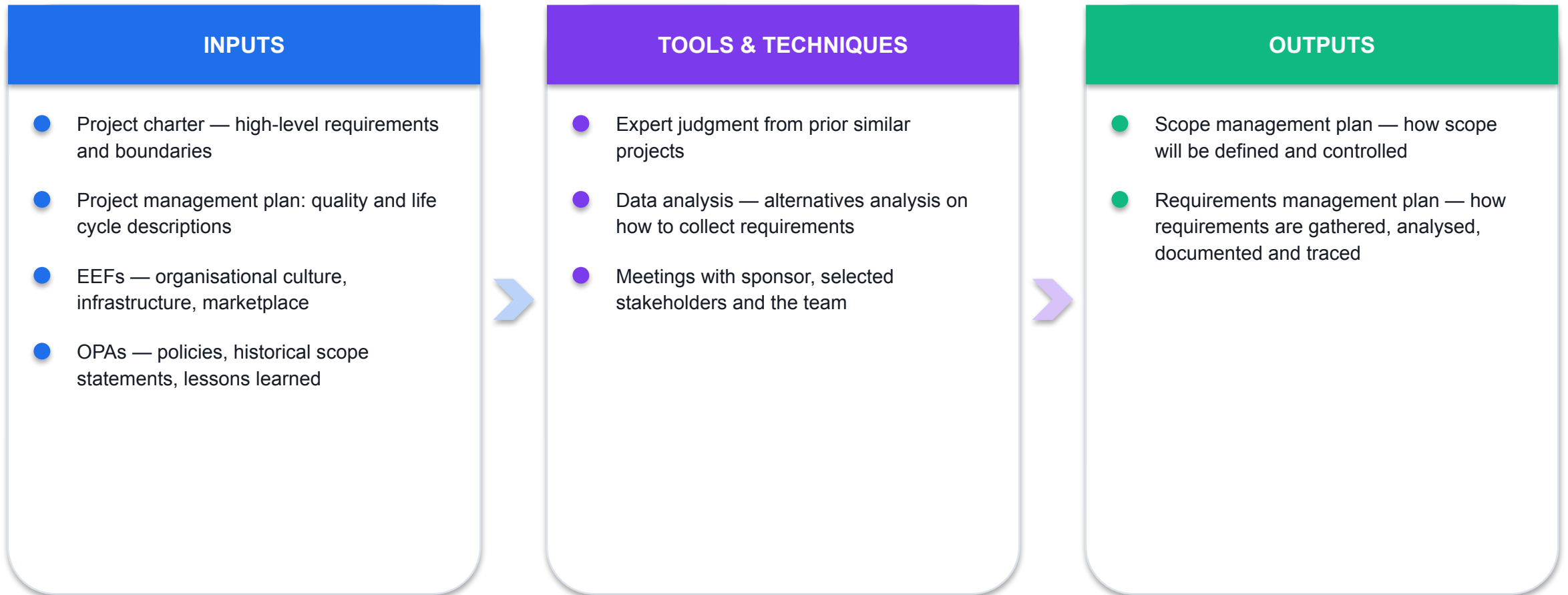
The work that must be performed to deliver that product with the specified features and functions.

Why it matters

Every schedule, cost, quality and risk number in the plan is derived from the agreed scope.

Plan Scope Management — ITTO

These two plans are process documents: they describe how you will do scope work, not what the scope actually is.



What the scope management plan must specify

1 Scope statement preparation

The process used to prepare a detailed project scope statement from the charter and requirements.

2 WBS creation

How the WBS will be created from the detailed scope statement, and the decomposition rules to apply.

3 Baseline approval

How the scope baseline will be reviewed, approved and subsequently maintained.

4 Deliverable acceptance

How formal acceptance of completed deliverables will be obtained and evidenced.

5 Change control

How change requests to the scope statement are raised, assessed, and routed for a decision.

6 Tailoring decisions

Which scope activities are simplified or omitted given this project's size and risk.

Scope planning analysis techniques

Technique	What it does	When you reach for it
Document analysis	Derives requirements from existing documents such as contracts, SOPs, regulations and prior project files	A predecessor system or process already exists on paper
Alternatives analysis	Evaluates possible options or approaches for executing and performing the project work	More than one credible way to deliver the same outcome
Product analysis	Asks questions about a product and forms answers describing its use, characteristics and value	The deliverable is a product rather than a service
Product breakdown	Splits a product and its requirements into components to gain a clear understanding of the work	Complex physical or software product with many subsystems
Systems engineering	Approaches design, integration, management and the life cycle of complex systems in a multidisciplinary way	Highly interdependent, safety- or mission-critical systems

Scope in predictive versus agile delivery

PREDICTIVE SCOPE

Fixed early, defended by change control

- Scope is planned in full at the start of the project
- Captured in a detailed project scope statement and WBS
- Baselined once approved by the sponsor or change board
- Any change during delivery requires a formal change request
- Time and cost flex; scope is held constant wherever possible

AGILE SCOPE

Fixed in the sprint, variable across the release

- Product owner determines and prioritises the product backlog
- Scope may legitimately change between every sprint
- Detail is added just in time, at backlog refinement
- Time (the sprint) and cost (the team) are fixed instead
- Value delivered per iteration is the control mechanism

Exam cue: 'requirements are expected to evolve' points to adaptive; 'requirements are stable and the contract is fixed price' points to predictive.

Agile scope artefacts

1 Product roadmap

Envisions and plans the big picture — product strategy, direction and the value to be delivered over time.

2 Themes and goals

Groups related features under overarching goals so structure and association are visible to stakeholders.

3 Milestones

Dates when major items will occur or be delivered — the fixed points a fluid backlog must still respect.

4 Story map

Organises user stories into functional groups within a narrative flow, so the whole product is visible at once.

5 Product backlog

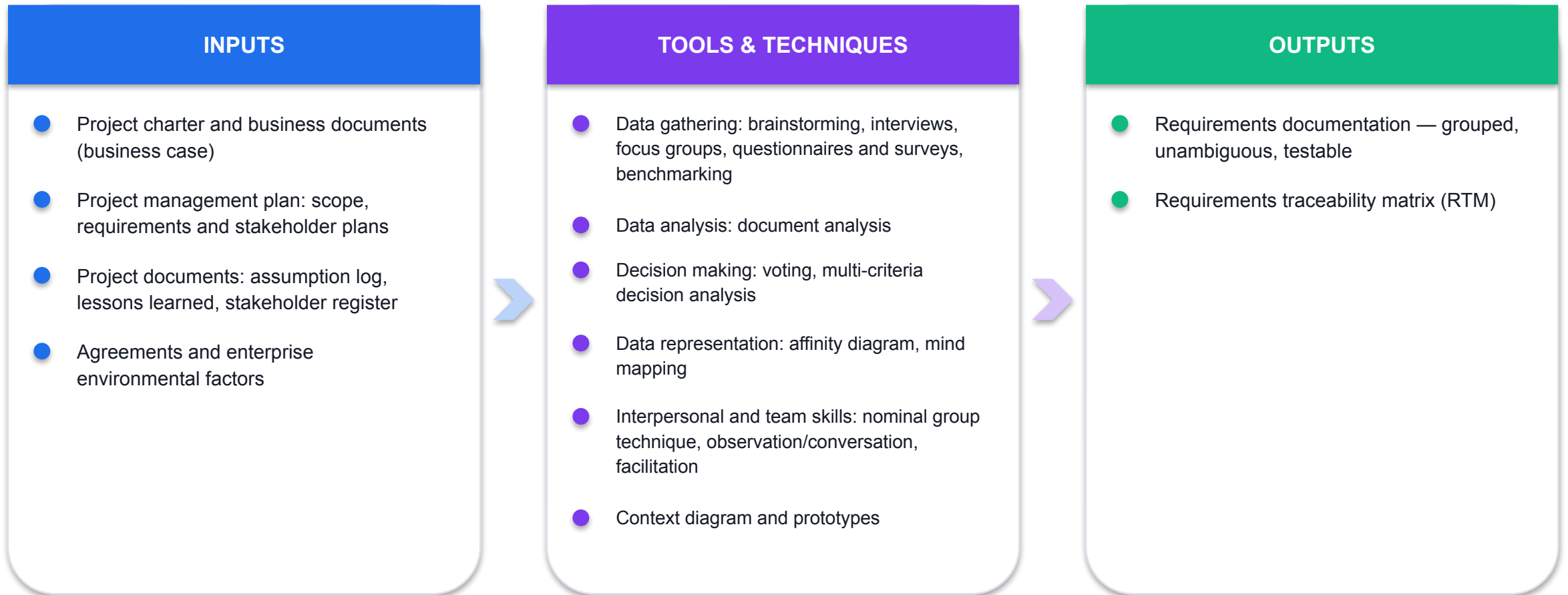
The single ordered list of everything that might be built, owned and prioritised by the product owner.

6 Increment

The working, potentially releasable slice of product produced by each iteration and inspected at review.

Collect Requirements — ITTO

Requirements are the bridge between stakeholder need and deliverable — every later baseline traces back to them.



The requirements management plan

1 What it is

The defined process for how requirements will be gathered, analysed, documented, managed and controlled.

2 Also called

The business analysis plan — in organisations with a formal BA function it is owned jointly with the BA.

3 Configuration

States how requirement changes are initiated, analysed, traced, tracked and approved.

4 Prioritisation

Records the prioritisation method that will be used, such as MoSCoW or a weighted scoring model.

5 Traceability

Defines the traceability structure — which attributes are recorded against each requirement.

6 Metrics

Defines the product metrics that will be used and the rationale for choosing them.

Requirements Traceability Matrix — what each column proves

Column	Contents	Question it answers
Requirement ID	Unique persistent identifier for the requirement	Can I refer to this unambiguously in every other artefact?
Business need	The business case objective or benefit it supports	Why are we building this at all?
Project objective	The charter objective the requirement serves	Is this in scope or is it somebody's pet feature?
WBS deliverable	The work package that produces it	Who is building it and where is it in the schedule?
Product design	The design element or component that realises it	Has anyone actually designed a solution for this?
Test case	The verification or acceptance test	How will we prove it works before handover?
Status	Active, cancelled, deferred, added, approved, assigned	Is this still live and who signed it off?

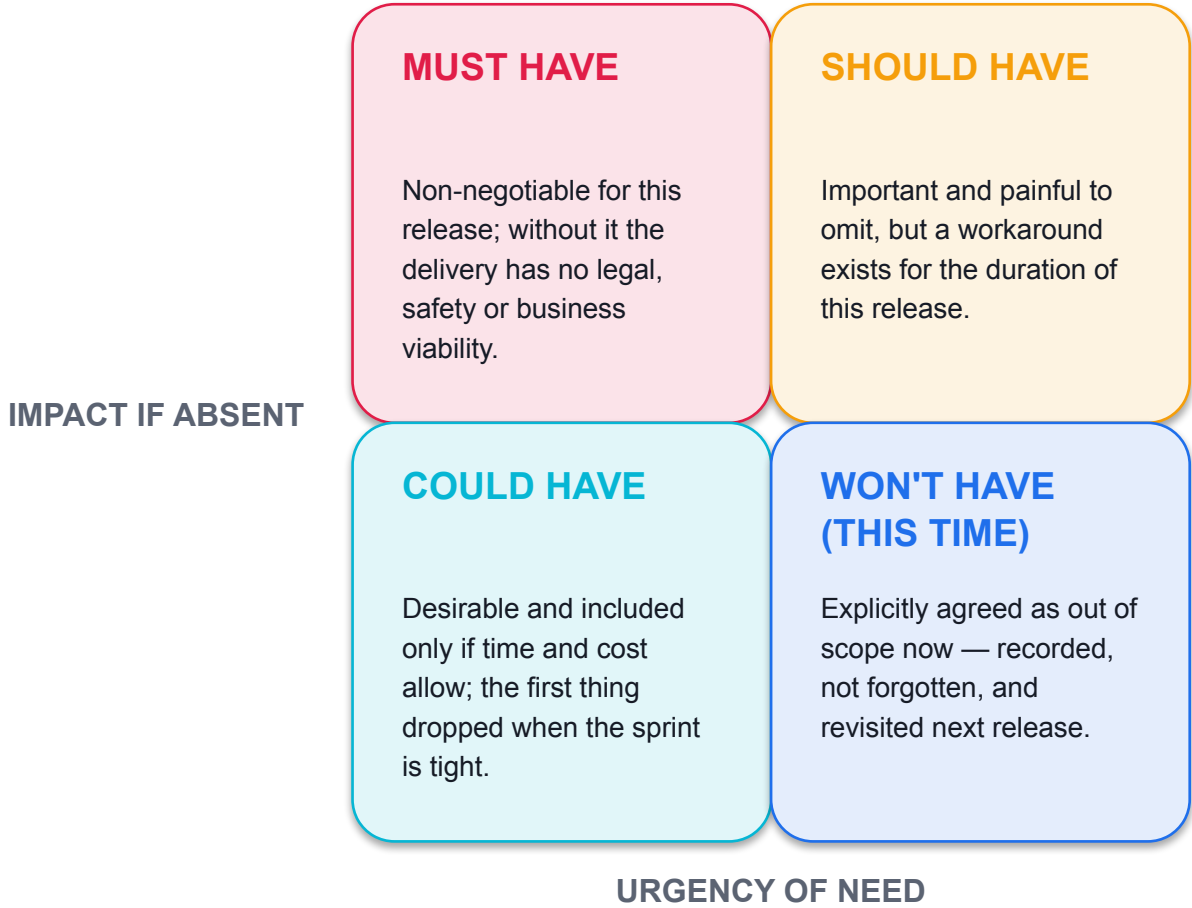
Requirement gathering techniques compared

Technique	How it works	Best for
Brainstorming	Relaxed informal group idea generation using lateral thinking; quantity first, judgement deferred	Wide divergent option generation early in discovery
Interviews	Structured or unstructured one-to-one questioning of stakeholders and SMEs	Sensitive, political or expert-held detail
Focus groups	Moderated discussion with pre-qualified stakeholders and subject matter experts	Attitudes, expectations and reactions
Questionnaires/surveys	Written question sets to large or geographically dispersed audiences	Statistically meaningful breadth
Benchmarking	Compares practices against other organisations to identify best practice	Setting a performance bar to meet or exceed
Nominal group technique	Brainstorm, then rank ideas by individual vote to produce a prioritised list	Converging a large idea pool
Observation (job shadowing)	Watch people do the work in situ	Tacit process detail people cannot articulate
Facilitation	Focused workshops (JAD, QFD) that cross functions	Reconciling conflicting cross-department requirements

Requirement prioritisation techniques

Technique	Mechanics	Benefit
MoSCoW (Dai Clegg)	Sort into Must have, Should have, Could have, Won't have this time	Reaches shared understanding of importance fast
Kano model	Classify features as Basic, Performance, Delighter, Indifferent or Reverse	Separates satisfiers from mere expectations
100-point method	Each stakeholder distributes 100 points across the candidate requirements	Forces trade-offs; exposes true preference
Weighted/multi-criteria	Score each requirement against weighted criteria and rank the totals	Defensible, auditable decision trail
Paired comparison	Compare requirements two at a time and tally the wins	Works when stakeholders cannot rank a long list
Value/effort matrix	Plot benefit against cost of delivery on a 2x2	Surfaces the quick wins and the money pits

MoSCoW as a decision grid



The discipline is in the W: writing down what you are NOT doing is what stops scope creep later. Cap Must-haves at roughly 60% of capacity so there is room to absorb reality.

Backlog and backlog refinement

1 Product backlog

A single ordered list of everything that might be built; the product owner owns its content and order.

2 User stories

Work items describing desired functionality from the user's point of view, with acceptance criteria attached.

3 Refinement (grooming)

Continual work by the product owner before iteration planning to add detail, estimates and order.

4 Joint refinement

Team and product owner refine together during the sprint so the next sprint starts with ready items.

5 Reprioritisation

Allows reorganisation so higher-value items are completed first as the market and feedback change.

6 Emergent detail

Only the top of the backlog needs full detail — the bottom stays coarse until it approaches delivery.

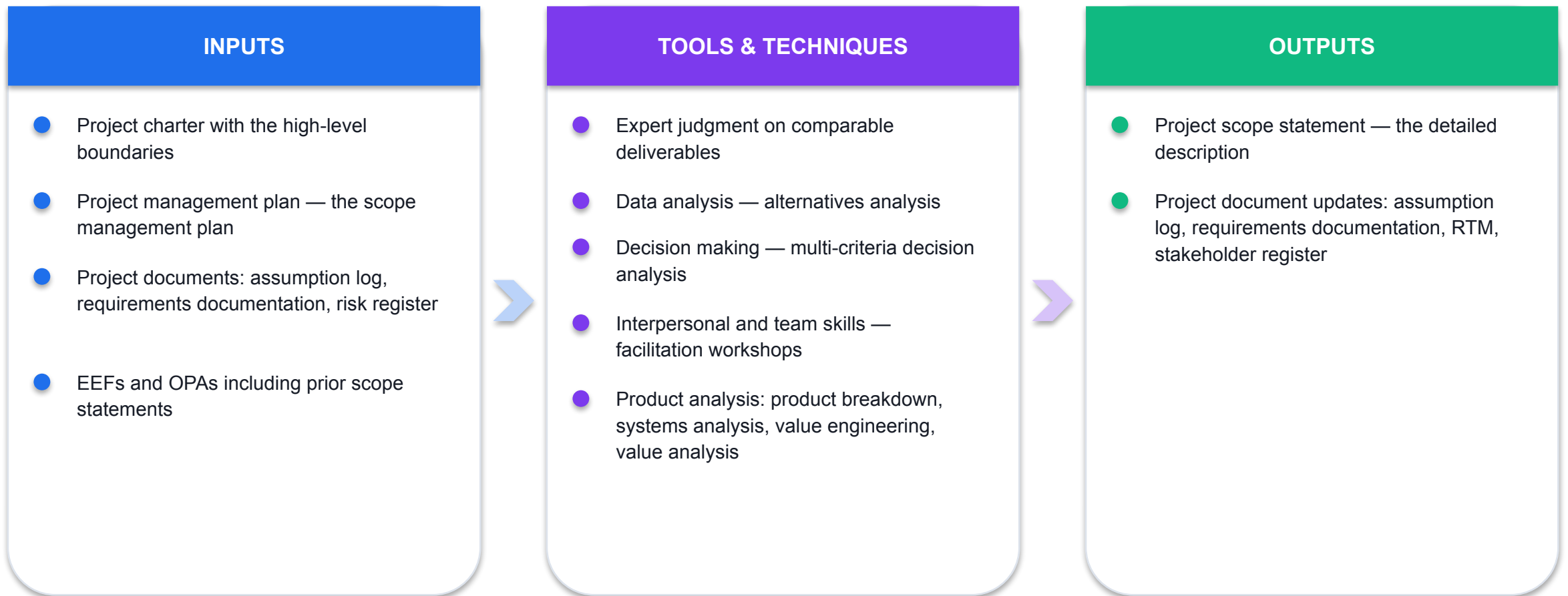
Relative sizing techniques

Technique	Scale	How to use it
T-shirt sizing	XS, S, M, L, XL	Fast, coarse sizing of a large backlog or a roadmap before any detail exists
Story points	Fibonacci: 1, 2, 3, 5, 8, 13, 21	Relative measure of difficulty, effort, volume and uncertainty combined
Planning poker	Simultaneous reveal of point cards	Removes anchoring bias; the discussion of outliers is the real value
Ideal days	Uninterrupted working days	Intuitive for new teams but invites confusion with calendar duration
Affinity estimation	Silent sorting of cards by size	Sizing 100+ backlog items in under an hour

Points are relative and team-specific: 8 points on one team is not 8 points on another. Never use velocity to compare teams.

Define Scope — ITTO

Define Scope converts a long list of requirements into a single agreed statement of what will and will not be delivered.



Anatomy of the project scope statement

1 Product scope description

Progressively elaborated description of the characteristics of the product, service or result.

2 Deliverables

Every unique verifiable product or capability required, including project management reports and documents.

3 Acceptance criteria

The conditions that must be met before the deliverables will be formally accepted by the customer.

4 Exclusions

Explicitly identifies what is out of scope, so stakeholder expectations are managed before delivery starts.

5 Constraints

Limiting factors — fixed budget, imposed dates, mandated technology, contractual provisions.

6 Assumptions

Factors treated as true without proof; each one is a risk with a friendlier name and belongs in the risk register.

Baselines — the approved versions you measure against

1 Scope baseline

Approved project scope statement + WBS + WBS dictionary; changed only via approved change request.

2 Schedule baseline

The approved version of the schedule model, with baseline start and finish dates for each activity.

3 Cost baseline

The approved time-phased budget excluding management reserve; the S-curve you track EV against.

4 Performance measurement baseline

Scope + schedule + cost combined; the single reference for earned value analysis.

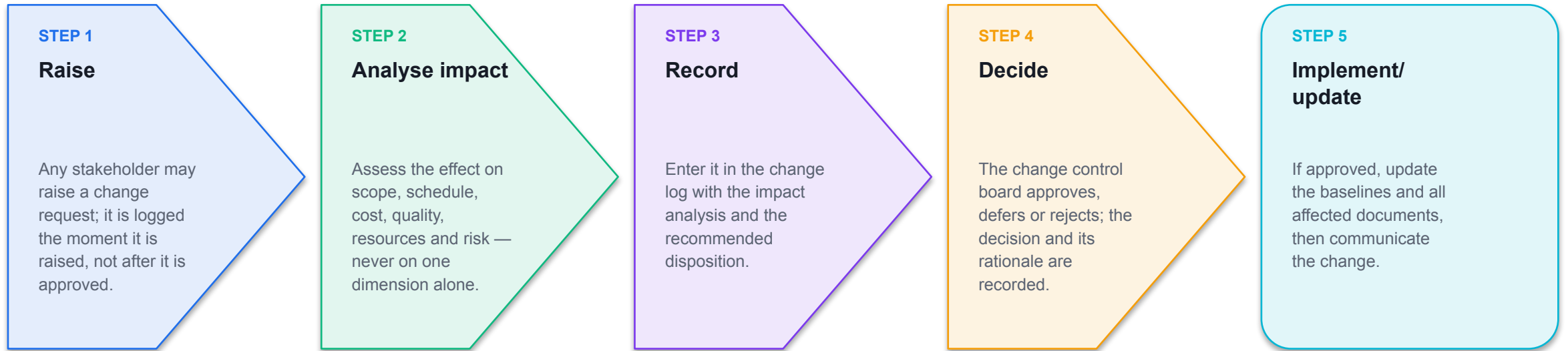
5 Why baseline at all

Variance is meaningless without a fixed reference; an un-baselined project cannot be controlled.

6 Changing a baseline

Only through integrated change control — never by quietly editing the file to match the actual.

Changing a baselined item — the change request route



A rejected change request is still a valuable artefact: it records that the request was considered, so it does not resurface as an informal expectation later in the project.

Scope creep versus gold plating

SCOPE CREEP

Uncontrolled expansion without approval

- Scope or features added that were not planned or approved
- Usually requested by stakeholders, accepted informally by the team
- Impacts cost and schedule that nobody has funded
- Root cause is a weak or bypassed change control process
- Cure: an explicit scope statement plus disciplined change control

GOLD PLATING

Extra work nobody asked for

- The team delivers features the customer did not need or request
- Motivated by pride, boredom or a wish to delight the customer
- Consumes budget and introduces untested, unrequested risk
- Never adds value the customer will pay for or thank you for
- Cure: definition of done, and acceptance criteria as the ceiling

Both destroy the baseline, but the direction differs: creep comes from outside the team, gold plating comes from inside.

Definition of Ready versus Definition of Done

DEFINITION OF READY (DoR)

What must be true before work may START

- Story is written from the user's perspective with clear value
- Acceptance criteria are written and agreed with the product owner
- Dependencies identified and either resolved or scheduled
- The story is small enough to complete inside one iteration
- The team has estimated it and understands what 'done' means

DEFINITION OF DONE (DoD)

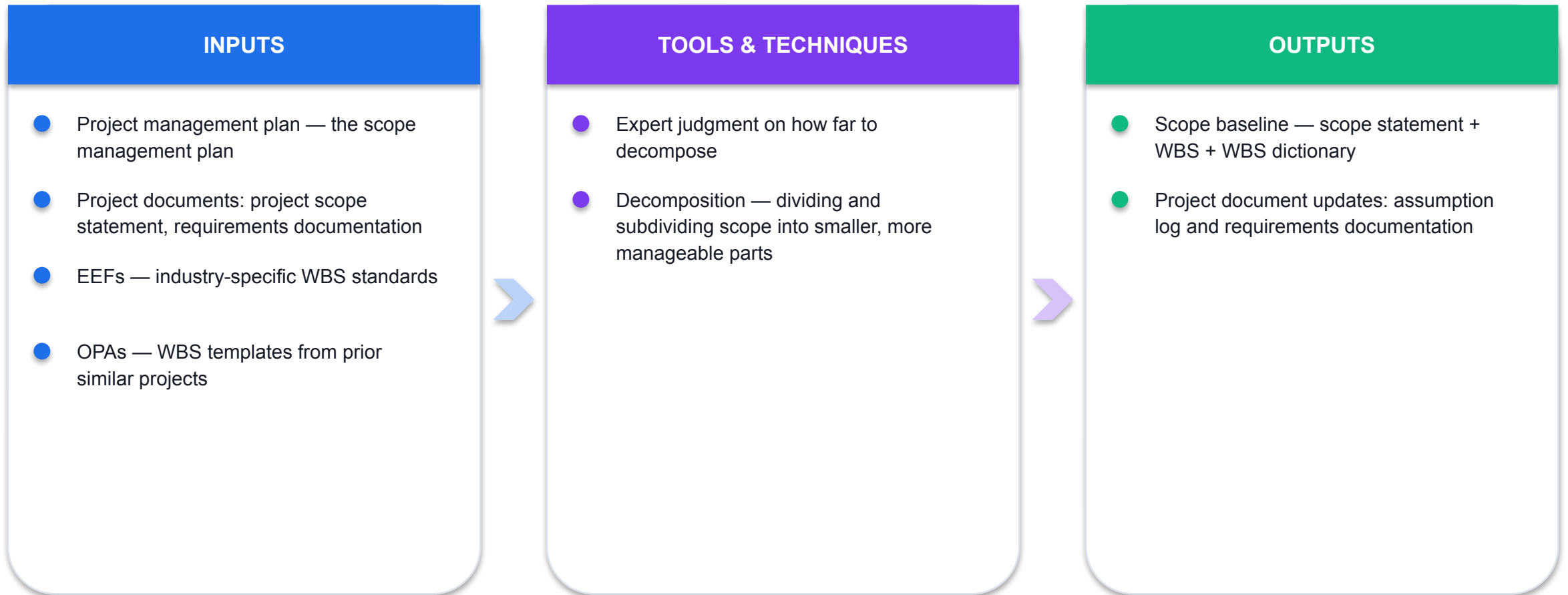
What must be true before work is COMPLETE

- Code written, peer reviewed and merged to the mainline
- Unit, integration and acceptance tests written and passing
- Documentation updated; no known defects above agreed severity
- Deployed to the agreed environment and demonstrated
- Accepted by the product owner against the acceptance criteria

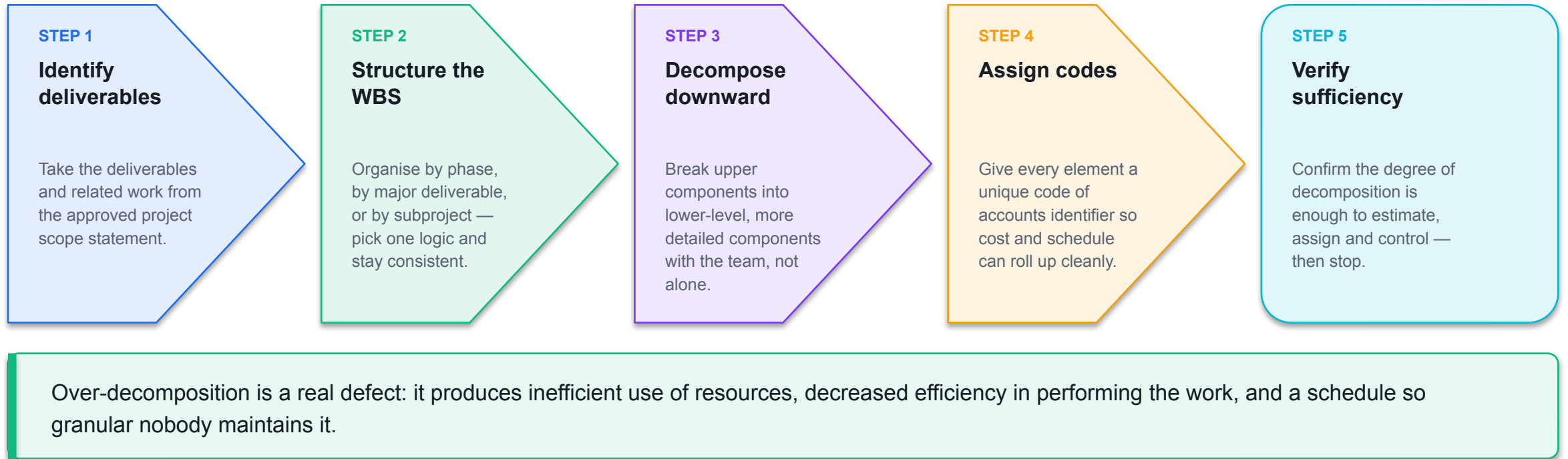
DoD is the team's quality contract with itself and is the single most effective agile defence against gold plating and against carrying unfinished work between sprints.

Create WBS — ITTO

The WBS is deliverable-oriented, not activity-oriented: its elements are nouns (things produced), not verbs.



Decomposition — from deliverable to work package



Work packages, planning packages and the 100% rule

1 Work package

The lowest level of the WBS — small enough to estimate cost and duration reliably and assign to one owner.

2 Control account

A management control point where scope, budget and schedule are integrated and compared to earned value.

3 Planning package

A control account component with known work but no detailed activities yet — rolling wave's placeholder.

4 The 100% rule

The WBS includes 100% of the work defined by the scope, and captures all deliverables — internal and external.

5 Nothing extra

Work not in the WBS is outside the project scope; this is what makes the WBS a scope-creep detector.

6 8/80 heuristic

A common rule of thumb: a work package should be between 8 and 80 hours of effort. It is guidance, not doctrine.

WBS dictionary — the detail behind each work package

Field	Why it is there
WBS code identifier	Links the package to the code of accounts so cost and schedule data roll up to the right control account
Description of work	Removes ambiguity about what the package includes and, critically, what it excludes
Assumptions and constraints	Records the conditions the estimate depends on so the estimate can be challenged fairly
Responsible organisation	Names the accountable unit or person for the package — no orphan work
Schedule milestones/activities	Connects the package to the schedule model and its associated activities
Resources required	States the people, equipment and materials the package consumes
Cost estimates	The package-level cost that bottom-up rolls into the project budget
Quality requirements and acceptance criteria	Defines what 'complete and correct' means for this package
Technical references / agreement information	Points to the standards, drawings or contract clauses that govern the work

Exam focus — scope

IF THE QUESTION SAYS

A stakeholder asks for a small extra feature mid-sprint



THE BEST ANSWER IS

Predictive: raise a change request. Agile: add it to the product backlog for the product owner to prioritise — never mid-sprint

IF THE QUESTION SAYS

The team added a feature the customer never asked for



THE BEST ANSWER IS

That is gold plating; it is not a benefit, it is an unapproved scope change consuming budget

IF THE QUESTION SAYS

You need to know if a requirement was delivered and tested



THE BEST ANSWER IS

Consult the requirements traceability matrix — it links need to deliverable to test to acceptance

IF THE QUESTION SAYS

A question mentions 'approved scope statement + WBS + dictionary'



THE BEST ANSWER IS

That combination is the definition of the scope baseline

IF THE QUESTION SAYS

Work is not shown anywhere in the WBS



THE BEST ANSWER IS

It is out of scope. The 100% rule means the WBS contains all of the work and only the work of the project

Recap — scope planning

1 Two plans first

Plan Scope Management yields the scope management plan and the requirements management plan — process, not content.

3 The RTM is the spine

It traces need to objective to deliverable to test to acceptance and status.

5 The WBS is deliverables

Decompose to work packages; the WBS dictionary carries the detail; obey the 100% rule.

2 Collect then define

Requirements documentation and the RTM come before the project scope statement can be written.

4 Prioritise explicitly

MoSCoW, Kano and the 100-point method turn 'everything is critical' into a defensible order.

6 Baseline and defend

Scope baseline changes only by approved change request; watch for creep from outside and gold plating from inside.

Knowledge area 2 of 8 in the planning block

Schedule: sequence the work, then face the maths

Derived, not chosen

A credible schedule is calculated from scope, dependencies and resources — it is not a date you were told.

The constraint

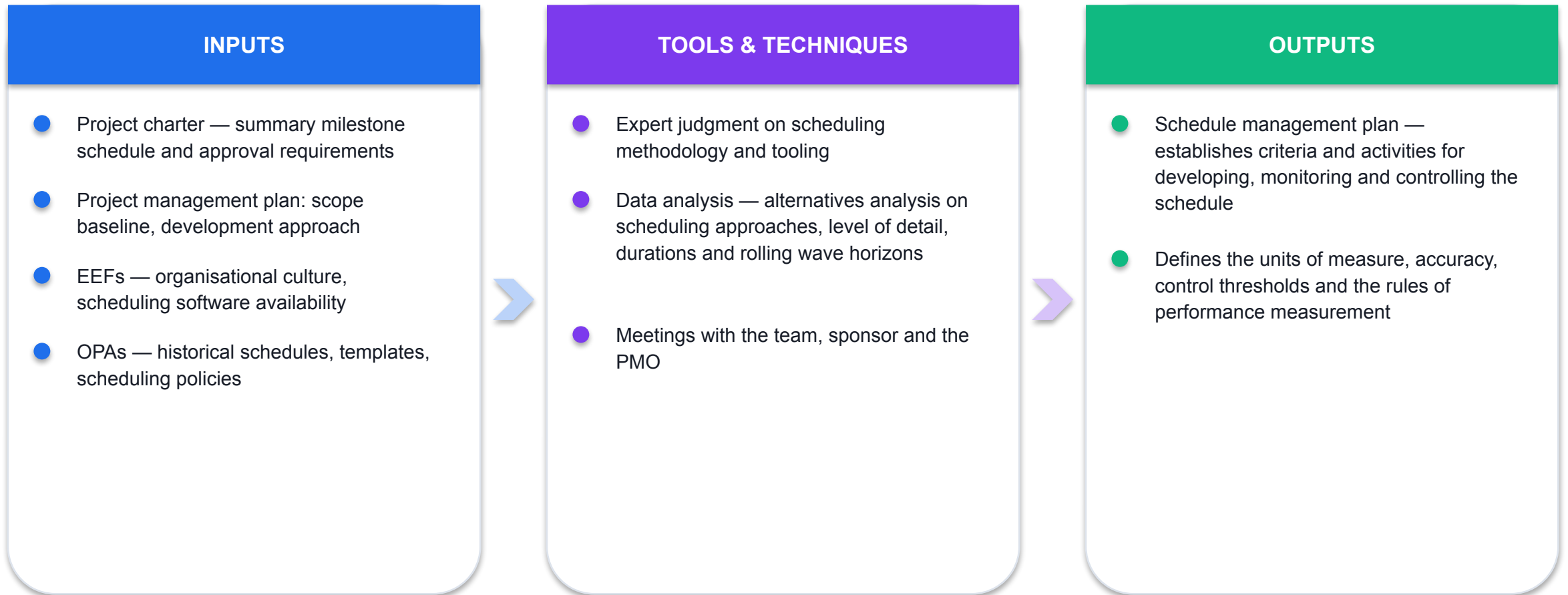
The critical path defines the shortest possible duration; everything else has float to spend.

Adaptive equivalent

Agile replaces the Gantt with cadence, velocity and a prioritised backlog — but still forecasts.

Plan Schedule Management — ITTO

This plan may be formal or informal, broadly framed or highly detailed — tailored to the needs of the project.



What the schedule management plan specifies

1 Scheduling method

Critical path method, critical chain, or an iteration-based cadence — chosen deliberately, stated explicitly.

2 Scheduling tool

The software of record; if two tools exist you will eventually have two conflicting truths.

3 Level of accuracy

The acceptable range for activity duration estimates, including a contingency amount.

4 Units of measure

Hours, days or weeks for time; staff hours for effort — mixing them silently is a classic estimating error.

5 Control thresholds

The variance that may be tolerated before an action is required — for example $\pm 10\%$ of planned duration.

6 Reporting formats

The formats and frequency of schedule reports, and the maintenance process for the schedule model.

Predictive versus adaptive scheduling

PREDICTIVE (WATERFALL) SCHEDULING

Activities decomposed from work packages, then sequenced

- Activities broken down from work packages and put in order
- Dependencies, durations and resources drive the calculated dates
- The critical path determines the shortest possible duration
- Once approved and baselined, changes need a change request
- Presented as a Gantt chart, milestone chart or network diagram

ADAPTIVE (AGILE) SCHEDULING

Fixed cadence, variable content, forecast by velocity

- Product owner owns the product backlog and the product roadmap
- The roadmap shows the order features are expected to be delivered
- Iteration length is fixed; the content of each iteration varies
- Forecasts come from measured velocity, not from imposed dates
- Presented as a roadmap, Kanban board, burndown or burnup chart

Both start from benchmarking and historical data: comparing against a similar product or service schedule gives a defensible starting point before detailed analysis.

Schedule presentation formats — choose to fit the audience

1 Roadmap

High-level, time-boxed view of themes and releases for executives who want direction rather than detail.

2 Gantt chart

Bar chart of activities against a calendar; shows duration, overlap and progress at a glance.

3 Milestone chart

Only zero-duration events on a timeline — the right artefact for a steering committee.

4 Network diagram

Nodes and arrows showing logical dependencies; the only format that reveals the critical path.

5 Kanban board

Columns per workflow stage with WIP limits — for continuous flow rather than time-boxed work.

6 Burndown/burnup

Work remaining or work completed over time, compared with the ideal line for the iteration or release.

Kanban board mechanics

1 Visualise

Organise work into tasks on cards so the whole flow is visible to everyone, including stakeholders.

2 Columns as stages

Each column is a workflow stage; tailor the stages to your actual process, not a textbook one.

3 Card information

Display owner, size, blocked status and age at every stage so the board answers questions without a meeting.

4 WIP limits

Cap the number of items allowed in a column; the cap is what converts a to-do list into a flow system.

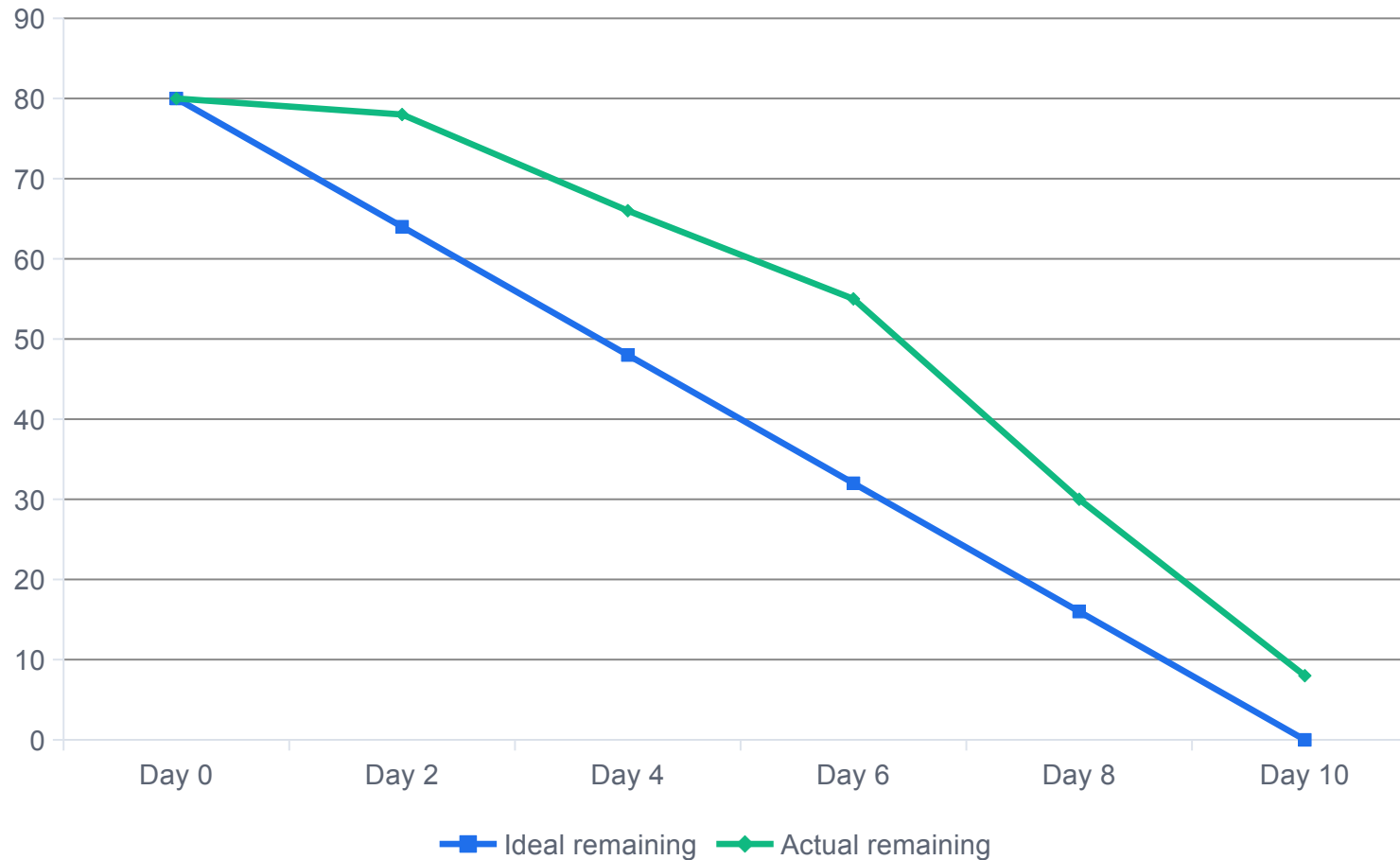
5 Pull, not push

Work is pulled when capacity frees up, rather than pushed onto a team already saturated.

6 Expose bottlenecks

The column that keeps hitting its WIP limit is the constraint — improve there, nowhere else.

Burndown chart — work remaining against the ideal line



WHAT IT TELLS YOU

- Tracks the work to be completed in the current iteration, measured in story points or hours.
- Used to analyse variance against the ideal burndown of work committed to during iteration planning.
- A flat start then a cliff usually means work was integrated late — a definition-of-done problem, not a capacity problem.
- A line above ideal at mid-sprint is the trigger for a scope conversation with the product owner, not overtime.

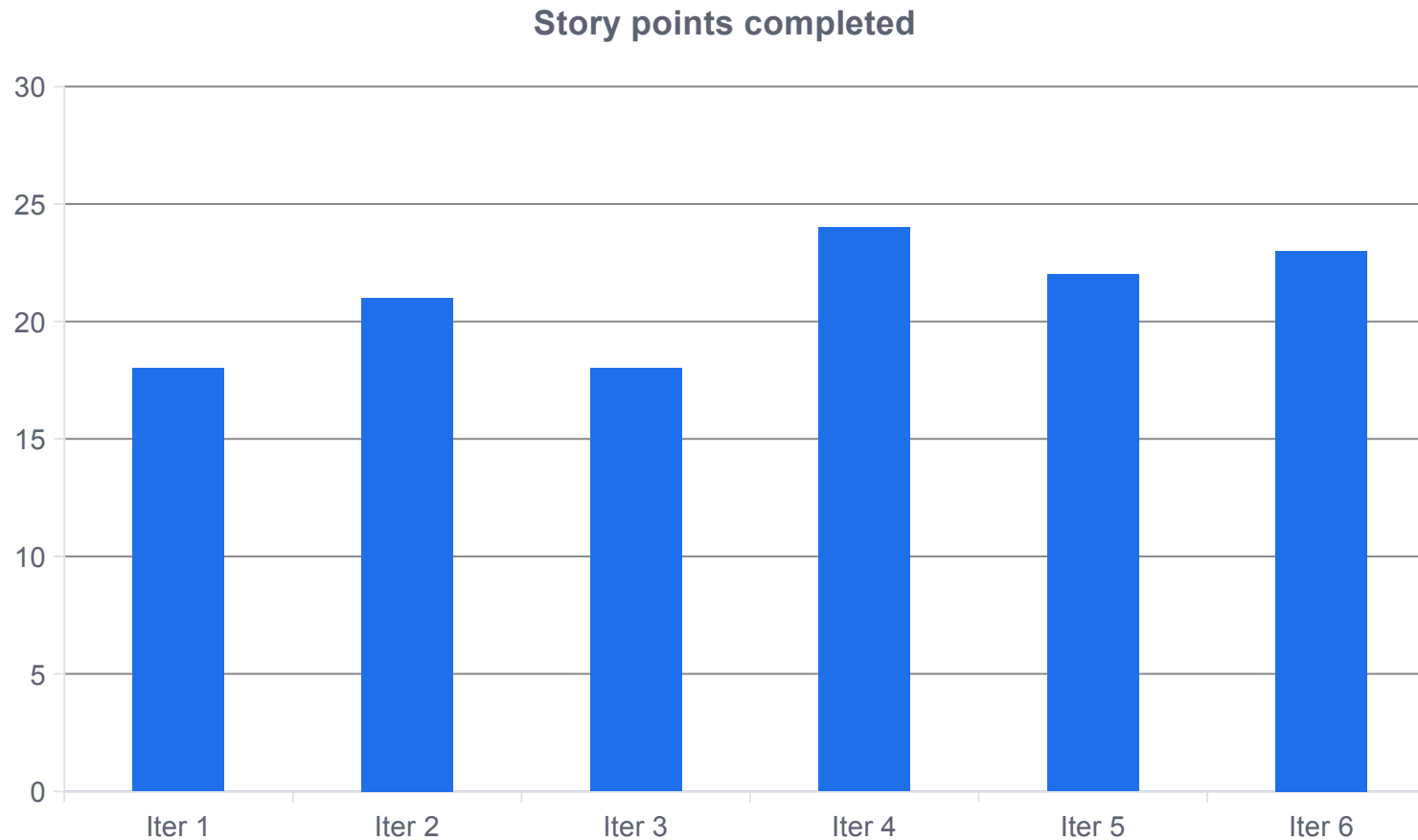
Burnup chart — completed work against total scope



WHAT IT TELLS YOU

- Shows accumulated progress of completed work, updated after each iteration.
- Unlike a burndown, it separates the two causes of a slipping date: slower delivery versus growing scope.
- A rising top line is scope growth made visible — the single most useful chart in a scope-creep conversation.
- The intersection of the two lines is the forecast completion point for the release.

Velocity — the team's measured rate of progress



WHAT IT TELLS YOU

- Velocity is the number of story points completed during an iteration, counted only for items meeting the definition of done.
- Estimate it initially, then modify it using actuals from subsequent iterations — never negotiate it upward.
- The goal is a constant, predictable velocity from one iteration to the next, not a rising one.
- Forecast: remaining backlog points divided by average velocity gives the number of iterations still required.

Continuous flow metrics

Metric	Definition	What it tells you
WIP (work in progress)	Measure of work started but not yet completed	High WIP means context switching and delayed value
Lead time	Length of time a work item takes to go through the entire process, from request to delivery	What the customer actually experiences as responsiveness
Cycle time	Length of time a work item is actively being worked on	Your team's execution speed once work truly starts
Throughput	Number of items entering or exiting the system per unit of time	Delivery capacity — the basis for flow forecasting
Queue time	Lead time minus cycle time	The waiting; usually the largest and most ignored part of lead time

Little's Law and flow arithmetic

Little's Law

$$L = \lambda \times W$$

Average work in progress equals arrival rate times average lead time — cut WIP and lead time falls.

Average cycle time

$$\text{WIP} \div \text{Throughput}$$

If 12 items are in progress and you finish 3 per week, each takes about 4 weeks.

Flow efficiency

$$\text{Cycle time} \div \text{Lead time}$$

Typically 5-15% in knowledge work; the rest is queueing, not working.

Iterations remaining

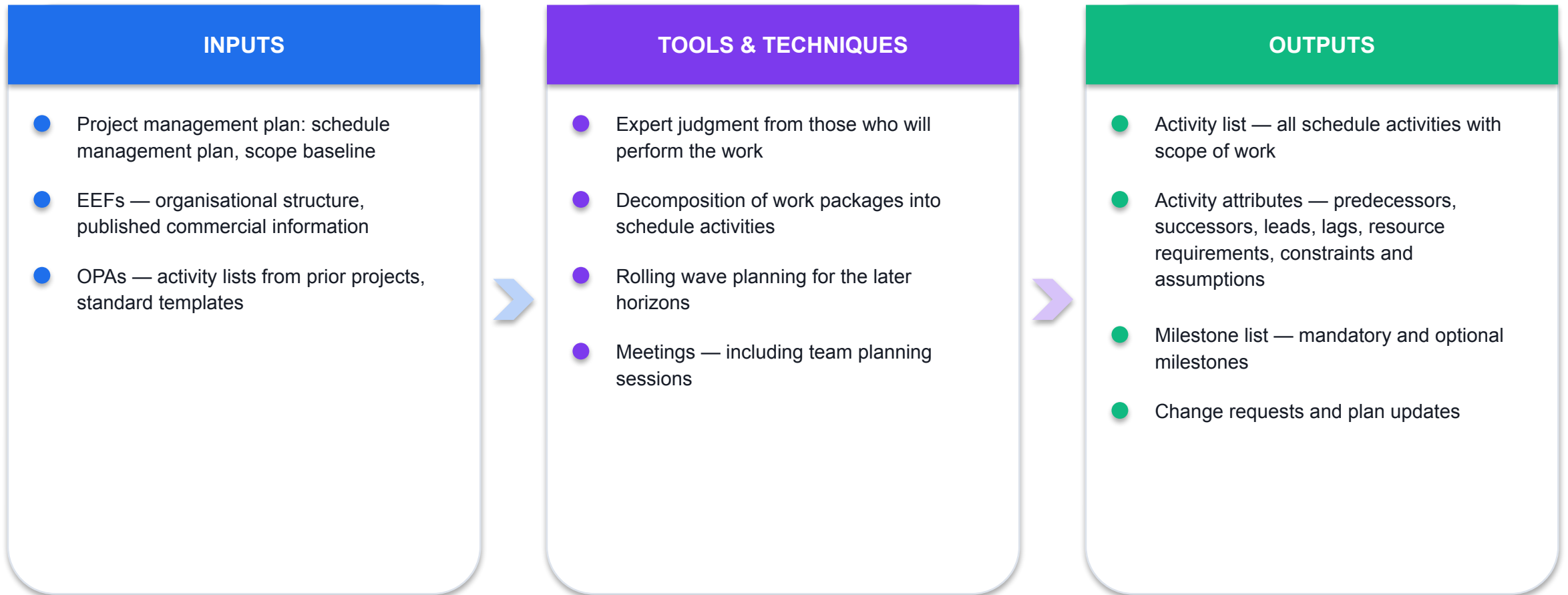
$$\text{Backlog points} \div \text{Velocity}$$

The agile forecast — a range, using best and worst recent velocity.

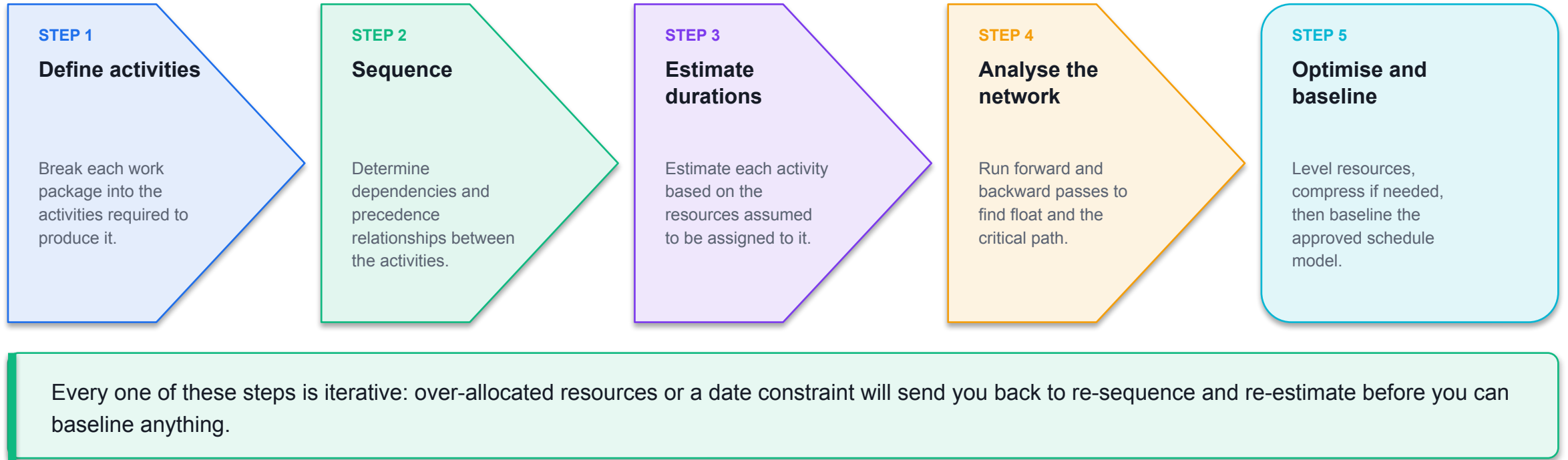
Little's Law is why WIP limits work: with throughput roughly fixed by team capacity, the only lever you control for faster delivery is starting less work at once.

Define Activities — ITTO

Work packages are nouns; activities are verbs. Define Activities is where the deliverable-oriented WBS becomes executable work.



The schedule development sequence



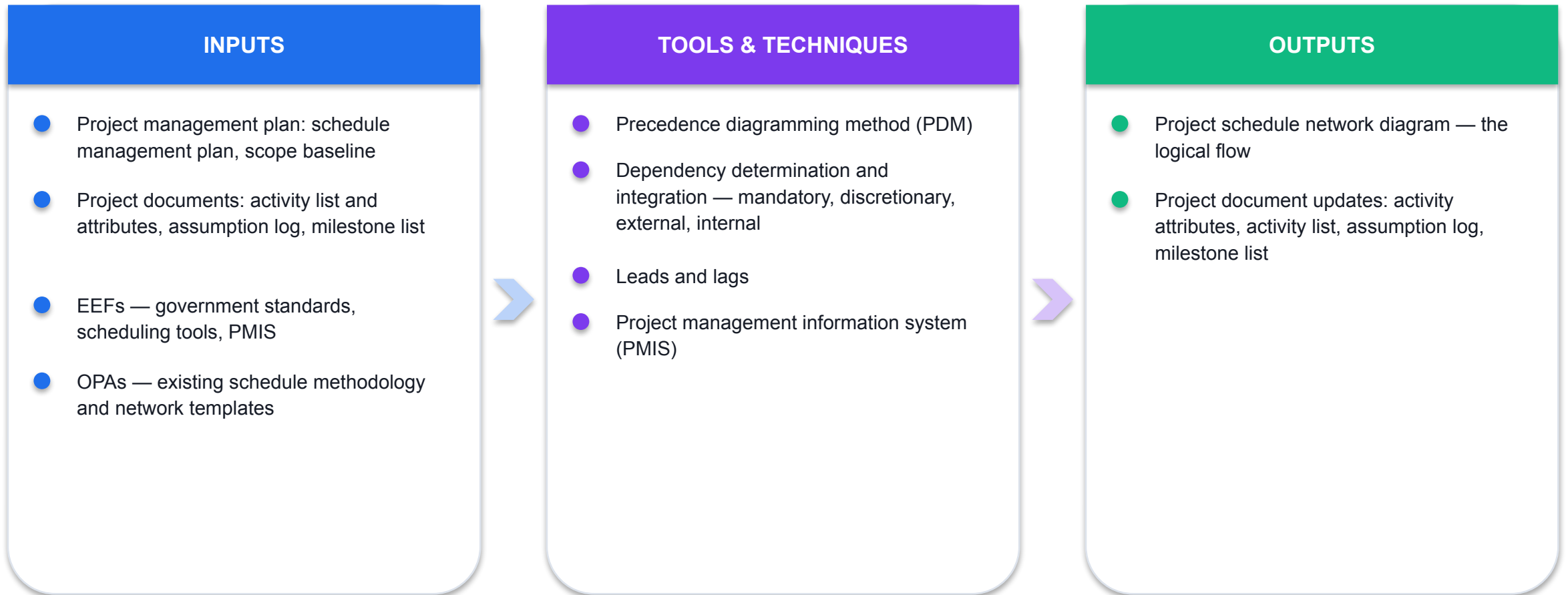
Activity dependency types

Type	Meaning	Action by the project manager
Mandatory (hard logic)	Contractually required or inherent in the nature of the work — concrete must cure before you build on it	Must schedule it; there is no way around this sequence
Discretionary (soft logic)	Established because of best practice or because a specific sequence is preferred	Can be changed; the first place to look when compressing
External	Relationship between a project activity and a non-project activity, such as a regulator's approval	Outside your control; manage as a risk and monitor closely
Internal	Relationship between two project activities, generally inside the team's control	Within the team's control; re-sequence freely if it helps

Dependencies combine: a dependency can be mandatory-external (a legally required inspection) or discretionary-internal (a preferred build order). Name both attributes.

Sequence Activities — ITTO

Sequencing produces logic, not dates. Dates only appear after durations are estimated and the network is analysed.



Precedence relationships in the PDM

Relationship	Rule	Everyday example
Finish-to-Start (FS)	The successor cannot start until the predecessor finishes	Testing cannot start until the build is finished — by far the most common
Start-to-Start (SS)	The successor cannot start until the predecessor starts	Pouring concrete starts once levelling starts, running in parallel behind it
Finish-to-Finish (FF)	The successor cannot finish until the predecessor finishes	Document writing cannot finish until testing finishes
Start-to-Finish (SF)	The successor cannot finish until the predecessor starts	The old shift cannot end until the new shift starts — rare, appears on the exam

Memory aid: name the PREDECESSOR's end first, then the SUCCESSOR's end. Start-to-Finish is the only one that feels backwards, which is exactly why it is examined.

Leads versus lags

LEAD — BRING IT FORWARD

Accelerates the successor; shown as a negative value

- The amount of time a successor activity can be advanced
- Example: start writing training material two weeks before build ends
- Written as FS minus 2 weeks on the network diagram
- Used to compress a schedule by overlapping activities
- Adds risk: you are working on an unfinished predecessor

LAG — DELAY IT

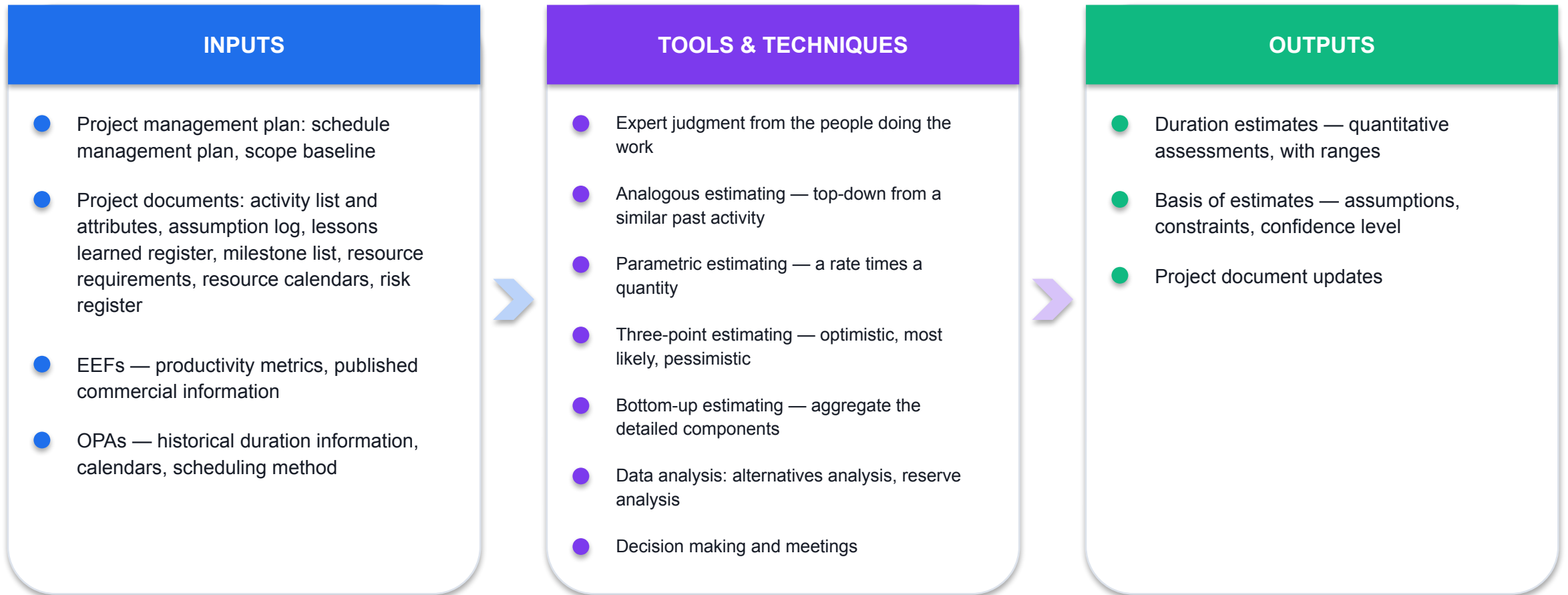
Delays the successor; shown as a positive value

- The amount of time a successor activity must be delayed
- Example: concrete must cure for 7 days before the next activity
- Written as FS plus 7 days on the network diagram
- Represents mandatory waiting, not slack or float
- Cannot usually be removed — it is physics or contract, not choice

Do not confuse lag with float. Lag is designed waiting inside the logic; float is calculated flexibility that falls out of the network analysis.

Estimate Activity Durations — ITTO

Duration is not effort: 40 hours of effort by one half-time person is a two-week duration.



Duration and cost estimating techniques compared

Technique	How it works	Accuracy and cost
Expert judgment	Ask the team or the people who will do the work, using their experience of similar tasks	Fast; quality depends entirely on the expert's relevance
Analogous (top-down)	Use a similar past project, product or system as the analogy and scale it	Fast and cheap; lowest accuracy — used early when little detail exists
Parametric	Multiply a unit rate by a quantity — \$55 per metre, 8 hours per drawing	More accurate than analogous if the historical data is genuinely comparable
Three-point (PERT)	Weighted average of optimistic, most likely and pessimistic estimates	Accounts for uncertainty and reduces single-point optimism bias
Bottom-up	Estimate each work package and aggregate upward through the WBS	Most accurate but the most time-consuming and costly

Three-point estimating — PERT arithmetic

Triangular (simple average)

$$(O + M + P) / 3$$

Equal weight to all three points; used when the distribution is genuinely unknown.

Beta / PERT

$$(O + 4M + P) / 6$$

Weights the most likely estimate four times; the PMI default for three-point estimating.

Standard deviation

$$(P - O) / 6$$

The spread of the estimate — a large sigma is a signal to decompose further.

Variance

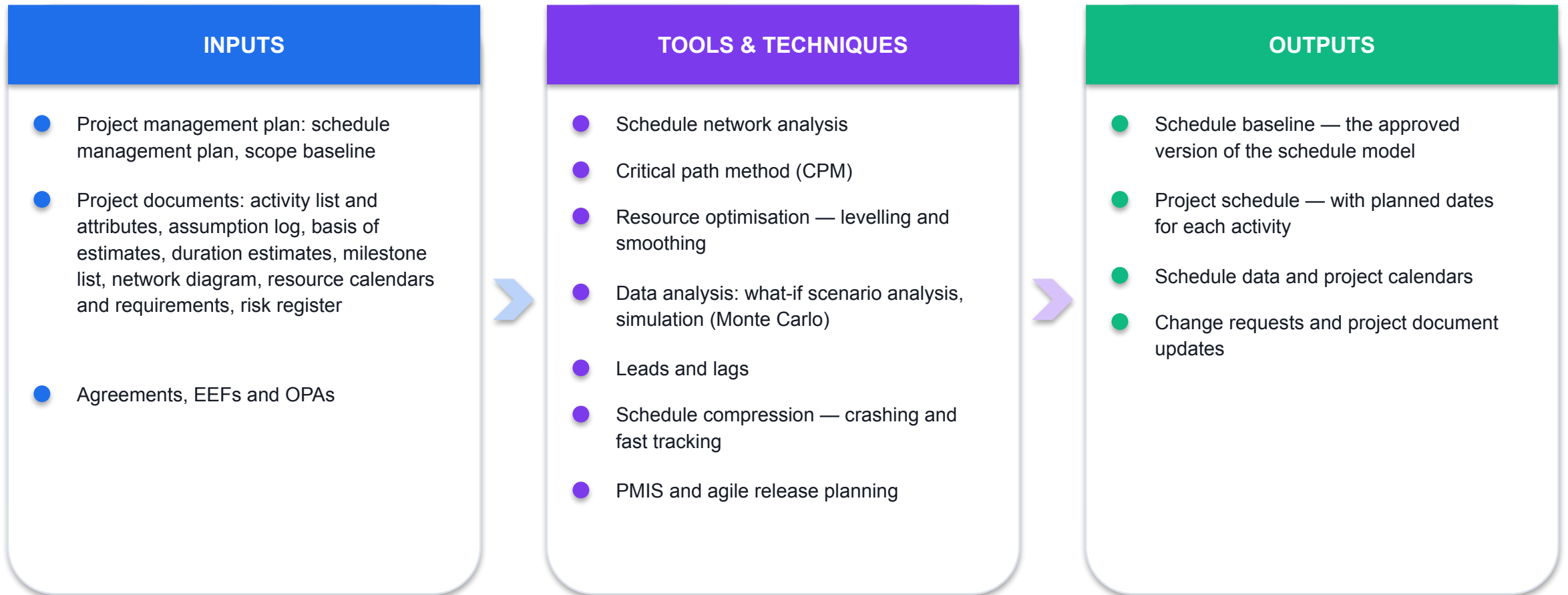
$$((P - O) / 6)^2$$

Variances are additive along a path; standard deviations are not. Sum variances, then square-root.

O = optimistic, M = most likely, P = pessimistic. Worked example: O=6, M=10, P=20 gives PERT = $(6 + 40 + 20)/6 = 11$ days, with sigma = $(20-6)/6 = 2.33$ days.

Develop Schedule — ITTO

Develop Schedule is iterative — you will loop through analysis, optimisation and compression before anything is baselined.



Float and critical path arithmetic

Total float

$$LS - ES \text{ (or } LF - EF)$$

How long an activity can slip without delaying the project end date.

Free float

$$ES(\text{successor}) - EF(\text{activity}) - 1$$

How long an activity can slip without delaying its earliest successor.

Critical path

The path with zero total float

The longest path through the network and therefore the shortest project duration.

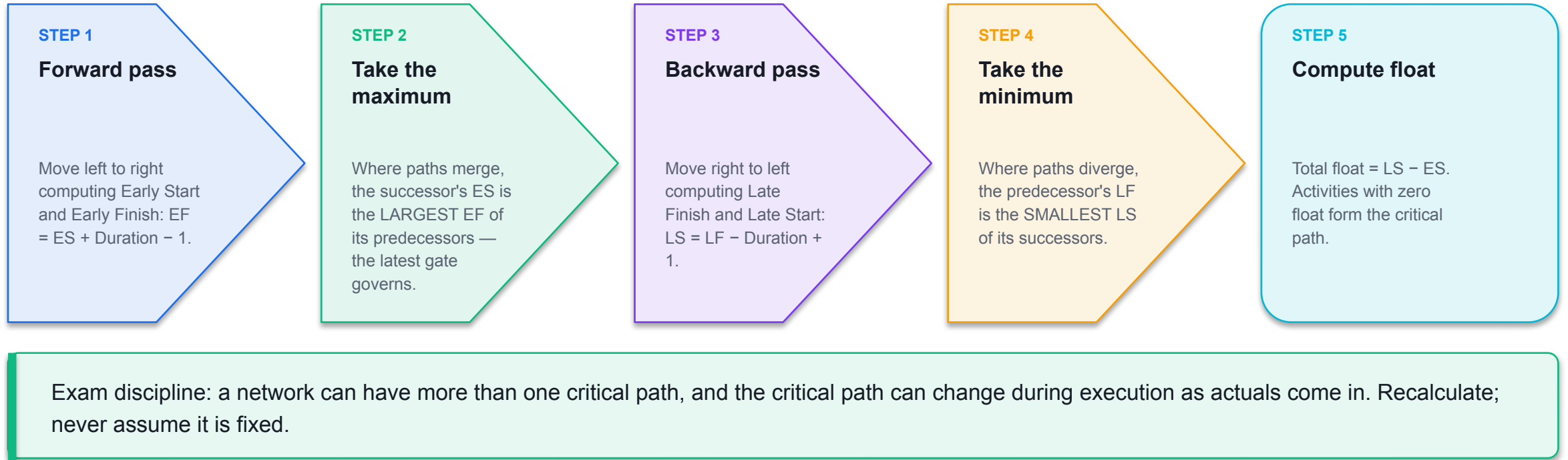
Project float

$$\text{Imposed date} - \text{Calculated finish}$$

Slack against an externally imposed deadline; can be negative.

Negative total float means the network cannot meet the imposed date as sequenced — the trigger for schedule compression, not for quietly changing the finish date in the tool.

Forward and backward pass — calculating the network



Schedule compression — crashing versus fast tracking

CRASHING

Add resources to critical path activities

- Adds cost to buy back duration on the critical path only
- Overtime, extra staff, paying to expedite deliveries
- Subject to diminishing returns — nine women, one month
- Always increases cost; may increase risk through fatigue
- Choose the activity with the lowest cost per day saved

FAST TRACKING

Overlap activities normally done in sequence

- Converts finish-to-start logic into overlap or start-to-start
- Costs nothing directly — you pay in risk instead
- Only discretionary dependencies can be fast tracked
- Increases rework risk: the predecessor may still change
- Try fast tracking first, then crash what remains

Neither technique changes scope. If compression cannot recover the date, the correct move is to escalate the trade-off, not to silently reduce quality.

Exam focus — schedule

IF THE QUESTION SAYS

The question gives a network diagram and asks for the shortest duration



THE BEST ANSWER IS

That is the critical path — the longest path through the network, with zero float

IF THE QUESTION SAYS

You need to shorten the schedule without adding cost



THE BEST ANSWER IS

Fast track by overlapping discretionary dependencies; crashing always costs money

IF THE QUESTION SAYS

An activity has 5 days of total float and slips 3 days



THE BEST ANSWER IS

The project end date is unaffected; only floats downstream change

IF THE QUESTION SAYS

Concrete must cure for 7 days before the next activity



THE BEST ANSWER IS

That is a mandatory dependency with a 7-day lag, not float

IF THE QUESTION SAYS

The team's velocity is 20 points and 100 points remain



THE BEST ANSWER IS

Forecast roughly 5 more iterations — express it as a range using best and worst recent velocity

Recap — schedule planning

1 Six processes

Plan Schedule Management, Define Activities, Sequence, Estimate Durations, Develop Schedule, Control Schedule.

3 Four dependencies

Mandatory, discretionary, external, internal — and they combine into pairs.

5 Float is calculated

Total float protects the end date; free float protects the successor; zero float is critical.

2 Logic before dates

Sequencing produces a network diagram; dates only exist after durations and network analysis.

4 Four relationships

FS is the default; SS and FF enable overlap; SF is rare and examined.

6 Agile still forecasts

Velocity, burndown, burnup and flow metrics replace the Gantt but do the same forecasting job.

Knowledge area 3 of 8 in the planning block

Cost: estimate honestly, then fund the uncertainty

Estimate

Determine the approximate monetary resources needed to complete each activity and work package.

Budget

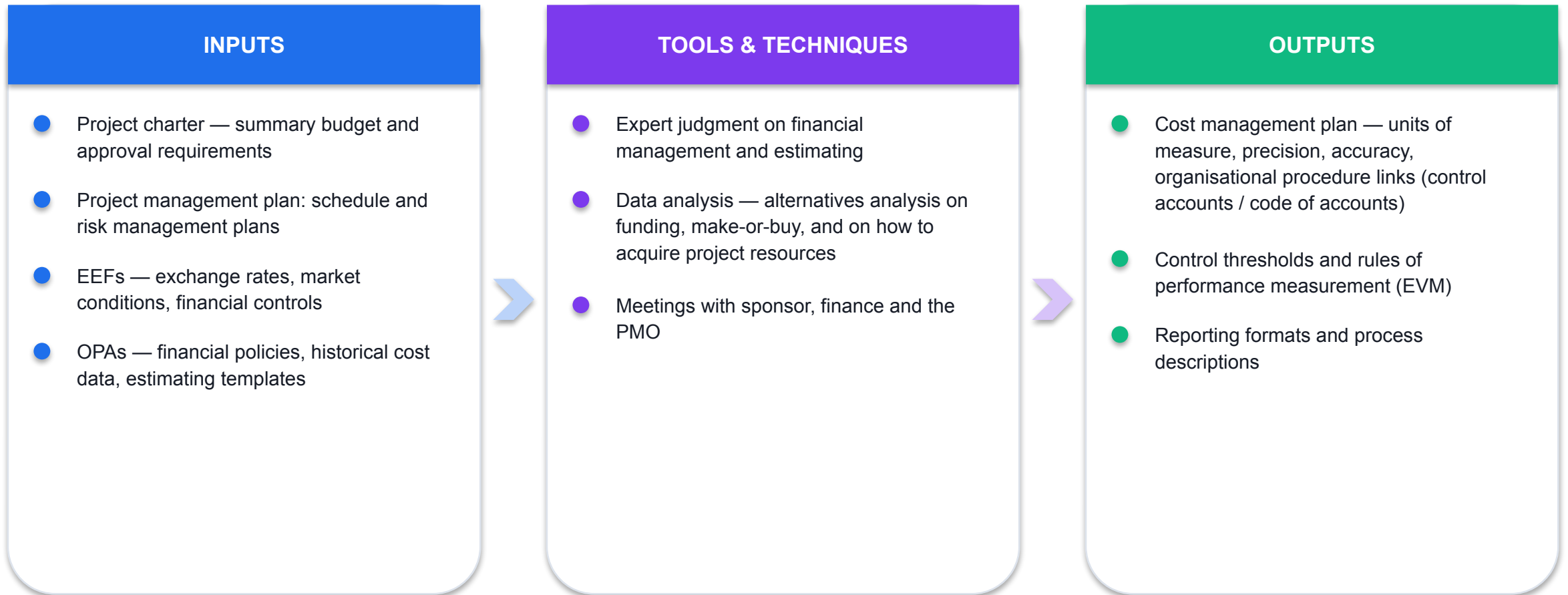
Aggregate those estimates into an authorised, time-phased cost baseline you can measure against.

Reserve

Fund the known-unknowns and the unknown-unknowns separately — they have different owners.

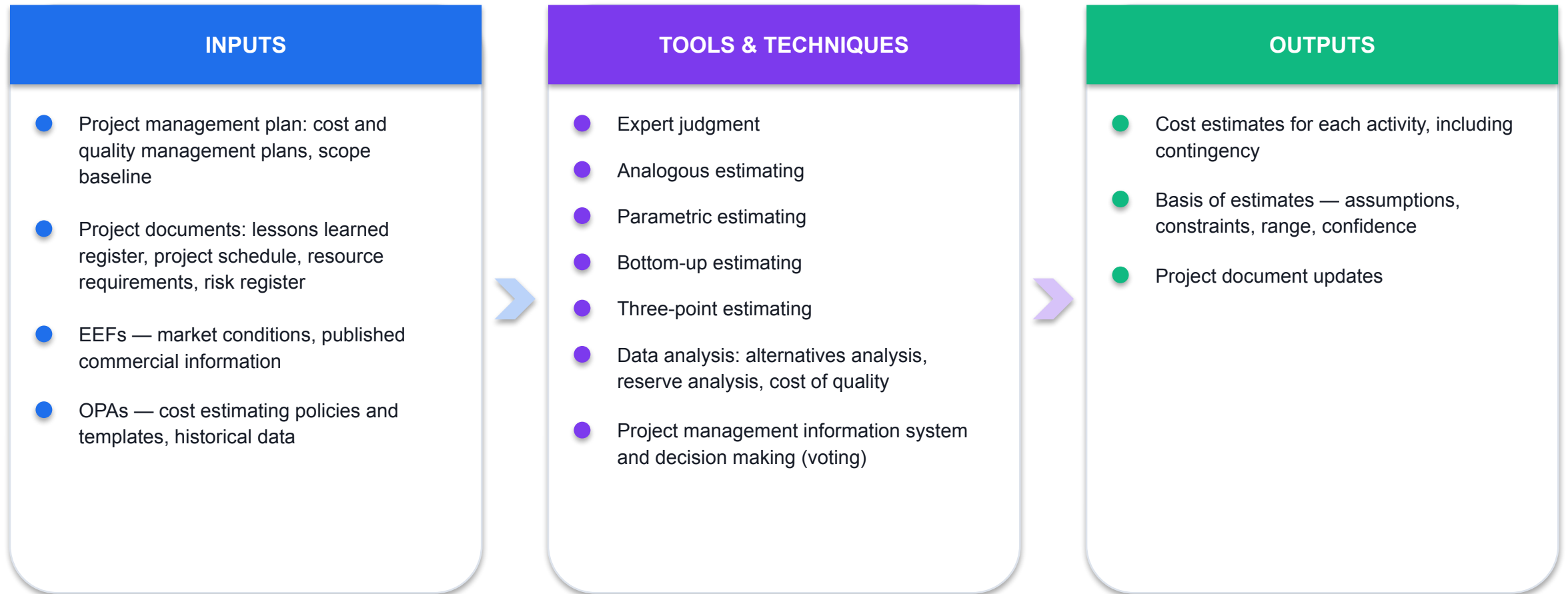
Plan Cost Management — ITTO

Set the rules of the money before you produce a number — otherwise nobody agrees what the number means.



Estimate Costs — ITTO

Cost estimates are a prediction with a range and a confidence level, never a single number stated as fact.



The four cost estimating methods in detail

1 Analogous estimating

Uses historical information and expert judgment from a previous similar project; pick the next closest project when an exact match does not exist.

2 When to use analogous

Low cost and quick — the right choice very early when detail is scarce and a rough order of magnitude is all that is needed.

3 Parametric estimating

Takes variables from historical data and applies them: $\text{cost} = \text{number of resistors} \times \text{cost per resistor}$, or $\text{rate per operator} \times \text{number of operators}$.

4 When to use parametric

When there is a genuine linear driver and credible historical data; scales far better than analogous.

5 Bottom-up estimating

Rolls costs up from the smallest activity upward, using the WBS to structure the estimates.

6 When to use bottom-up

The most accurate method but the most time consuming and costly; used once the WBS is complete.

7 Three-point estimating

Uses a weighted average of three estimates to reduce bias and represent uncertainty explicitly.

8 When to use three-point

Whenever a single number would hide real uncertainty — especially for novel or high-risk work.

Three-point cost estimating

Beta / PERT cost

$$C_e = (C_o + 4C_m + C_p) / 6$$

C_e estimated cost, C_o optimistic, C_m most likely, C_p pessimistic.

Triangular cost

$$C_e = (C_o + C_m + C_p) / 3$$

Used when there is no reason to weight the most likely value more heavily.

Cost standard deviation

$$\sigma = (C_p - C_o) / 6$$

The spread; quote C_e ± 2σ to give the sponsor a 95% range.

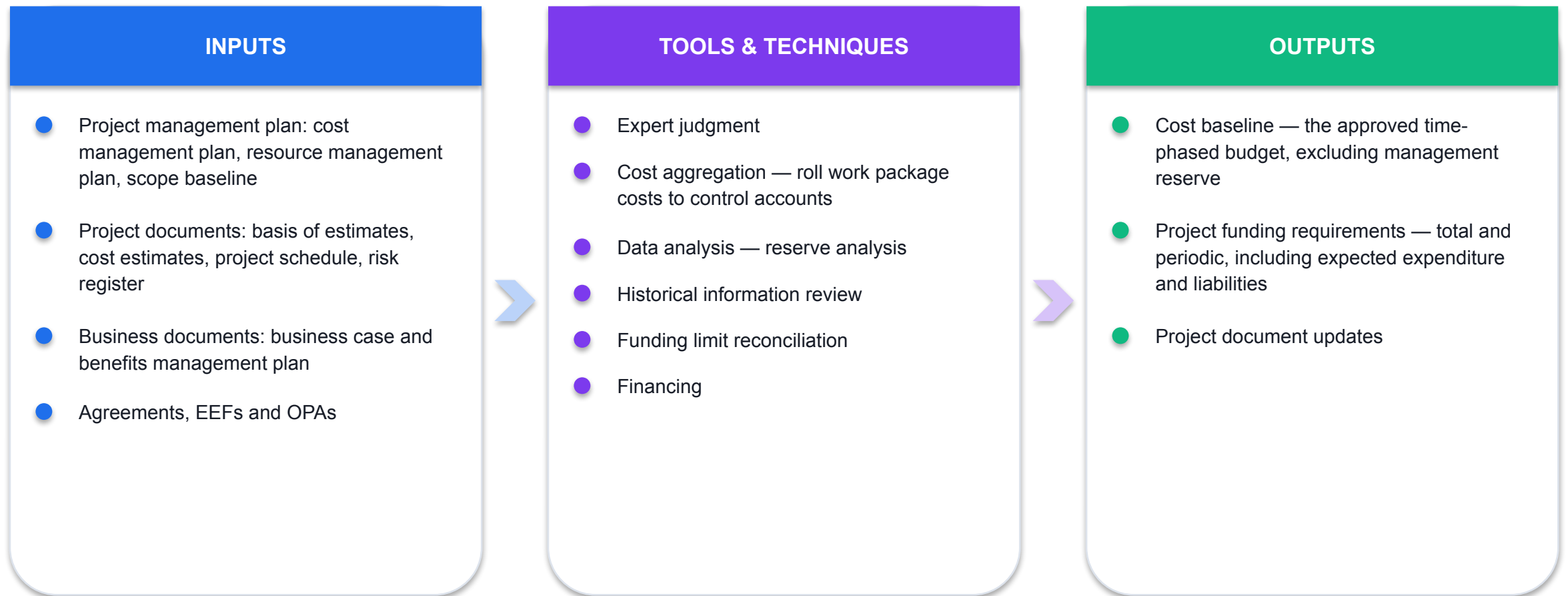
Range example

$$O=80k, M=100k, P=180k$$

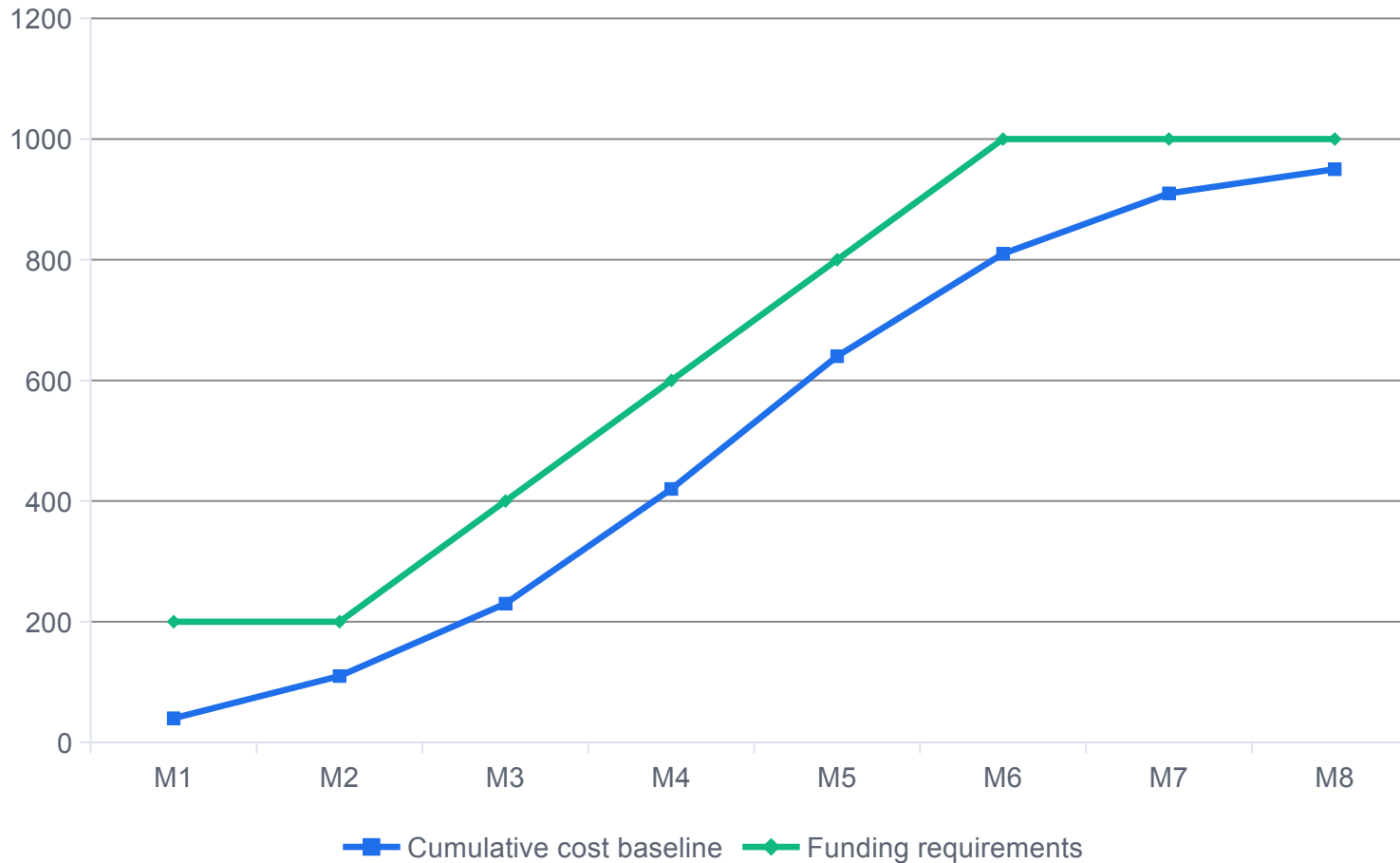
PERT = (80 + 400 + 180)/6 = \$110k, σ = \$16.7k, so 95% range is roughly \$77k to \$143k.

Determine Budget — ITTO

The cost baseline is the S-curve. Management reserve sits above it and forms the total project budget.



The cost baseline S-curve and the funding steps



WHAT IT TELLS YOU

- The cost baseline is time-phased: it says not only how much, but when the money is expected to be spent.
- Cumulative spend follows an S-curve — slow at first, steep in execution, flattening at closure.
- Funding arrives in steps, usually quarterly; funding limit reconciliation aligns the smooth curve to the steps.
- The gap between the funding line and the baseline is the management reserve plus the funding buffer.

Contingency reserve versus management reserve

CONTINGENCY RESERVE

For identified risks — the known-unknowns

- Set aside to deal with identified risks should they occur
- Calculated from quantitative risk analysis or EMV of the register
- INCLUDED in the cost baseline and in the schedule baseline
- The project manager may spend it without a change request
- Reserve analysis reports risk remaining versus contingency remaining

MANAGEMENT RESERVE

For unidentified risks — the unknown-unknowns

- Set aside for unforeseen work that is still within project scope
- Typically a management-set percentage of the total baseline
- EXCLUDED from the cost baseline; part of the total project budget
- Requires management approval to access — usually a change request
- Using it changes the cost baseline; using contingency does not

Total project budget = cost baseline + management reserve. Cost baseline = work package costs + contingency reserve. This distinction is examined heavily.

ECO 2026 emphasis — managing financial reserves

1 Quantify allocations

Quantify risk and contingency financial allocations explicitly rather than adding a round percentage.

2 Plan spend tracking

Design how spend will be tracked across the whole life cycle before the first invoice arrives.

3 Plan financial reporting

Agree the format, frequency and audience of financial reporting with finance and the sponsor.

4 Anticipate challenges

Look ahead for funding gaps, exchange rate exposure, and cost inflation on long-lead procurement.

5 Monitor variation

Work with the governance process when financial variations exceed the agreed control threshold.

6 Release unused reserve

Reserves for risks that have passed should be returned, not quietly absorbed into the project.

Exam focus — cost

IF THE QUESTION SAYS

A risk in the register occurs and you need funds



THE BEST ANSWER IS

Use the contingency reserve — no change request needed, it is already inside the cost baseline

IF THE QUESTION SAYS

Unforeseen in-scope work appears that nobody identified



THE BEST ANSWER IS

Request management reserve through management approval; this changes the cost baseline

IF THE QUESTION SAYS

You need a quick figure with almost no detail available



THE BEST ANSWER IS

Analogous estimating — fast and cheap, lowest accuracy

IF THE QUESTION SAYS

The question gives O, M and P values



THE BEST ANSWER IS

Apply $(O + 4M + P) / 6$ unless the question explicitly says triangular or simple average

IF THE QUESTION SAYS

You are asked for the total project budget



THE BEST ANSWER IS

Cost baseline plus management reserve — the baseline alone is not the budget

Recap — cost planning

1 Rules first

The cost management plan sets units, precision, thresholds and the EVM rules before any number is produced.

3 PERT

$(O + 4M + P) / 6$ with $\sigma = (P - O) / 6$ turns a guess into a range with a stated confidence.

5 Two reserves

Contingency is inside the baseline for known risks; management reserve sits outside it for unknown ones.

2 Four methods

Analogous, parametric, bottom-up and three-point — trading speed against accuracy.

4 Aggregate to baseline

Work packages roll to control accounts, then to the time-phased cost baseline S-curve.

6 Reconcile funding

Funding limit reconciliation aligns the smooth spend curve to the stepped release of actual funds.

Knowledge area 4 of 8 in the planning block

Quality: how well deliverables meet requirements

Quality vs grade

Quality is conformance to requirements; grade is the category of features. Low grade can be acceptable; low quality never is.

Prevention over inspection

It is always cheaper to design defects out than to inspect and rework them later.

Everyone's job

In agile the whole team owns quality, from writing correct requirements to automated testing.

Quality in predictive versus agile delivery

PREDICTIVE QUALITY

Build, then test — usually in large batches

- The product is developed and then tested in sequential order
- Testing is a distinct phase performed by a separate QA function
- Defects are found late, when they are most expensive to fix
- Formal quality audits and phase gate reviews provide assurance
- Acceptance is a single formal event near the end

AGILE QUALITY

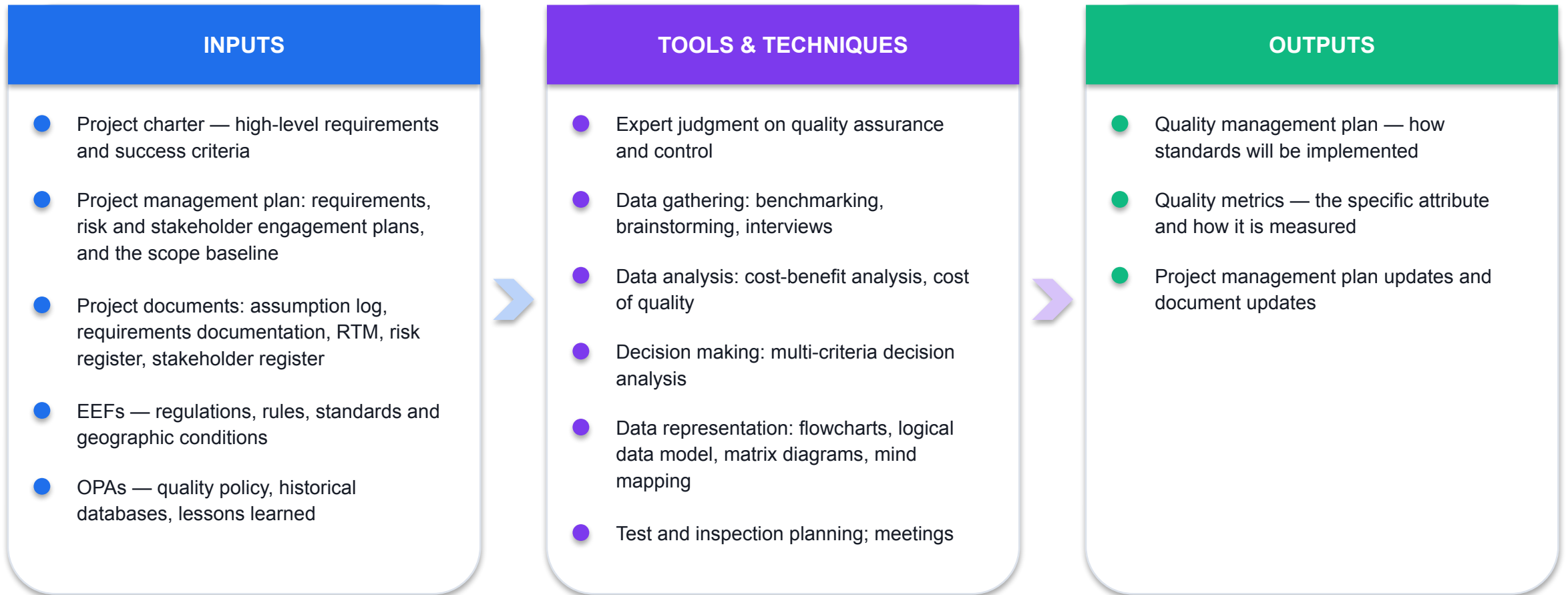
Build quality in, continuously

- Quality is everyone's responsibility, from requirement to release
- Test-driven development and automated regression on every commit
- Definition of done enforces quality before a story is accepted
- Continuous integration finds integration defects within hours
- Acceptance happens every iteration at the review

Both approaches tie quality to the product acceptance criteria described in the statement of work or design documents — update those criteria as experimentation and prioritisation occur, then validate them at acceptance.

Plan Quality Management — ITTO

Plan Quality Management defines what good looks like and how it will be proven, before any deliverable exists.



The quality management plan and its benefits

1 What it contains

The activities and resources that will achieve the quality objectives, and who is accountable for each.

2 How formal

May be formal or informal, highly detailed or broadly framed — tailored to the project's risk and regulatory context.

3 Living document

Reviewed throughout the project as the product and the risk profile change.

4 Sharper value focus

Forces the team to state what the customer actually values, rather than what is easy to measure.

5 Cost reduction

Prevention and appraisal spend consistently costs less than the failure costs it avoids.

6 Fewer schedule overruns

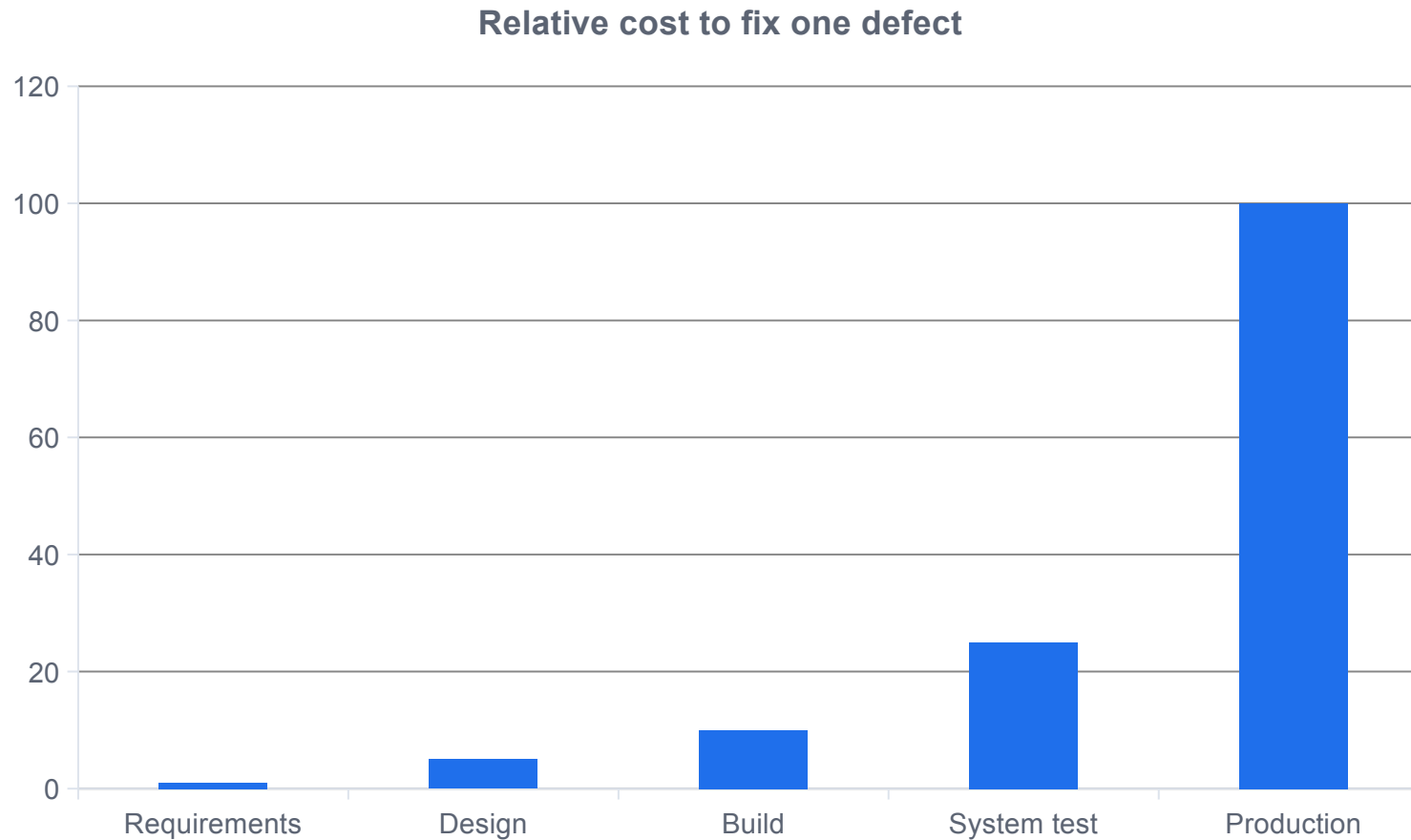
Most predictive schedule overrun is rework; designing rework out protects the date.

Cost of Quality (CoQ) — the four categories

Category	Definition	Typical spend
Prevention (conformance)	The cost to prevent future defects or errors in the product	Training, prototypes, code refactoring, design reviews, standards, tooling
Appraisal (conformance)	The cost to find defects or errors in the product	Testing, quality audits, inspections, the sprint review, destructive test loss
Internal failure (non-conformance)	Failure found by the project, before the customer sees it	Rework, scrap, re-testing, defect triage, schedule slip
External failure (non-conformance)	Failure found by the customer, after delivery	Warranty work, recalls, liability, lost business, reputational damage

Cost of conformance (prevention + appraisal) is money spent during the project to avoid failure. Cost of non-conformance is money spent during and after the project because you did not.

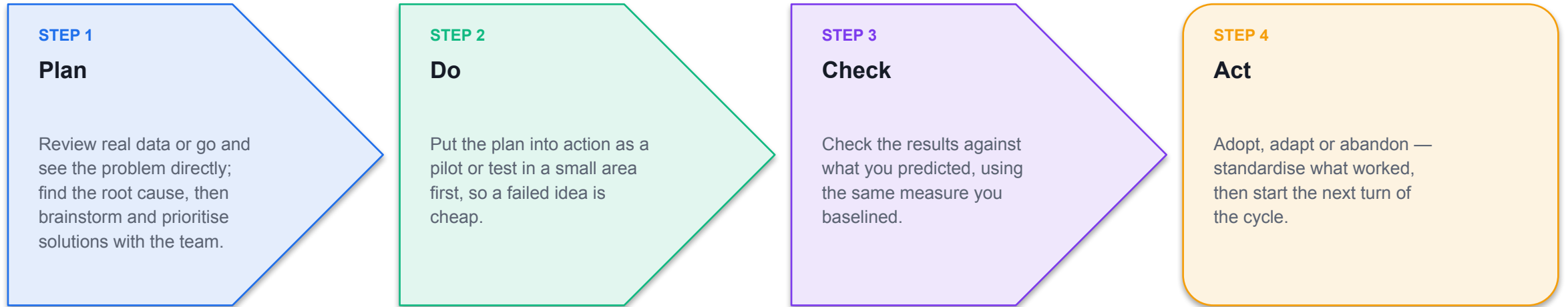
The cost of change curve — quality is cheapest early



WHAT IT TELLS YOU

- Quality is more expensive to fix the closer it gets to the customer — the classic 1:100 relationship.
- This curve is the economic justification for prevention spend, code review and test automation.
- It is also the argument for short iterations: shorter feedback loops keep you at the cheap end of the curve.
- Project risk follows exactly the same curve — early treatment is dramatically cheaper than late response.

PDCA — the Deming/Shewhart improvement cycle



PDCA is a cycle, not a project: the value comes from turning it repeatedly and quickly, not from executing it perfectly once.

DMAIC — the Six Sigma improvement framework



Six Sigma focuses on removing variation and waste. Kaizen — 'change for better' — is the complementary philosophy of many small continuous improvements made by the people doing the work.

Root cause analysis toolkit

1 Fishbone / Ishikawa

A cause-and-effect diagram grouping candidate causes into categories around a single problem statement.

2 The 6M categories

Manpower, Method, Machine, Material, Measurement, Mother Nature — the classic manufacturing spine.

3 Five Whys

Ask 'why' repeatedly to drill from the visible symptom down to the systemic cause that actually needs fixing.

4 Pareto chart

Ranks defect categories by frequency to find the vital few driving most of the pain.

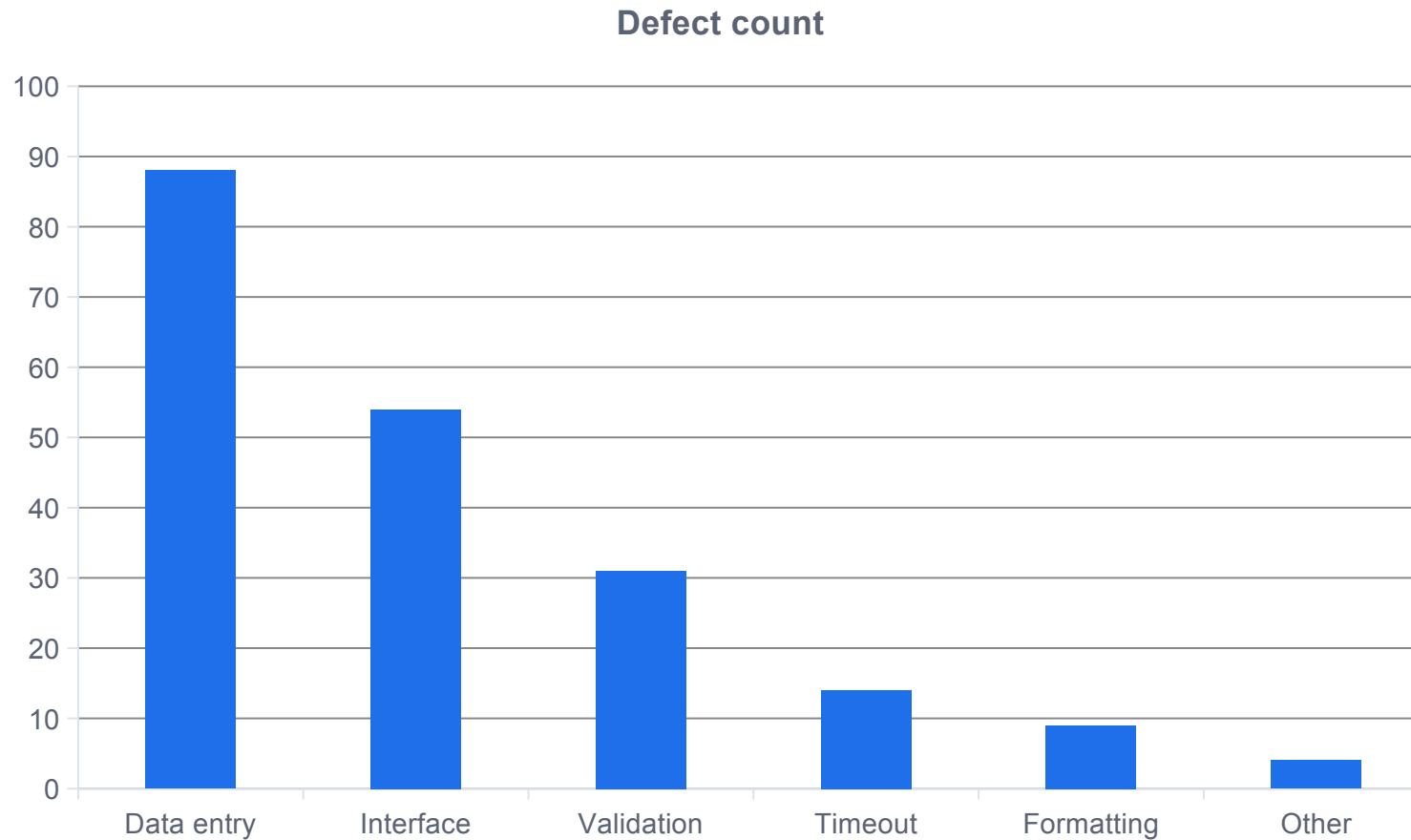
5 Six Thinking Hats

De Bono's method: examine a situation deliberately from six distinct thinking modes in turn.

6 Control chart

Plots a process against control limits over time to separate special-cause signal from common-cause noise.

Pareto chart — the vital few defect categories



WHAT IT TELLS YOU

- The 80/20 rule: roughly 80% of the defects come from roughly 20% of the causes.
- Directs improvement effort where it makes the biggest impact, instead of spreading it evenly across every category.
- Takes a big vague problem ('quality is poor') and breaks it into actionable, countable pieces.
- Re-run after the improvement: the bars should shrink and the ranking should change.

Quality standards, regulations and compliance

1 ASQ

The American Society for Quality — steward of the ISO 9000 series quality management standards.

2 Chartered Quality Institute

The UK professional body for quality practitioners and quality management systems.

3 ASTM International

Publishes technical standards for materials, products, systems and services across industries.

4 Government regulation

Appropriate government regulations are mandatory inputs, not optional quality aspirations.

5 Organisational policies

Internal quality policy and product standards constrain what 'acceptable' can mean.

6 Compliance actions

Classify compliance categories, determine threats to compliance, analyse the consequences of non-compliance.

ECO 2026 emphasis — sustainability in quality planning

1 Sustainability as a requirement

Environmental and social criteria are gathered as quality requirements, not treated as PR.

2 Measure it

Define metrics — energy per transaction, waste diverted, carbon per deliverable — and baseline them like any other.

3 Cost of quality extension

Environmental failure cost (remediation, penalties, disposal) belongs in the non-conformance column.

4 Design out waste

Prevention applies to material and energy waste exactly as it applies to defects.

5 Regulatory exposure

Sustainability reporting obligations are compliance requirements with real legal consequence.

6 AI-assisted review

AI tools can scan designs and specifications for waste and compliance gaps — the human still decides.

Exam focus — quality

IF THE QUESTION SAYS

The team wants to inspect more to raise quality



THE BEST ANSWER IS

Prefer prevention over inspection — the cheapest defect is the one never created

IF THE QUESTION SAYS

A customer reports a defect after handover



THE BEST ANSWER IS

That is an external failure cost — the most expensive category of cost of non-conformance

IF THE QUESTION SAYS

You must find why a defect keeps recurring



THE BEST ANSWER IS

Root cause analysis: five whys or an Ishikawa diagram — treat the cause, not the symptom

IF THE QUESTION SAYS

Many defect categories; limited improvement budget



THE BEST ANSWER IS

Build a Pareto chart and attack the vital few that produce most of the defects

IF THE QUESTION SAYS

The product meets all specifications but is low-featured



THE BEST ANSWER IS

That is low grade with high quality — acceptable if agreed; low quality never is

Recap — quality planning

1 Quality is conformance

How well the deliverable meets the requirement — distinct from grade, which is feature category.

3 Four CoQ buckets

Prevention and appraisal are conformance; internal and external failure are non-conformance.

5 Improvement cycles

PDCA and DMAIC give a repeatable structure; Kaizen supplies the culture of small continuous change.

2 Plan it

The quality management plan states standards, metrics, roles and review activity before delivery starts.

4 Cost of change curve

Defect fix cost rises roughly 1:100 from requirements to production — buy prevention.

6 RCA toolkit

Fishbone, five whys, Pareto, Six Thinking Hats and control charts — pick the tool that fits the data you have.

Knowledge area 5 of 8 in the planning block

Resources: people and things, planned deliberately

Team resources

The people who will do the work, their roles, authority, responsibility and required competency.

Physical resources

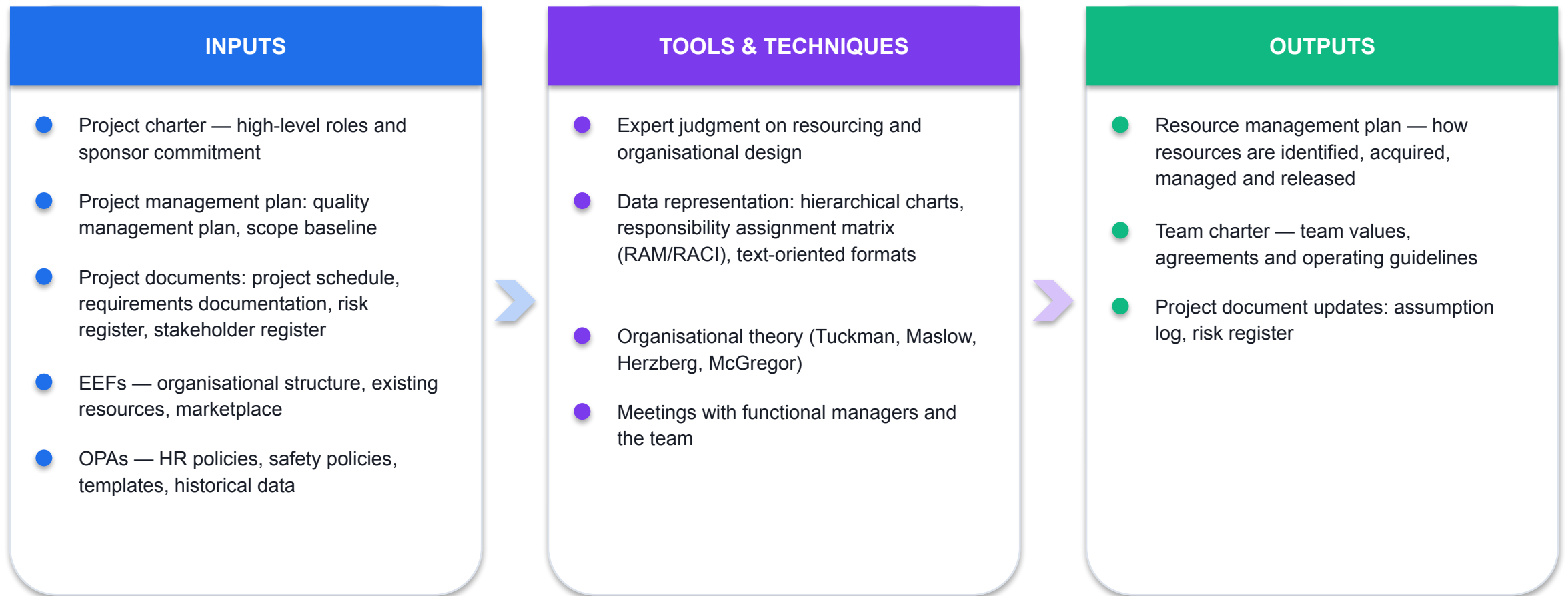
Materials, equipment, facilities and infrastructure that must be acquired and released.

The constraint

Resource availability, not activity duration, is what most often determines the real schedule.

Plan Resource Management — ITTO

This plan covers physical resources and team resources — most exam errors come from forgetting the physical half.



Role definition — the four attributes

Attribute	Definition	What goes wrong if it is vague
Role	A person's designated function on the project — business analyst, test lead, architect	People duplicate work or assume someone else is covering a gap
Authority	The right to apply resources, make decisions, sign approvals and accept deliverables	Decisions stall waiting for someone who never had the mandate
Responsibility	The assigned duties a role must perform to complete project activities	Work is 'everybody's job' and therefore nobody's job
Competency	The skill and capacity required to complete the assigned activities within the constraints	Work is assigned to people who cannot succeed at it — a schedule risk, not a people problem

Resource optimisation — levelling versus smoothing

RESOURCE LEVELLING

Fix over-allocation; the end date may move

- Applied when shared resources are over-allocated
- Or when a resource is assigned to two or more activities in the same period
- Adjusts start and finish dates based on resource constraints
- CAN change the critical path — usually extending it
- Use when the resource limit is genuinely hard (one specialist)

RESOURCE SMOOTHING

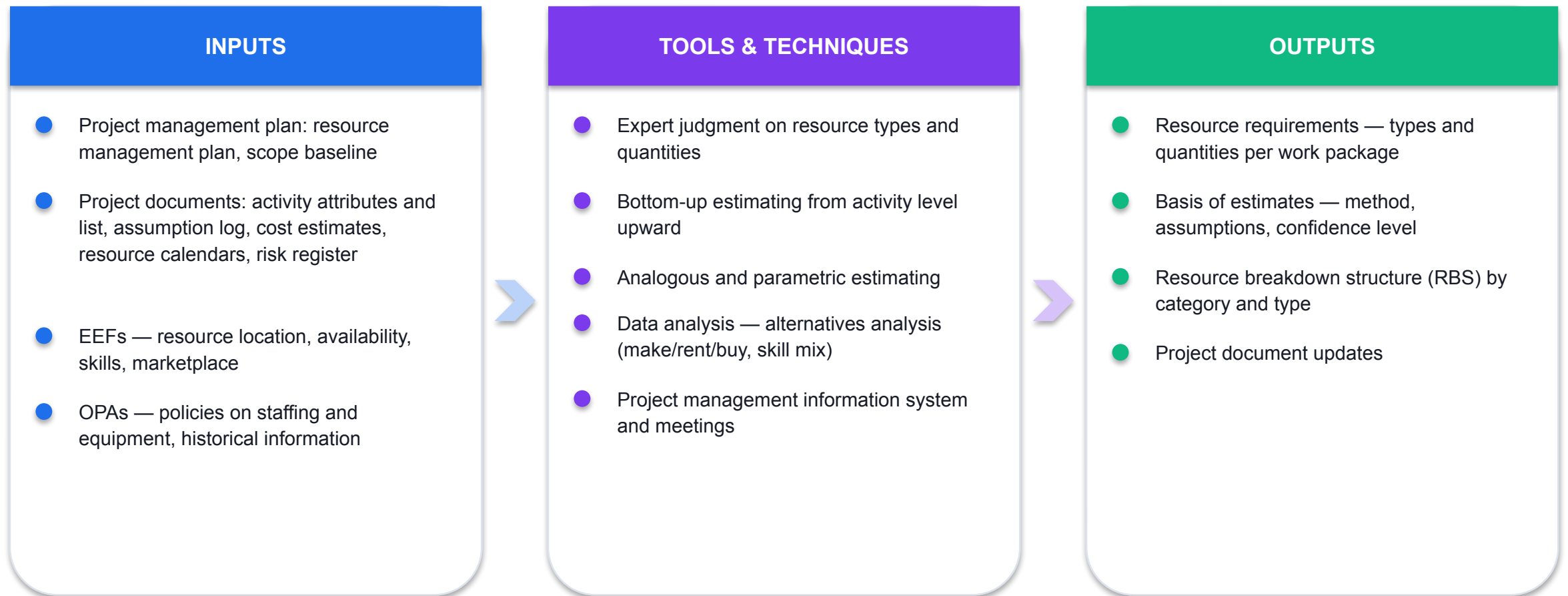
Even out demand; the end date is protected

- Adjusts activities only within their free and total float
- The critical path is NOT changed by smoothing
- Cannot fully resolve a genuine over-allocation
- Produces a more even resource demand profile over time
- Use when the date is fixed and float is available to spend

Memory hook: LEVELLING may LENGTHEN the project; SMOOTHING Stays inside Slack. If the question says the end date must not move, the answer is smoothing.

Estimate Activity Resources — ITTO

Estimate Activity Resources runs closely with Estimate Activity Durations — the two are mutually dependent.



RACI — the responsibility assignment matrix

Letter	Meaning	Rule
R — Responsible	The person or people who actually perform the work	At least one per activity; more than one is fine
A — Accountable	The single person answerable for the outcome, who approves the work	EXACTLY ONE per activity — two means nobody
C — Consulted	People whose input is sought before the work is done; two-way communication	Keep the list short; every C is a delay
I — Informed	People told about progress or outcomes; one-way communication	Cheap to add, so it is where over-communication usually hides

A responsibility assignment matrix shows which team members are responsible for which work packages. Read the ROWS to find unassigned work; read the COLUMNS to find overloaded people.

Co-location, osmotic communication and the team charter

1 Co-location

Agile teams prefer to sit together so that coordination happens continuously rather than in scheduled meetings.

2 Osmotic communication

Team members absorb useful information from overheard conversations without anyone having to brief them.

3 Information radiator

A large visible display of project information in the team area — status without a status meeting.

4 Team charter

Created WITH the team; it establishes values, agreements and operating guidelines the team owns.

5 Ground rules

Explicit agreements on meetings, decisions, conflict and communication — enforced by the team, not just the PM.

6 Virtual co-location

Persistent video rooms and shared boards recreate some osmotic benefit for distributed teams.

Exam focus — resources

IF THE QUESTION SAYS

A key specialist is booked on two activities at the same time



THE BEST ANSWER IS

Resource levelling — and accept that the end date may move

IF THE QUESTION SAYS

Demand is uneven but the finish date is fixed and float exists



THE BEST ANSWER IS

Resource smoothing — adjust only within free and total float

IF THE QUESTION SAYS

Two people are marked Accountable for the same deliverable



THE BEST ANSWER IS

That is the defect: RACI allows exactly one Accountable per row

IF THE QUESTION SAYS

The team keeps rediscovering the same information late



THE BEST ANSWER IS

Co-locate or add an information radiator; osmotic communication solves this without adding meetings

Knowledge area 6 of 8 in the planning block

Communication: the PM's largest single time cost

The 90% figure

Project managers spend the overwhelming majority of their time communicating — plan it like any other work.

Planned, not ad hoc

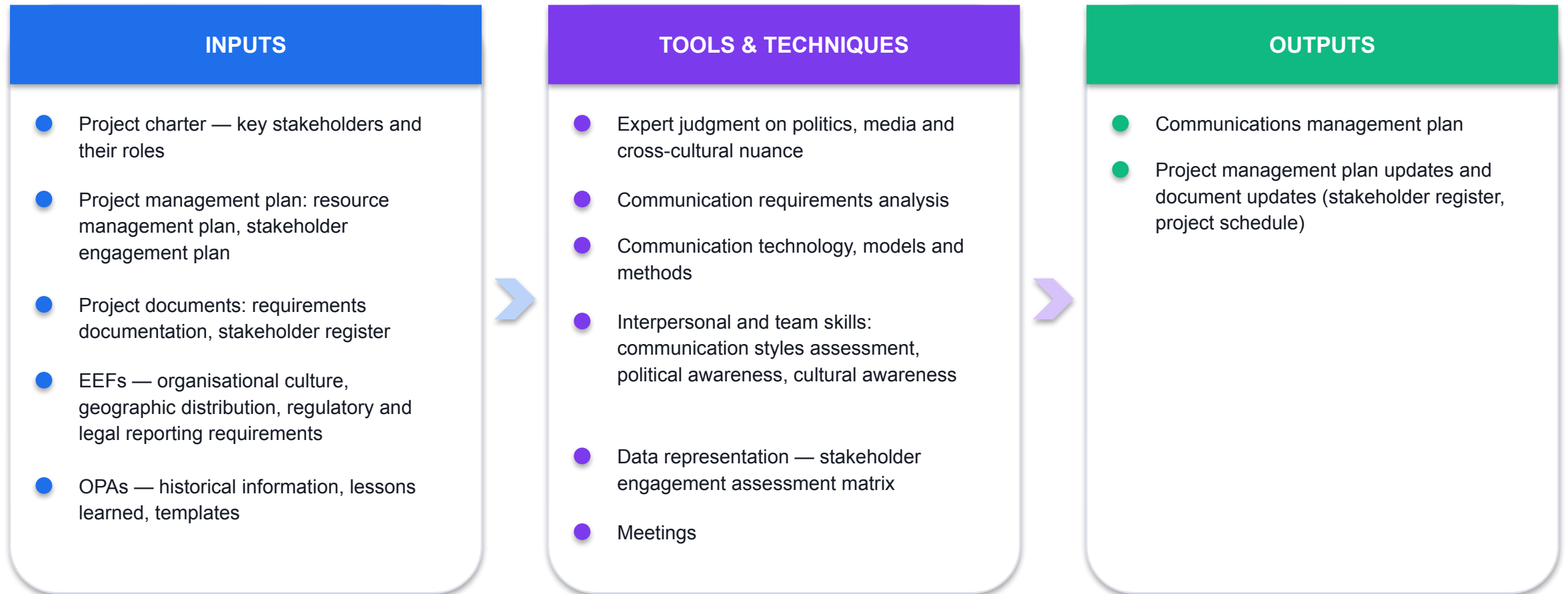
Who needs what, in what format, how often, through which channel, and why.

Two-way by design

Communication without a feedback loop is broadcasting, and broadcasting is not engagement.

Plan Communications Management — ITTO

The plan defines the strategy; execution is Manage Communications and Monitor Communications.



What the communications management plan records

1 Roles

Identifies team members and stakeholders as senders, receivers, and the authorising person for confidential information.

2 Type of information

What is communicated — status, decisions, risks, escalations, financials.

3 Reason for communication

Why the communication exists; a report nobody can justify is a report you can stop producing.

4 Language, format, detail

Written in the audience's language and at the level of detail their decision actually requires.

5 Time frame and frequency

When and how often, including escalation-triggered rather than calendar-triggered messages.

6 Method and technology

The channel: meeting, dashboard, email, report — chosen for urgency and richness, not habit.

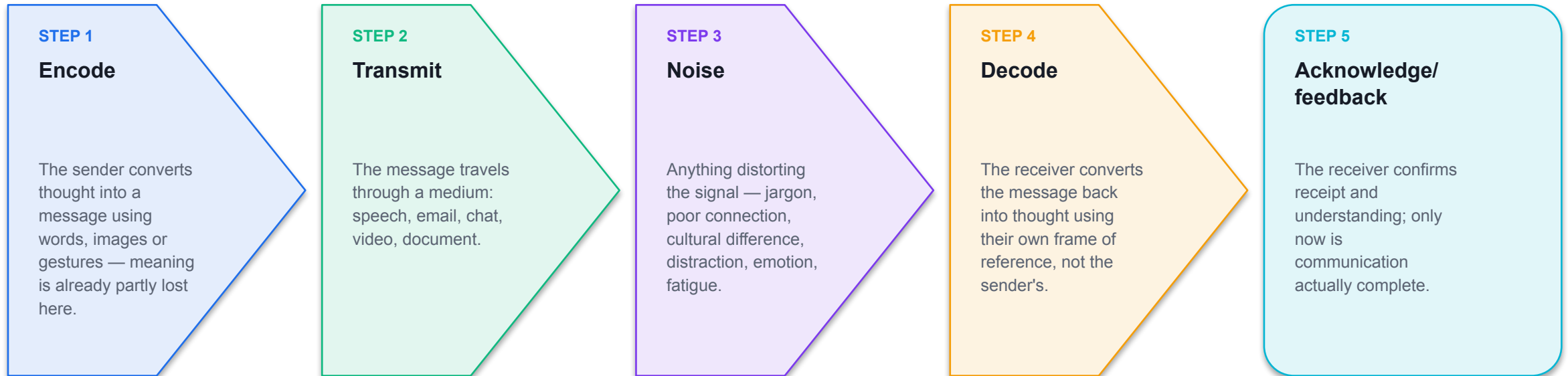
7 Escalation process

The route and the thresholds for escalating an issue when the normal channel fails.

8 Glossary and constraints

Common terminology, plus any legal or regulatory reporting constraints that apply.

The communication model — where meaning goes missing



On the exam, if a message was sent but not confirmed, communication has NOT occurred. The sender is responsible for making the message clear, complete and understood.

Communication channels

Channels formula

$$N \times (N - 1) / 2$$

N is the number of people; the result is the number of potential communication paths.

Team of 10

$$10 \times 9 / 2 = 45$$

Ten people generate forty-five possible two-way channels to keep coherent.

Adding one person

$$\text{From 10 to 11: } 45 \rightarrow 55$$

One extra person adds ten new channels — complexity grows quadratically.

Team of 50

$$50 \times 49 / 2 = 1225$$

Why large teams need structure, sub-teams and a plan rather than goodwill.

Exam trap: read carefully whether the question includes the project manager in N. If it says 'you have a team of 8', N is usually 9. If it asks for ADDITIONAL channels, subtract the original count.

Push, pull and interactive communication

Method	How it works	Use it for
Interactive	Multi-directional real-time exchange between two or more parties — meetings, calls, video, workshops	Complex, sensitive, ambiguous or urgent topics needing agreement
Push	Sent to specific recipients who need the information — email, memos, reports, letters, voicemail	Distribution where receipt matters but discussion does not
Pull	Recipients access the information at their discretion — intranets, dashboards, knowledge repositories, wikis	Large volumes of information for large, self-serving audiences

Push guarantees distribution but NOT understanding. Only interactive communication closes the loop, which is why bad news is always delivered interactively first.

The Five Cs of written communication

1 Correct

Correct grammar and spelling — errors undermine credibility before the content is even read.

2 Concise

Concise expression with excess words eliminated; length is not a measure of thoroughness.

3 Clear

Clear purpose and expression directed towards the needs of the reader, not the convenience of the writer.

4 Coherent

A logical flow of ideas so the reader can follow the argument without re-reading.

5 Controlling

Controlled flow of words and ideas — structure, headings and emphasis that guide attention.

6 Applied where

Status reports, escalations, meeting minutes, change requests and every stakeholder email.

Active listening, feedback and non-verbal signals

1 Listen actively

Stay engaged with the speaker, show interest, and summarise or repeat the message back to check understanding.

2 Seek feedback

Listen to written, verbal and non-verbal feedback to ensure your message and others' were heard correctly.

3 Read non-verbals

Body language, tone and gesture carry a large share of the meaning, especially in conflict.

4 Ask open questions

Questions that cannot be answered yes or no surface the concern behind the objection.

5 Paraphrase, don't parrot

Restating in your own words proves you understood; repeating words only proves you heard.

6 Manage your own bias

The receiver decodes through their own frame of reference; assume misunderstanding, then verify.

Virtual versus co-located teams

VIRTUAL TEAM

Now normal in most workplaces

- Access to better skills at lower cost across geographies
- Avoids relocation expense and widens the talent pool
- Supports work/life balance and inclusion of remote members
- Relies entirely on communication technology to function
- May have bonding challenges and slower trust formation

CO-LOCATED TEAM

Everyone in one team space

- Interaction is easy and unplanned — questions cost seconds
- Bonding and trust are facilitated by informal contact
- Osmotic communication and information radiators work naturally
- Non-verbal signals are visible, so conflict surfaces earlier
- Constrained to the talent available in one location

For virtual teams, invest deliberately in what co-location gives you free: explicit working agreements, a persistent shared board, video-on norms and scheduled social contact.

Exam focus — communication

IF THE QUESTION SAYS

You have 8 team members and 3 new people join



THE BEST ANSWER IS

Channels go from $9 \times 8/2 = 36$ to $12 \times 11/2 = 66$ — read whether the PM is counted in N

IF THE QUESTION SAYS

A stakeholder says they were never informed of a decision



THE BEST ANSWER IS

Review the communications management plan; the omission is a planning defect, not a personal failing

IF THE QUESTION SAYS

You must deliver bad news to a sponsor



THE BEST ANSWER IS

Use interactive communication — face-to-face or a call — never a push email as the first contact

IF THE QUESTION SAYS

Large volumes of reference material for many stakeholders



THE BEST ANSWER IS

Pull communication: publish to a repository or dashboard they access at their own discretion

Knowledge area 7 of 8 in the planning block

Risk: uncertainty that matters, managed on purpose

Both directions

Risks can be negative (threats) or positive (opportunities) — PMI expects you to plan for both.

Known causes

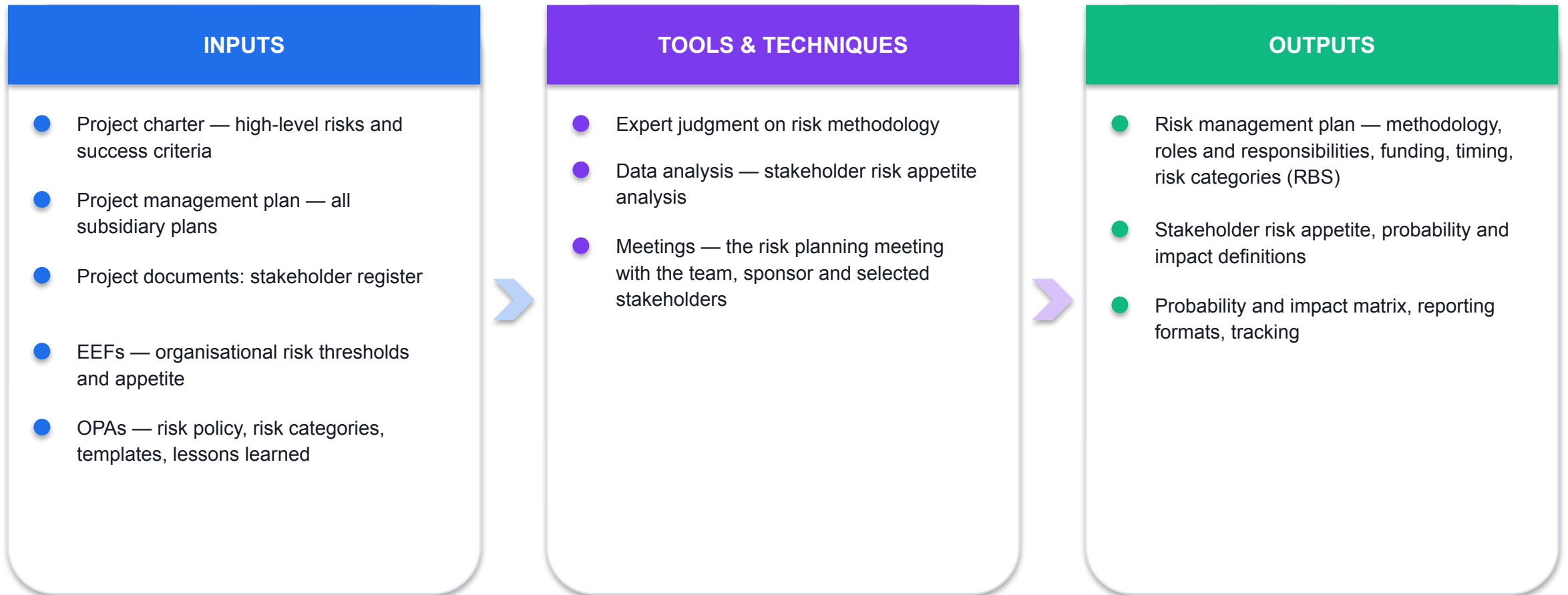
Risk originates from known and unknown causes inside and outside the business environment.

Trigger conditions

Risk development is signalled by a trigger — identify the trigger, not just the risk.

Plan Risk Management — ITTO

Plan Risk Management defines HOW risk work will be done — the definitions of 'high probability' and 'high impact' are set here.



Risk versus issue

RISK

An uncertain future event

- Focused on the FUTURE — it may or may not happen
- Can be positive (opportunity) or negative (threat)
- Documented in the RISK REGISTER
- The reaction is called a RISK RESPONSE, planned in advance
- Has a probability, an impact, an owner and a trigger condition

ISSUE

A present condition already affecting the project

- Focused on the PRESENT — it has already occurred
- Will always be negative
- Documented in the ISSUE LOG
- The reaction is called a WORKAROUND, produced reactively
- Has an owner, a target resolution date and an escalation path

When a risk becomes an issue you must act: implement the planned response, log it in the issue log, and update the risk register status. Recognising that transition is an explicit ECO enabler.

Risk attitude, appetite, tolerance and threshold

1 Risk attitude

The organisation's or individual's disposition toward uncertainty — averse, neutral or seeking.

2 Risk appetite

The degree of uncertainty the organisation is willing to accept in anticipation of a reward.

3 Risk tolerance

The measurable degree of variation around an objective that is acceptable — for example $\pm 10\%$ of budget.

4 Risk threshold

The specific point at which a risk becomes unacceptable and a response or escalation is mandatory.

5 Why it matters

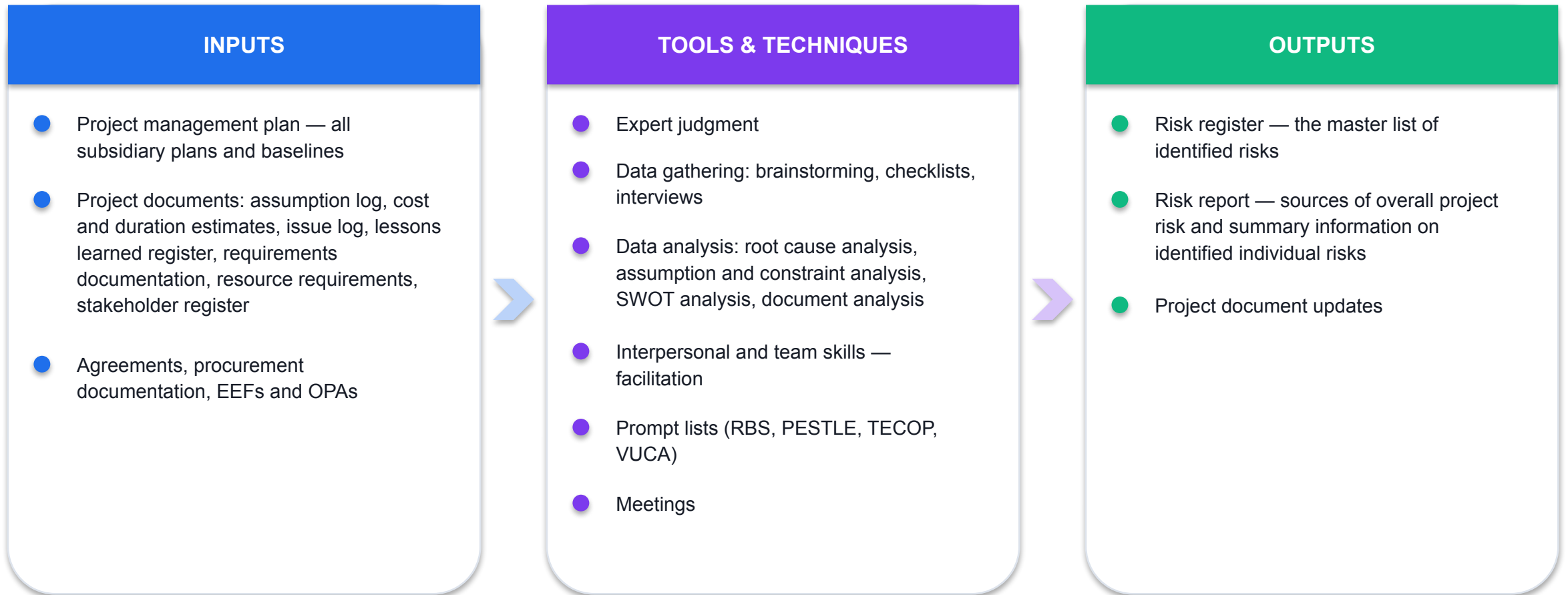
Thresholds turn a subjective conversation into an automatic, auditable trigger for action.

6 Where captured

Stakeholder risk appetite is analysed in Plan Risk Management and recorded in the risk management plan.

Identify Risks — ITTO

Identify Risks is iterative — new risks emerge throughout the project, so the register is never finished.



Risk identification techniques

Technique	How it works	Strength
Risk breakdown structure	Hierarchical list of risk categories used as a checklist when brainstorming	Systematic coverage; exposes categories the team never thought about
Brainstorming	Facilitated group generation of candidate risks with judgement deferred	Volume and diversity of ideas early
SWOT analysis	Examine strengths, weaknesses, opportunities and threats of the project and organisation	Systematically surfaces OPPORTUNITIES, not just threats
Assumption analysis	Test the validity and stability of each documented assumption and constraint	Every assumption is an unregistered risk until tested
Delphi technique	Anonymous rounds of expert input, summarised and recirculated until consensus	Removes dominance and groupthink from expert judgement
Document review	Structured review of plans, contracts, prior project files and lessons learned	Cheap and finds risks the organisation has already met before
Nominal group technique	Brainstorm then vote to rank the risks	Produces a prioritised list in one session
Interviews	One-to-one questioning of experienced participants	Surfaces political or commercially sensitive risks

Common project risk categories

1 Financial

Benefit shortfall, budget overrun, exchange rate movement, funding withdrawal, cost inflation on long-lead items.

2 Technical

Design risk, project complexity, integration failure, unproven technology, technical debt accumulation.

3 Commercial

Supplier failure, procurement delay, contract dispute, vendor lock-in, insolvency of a key partner.

4 Compliance

Regulatory change, legal risk, health and safety, data protection, security and IT security exposure.

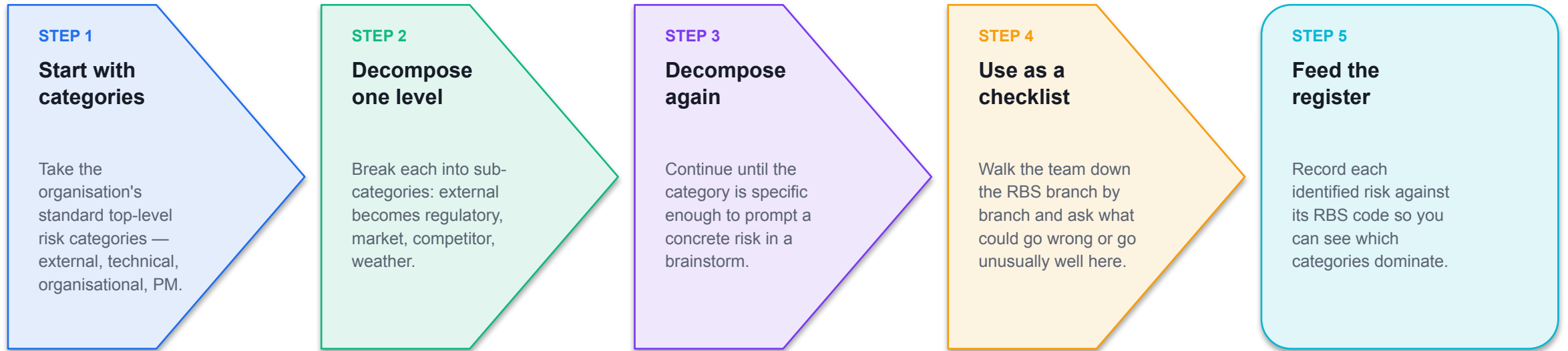
5 Requirements

Scope creep, requirement volatility, poor estimates, unclear acceptance criteria, quality risk.

6 Organisational

Resistance to change, resource unavailability, programme dependency, reputation risk, key person dependency.

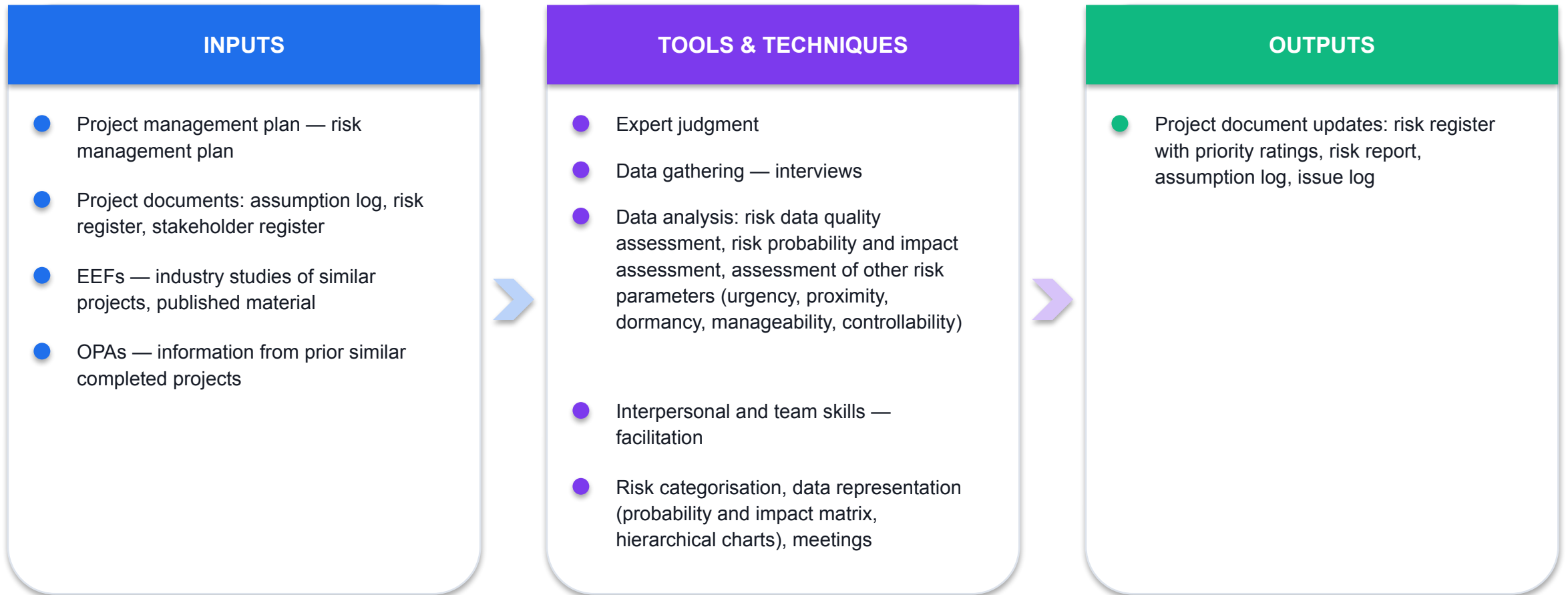
Building a Risk Breakdown Structure (RBS)



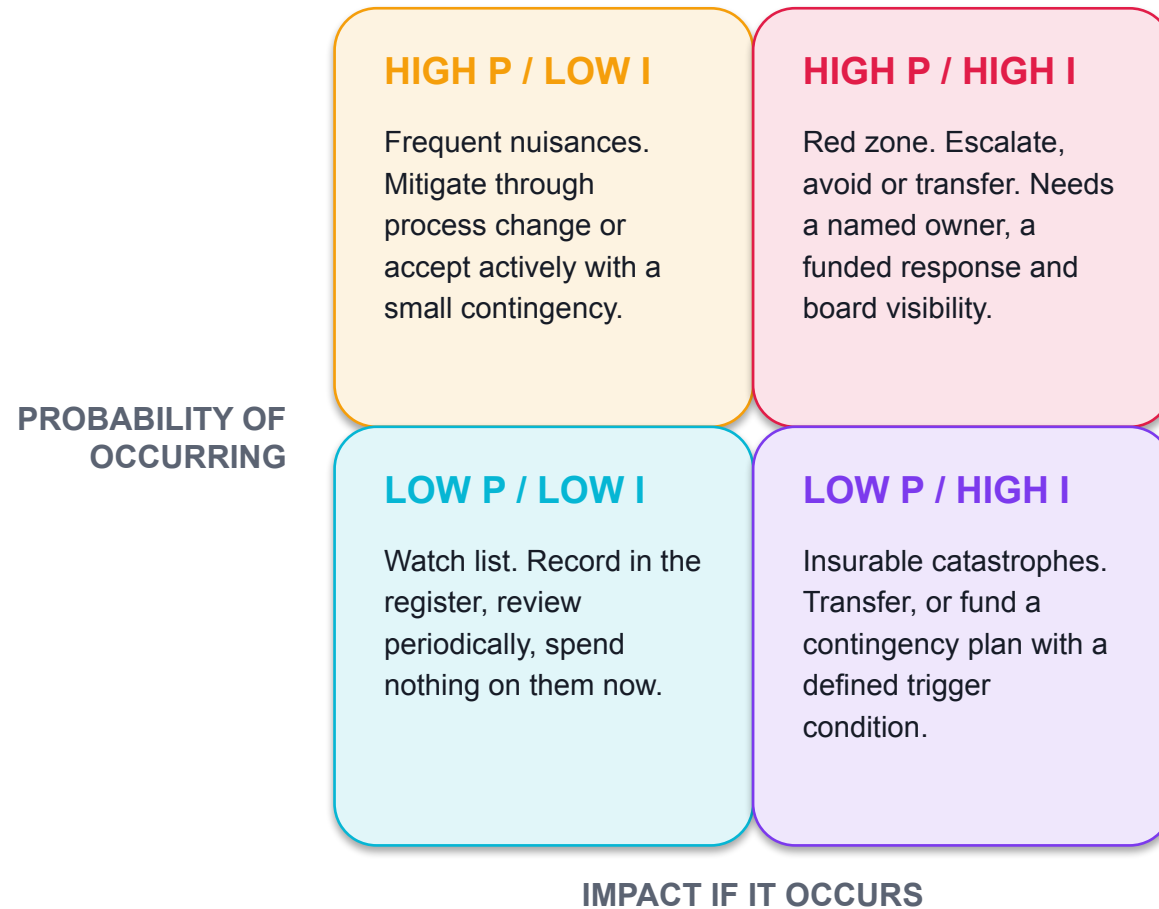
The RBS also answers a management question the register cannot: which category of risk carries most of our exposure, and is that where our mitigation money is actually going?

Perform Qualitative Risk Analysis — ITTO

Perform the subjective qualitative assessment FIRST to prioritise risks; only escalate to quantitative analysis if further support is required.



Probability and impact matrix



Priority score = probability × impact, using the numeric scales defined in the risk management plan. The matrix converts subjective judgement into a consistent, comparable ranking.

Other risk parameters beyond probability and impact

1 Urgency

How soon a response must be implemented for it to be effective — a short period indicates high urgency.

2 Proximity

How soon the risk itself is likely to occur — a short period means the risk is imminent.

3 Dormancy

The period between the risk occurring and the project discovering it — long dormancy is dangerous.

4 Manageability

How easily the risk owner can actually manage the occurrence or impact of the risk.

5 Controllability

The degree to which the organisation is able to control the risk outcome at all.

6 Connectivity

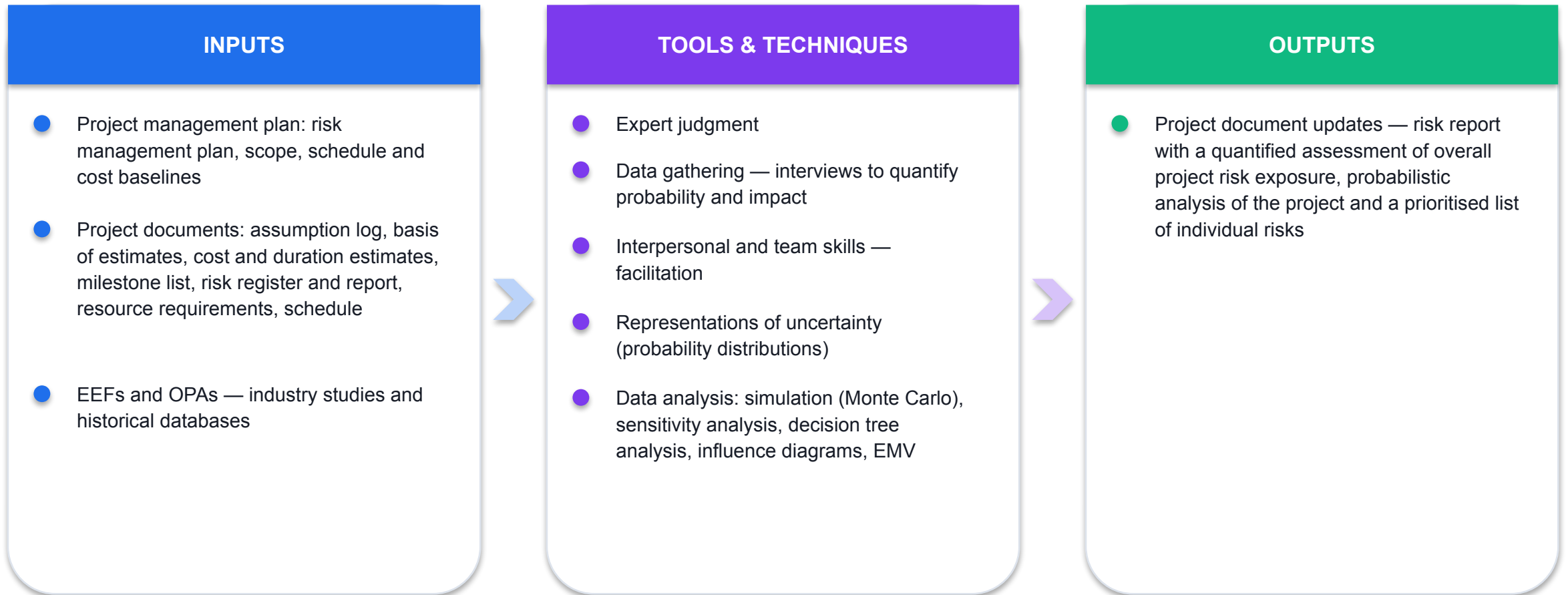
How strongly the risk is linked to other risks — clustered risks fail together.

Risk register — the fields that matter

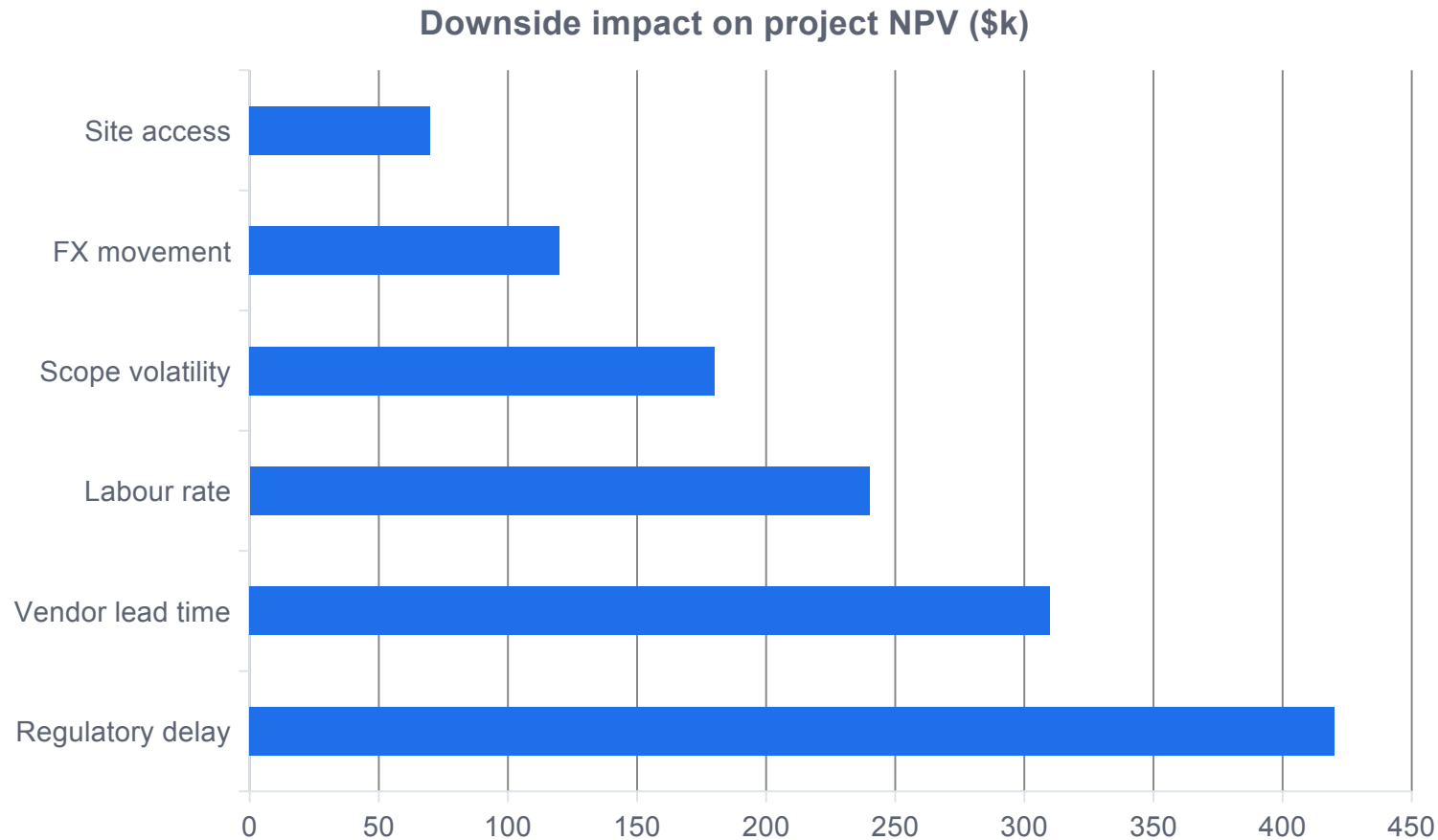
Field	Why it is there
Risk ID and title	A unique persistent reference used in reporting and in every escalation
Category (RBS code)	Lets you analyse where exposure is concentrated rather than only listing items
Risk description	Written as cause → event → effect, so the response can target the cause
Probability and impact	Rated on the scales defined in the risk management plan; produces the priority score
Risk score / priority	The ranking that decides where analysis and money go next
Risk owner	One named person accountable for monitoring and implementing the response — never a team
Planned response	The chosen strategy and the specific actions, with cost and schedule impact
Trigger condition	The observable signal that the risk is materialising and the response should be executed
Residual and secondary risk	What remains after the response, and what the response itself creates
Status	Open, occurred, closed, or retired — with the date

Perform Quantitative Risk Analysis — ITTO

Quantitative analysis is optional and usually reserved for large, complex or high-value projects where the modelling pays for itself.



Tornado chart — sensitivity analysis



WHAT IT TELLS YOU

- Sensitivity analysis shows which individual risks have the greatest potential impact on the project outcome.
- Bars are sorted longest at the top, producing the tornado shape that gives the chart its name.
- Bars show the size of the downside swing in NPV when that one input is varied and all others are held constant.
- Management action: fund mitigation for the top two or three bars; the tail rarely justifies the spend.

Expected Monetary Value and decision trees

EMV of a risk

$$EMV = P \times I$$

Probability times impact. Threats are negative, opportunities positive.

EMV of a branch

$$\Sigma (P \times I) \text{ for all outcomes}$$

Sum the expected values of every outcome on that decision branch.

Decision value

$$\text{Branch EMV} - \text{Cost of branch}$$

Choose the path with the best net expected value, not the best best-case.

Worked example

$$0.3 \times (-\$200k) = -\$60k$$

A 30% chance of a \$200k loss carries \$60k of contingency, not \$200k.

Decision trees show the decisions and their impacts (costs or benefits) along each path, together with their probabilities, producing an expected monetary value for each branch outcome.

Monte Carlo simulation

1 What it does

Runs the schedule or cost model thousands of times, sampling each uncertain input from its distribution.

2 What you supply

Three-point estimates or distributions for the uncertain activities, plus the network logic.

3 What you get

An S-curve of possible outcomes with a confidence level attached to every date or cost.

4 The P-values

P50 is the median outcome; P80 is the value you have an 80% chance of meeting — commit to P80, not P50.

5 Criticality index

How often each activity lands on the critical path across all runs — a better target list than one static CPM.

6 The limitation

Output quality is bounded by input quality; precise-looking numbers from guessed inputs are still guesses.

Plan Risk Responses — ITTO

Every response must be timely, cost-effective, agreed by all parties and owned by one named person.



Threat response strategies

Strategy	What it means	Example
Escalate	The threat is outside the project's scope or authority; hand it to the programme or portfolio level	A regulatory change affecting the whole business unit
Avoid	Eliminate the threat entirely by removing its cause or changing the plan	Drop the risky feature, or change to proven technology
Transfer	Shift the impact and ownership of the threat to a third party — the threat still exists	Insurance, warranties, performance bonds, fixed-price contracts
Mitigate	Reduce the probability or impact of the threat to an acceptable level	Prototype early, add redundancy, run more tests
Accept	Acknowledge the threat and take no proactive action. ACTIVE acceptance funds a contingency reserve; PASSIVE does nothing but document it	Minor weather delay risk with a small time buffer

Opportunity response strategies

Strategy	What it means	Example
Escalate	The opportunity is outside the project's authority to pursue; raise it to programme or portfolio	A reusable component that benefits the entire portfolio
Exploit	Ensure the opportunity definitely happens — remove all uncertainty from it	Assign your best people to guarantee an early finish that unlocks a bonus
Share	Allocate ownership to a third party best able to capture the benefit	Joint venture, partnership, risk-sharing consortium
Enhance	Increase the probability or the positive impact of the opportunity	Add resource to a task whose early finish would pull in the whole schedule
Accept	Take advantage of it if it arises but do not actively pursue it	Note a possible favourable exchange rate movement, take no action

Mirror pairs: Avoid ↔ Exploit, Transfer ↔ Share, Mitigate ↔ Enhance, and Escalate and Accept appear on both sides. Learning them as pairs halves the memorisation.

Secondary risk, residual risk and contingency plans

1 Trigger condition

The observable signal that a risk is about to develop — the cue for the team to implement the response.

2 Secondary risk

A new risk that arises as a direct result of implementing a risk response — always re-assess after responding.

3 Residual risk

The risk that remains after responses have been implemented; it is accepted knowingly and stays in the register.

4 Contingency plan

The prepared response executed if the risk occurs, funded from the contingency reserve.

5 Fallback plan

The backup executed if the contingency plan itself proves inadequate — plan B for plan B.

6 Risk owner

One named individual accountable for watching the trigger and executing the response — never a committee.

ECO 2026 emphasis — AI-assisted risk detection

1 Pattern detection

AI can scan issue logs, commit history and supplier data to surface risk signals a human would not spot.

2 Estimation support

Models trained on historical project data produce range estimates and challenge optimistic single points.

3 Register hygiene

Language models can cluster duplicate risks and rewrite vague entries as cause–event–effect statements.

4 Early warning

Sentiment and cadence signals from team communication can flag emerging delivery risk earlier than reports.

5 Human accountability

The risk owner remains a named human; AI output is an input to judgement, never the decision itself.

6 New risk category

AI use introduces its own risks: data privacy, hallucinated output, model bias and vendor dependency.

Exam focus — risk

IF THE QUESTION SAYS

A risk has occurred and is now affecting the project



THE BEST ANSWER IS

It is an issue: implement the planned response, record it in the issue log and update the risk register

IF THE QUESTION SAYS

You buy insurance against a possible loss



THE BEST ANSWER IS

That is transfer, not mitigate — the risk still exists, only the impact ownership moved

IF THE QUESTION SAYS

You cancel the risky component of the scope entirely



THE BEST ANSWER IS

That is avoid — the only strategy that removes the risk rather than reducing or moving it

IF THE QUESTION SAYS

A new risk appears because of the response you implemented



THE BEST ANSWER IS

That is a secondary risk; add it to the register and analyse it like any other

IF THE QUESTION SAYS

The question asks what remains after you respond



THE BEST ANSWER IS

Residual risk — knowingly accepted and still tracked in the register

Recap — risk planning

1 Five planning processes

Plan Risk Management, Identify, Qualitative, Quantitative, Plan Responses — in that order.

3 Qualitative before quantitative

Prioritise subjectively first; model numerically only where it pays for itself.

5 Register discipline

Cause—event—effect wording, one named owner, a trigger condition and a status on every row.

2 Definitions come first

The risk management plan sets the probability and impact scales everything else uses.

4 Ten strategies

Five for threats and five for opportunities, learned as mirror pairs.

6 Never finished

Identification is iterative; the register is reviewed at every status cycle for the life of the project.

Knowledge area 8 of 8 in the planning block

Procurement: what you buy, and who carries the risk

Make or buy

The first decision is whether the work should be done internally at all.

Agreement vs contract

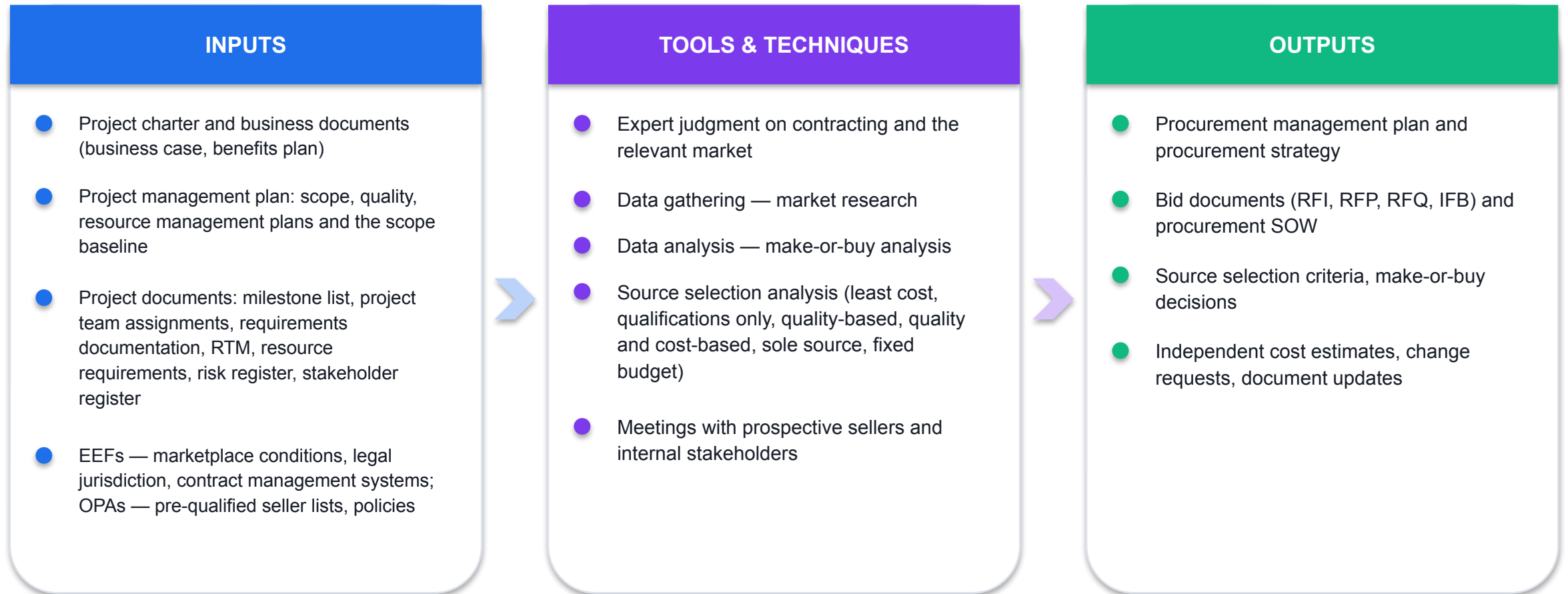
You can have an agreement without a contract, but you cannot have a contract without an agreement.

Risk allocation

Contract type is not an administrative detail — it decides who absorbs cost overrun.

Plan Procurement Management — ITTO

Procurement planning decides what to buy, how to buy it, how much to buy and when — before any seller is contacted.



Procurement strategy fundamentals

1 What procurement covers

Any resource external to your company — a vendor, contractors, consultants or any other outside resource.

2 Agreement without contract

The project charter can agree the use of internal resources without any contract existing.

3 Contract needs agreement

You cannot have a contract without an underlying agreement between the parties.

4 Delivery method

Decide turnkey, design-build, design-bid-build or staff augmentation before writing the bid documents.

5 Payment approach

Choose the contract payment type that matches how well the scope is actually defined.

6 Procurement phases

Sequence the procurement across the life cycle so long-lead items are ordered early enough to arrive.

The procurement process end to end



Bidder conferences (also vendor or pre-bid conferences) exist to ensure no seller receives preferential treatment — every question and answer is shared with all prospective sellers.

Bid documents — pick the right instrument

Document	Use it when	What you get back
Request for Information (RFI)	You need more information about what sellers can provide before you can even specify	Capability statements that shape your requirements
Request for Proposal (RFP)	There is a problem and the solution is not obvious — you need someone to propose an approach	Technical solutions with methodology, team and price
Request for Quote (RFQ)	You know exactly what you need and just require the cost from the seller	Prices for a clearly specified item or service
Invitation for Bid (IFB)	A well-defined scope where price is the deciding factor, often in public sector	Sealed price bids against an identical specification

The rule of thumb: RFI when you do not know what to ask for, RFP when you know the problem but not the solution, RFQ/IFB when you know exactly what you want and only need a number.

Why contracts exist, and what they must contain

1 Legalise agreements

A contract turns a working understanding into a legally enforceable obligation for both parties.

2 Structure relationships

Defines who does what, by when, to what standard, and what happens when that fails.

3 Description of work

The deliverables and scope, usually via an attached procurement statement of work.

4 Schedule and delivery

Delivery dates, milestones, and the schedule information both parties are bound to.

5 Authority and responsibility

Identification of authority where appropriate, and the responsibilities of both parties.

6 Price and payment terms

The amount, the milestones that trigger payment, and the invoicing and approval route.

7 Termination and disputes

Provisions for termination, alternative dispute resolution and governing jurisdiction.

8 Change and management

How technical and business aspects are managed, including how contract changes are agreed.

Contract types and who carries the cost risk

Type	How it works	Risk sits with
Firm Fixed Price (FFP)	One price for a well-defined scope, regardless of the seller's actual cost	SELLER — the most common and simplest fixed-price form
Fixed Price Incentive Fee (FPIF)	Fixed price plus an incentive for meeting agreed cost or performance targets	Mostly seller, shared above the target through a share ratio
Fixed Price with Economic Price Adjustment (FP-EPA)	Fixed price with a defined adjustment for inflation or currency	Seller, with defined external exposure shared — for multi-year deals
Cost Plus Fixed Fee (CPFF)	Actual allowable costs reimbursed plus a fixed fee that does not vary with cost	BUYER — the seller has no incentive to control cost
Cost Plus Incentive Fee (CPIF)	Costs reimbursed plus a fee that varies with performance against targets	Buyer, partially shared through the incentive formula
Cost Plus Award Fee (CPAF)	Costs reimbursed plus an award fee at the buyer's subjective judgement	Buyer; award is usually not subject to appeal
Time and Materials (T&M)	Hourly or daily rates plus materials; a hybrid of fixed unit price and cost reimbursable	Buyer — cap it with a not-to-exceed clause

Fixed price versus cost reimbursable — when to choose

FIXED PRICE

Choose when the scope is well defined

- Requirements are clear, stable and fully specified
- The seller carries the cost risk and prices it in
- Buyer's cost is predictable from the day of signature
- Change is expensive: every variation is a renegotiation
- Sellers may cut corners if their margin disappears

COST REIMBURSABLE

Choose when the scope is uncertain

- Requirements are expected to evolve, or the work is R&D
- The buyer carries the cost risk and must actively monitor it
- Buyer's final cost is unknown until the work is complete
- Change is cheap and flexible to accommodate
- Requires strong buyer-side cost control and audit rights

Time and materials sits between the two: use it for small amounts of work, staff augmentation, or when you need to start before the scope can be written — always with a ceiling.

Make-or-buy analysis and vendor qualification

1 Make-or-buy analysis

Compare the total cost, capability, capacity and risk of doing the work internally versus procuring it.

2 Beyond price

Consider strategic capability retention, intellectual property and long-term supportability, not just cost.

3 Independent estimate

Produce your own cost estimate before you see any bid, so you can judge whether a bid is realistic.

4 Qualified vendors

Pre-approved by the organisation, with a history of work and accounts already set up.

5 Preferred vendors

Often preferred precisely because they are proven — the procurement cycle is shorter and less risky.

6 Source selection criteria

Published in advance and applied consistently: capability, capacity, price, references, risk.

ECO 2026 emphasis — sustainable procurement

1 Sustainability criteria

Environmental and social performance sit in the source selection criteria alongside price and capability.

2 Supply chain due diligence

Assess labour practice, provenance and modern slavery exposure across the seller's own supply chain.

3 Whole-life cost

Evaluate energy, maintenance and disposal cost over the asset life, not just the purchase price.

4 Contractual commitments

Write sustainability obligations and reporting duties into the contract, with measurable targets.

5 Local and inclusive sourcing

Many jurisdictions now weight local content and supplier diversity in public procurement.

6 Reporting obligation

Sustainability disclosure is increasingly a regulatory requirement flowing down to project suppliers.

Exam focus — procurement

IF THE QUESTION SAYS

The scope is unclear and likely to change



THE BEST ANSWER IS

Choose a cost-reimbursable or T&M contract; a fixed price forces renegotiation on every variation

IF THE QUESTION SAYS

The buyer wants maximum cost certainty



THE BEST ANSWER IS

Firm Fixed Price — the seller carries the cost risk entirely

IF THE QUESTION SAYS

All prospective sellers must receive the same information



THE BEST ANSWER IS

Hold a bidder conference; share every question and answer with all sellers

IF THE QUESTION SAYS

You need to know what solutions exist before specifying



THE BEST ANSWER IS

Issue an RFI — an RFP presumes you can already state the problem

IF THE QUESTION SAYS

A T&M contract is running over budget



THE BEST ANSWER IS

T&M should carry a not-to-exceed ceiling; the buyer holds the cost risk and must monitor actively

Recap — procurement planning

1 Make or buy first

Decide internally versus externally on total cost, capability, capacity and risk — not price alone.

3 Right instrument

RFI to learn, RFP to solve, RFQ and IFB to price a known specification.

5 Fair process

Bidder conferences and published source selection criteria keep the process defensible.

2 Strategy then documents

The procurement strategy sets delivery method and payment type before bid documents are written.

4 Contract type allocates risk

Fixed price puts cost risk on the seller; cost reimbursable puts it on the buyer.

6 Sustainability counts

ECO 2026 expects environmental and social criteria in selection and in the contract itself.

Bringing the eight plans back together

The integrated plan is a system, not a folder

Dependencies

Assess the consolidated plans for dependencies, gaps and continued business value before you baseline anything.

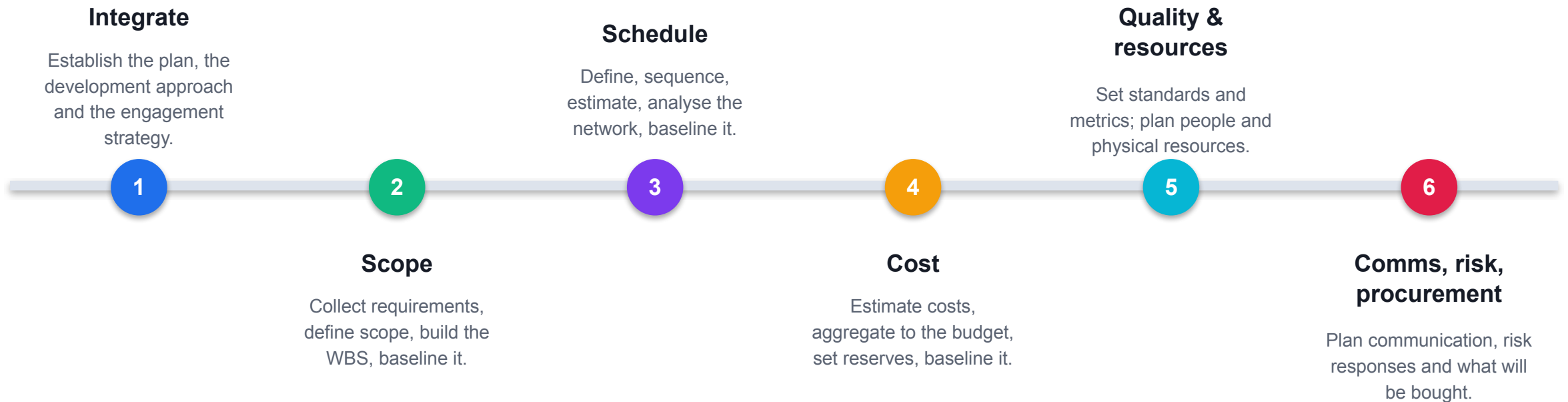
Consistency

The schedule, budget and resource plan must all describe the same project or none of them is true.

Maintenance

The integrated plan is maintained throughout, not written once and archived.

The planning sequence at a glance



The order matters because the outputs cascade: you cannot estimate a duration without a work package, or a budget without a duration and a resource. Late scope change ripples through all of it.

Topic 3 recap — plan the project

1 Integration

The project management plan reconciles all subsidiary plans and the three baselines into one coherent system.

3 Schedule

Activities, dependencies, durations and network analysis produce a calculated, defensible date.

5 Quality

Standards, metrics, cost of quality and improvement cycles build quality in instead of inspecting it in.

7 Communication

A planned strategy with channels, methods, cadence and a feedback loop for every stakeholder.

2 Scope

Requirements, RTM, scope statement, WBS and WBS dictionary define and defend what will be delivered.

4 Cost

Estimating methods and reserve strategy produce a time-phased cost baseline plus management reserve.

6 Resources

Roles, authority, responsibility, competency, RACI, levelling and smoothing make the plan executable.

8 Risk & procurement

Ten response strategies, a disciplined register, and contract types that allocate risk deliberately.

Where Topic 3 leads next

A plan is only a hypothesis until it meets reality

Topic 4 — lead

Take the resource and communication plans into practice: leadership style, motivation, conflict and coaching.

Topic 5 — control

Measure actuals against these baselines with earned value, quality control and integrated change control.

Topic 6 — close

Confirm the acceptance criteria written here were actually met, and harvest the lessons.

Hands-on Labs — Topic 3

1 Lab 09. Requirements Elicitation and Traceability Matrix

Process T2 · LO1 · 30 min in class + 20 min self-study

Produces: Requirements list, RTM

2 Lab 10. MoSCoW and Kano Prioritisation with Product Backlog

Process T3 · LO1 · 30 min in class + 20 min self-study

Produces: Prioritised backlog, Story point estimates

3 Lab 11. Work Breakdown Structure and WBS Dictionary

Process T2 · LO1 · 30 min in class + 20 min self-study

Produces: WBS, WBS dictionary, Scope baseline

4 Lab 12. Network Diagram, PERT Estimating and Critical Path

Process T8 · LO2 · 35 min in class + 25 min self-study

Produces: Network diagram, Critical path, Float table

5 Lab 13. Cost Estimating, Budget and Reserve Analysis

Process T6 · LO2 · 25 min in class + 25 min self-study

Produces: Cost estimates, Cost baseline, Reserve plan

6 Lab 14. Risk Register, Probability/Impact Matrix and EMV Decision Tree

BE T5 · LO4 · 30 min in class + 30 min self-study

Produces: Risk register, P/I matrix, EMV decision tree

Lab 09. Requirements Elicitation and Traceability Matrix

Case study: Contoso Training Portal Upgrade. Process T2 · LO1. Produces Requirements list, RTM.

30 min in class

1 Run a requirements elicitation session using the Mind Mapping tool, branching from the portal's core capabilities.

2 Convert each branch into a numbered requirement written as a testable statement, not a wish.

3 Classify each requirement as functional, non-functional, business, stakeholder, transition or quality.

4 Build the Requirements Traceability Matrix linking requirement to business need, deliverable, test and acceptance criterion.

5 Identify gaps: requirements with no test, and deliverables with no originating requirement.

6 Have a nominated 'business owner' review and sign off the matrix.

DELIVERABLE

Requirements list, RTM

TOOL

Mind Mapping

alfredang.github.io/mindmapping/

Lab 10. MoSCoW and Kano Prioritisation with Product Backlog

Case study: Contoso Training Portal Upgrade. Process T3 · LO1. Produces Prioritised backlog, Story point estimates.

30 min in class

1 Take the requirements from Lab 09 and apply MoSCoW: Must, Should, Could, Won't have this release.

2 Challenge the Musts — if more than 60% of effort is Must, the prioritisation has failed and must be redone.

3 Apply Kano classification: basic, performance and delight features, and note which Musts are merely basic expectations.

4 Write the top fifteen items as user stories in the standard 'As a... I want... so that...' form with acceptance criteria.

5 Estimate each story in story points using relative sizing against a chosen reference story.

6 Order the backlog by value per point and confirm the first sprint's capacity is realistic.

DELIVERABLE

Prioritised backlog, Story point estimates

TOOL

Scrum Simulator

alfredang.github.io/scrum/

Lab 11. Work Breakdown Structure and WBS Dictionary

Case study: Contoso Training Portal Upgrade. Process T2 · LO1. Produces WBS, WBS dictionary, Scope baseline.

30 min in class

1 Decompose the portal scope into major deliverables — not activities — at WBS level 2.

2 Break each deliverable down until every work package is estimable and assignable, typically 8-80 hours.

3 Apply the 100% rule: verify the children of every parent sum to exactly the parent's scope, no more and no less.

4 Assign a unique WBS code identifier to every element.

5 Write the WBS dictionary entry for five work packages: description, resources, cost, quality requirements and acceptance criteria.

6 Assemble the scope baseline — scope statement plus WBS plus dictionary — and record who approves it.

DELIVERABLE

WBS, WBS dictionary, Scope baseline

Lab 12. Network Diagram, PERT Estimating and Critical Path

Case study: Contoso Training Portal Upgrade. Process T8 · LO2. Produces Network diagram, Critical path, Float table.

35 min in class

1 List the activities derived from the Lab 11 work packages with their durations.

2 Determine each dependency and classify it as mandatory, discretionary, external or internal.

3 Draw the precedence network using finish-to-start relationships, adding leads and lags where genuinely required.

4 Apply three-point PERT estimating to the five most uncertain activities: $(O + 4M + P) / 6$, with $\sigma = (P - O) / 6$.

5 Run the forward and backward pass to compute early and late start and finish for every activity.

6 Identify the critical path, state the project duration, and list the total and free float for every non-critical activity.

DELIVERABLE

Network diagram, Critical path, Float table

Lab 13. Cost Estimating, Budget and Reserve Analysis

Case study: Contoso Training Portal Upgrade. Process T6 · LO2. Produces Cost estimates, Cost baseline, Reserve plan.

25 min in class

1 Estimate each work package by the most defensible method — analogous, parametric or bottom-up — and record which you used and why.

2 Apply three-point estimating to the packages with the widest uncertainty.

3 Roll the estimates up through the WBS to a total direct cost.

4 Calculate contingency reserve from the Lab 14 risk EMVs and add it to form the cost baseline.

5 Add management reserve above the baseline and state who must approve its release.

6 Plot the cumulative spend as an S-curve and identify the periods of highest cash demand.

DELIVERABLE

Cost estimates, Cost baseline, Reserve plan

TOOL

Statistics

alfredang.github.io/novastats/

HANDS-ON ACTIVITY

Lab 14. Risk Register, Probability/Impact Matrix and EMV Decision Tree

Case study: Contoso Training Portal Upgrade. BE T5 · LO4. Produces Risk register, P/I matrix, EMV decision tree.

30 min in class

1 Build a risk breakdown structure grouping risks into technical, external, organisational and project-management categories.

2 Identify at least fifteen risks, writing each in cause-event-effect form so the response is targetable.

3 Score probability and impact on 1-5 scales and plot every risk on the probability/impact matrix.

4 Use the System Thinking tool to map the reinforcing loops behind your top three risks — most schedule risk is systemic, not incidental.

5 Build an EMV decision tree for the highest-value decision and calculate the expected monetary value of each branch.

6 Assign a response strategy and a named owner to every red and amber risk, then note the secondary risks your responses create.

DELIVERABLE

Risk register, P/I matrix, EMV decision tree

TOOL

System Thinking

alfredang.github.io/systemloop/

04

TOPIC

Lead the Project Team

The People domain in practice. Execution is where the plan meets human beings — this topic covers how you direct the work, grow the team, resolve conflict and communicate so the plan actually lands.

- 1 Direct and manage project work, artifacts and knowledge
- 2 Develop the team: Tuckman, motivation and team-building
- 3 Craft your leadership style and emotional intelligence
- 4 Train, coach and mentor to close capability gaps
- 5 Manage conflict, decisions and negotiation
- 6 Plan and manage communication across a distributed team

Why this topic carries the most exam weight

Plans do not deliver projects. People do.

33% of the exam

The People domain is the single largest cluster of ECO tasks and is tested across every question type.

Situational questions

Most People items are scenario-based: you pick the PM's next action, not a definition.

Servant-leader default

PMI expects you to facilitate, coach and empower before you ever direct or escalate.

What 'leading the project team' actually means

1 Set expectations

Agree what good looks like at team level — quality bars, response times, definition of done, escalation rules.

2 Empower decisions

Push authority to the level closest to the work so the team decides without waiting on the project manager.

3 Solve problems

Break impediments down to root cause, then remove them or escalate with a recommended option, never just a complaint.

4 Represent the voice of the team

Carry the team's constraints, capacity and concerns upward to sponsors honestly and without filter.

5 Support varied experience

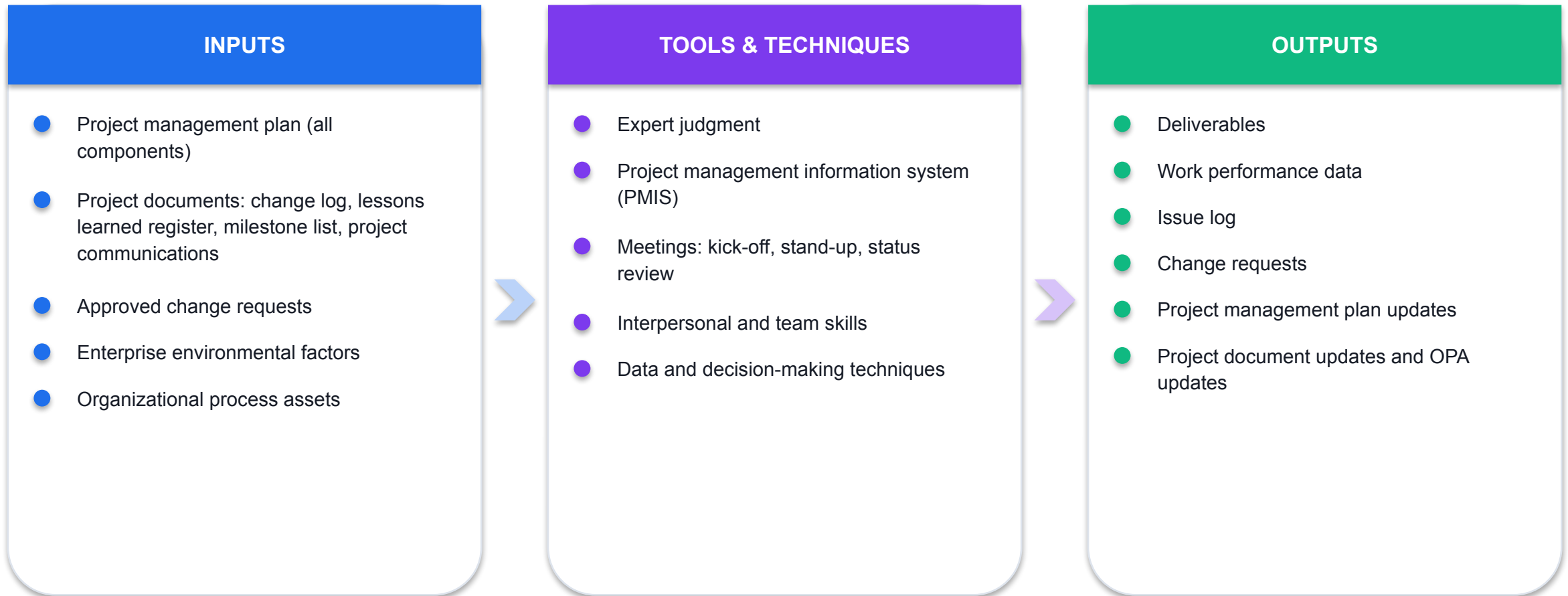
Deliberately use the differing skills, cultures and perspectives on the team as a design advantage.

6 Clarify roles

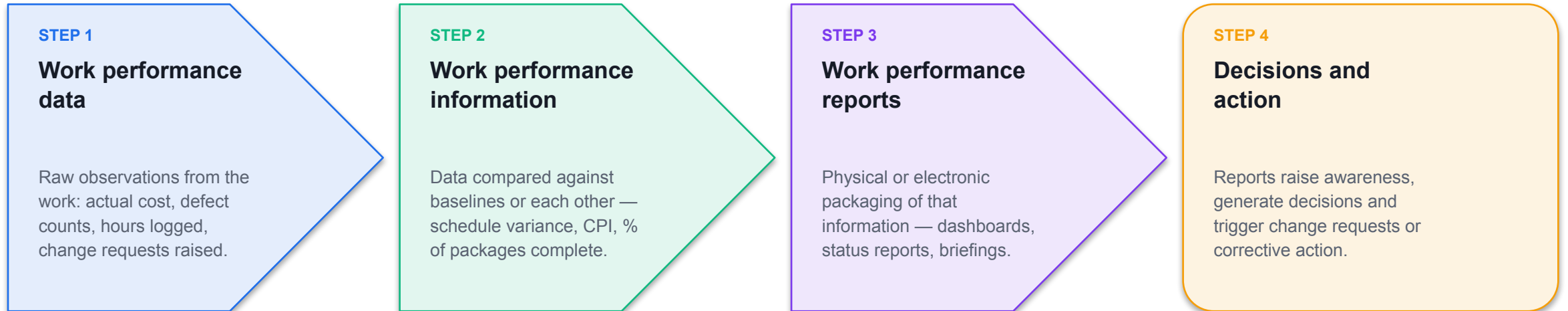
Every work package has one accountable owner; ambiguity about ownership is the number-one source of team conflict.

Direct and Manage Project Work

Executing process: lead and perform the work defined in the plan and implement approved changes to achieve the objectives.



Data becomes information becomes reports



Exam cue: data is produced by Direct and Manage Project Work; information by the Control processes; reports by Monitor and Control Project Work. The verb in the stem tells you which.

Data, information or report?

IF THE QUESTION SAYS

'The team logs 14 defects this sprint' — what is this?



THE BEST ANSWER IS

Work performance data. Raw, unprocessed measurement straight from the work itself.

IF THE QUESTION SAYS

'Defect rate is 40% above the baseline' — what is this?



THE BEST ANSWER IS

Work performance information. Data has been compared to a baseline and now carries meaning.

IF THE QUESTION SAYS

'The sponsor asks for the monthly status pack' — what is this?



THE BEST ANSWER IS

A work performance report — the communication artifact built from work performance information.

IF THE QUESTION SAYS

Question asks where deliverables are produced



THE BEST ANSWER IS

Direct and Manage Project Work. Deliverables are its primary output; verified deliverables come later from Control Quality.

Standardise your project artifacts

1 Simple production route

One agreed way to create a document so the team spends effort on content rather than on formatting.

2 Templates and formats

Standard formats for status reports, change requests and minutes make information comparable across periods.

3 Review and approval

A defined path for who drafts, who reviews and who signs off before an artifact becomes authoritative.

4 Version control

Every artifact carries a version number so people can tell instantly whether they hold the current truth.

5 Security and access

Classify artifacts and restrict who can edit versus who can only read, especially for contracts and HR data.

6 Timely distribution

An artifact that arrives after the decision has been taken has zero value regardless of its quality.

Configuration management plan vs configuration management system

Configuration management plan

A component of the project management plan

- States which items are placed under configuration control
- Defines how project and product information is recorded
- Says who may request, review and approve a configuration change
- Facilitates consistency of the product, service or result
- Written during planning; maintained across the whole life cycle

Configuration management system

The tooling and procedures that enforce the plan

- How the PM tracks project artifacts day to day
- Monitors and controls changes to baselined items
- Holds the audit trail of what changed, when and by whom
- Usually a subsystem of the wider PMIS
- Includes document control and version control mechanisms

Configuration management controls the PRODUCT; change control controls the PLAN. Exam items often test that distinction directly.

Version control and document control in practice

1 A subset of configuration management

Document control is the part of configuration management dealing with documents and digital records.

2 New version number

Increment on every substantive change so two people can confirm they are debating the same content.

3 Date and time stamp

Establishes sequence when several edits land on the same working day from different contributors.

4 Author of the change

Names the person accountable, which makes it possible to go back and ask why a change was made.

5 Change summary line

One sentence describing what moved lets a reader decide whether they need to re-read the whole document.

6 Single source of truth

Only one location is authoritative; all other copies are explicitly labelled as reference copies.

Work information management systems

Project Management Information System (PMIS)

Gathers, integrates and shares project data

- Automated collection of schedule, cost and resource data
- Ensures consistency in how data is collected and reported
- Feeds dashboards, earned value and status reporting
- Includes scheduling tools, work authorisation and configuration systems
- An input to almost every executing and controlling process

Artifact management systems

Store and maintain the project's documents

- Repository for plans, registers, logs and specifications
- Enforces access rights, retention and version history
- Makes artifacts findable — accessibility is an ECO enabler
- Supports handover of records at project or phase closure
- Continually assessed for effectiveness, per ECO Process T9

ECO Process T9 asks you to 'continually assess the effectiveness of artifact management' — a new emphasis in the 2026 outline.

Manage Project Knowledge

Use existing knowledge and create new knowledge to achieve the project's objectives and contribute to organisational learning.



Explicit vs tacit knowledge

Explicit knowledge

Codified — it can be written down and sent

- Facts, numbers, procedures, diagrams, templates
- Easily stored in a repository and searched later
- Transferred by documents, training material and databases
- Risk: people assume that documenting equals transferring
- Managed mainly through information management techniques

Tacit knowledge

Personal — beliefs, insight, experience, know-how

- Judgement, context, 'why we did it that way', relationships
- Hard to articulate; often the person does not know they hold it
- Transferred by conversation, shadowing, pairing and storytelling
- Lost permanently when a key person leaves without handover
- Managed mainly through knowledge management techniques

The hardest part of knowledge management is creating an atmosphere of trust so people are willing to share the tacit knowledge that makes them individually valuable.

Information radiators

1 Big visible charts

Large, low-tech displays placed where the team works so status is absorbed passively rather than requested.

2 Task or Kanban board

Shows every work item and its state; makes work-in-progress and bottlenecks obvious without a status meeting.

3 Burndown / burnup chart

Plots remaining or completed work against time so the team sees its own trend against the commitment.

4 Impediment board

Lists blockers, owner and age — an ageing blocker on a public wall creates the pressure to resolve it.

5 Team calendar and capacity

Absences, holidays and shared commitments visible so the team plans realistically together.

6 Radiate, don't report

Radiators push information out continuously; reports pull it out periodically and are always slightly stale.

The lessons learned register

1 Created early, not at the end

Opened during Manage Project Knowledge in the first phase and updated throughout execution.

2 What it records

The situation, what was done, the impact, and the recommendation for anyone facing that situation again.

3 Categories

Lessons on requirements, estimating, risk, procurement, communication, quality and team dynamics.

4 Feeds later phases

A live register lets the current project fix itself, not just help the next project.

5 Becomes the repository

At closure the register is folded into the organisational lessons learned repository, an OPA.

6 Retrospectives supply it

In adaptive work every iteration retrospective is a scheduled input to the register.

Foster a knowledge-sharing culture

1 Everyone teaches and learns

Knowledge transfer is not a one-way flow from senior to junior; every role holds transferable know-how.

2 Include stakeholders and customers

Suppliers, users and operations staff hold context the delivery team cannot generate on its own.

3 Dedicated sharing roles

Agile coaches and Scrum Masters exist partly to make knowledge move across team boundaries.

4 Essential at transition

Product handover to operations fails when only explicit knowledge crosses and tacit knowledge does not.

5 Communities of practice

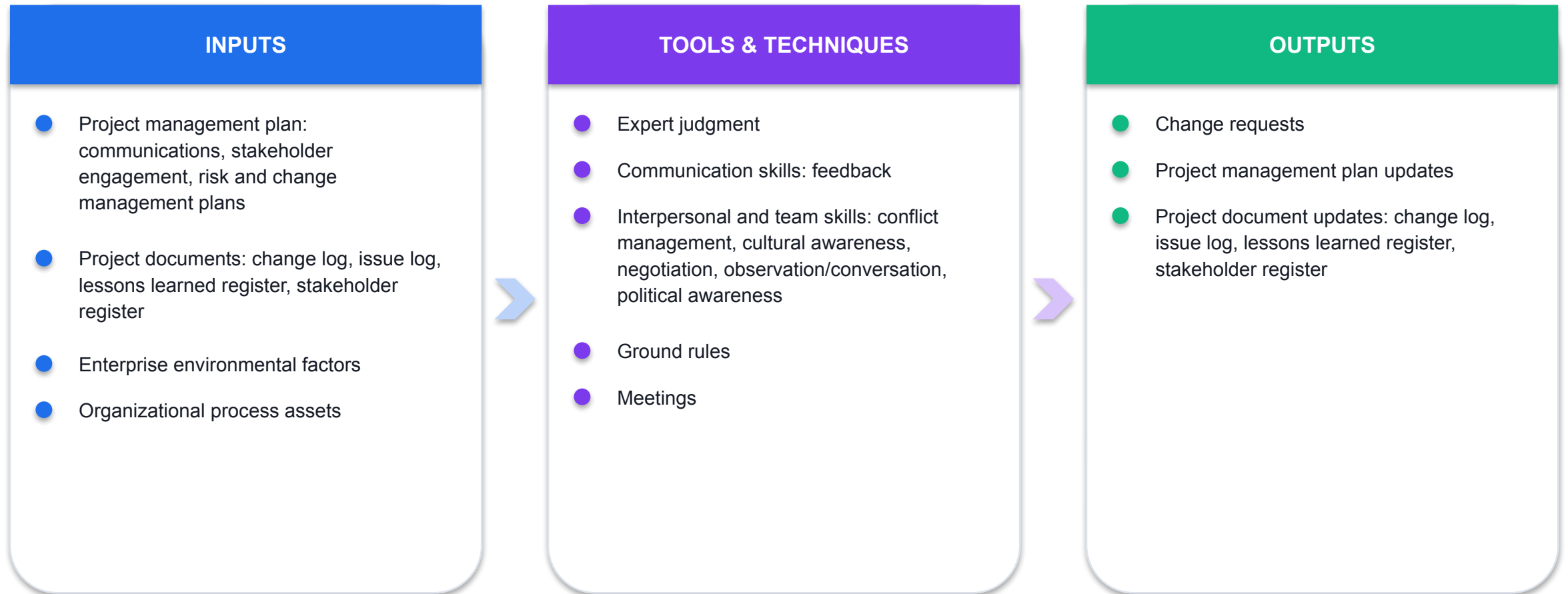
Cross-project guilds keep specialists connected so a solution found once is not re-invented.

6 Psychological safety first

People share what went wrong only when they are confident the disclosure will not be used against them.

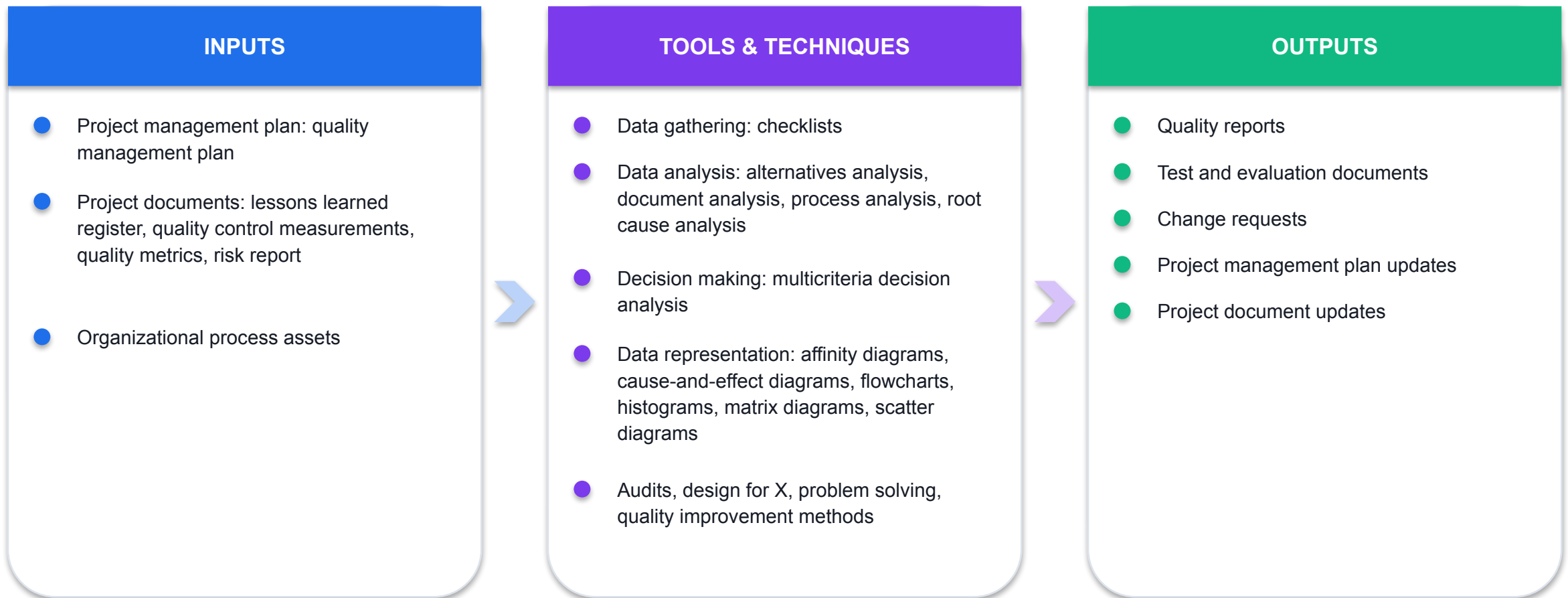
Manage Stakeholder Engagement

Communicate and work with stakeholders to meet their needs, address issues and foster appropriate engagement.



Manage Quality

Translate the quality management plan into executable quality activities — this is quality ASSURANCE, about the process.



Manage Quality vs Control Quality

Manage Quality (Executing)

Are we using the right process?

- Quality assurance — audits the process, not the product
- Outputs quality reports and test/evaluation documents
- Uses process analysis and root cause analysis
- Drives continuous improvement of how work is done
- Owned by the PM, quality department and often the whole team

Control Quality (Monitoring & Controlling)

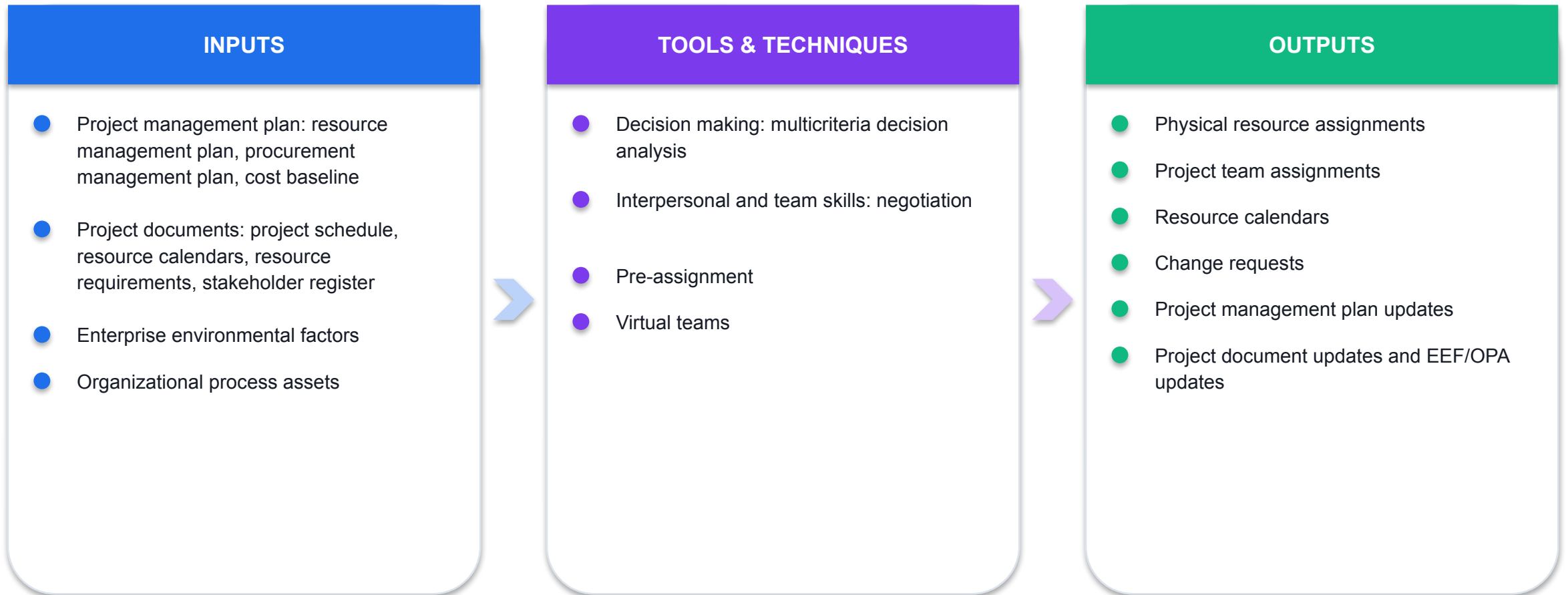
Did we build the right thing correctly?

- Inspects deliverables against acceptance criteria
- Outputs quality control measurements and verified deliverables
- Uses inspection, testing, control charts and checksheets
- Determines correctness of the output, one deliverable at a time
- Feeds verified deliverables into Validate Scope for acceptance

Mnemonic: Manage Quality is about the PROCESS and prevention; Control Quality is about the PRODUCT and inspection.

Acquire Resources

Obtain team members, facilities, equipment, materials and supplies necessary to complete the project work.



Four ways you obtain project resources

Route	How it works	When to use it	Watch out for
Pre-assignment	People are named in the charter or promised in the proposal	Competitive bids won on named experts; unique specialists	The named person may no longer be available when you start
Negotiation	You bargain with functional managers, other PMs or the PMO	Matrix organisations where resources are shared	You win only what your influence and the project's priority allow
Acquisition	Hire externally or contract in the skill from a supplier	The skill genuinely does not exist inside the organisation	Cost, lead time, onboarding and knowledge leaving at contract end
Virtual teams	People contribute from different locations or time zones	Scarce skills, remote workers, cost or mobility constraints	Isolation, communication lag, weaker informal knowledge flow

If the question describes a resource you cannot get, negotiate first, then escalate to the sponsor — do not simply re-plan the schedule around the gap.

Develop Team

Improve competencies, team member interaction and the overall team environment to enhance project performance.



Develop Team in one line

An effective team is the deliverable behind every deliverable.

Cohesion

Solidarity and mutual reliance measurably raise throughput and lower defect rates.

Bonding needs leadership

Teams do not gel by accident; the leader engineers the conditions that let it happen.

Team over individual

Team-building shifts identity from 'my task is done' to 'our increment is done'.

Team-building activities

1 Formal

Facilitated offsites, chartering workshops and structured team assessments with a defined agenda and outcome.

2 Informal

Five minutes at stand-up, shared meals, coffee walks — low-cost, high-frequency contact that builds trust.

3 Brief

A short check-in ritual or icebreaker at the start of every meeting keeps connection alive between bigger events.

4 Extended

Multi-day workshops used at project start or after a major team change to reset norms and relationships.

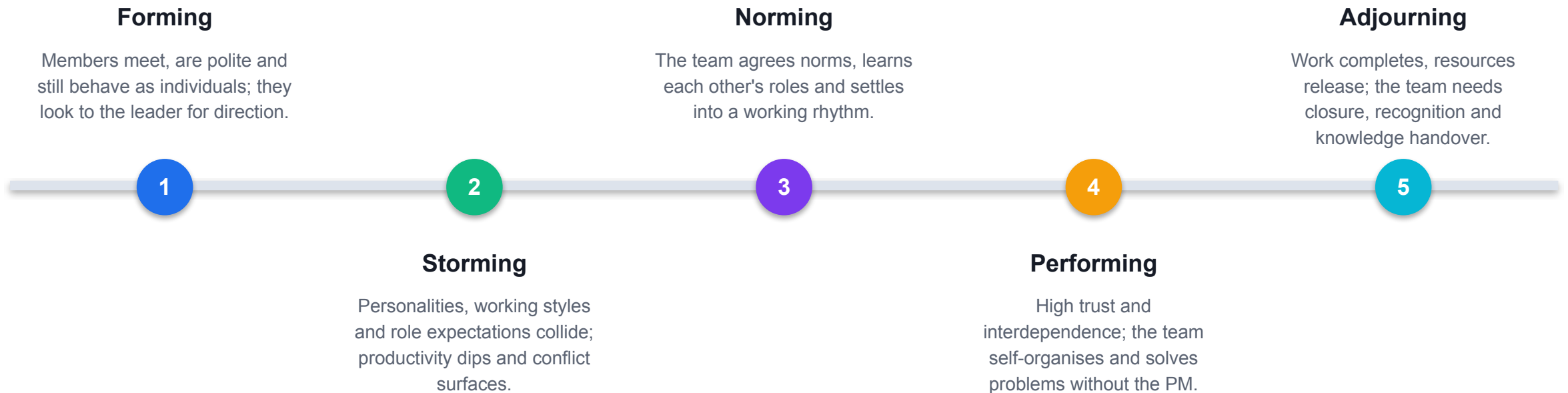
5 Self-facilitated

The PM runs it — cheap and fast, but the PM cannot both facilitate and participate honestly.

6 Professionally facilitated

Neutral facilitator needed when the PM is part of the problem or the conflict level is high.

Tuckman's ladder of team development



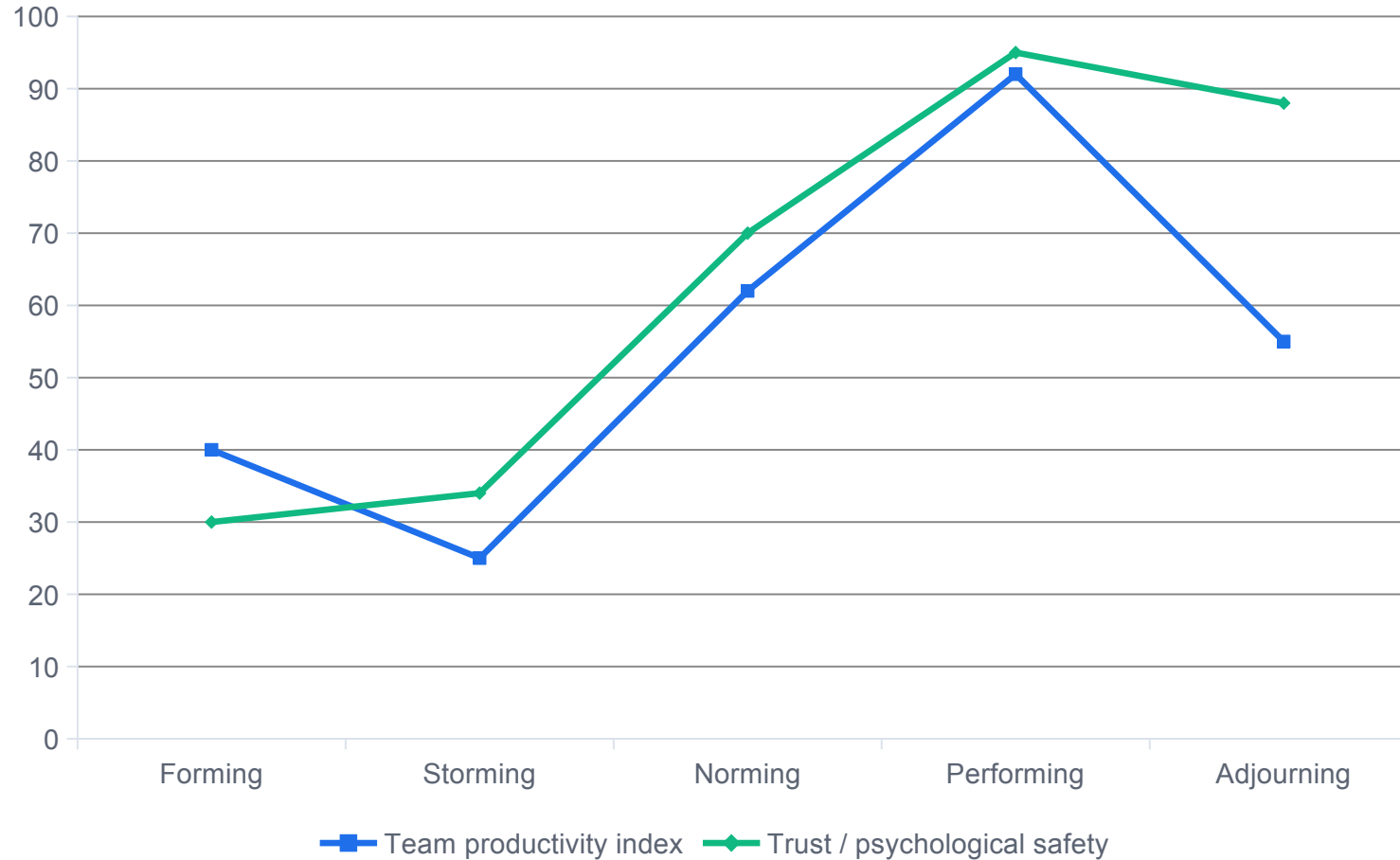
Teams can regress: a new member, a change of scope or a change of PM can drop a performing team back into storming.

What the project manager does at each Tuckman stage

Stage	What the team looks like	Your leadership move
Forming	Polite, cautious, dependent, unclear on roles	Be DIRECTIVE — set purpose, roles, ground rules and the team charter
Storming	Disagreement on approach, resistance, low morale	COACH — surface conflict, facilitate, do not suppress the debate
Norming	Norms accepted, trust building, rhythm forming	SUPPORT — let them own decisions, reinforce the norms they set
Performing	Interdependent, self-correcting, high output	DELEGATE — remove impediments, protect the team, stay out of the way
Adjourning	Anxiety about what is next, energy drops	CLOSE — recognise contribution, capture lessons, plan the transition

Exam pattern: 'the team is arguing about approach' = storming, and the best answer is almost always to facilitate the discussion, not to decide for them or to reassign someone.

The performance dip you must expect at storming



WHAT IT TELLS YOU

- Productivity falls at storming — this is normal, not a sign your team is failing.
- Trust keeps rising through storming if conflict is handled openly; suppressing it flattens the whole curve.
- Teams left in storming never reach the performing payoff, which is where most of the value is created.
- At adjourning output falls but trust stays high — that is exactly when knowledge transfer is easiest.

Tuckman traps on the exam

IF THE QUESTION SAYS

A new member joins a high-performing team



THE BEST ANSWER IS

The team regresses toward forming/storming; re-establish roles and ground rules rather than assuming performance continues.

IF THE QUESTION SAYS

Team members openly disagree about the technical approach



THE BEST ANSWER IS

This is storming. Facilitate a structured discussion — conflict here is productive and expected.

IF THE QUESTION SAYS

The team resolves issues without involving you



THE BEST ANSWER IS

Performing. Your job is removing impediments and shielding the team, not reasserting control.

IF THE QUESTION SAYS

Project ends and the team seems demotivated



THE BEST ANSWER IS

Adjourning. Recognise contributions, complete lessons learned and manage the release/transition of each member.

The motivational theories PMI expects you to know

Theory	Core idea	What it tells a project manager
Maslow's hierarchy of needs	Physiological, safety, belonging, esteem, self-actualisation	You cannot motivate with growth opportunities if job security is in doubt — lower needs dominate first
Herzberg's motivation-hygiene	Hygiene factors prevent dissatisfaction; motivators create satisfaction	Fixing salary or the office removes complaints but never creates engagement — that needs achievement and recognition
McGregor's Theory X / Theory Y	X assumes people avoid work; Y assumes people seek responsibility	Your assumption becomes self-fulfilling; Theory Y underpins agile and servant leadership
McClelland's achievement motivation	Achievement, affiliation or power becomes dominant per person	Match the assignment and the reward to the individual's dominant need, not to your own
Vroom's expectancy theory	Motivation = expectancy x instrumentality x valence	If any of the three is zero, motivation is zero — the reward must be attainable, linked and actually wanted

Maslow's hierarchy of needs



Practical read: announcing a restructure destroys the safety level, and every motivational lever above it stops working until safety is restored.

Herzberg: hygiene factors vs motivators

Hygiene factors

Their absence causes dissatisfaction; their presence does not motivate

- Salary, benefits and job security
- Working conditions and equipment
- Company policy and administration
- Quality of supervision and interpersonal relations
- Fix these to get to neutral — never to get to engaged

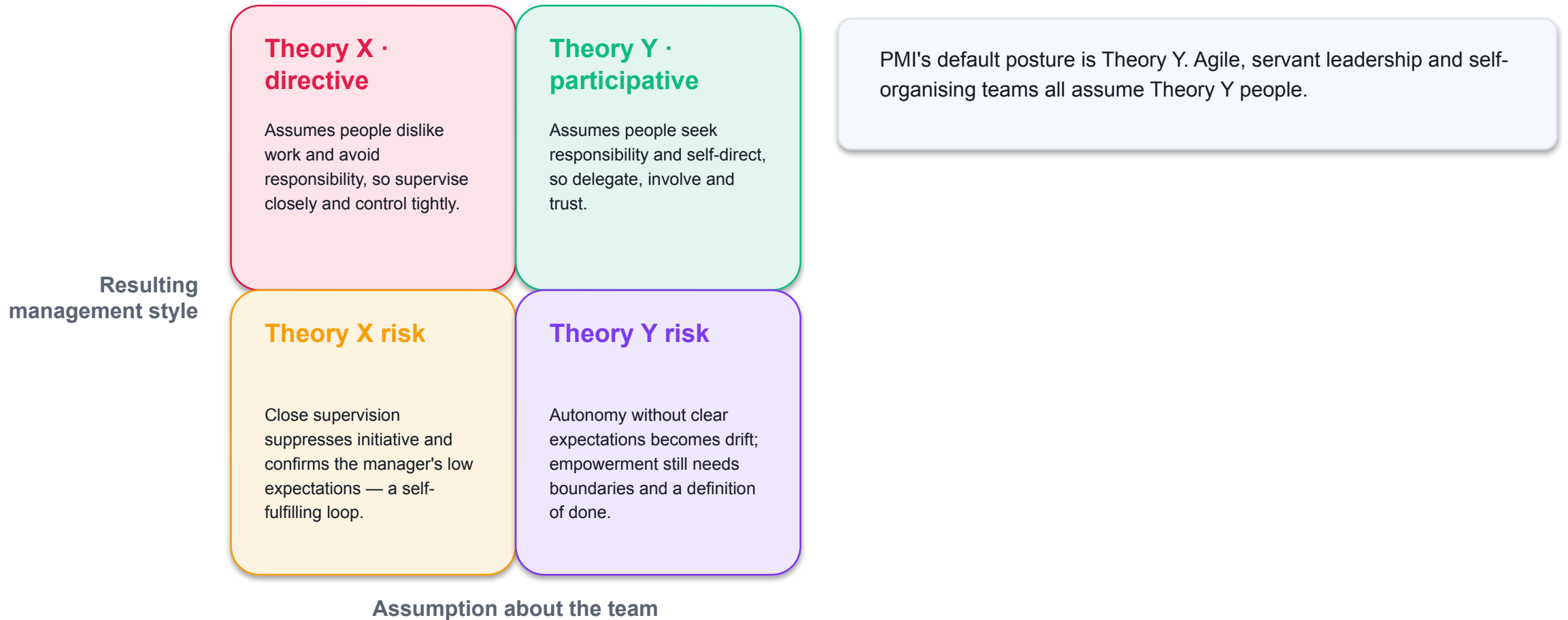
Motivators

Their presence creates genuine satisfaction and effort

- Achievement — visibly completing something difficult
- Recognition for the work done
- The work itself being interesting and meaningful
- Responsibility and autonomy over how the work is done
- Advancement and personal growth

Exam cue: 'the team got a bonus but morale is still low' — you addressed a hygiene factor. Look for a motivator such as recognition, autonomy or growth.

McGregor's Theory X and Theory Y in practice



McClelland's achievement motivation theory

1 Need for achievement (nAch)

Driven by challenging but attainable goals and concrete feedback; dislikes vague or trivial work.

2 Need for affiliation (nAff)

Driven by belonging and harmony; give collaborative work and avoid isolating or adversarial roles.

3 Need for power (nPow)

Driven by influence and impact; give coordination, negotiation and representation responsibilities.

4 Shaped by life experience

One of the three becomes dominant based on a person's history, not on their job title.

5 Use it to design roles

Craft assignments and rewards around each member's dominant need rather than a single team-wide incentive.

6 Build T-shaped people

Balance the needs across the team so it can self-organise instead of depending on one dominant personality.

Vroom's expectancy theory

Motivational force

$$M = E \times I \times V$$

Motivation is the product of three beliefs — if any one is zero, the whole motivation is zero.

Expectancy (E)

Effort → Performance

'If I work hard, can I actually succeed?' Undermined by unrealistic targets, missing skills or missing tools.

Instrumentality (I)

Performance → Outcome

'If I succeed, will the promised reward really arrive?' Destroyed by one broken promise.

Valence (V)

Value of the outcome

'Do I actually want that reward?' A team dinner has zero valence for someone who wants a promotion.

Diagnostic use: when motivation is low, test all three terms before assuming the problem is the size of the reward.

Two behaviours that quietly destroy your schedule

Parkinson's Law

Work expands to fill the time allotted to it

- A five-day task given ten days will take ten days
- Generous estimates get consumed, never returned as savings
- Counter with short timeboxes and fixed iteration lengths
- Counter by making remaining work visible on a burndown
- Explains why padding every estimate does not create real buffer

Student syndrome

People start only at the last possible moment

- Safety in the estimate is spent at the end, not at the start
- Any early problem then hits the deadline with no buffer left
- Counter with early checkpoints and definition-of-done gates
- Counter by decomposing into small items that finish continuously
- Together with Parkinson's Law, the core argument for timeboxing

Critical chain scheduling responds to both by stripping individual task padding and pooling it into a shared project buffer the PM controls.

Servant leadership behaviours

1 Listening

Understand the problem as the team experiences it before offering your own solution or judgement.

2 Helping people grow

Treat capability building as a deliverable; grow the team's role capability, not just this project's output.

3 Safe to fail

Make it safe to make mistakes and raise concerns early, when they are still cheap to fix.

4 Coaching, not controlling

Ask questions that let the team find the answer instead of issuing the answer yourself.

5 Remove impediments

Your primary daily job: clear the blockers the team cannot clear from inside the team.

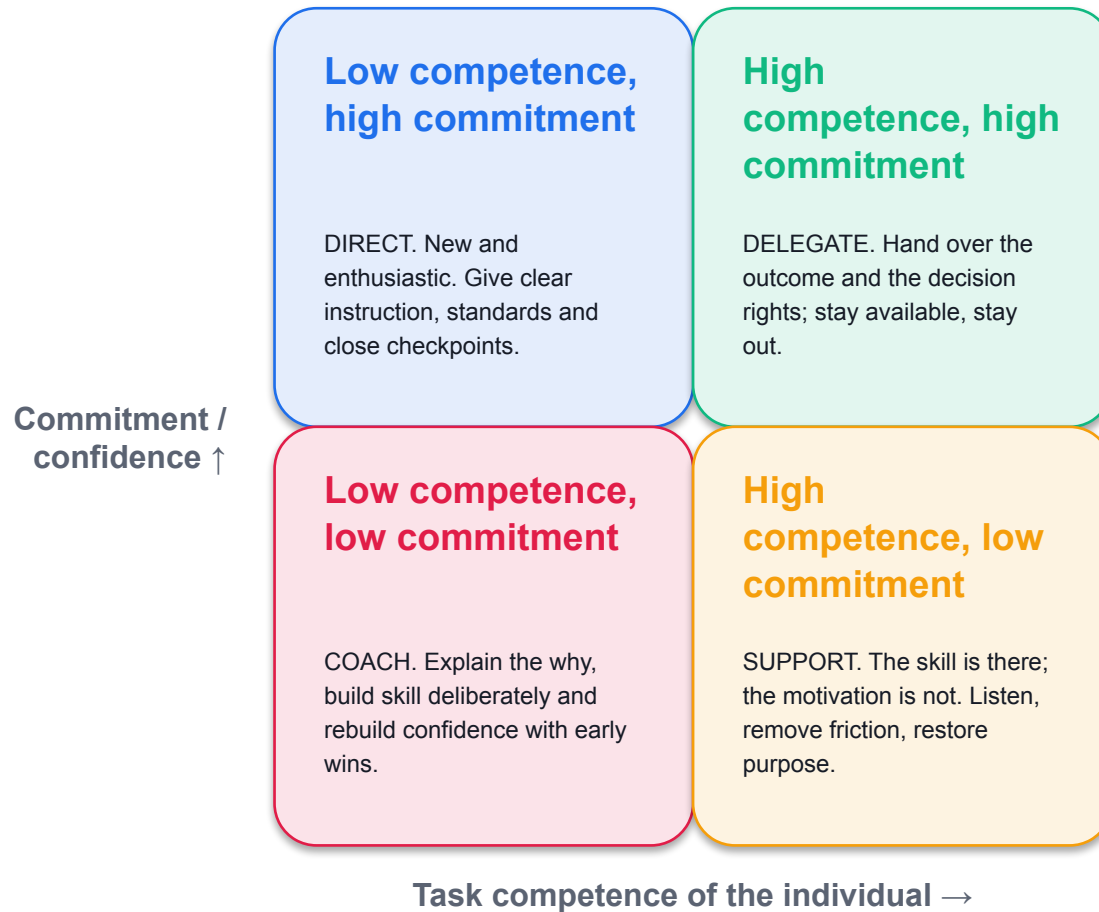
6 Pave the way for others

Promote the energy and intelligence of others; take the credit last and the blame first.

Leadership styles you must recognise

Style	Behaviour	Best fit
Servant	Focuses on others' growth, learning, autonomy and well-being	Agile and knowledge work; PMI's default posture
Laissez-faire	Hands-off; the team makes its own decisions with little input	Highly mature, expert, self-organising teams
Transactional	Rewards for meeting goals; manages by exception when things go wrong	Routine, repeatable work with clear measurable targets
Transformational	Inspirational motivation, intellectual stimulation, individual attention	Change programmes needing energy and a compelling vision
Charismatic	High energy, self-confident, strong convictions, persuasive presence	Rallying stakeholders; risky if it substitutes for substance
Interactional	Blend of transactional, transformational and charismatic	Most real projects — deliberately mixed as the situation shifts
Directive	Tells people what to do and how; low participation	Crisis, safety incidents, and brand-new teams at forming

Situational leadership: match style to readiness



There is no single correct leadership style. The ECO enabler is literally 'determine an appropriate leadership style' — appropriate to this person, this task, this moment.

Leading vs managing

Leading

Direction, alignment, commitment — about people

- Sets and communicates the vision and the why
- Aligns people around a shared goal
- Inspires, motivates and builds commitment
- Develops capability and challenges the status quo
- Asks: are we doing the right thing?

Managing

Planning, organising, controlling — about work

- Plans, budgets and schedules the work
- Organises structure, roles and staffing
- Monitors variance and applies corrective action
- Maintains stability, predictability and control
- Asks: are we doing the thing right?

A project manager does both every day. The exam usually wants leading behaviour in People-domain scenarios and managing behaviour in Process-domain scenarios.

Emotional intelligence — the four domains

1 Self-awareness

Recognising your own emotions in the moment and understanding how your actions land on other people.

2 Self-management

Thinking before you act; regulating your response so pressure does not leak into the team as anxiety.

3 Social awareness

Reading the emotional state of the room; empathy for what others are experiencing but not saying.

4 Social skill

Establishing rapport, managing your own attitude and steering relationships toward a productive outcome.

5 Why it matters here

EI is the mechanism behind conflict resolution, influencing, negotiation and stakeholder trust.

6 It is trainable

Unlike personality, EI is a learnable practice — pause, name the emotion, choose the response.

HANDS-ON ACTIVITY

Role play: practising emotional intelligence under pressure

In trios, run a coaching conversation about a stressful workplace moment. Rotate roles: team leader, team member, observer. The full script and prompts are in the Learner Guide.

25 MIN

1 Team member recalls a recent moment when they reacted emotionally at work and describes it factually.

2 Leader coaches self-awareness: 'what were you feeling, and what did you notice in yourself first?'

3 Leader models self-management by describing a pause-and-clarify response instead of an immediate reaction.

4 Member re-runs the scenario applying the pause, then asks a clarifying question rather than defending.

5 Leader coaches empathy: what might the other person have been experiencing that you did not see?

6 Observer records which of the four EI domains appeared and where the conversation slipped into advice-giving.

DELIVERABLE

One observation sheet per trio mapping the conversation to the four EI domains, plus one improvement each person will apply this week.

Interpersonal and team skills

1 Active listening

Listen for content, tone and what is being avoided; confirm understanding before responding.

2 Communication styles assessment

Identify preferred styles so you can tailor how information is delivered per stakeholder.

3 Influencing

Persuade without authority using evidence, relationships and alignment to the other party's interests.

4 Motivation

Apply the right theory to the right person to sustain discretionary effort over a long project.

5 Nominal group technique

Silent idea generation then structured ranking, so loud voices do not dominate the outcome.

6 Political awareness

Understand where real power and informal influence sit, and work with the organisation as it actually is.

Rewards vs recognition

Rewards

Tangible, consumable, tied to an outcome

- A bonus, a voucher, a gift, a paid course
- Given for a specific achievement or measurable outcome
- Used to motivate toward a defined result
- Must be attainable, visible and fairly applied
- Never give a reward without recognition attached to it

Recognition

Intangible, experiential, tied to behaviour

- Public thanks, a note to a manager, visible credit
- Acknowledges the person's behaviour, not just the outcome
- Increases the recipient's sense of being valued
- Costs nothing and can be given continuously
- Can absolutely be given on its own without any reward

Cultural caution: public individual recognition motivates in some cultures and embarrasses in others. Ask before you design the ritual.

Adopt a growth mindset as a leader

1 Experience guides, not dictates

Let past processes inform your action without locking you into last project's answer.

2 Commit to improving

Treat your own leadership practice as something to iterate on, with feedback and deliberate change.

3 Discover through discussion

Find the best approach through debate and introspection rather than asserting it up front.

4 Avoid complacency

Blind acceptance of 'how we do things' is how an organisation stops learning.

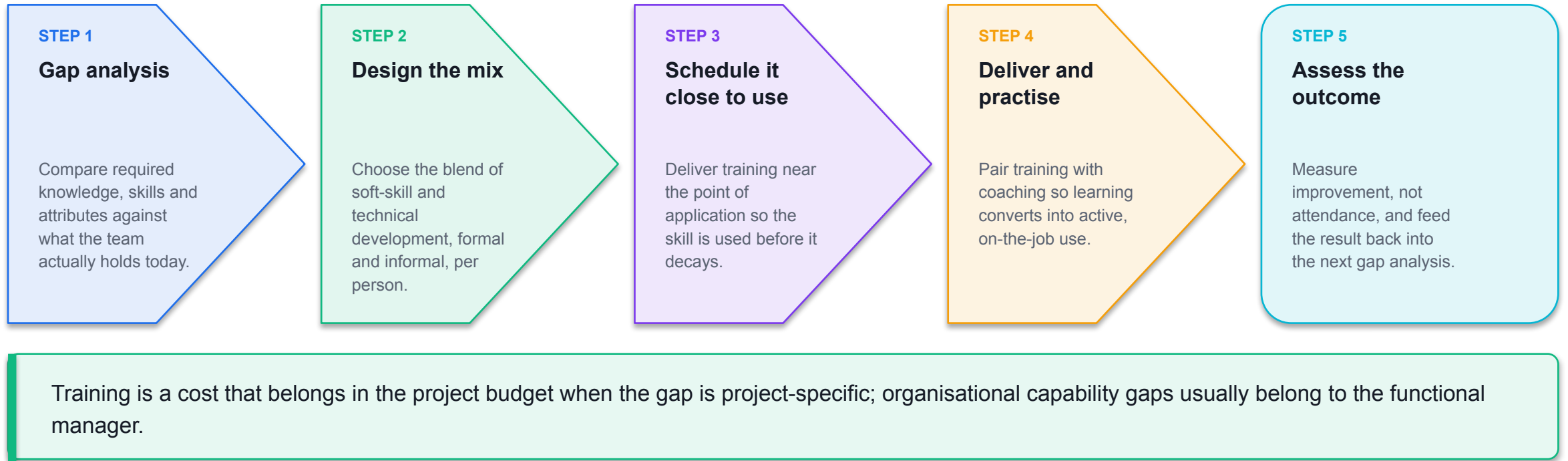
5 Model fallibility

A leader who says 'I got that wrong' licenses the whole team to surface their own errors early.

6 Inclusive competence

Lead with empathy, assume motivations vary, and deliberately include external and part-time resources.

Plan training, coaching and mentoring



Elements of training and delivery models

1 Instructor-led classroom

Highest interaction and fastest clarification; highest cost in time and travel.

2 Virtual classroom

Live but distributed; needs deliberate facilitation to keep participation genuine rather than passive.

3 Self-paced e-learning

Cheap and scalable for explicit knowledge; weak for tacit knowledge and judgement.

4 Document reviews

Structured reading with a comprehension checkpoint; good for standards, policies and compliance content.

5 Interactive simulation

Safe rehearsal of a high-stakes task; the only realistic option for rare, dangerous or costly scenarios.

6 On-the-job training

Real work with a safety net; highest transfer rate but consumes the time of an experienced person.

Training vs coaching vs mentoring

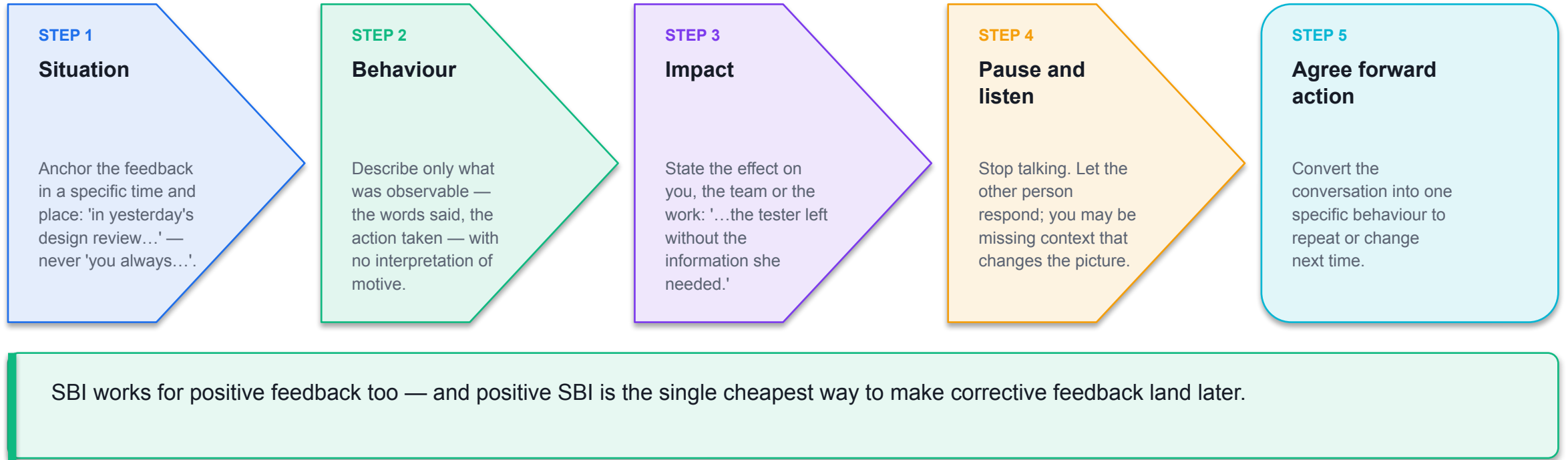
Dimension	Training	Coaching	Mentoring
Purpose	Acquire a skill for use now	Apply or improve an existing skill	Long-term personal and professional growth
Horizon	Short — a session or a course	Medium — weeks around a performance goal	Long — months or years
Relationship	Trainer to learners	Coach to individual or team	Experienced person to a less experienced person
Content owner	The trainer sets the curriculum	The coachee sets the goal; coach asks the questions	The mentee sets the direction; mentor shares experience
Also called	Upskilling	Performance coaching	Sponsorship, job shadowing, professional guidance
Typical output	Certification, badge, new capability	Changed behaviour on a specific task	Career progression, transferred tacit knowledge

Steps to coach performance



Coaching is asking, not telling. If you find yourself giving the answer in the first two minutes, you have switched from coaching to instructing.

SBI — a technique for giving feedback



Coach teams and individuals in project management

1 Delegate then observe

Hand over a real task, watch how it is done and feed back — the most common informal coaching opportunity.

2 Encourage others to lead

Let a team member run the stand-up, the demo or the risk review while you sit in the room.

3 Collaborate on a PM task

Build the schedule or the risk register together so the reasoning is transferred, not just the artifact.

4 Facilitate sessions

Formalise it: teach facilitation by co-facilitating, then handing the session over.

5 Pair individuals

Deliberately pair a strong and a developing member on the same work item to transfer tacit knowledge.

6 Model behaviours

The team copies what you do under pressure, not what you wrote in the team charter.

HANDS-ON ACTIVITY

Role play: coaching a struggling team member

In pairs, run a one-to-one coaching conversation about a performance gap (e.g. low customer satisfaction scores). Swap roles after the first round. Full script in the Learner Guide.

25 MIN

1 Leader opens with a genuine strength observed before naming the gap.

2 Leader uses SBI to describe one specific, observed behaviour and its impact — no character judgements.

3 Leader asks open questions to let the coachee diagnose the cause themselves rather than supplying it.

4 Leader demonstrates the desired behaviour once, briefly, then hands the conversation back.

5 Coachee rehearses the behaviour; leader gives one reinforcing and one adjusting piece of feedback.

6 Both agree one dated action and the date of the follow-up check-in.

DELIVERABLE

A completed coaching note per pair: the observed behaviour, the impact, the agreed action and the follow-up date.

Measure training and coaching outcomes

1 Post-training assessment

Test whether the skill can be performed, not merely whether the session was enjoyed.

2 Observation on the job

The real proof: does the behaviour appear in actual work two weeks after the course?

3 Certification evidence

Badges or a letter from the awarding body give externally verifiable proof of a capability.

4 Retrospective discussion

Share what was learned in a team retrospective so one person's training benefits everyone.

5 Augment with coaching

Training alone decays fast; coaching after training converts knowledge into active use.

6 Knowledge-sharing pairs

Pair the trained person with a peer so they must explain it — teaching consolidates learning.

Maintain mentorships and job shadowing

1 Longer-term partnerships

Mentoring is a relationship, not an event; value accrues over months of repeated contact.

2 Job shadowing

Observing an expert at work transfers both explicit procedure and the tacit judgement behind it.

3 Tailor to the context

Some organisations run formal mentoring with reporting; others keep it deliberately informal.

4 Use during transitions

Shadowing at handover moves skills from the project team into the receiving operations team.

5 Recruit stakeholders

Valued stakeholders make excellent mentors and their involvement deepens their engagement.

6 Mentor project managers

Many organisations grow PMs specifically through mentorship rather than formal training.

Assess team member performance

1 Formal and informal methods

Combine structured assessments with continuous observation; neither alone gives a fair picture.

2 Baseline on joining

Assess when a member joins so you can measure development rather than guess at it later.

3 Monitor progress

Repeat the assessment so the conversation is about trajectory, not a single snapshot.

4 Self-regulating teams

Psychologically safe agile teams assess and correct their own performance in retrospectives.

5 Focus on the team

Team performance assessments target the team's capability; individual appraisal usually sits with line management.

6 Feed Develop Team

Assessment results identify the training, coaching and team-building the team actually needs next.

Self-organising teams collaborate and learn

1 Encourage initiative

Expect team members to pick up work and raise problems without waiting for an assignment.

2 Cross-role coaching

Coach people on how to contribute to other project roles so the team is resilient to absence.

3 Surface tacit knowledge

Coach the individual who holds undocumented know-how into explaining and demonstrating it.

4 Use servant leadership

Support and enable rather than assign and check; self-organisation dies under close supervision.

5 Shadow during transitions

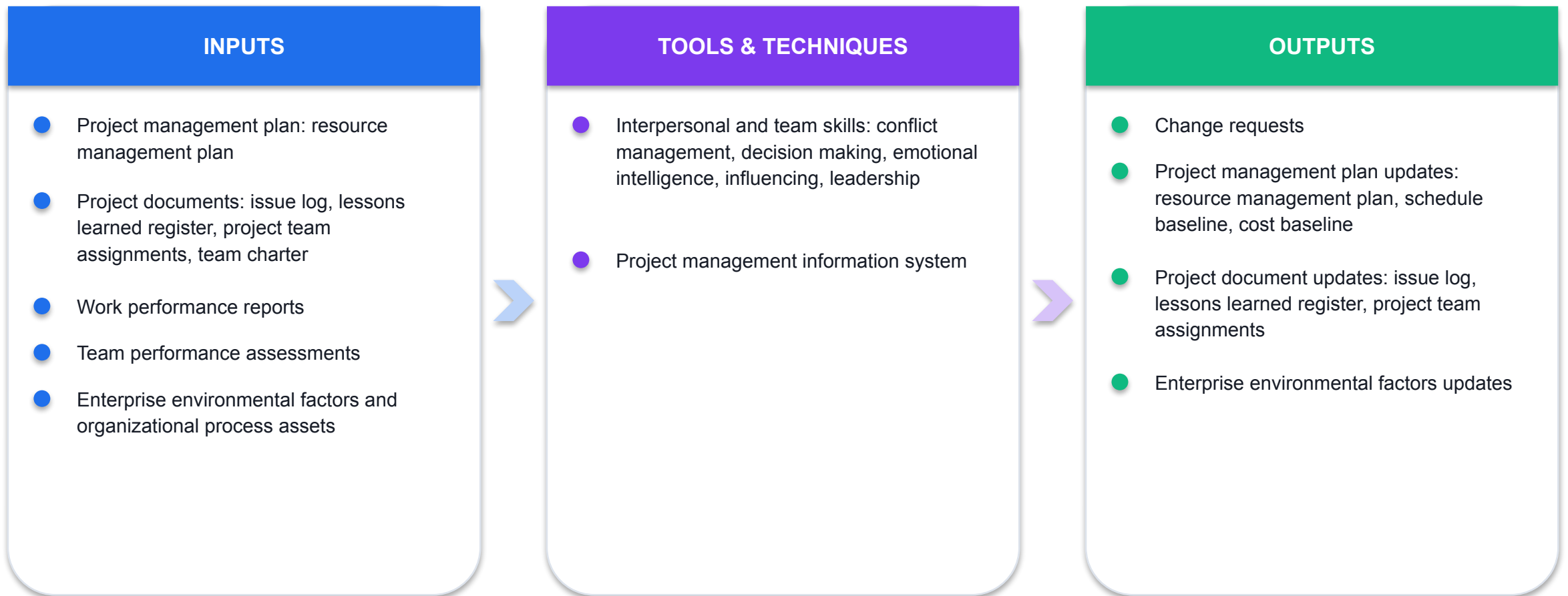
Use shadowing, coaching and mentoring to move skills from the project team to the organisation.

6 Ground rules still apply

Self-organising is not unbounded — the team charter defines the boundary they organise within.

Manage Team

Track team performance, provide feedback, resolve issues and manage team changes to optimise project performance.



Who owns conflict on a project

Everyone owns it. You influence how it is handled.

Shared responsibility

All team members and stakeholders are responsible for managing conflict, not just the PM.

PM influences direction

You shape the handling through interpersonal skills and servant leadership, not authority.

Team resolves it

The team is empowered to resolve its own conflicts; the leader facilitates when they cannot.

Sources of conflict on projects

1 Competition for resources

Scarce people, budget and equipment force zero-sum choices between work streams.

2 Differing objectives and values

Goals that conflict at the organisational level surface as friction at the team level.

3 Role and responsibility disputes

Ambiguity about who decides and who delivers is the most common and most fixable source.

4 Differing work approaches

Two competent people with different methods each believe the other is doing it wrong.

5 Communication breakdown

Missing, late or ambiguous information gets filled in with assumption, then with blame.

6 Novelty of the project

Projects are unique and members often have not worked together before — no established norms exist yet.

Effective vs ineffective conflict management

Ineffective conflict management

Conflict suppressed, avoided or escalated badly

- Destructive behaviour and personal attacks
- Animosity that outlasts the disagreement itself
- Poor performance as energy diverts from the work
- Reduced productivity and lost calendar time
- Attrition — good people leave dysfunctional teams

Effective conflict management

Conflict surfaced early and worked through openly

- Improved understanding of the other position
- Better decisions because assumptions were challenged
- Higher performance from genuine, not artificial, agreement
- Higher productivity as issues stop recurring
- Stronger relationships that survive the next disagreement

Zero conflict is not the target. Zero conflict usually means people have stopped saying what they think.

Conflict as part of a psychologically safe culture

1 Disruption and innovation connect

The friction that produces new ideas is the same friction that feels uncomfortable.

2 Encourage disagreement

Actively invite dissent so risks are named while there is still time to act on them.

3 Separate people from positions

Attack the problem, not the person holding the opposing view.

4 Prevent escalation

Address friction at the level of a differing opinion, before it hardens into a personal contest.

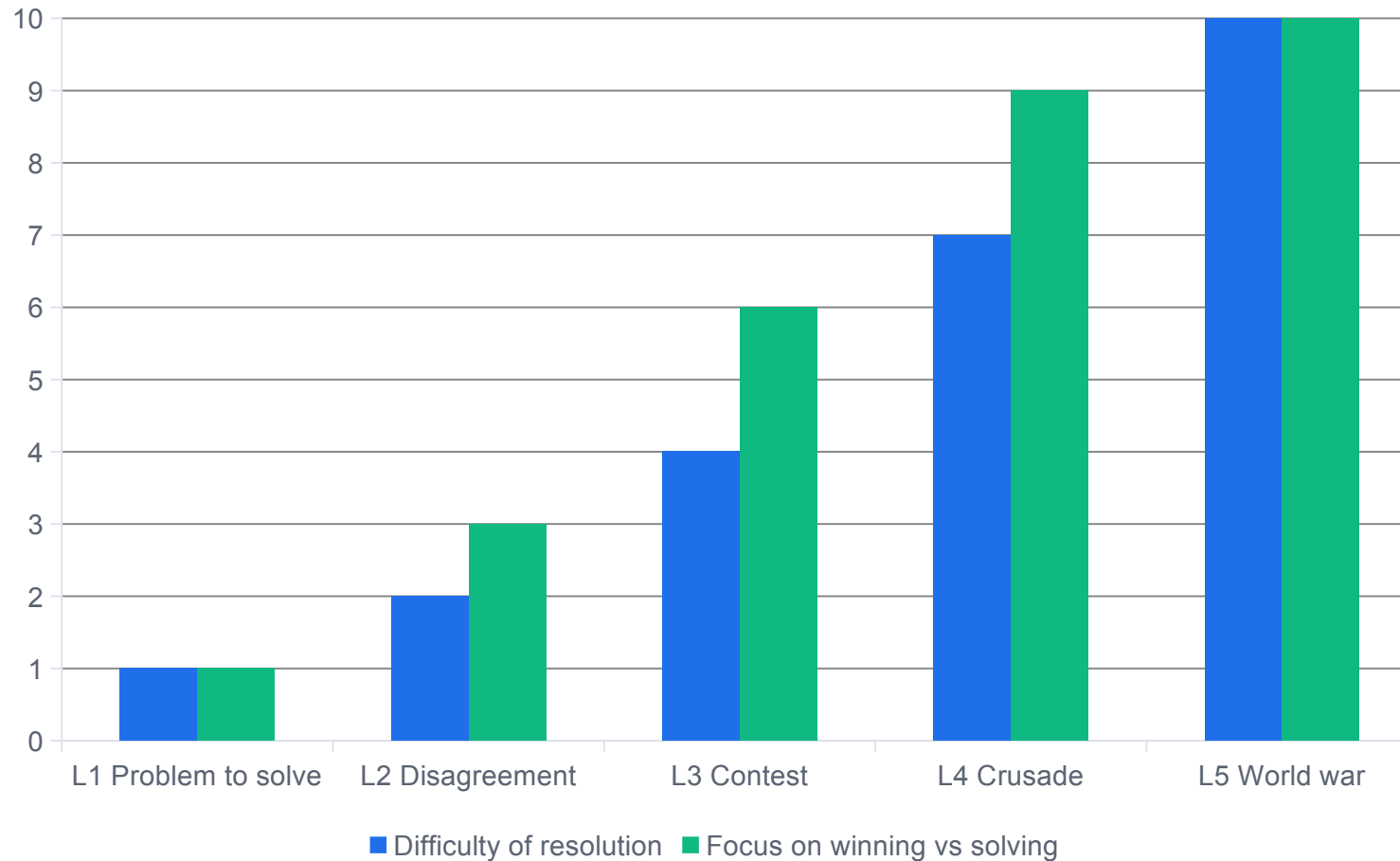
5 Ground rules make it safe

Agreed norms tell people what disagreement is allowed to look like on this team.

6 Leader goes first

Safety is created when the leader publicly accepts challenge without punishing the challenger.

Leas' five levels of conflict — escalation



WHAT IT TELLS YOU

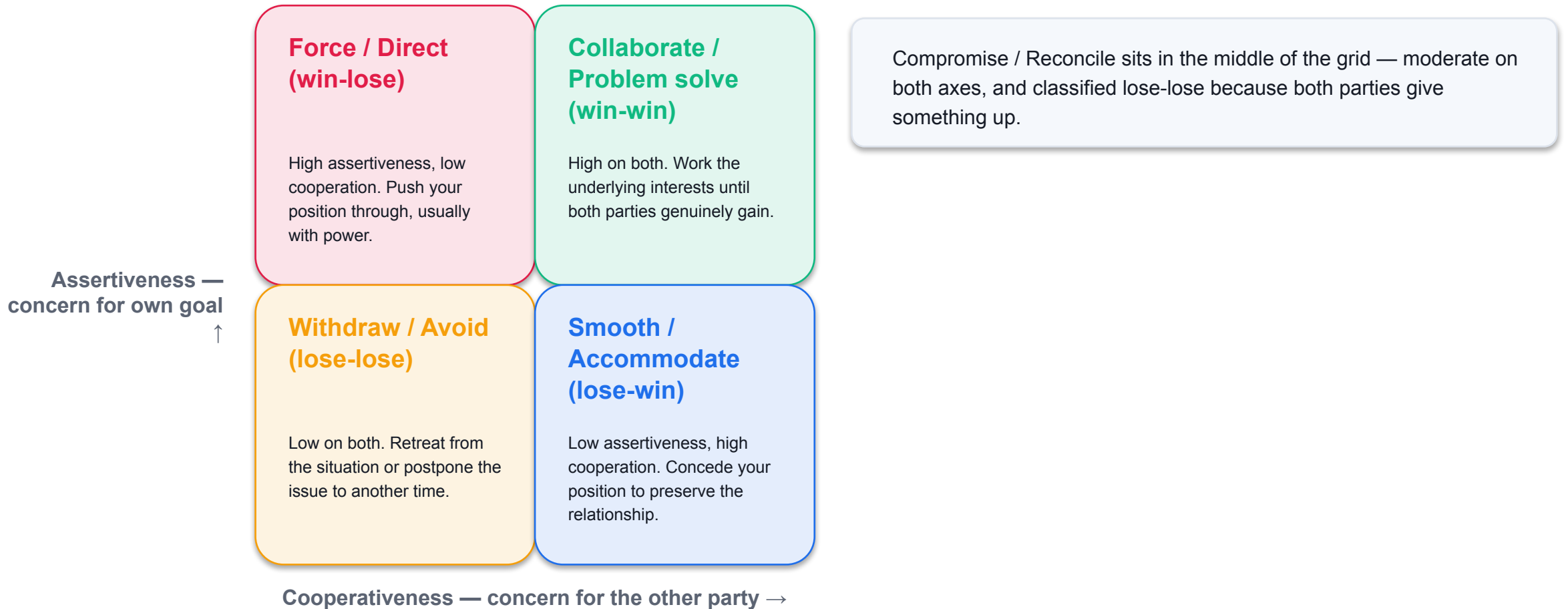
- L1: information sharing and collaboration still work; the parties want a solution.
- L2: self-protection begins — language becomes guarded and positions get defended.
- L3: the goal shifts from solving the problem to winning against the other party.
- L4: protecting one's own group becomes the point; the original issue is almost irrelevant.
- L5: destroying the other party is the objective — separate the parties, escalate, do not mediate.

Leas' levels of conflict and your intervention

Level	What you hear	What to do
1 · Problem to solve	'How should we handle this?' — focus on the issue, open language	Facilitate collaboratively; the team can resolve this themselves
2 · Disagreement	Guarded language, self-protection, humour used to deflect	Encourage direct dialogue; make it safe to state the real position
3 · Contest	'I' vs 'you', winning matters, positions distorted to score points	Structure the conversation; refocus on shared objectives and data
4 · Crusade	'Us' vs 'them', factions form, the issue becomes ideological	Bring in a neutral facilitator; the PM is likely seen as partisan
5 · World war	Destroying the other party is the goal; no resolution is acceptable	Separate the parties, escalate to sponsor/HR — mediation will fail

Intervention must match the level. Collaboration attempted at level 4 or 5 does not just fail — it burns your credibility as a neutral party.

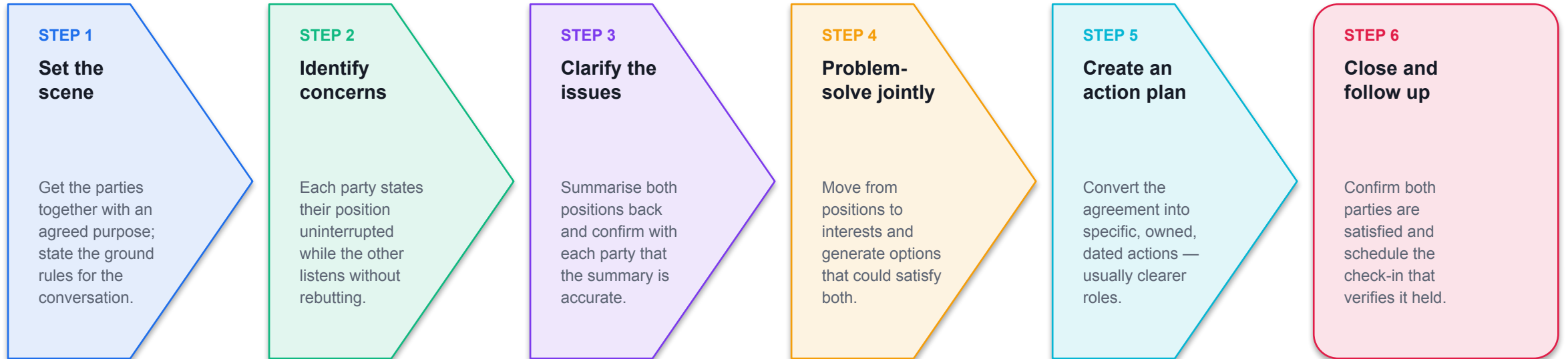
The five conflict-resolution modes



The five conflict approaches and when each is appropriate

Approach	Outcome	What it looks like	When it is right
Collaborate / Problem solve	Win-win	Incorporate multiple viewpoints; open dialogue to real consensus	PMI's default. Time exists, relationship matters, stakes are high
Compromise / Reconcile	Lose-lose	Both parties give something up to reach a workable middle	Equal power, moderate stakes, a temporary fix under time pressure
Smooth / Accommodate	Lose-win	Emphasise areas of agreement; concede to keep harmony	You are wrong, the issue matters far more to them, or goodwill is the real objective
Force / Direct	Win-lose	Pursue your view at the other party's expense; use position power	Safety, legal or ethical breach; genuine emergency; no time
Withdraw / Avoid	Lose-lose	Retreat from the situation or postpone the issue	Emotions too high to be productive; you need facts; the issue is trivial or will resolve itself

A repeatable conflict-resolution sequence



The four behaviours that make this work: open communication, active listening, joint problem solving, and a clear action plan that neither party can quietly ignore.

Role play: a conflict over project responsibilities

In trios, work a real conflict: a senior member feels overloaded and over-responsible; a newer member feels distrusted and blocked from contributing. Third person facilitates. Full script in the Learner Guide.

30 MIN

1 Facilitator sets the scene, states the purpose and agrees ground rules for the conversation.

2 Each party states their concern for two minutes, uninterrupted, using 'I feel / I observe' language.

3 Facilitator summarises both concerns and gets explicit agreement that the summary is accurate.

4 Both parties generate options together; the facilitator captures them without evaluating.

5 Agree a concrete action plan — clear task ownership and a weekly check-in — and name the owner of each item.

6 Debrief: which of the five conflict modes was actually used, and which of Leas' levels did this conflict sit at?

DELIVERABLE

One completed conflict resolution record per trio: sources, level, mode used, agreed actions with owners, and the follow-up date.

Interpersonal skills that resolve conflict

1 Emotional intelligence

Use empathy to understand what is driving each party and to defuse the emotional charge first.

2 Influencing

Persuade parties to reconsider their tone, approach or mindset when direct authority would only harden positions.

3 Leadership

Steer the exchange toward a constructive direction and hold the group to the agreed ground rules.

4 Decision making

When the parties genuinely deadlock, offer a decision method or a solution that moves the situation forward.

5 Active listening

Listen for accusing language, caustic tone and defensive posture — these tell you the escalation level.

6 Political awareness

Understand what each party stands to lose organisationally, which is often the real driver.

Ground rules: establish, adhere, rectify

1 Where they live

Ground rules are documented in the team charter, created by the team rather than imposed by the PM.

2 What they cover

Meeting conduct, response times, decision rights, how disagreement is raised, escalation paths.

3 Foster adherence

Reference them regularly in real situations so they stay a live agreement rather than a forgotten document.

4 Detect violations

Watch for the quiet ones: interruptions, decisions made outside the agreed forum, missed commitments.

5 Rectify violations

Address privately and early using SBI; the whole team is watching whether the rule is real.

6 Revisit and update

When a rule stops fitting, the team changes it deliberately — it does not simply erode.

Conflict questions — the traps PMI sets

IF THE QUESTION SAYS

Two team members disagree; the PM should...



THE BEST ANSWER IS

Let them resolve it first. The team is empowered; you facilitate only if they cannot reach resolution.

IF THE QUESTION SAYS

'What is the BEST conflict resolution technique?'



THE BEST ANSWER IS

Collaborate / problem solve. It is the only genuine win-win and is PMI's default answer absent other constraints.

IF THE QUESTION SAYS

A safety or compliance breach is happening now



THE BEST ANSWER IS

Force / direct. Emergencies and ethical breaches are the legitimate case for asserting authority.

IF THE QUESTION SAYS

Emotions are running too high to discuss the issue



THE BEST ANSWER IS

Withdraw / avoid temporarily — cool down, gather facts, then return and collaborate.

More conflict and team traps

IF THE QUESTION SAYS

Conflict recurs on the same topic every week



THE BEST ANSWER IS

Go to root cause. Recurring conflict usually signals unclear roles or an unresolved structural constraint, not personalities.

IF THE QUESTION SAYS

A team member repeatedly breaks an agreed ground rule



THE BEST ANSWER IS

Address it directly and privately with the individual first, referencing the team charter the team itself created.

IF THE QUESTION SAYS

The sponsor asks you to overrule the team's technical decision



THE BEST ANSWER IS

Represent the voice of the team: present their reasoning and the impact, then negotiate rather than simply comply or refuse.

IF THE QUESTION SAYS

Conflict is at Leas' level 5 between two departments



THE BEST ANSWER IS

Separate the parties and escalate. Collaboration and mediation are not viable at that level.

How project teams decide

1 Team charter sets the rules

The charter identifies the decision-making and conflict-resolution criteria in advance.

2 Teams define their Way of Working

Mature teams establish their own norms for which decisions need which method.

3 Consensus is the aim

Teams try for genuine agreement first; a vote is what you use when consensus is not reachable.

4 Decide who decides

The most common decision failure is ambiguity about whether a group is advising or deciding.

5 Timebox the decision

An undecided decision has a cost; set a deadline and a fallback method before you start.

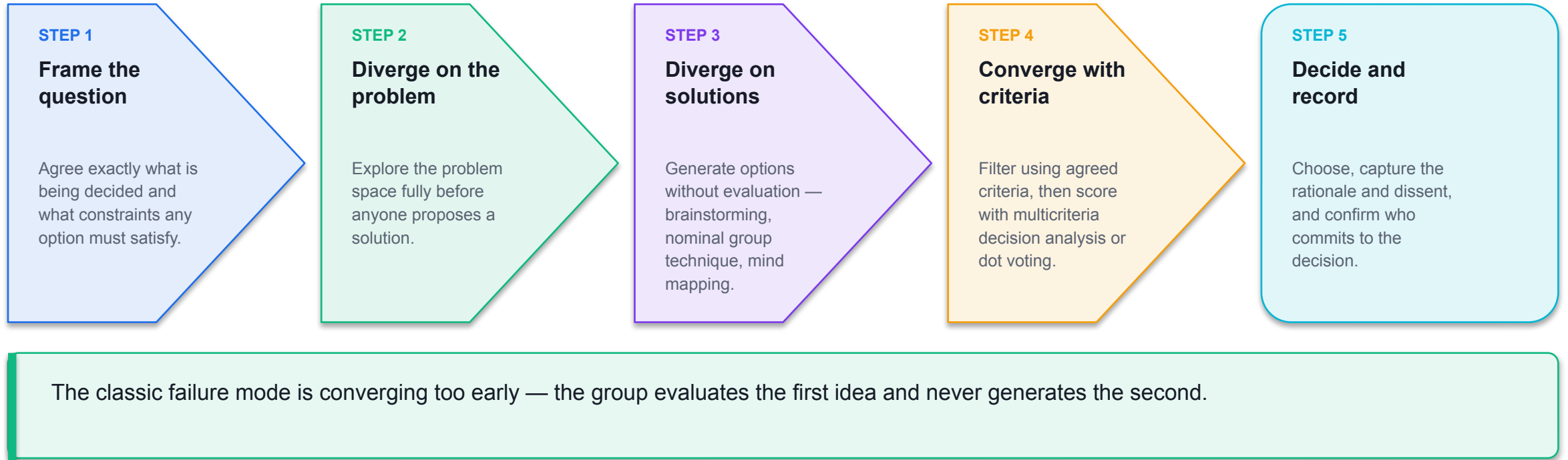
6 Record the rationale

Capture why, not just what — it prevents the same debate reopening three months later.

Decision-making methods

Method	How it works	When to use it
Unanimity	Everyone agrees on a single course of action	High-cohesion teams; irreversible decisions; the Delphi technique converges toward it
Majority	More than 50% of the group supports the option	Use odd-numbered groups so a tie cannot block the decision
Plurality	The largest single block decides, even without 50%	More than two options on the table and no majority emerges
Autocratic	One individual decides for the whole group	Emergency, or the decision is genuinely one person's accountability
Delphi technique	Anonymous expert rounds with feedback until convergence	Expert estimates where seniority or politics would distort an open discussion
Multicriteria decision analysis	Score options against weighted criteria in a matrix	Comparing suppliers, designs or candidates on several dimensions at once
Fist of five	Each person shows 0-5 fingers to signal support level	Fast consensus check; anything below three triggers a discussion
Dot voting	Each participant places a fixed number of dots on options	Prioritising a long list of ideas quickly and visibly

Diverge then converge



Practice: multi-criteria decision analysis

In table groups, choose between three candidate vendors for a project component using a weighted scoring matrix, then compare how the group's decision method changed the outcome.

25 MIN

1 List the decision criteria that genuinely matter (cost, capability, risk, delivery time, support).

2 Weight each criterion by importance and get the group to agree the weights before seeing the options.

3 Score all three vendors against each criterion, using evidence rather than preference.

4 Compute the weighted totals and identify the top-ranked option.

5 Re-run the same choice using dot voting alone and note whether the answer changes.

6 Debrief which method you would defend to a sponsor and why.

DELIVERABLE

A completed weighted scoring matrix with agreed weights, scores, the recommended vendor and a one-paragraph rationale.

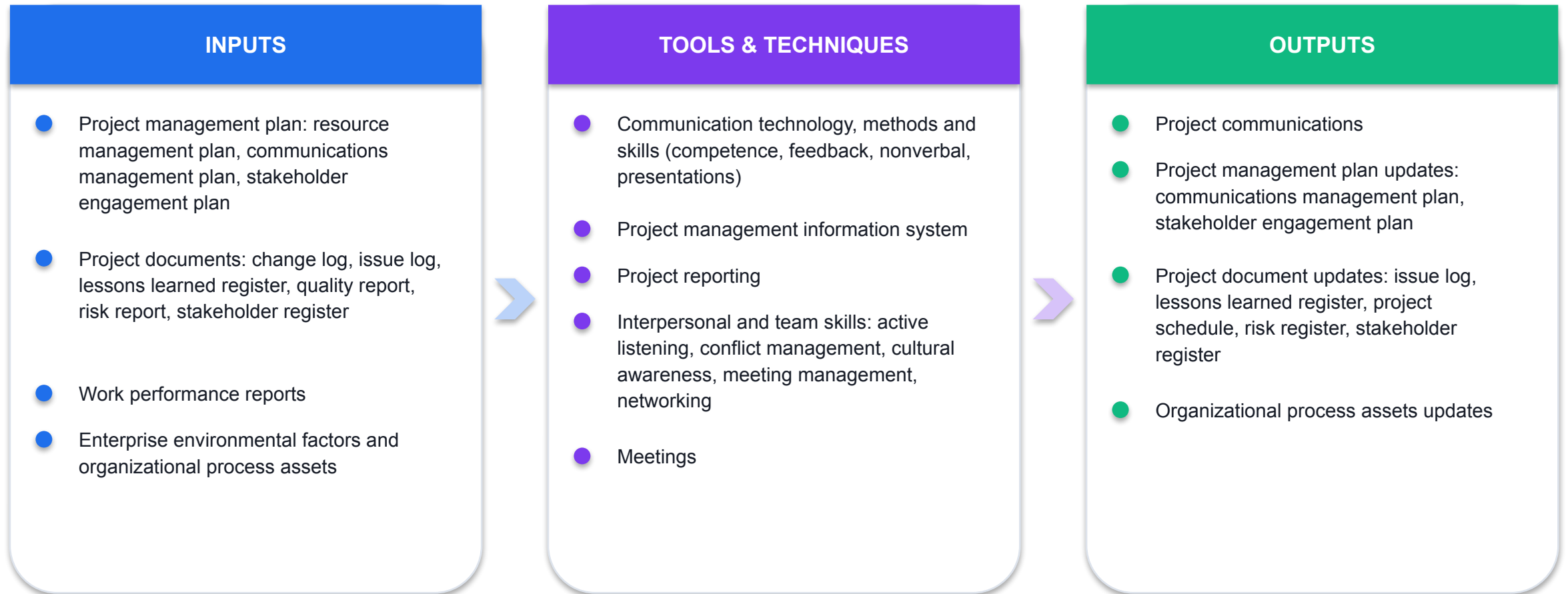
TOOL

Pivot Analysis

<https://alfredang.github.io/novapivot/>

Manage Communications

Ensure timely and appropriate collection, creation, distribution, storage, retrieval, management, monitoring and disposition of project information.



Barriers to communication

1 Language differences

Misunderstanding from differences in language, dialect or fluency — including polite over-agreement.

2 Cultural differences

Norms about directness, hierarchy and disagreement vary; the same words carry different meaning.

3 Technical jargon

Insider vocabulary excludes the very stakeholders whose decision you need.

4 Physical barriers and noise

Distance, poor connections, background noise and time-zone lag degrade the signal.

5 Psychological barriers

Stress, fear of blame or fatigue stop people hearing and stop people speaking.

6 Assumptions and perception

Filling gaps with what you expect to be true is the fastest route to a costly rework.

Assertive vs aggressive communication

Assertive

Direct, respectful, confident — the target style

- States a position clearly without attacking the other person
- Uses 'I' language: 'I need the spec by Friday because...'
- Says no to the request without rejecting the person
- Invites the other view and genuinely listens to it
- Produces clarity and preserves the relationship

Aggressive

Dominating, confrontational, disrespectful

- Wins the exchange at the other person's expense
- Uses 'you' accusations: 'you never deliver anything on time'
- Interrupts, raises voice, uses status to close down debate
- Treats disagreement as disloyalty
- Produces short-term compliance and long-term silence

The two failure modes below assertive: PASSIVE (avoids confrontation, agrees then does not deliver) and PASSIVE-AGGRESSIVE (disguised hostility, indirect sabotage, sarcasm).

The communication styles

Style	Behaviour	Effect on the team
Assertive	Direct, respectful, confident; states needs clearly	Issues get raised early and resolved; the target style
Aggressive	Dominating, confrontational, disrespectful	Compliance in the room, silence afterwards; risks go unreported
Passive	Submissive, avoids confrontation, indirect	Agreement that is not real; commitments quietly missed
Passive-aggressive	Indirect, disguised hostility, sarcasm	Corrosive; the conflict continues underground and cannot be resolved
Manipulative	Controlling and deceitful; uses influence covertly	Destroys trust permanently once detected; an ethics issue

Communication styles assessment is an explicit interpersonal and team skill in Plan Communications Management — use it to tailor your approach per stakeholder.

HANDS-ON ACTIVITY

Role play: overcoming a communication barrier

In pairs, run a conversation in which one party uses heavy technical jargon and unstated assumptions with a non-technical counterpart. Full script in the Learner Guide.

20 MIN

1 Speaker deliberately explains a solution using acronyms and internal jargon for ninety seconds.

2 Listener notes every point at which they lost the thread, without interrupting.

3 Speaker restarts, replacing every technical term with a plain outcome the listener cares about.

4 Listener asks clarifying questions; speaker answers without reintroducing jargon.

5 Both identify the assumptions the speaker made about the listener's prior knowledge.

6 Swap roles and repeat with a different topic.

DELIVERABLE

A short 'plain-language rewrite': the original jargon-heavy explanation beside the version a sponsor could act on.

Where and how the team works

1 Colocation is best where possible

A shared physical space maximises informal communication and tacit knowledge transfer.

2 Tight matrix / war room

Deliberately putting the team in one room for a critical phase measurably raises throughput.

3 Environment affects performance

Noise, lighting, screens and meeting-room access are real performance variables, not comforts.

4 Foster meaningful interaction

Design contact points that support autonomy rather than surveillance.

5 Respect agreed working hours

Ground rules about core hours and response times protect focus and prevent burnout.

6 Hybrid is the normal case

Most teams are now partly distributed; plan for it rather than treating it as an exception.

Managing virtual team members

1 Watch team dynamics

Distance hides friction; you must actively look for it rather than waiting for it to be reported.

2 Enforce transparency

Make work and status visible with Kanban-style boards so nobody has to ask what is happening.

3 Hold accountability

Clear ownership matters more remotely, where informal reminders do not happen naturally.

4 Use video deliberately

Cameras on for discussion so you can read body language, tone and hesitation.

5 Check active participation

Silence on a call is ambiguous — invite each person by name rather than asking 'any questions?'.

6 Overlap working hours

Protect a common window across time zones for the interaction that genuinely needs to be synchronous.

Virtual team best practices

1 Manage the isolation risk

Remote members drift out of the informal loop; build in deliberate social contact.

2 Shared goals over individual output

Focus on team commitments so distributed members feel part of one outcome.

3 Instil shared commitment

A visible, common definition of done does more for cohesion than any social event.

4 Standardise the tooling

One agreed channel per purpose prevents information fragmenting across five tools.

5 Write more, assume less

Distributed teams need decisions recorded in writing because corridor clarification is unavailable.

6 Rotate meeting inconvenience

Do not always make the same time zone take the 11pm call.

Negotiate and run effective meetings

1 Negotiation is a conversation

Treat it as a conversation with internal or external parties aimed at reaching agreement.

2 Aim for consensus

Use effective communication so collaboration, not victory, is the objective of the exchange.

3 Stay positive

Keeping the tone constructive materially increases the likelihood of a durable agreement.

4 Meetings: publish an agenda

State the purpose and the desired outcome; if you cannot, cancel the meeting.

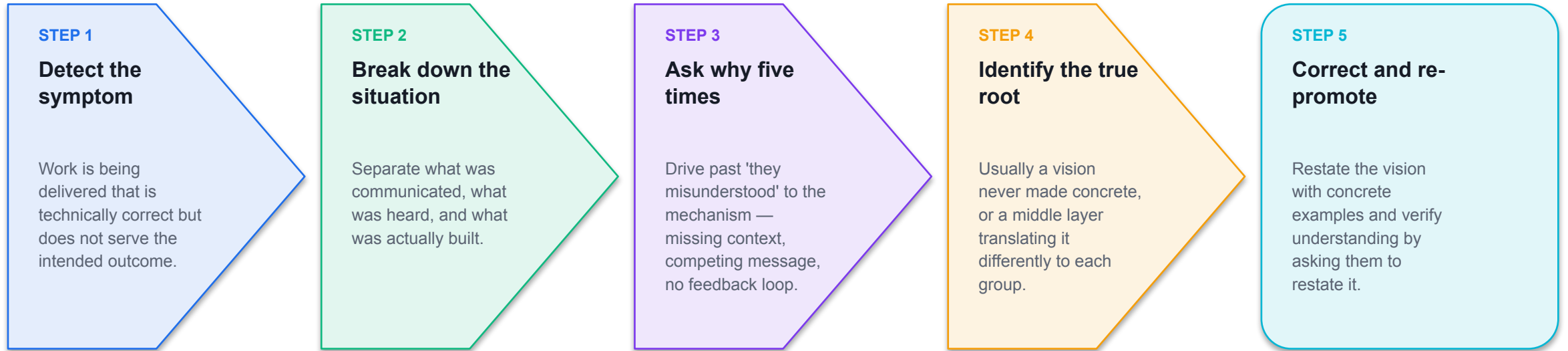
5 Timebox discussion

Everyone's time is a project cost; protect it with explicit time limits per item.

6 Close with actions

End every meeting with decisions, owners and dates — otherwise it produced nothing.

Root-cause analysis of a misunderstanding of the vision



ECO People T1 explicitly requires you to 'break down situations to identify the root cause of a misunderstanding of the vision' — a new 2026 enabler.

Keep the common vision alive

1 Co-create it

A vision the key stakeholders helped write is defended; one they were shown is merely tolerated.

2 Promote it continuously

Repeat it at kick-off, at every review and in every decision rationale until it is boring to you.

3 Make it decidable

A useful vision resolves trade-offs — if it cannot settle an argument it is a slogan, not a vision.

4 Keep it current

Business environments move; a vision not revisited quietly becomes false and the team notices before you do.

5 Test understanding

Ask team members to state the vision in their own words; divergence is your early warning.

6 Connect daily work to it

Show each person how their work package moves the outcome — this is the strongest motivator you have.

AI's effect on team roles and upskilling

1 Roles shift, not vanish

Routine drafting, summarising and reporting move to AI; judgement, context and accountability stay human.

2 New capability gaps

Prompting, output verification and data handling become skills your gap analysis must now include.

3 Verification is a task

AI output must be reviewed by a competent human; build that review step explicitly into the workflow.

4 Upskill, do not displace

Plan training so existing members move up the value chain rather than being replaced by tooling.

5 Address the anxiety

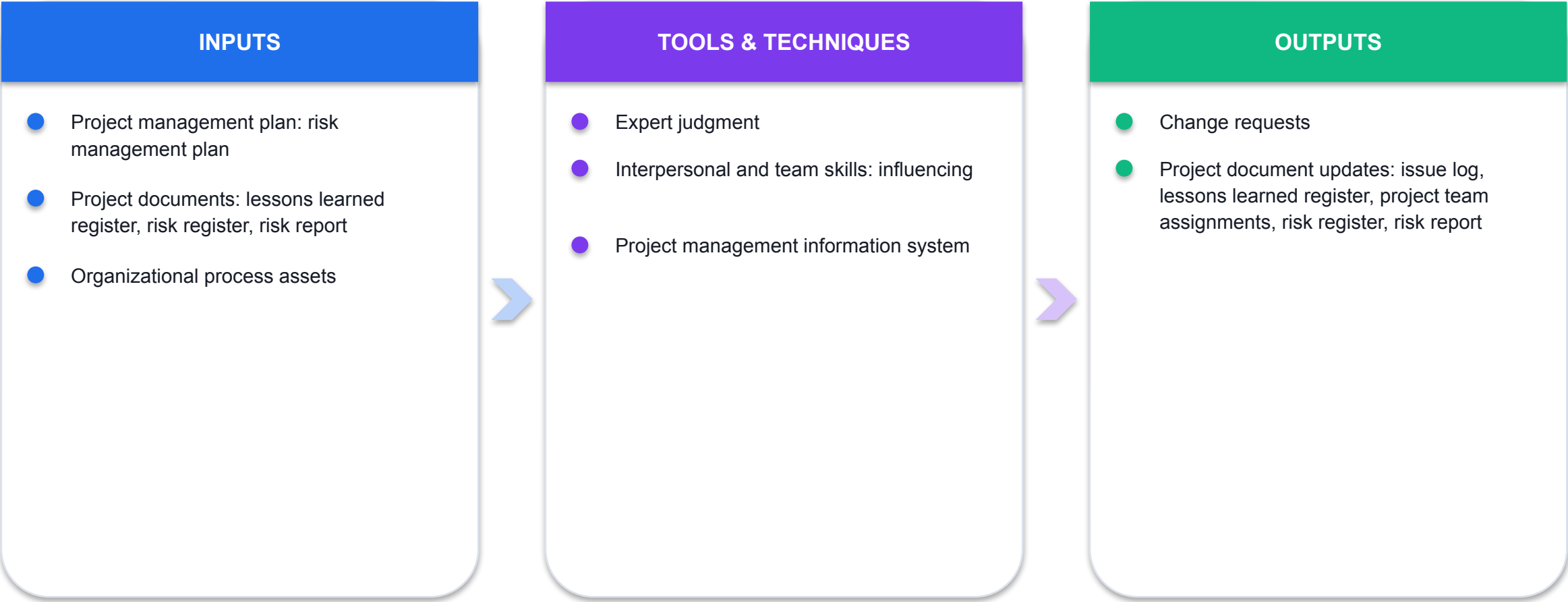
Fear of replacement is a safety-level need in Maslow's terms and will block every other motivator.

6 Govern the use

Agree ground rules on confidentiality, attribution and permitted uses before shadow adoption sets them.

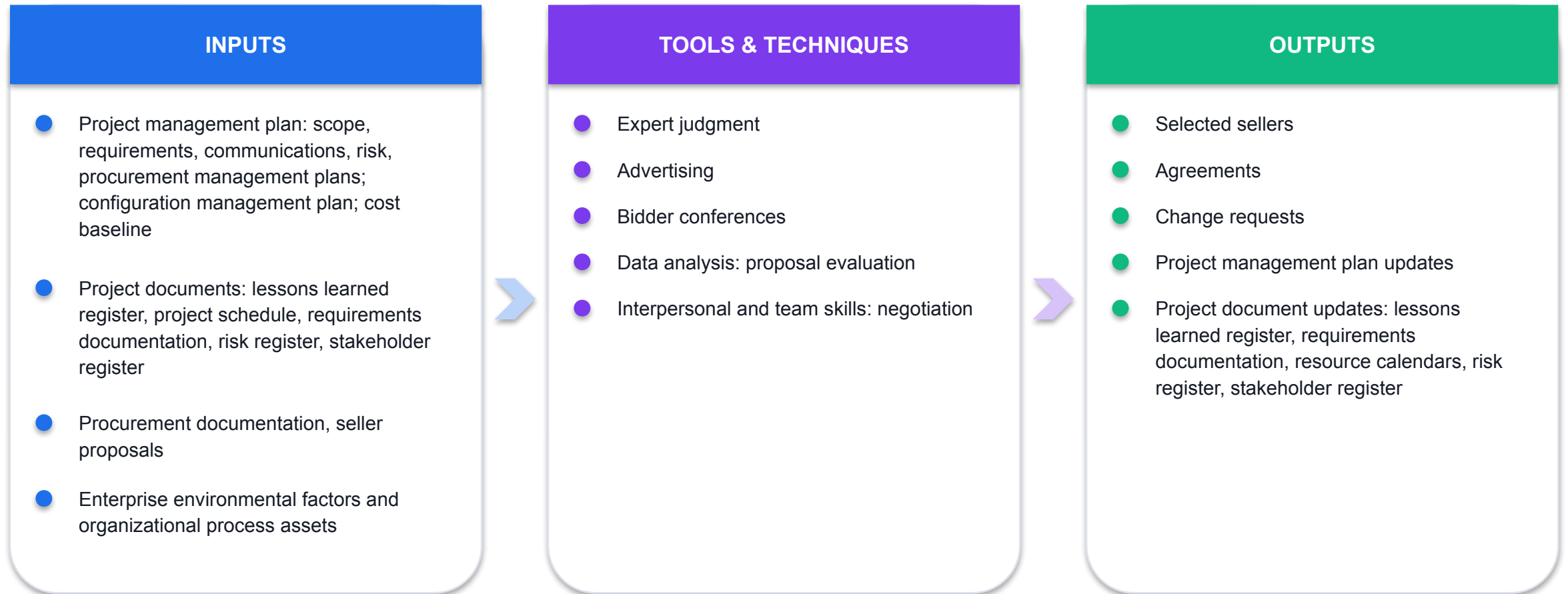
Implement Risk Responses

Execute agreed-upon risk response plans. Planning responses without implementing them is a classic exam trap.



Conduct Procurements

Obtain seller responses, select a seller and award a contract — the executing half of procurement management.



The habit that carries you through the People domain

Facilitate first. Direct only when you must.

Empower before deciding

Ask whether the team can resolve it before you resolve it for them.

Understand before acting

Gather facts and hear the parties before choosing an intervention or escalating.

Escalate with a recommendation

When you must go up, bring the impact, the options and your recommended action.

Topic 4 recap — Lead the Project Team

1 Execution processes

Direct and Manage Project Work produces deliverables, work performance data, issues and change requests.

3 Artifacts under control

Configuration management governs the product; version and document control keep the record honest.

5 Team development

Tuckman: forming, storming, norming, performing, adjourning — direct, coach, support, delegate, close.

7 Leadership style

Servant leadership is the default; the right style is situational and deliberately chosen.

9 Conflict

Collaborate is the best mode; Leas' levels tell you when collaboration is no longer viable and escalation is required.

2 Data → information → reports

Raw measurement becomes meaningful by comparison, then becomes a communication artifact.

4 Knowledge

Explicit knowledge is codified and shared; tacit knowledge needs trust, conversation and shadowing to move.

6 Motivation

Maslow, Herzberg, McGregor, McClelland and Vroom each answer a different question about why effort appears.

8 Emotional intelligence

Self-awareness, self-management, social awareness and social skill underpin every People-domain task.

10 Communication

Tailor to the stakeholder, remove barriers, be assertive, and make virtual work visible and written down.

Hands-on Labs — Topic 4

1 Lab 15. RACI Responsibility Assignment Matrix

People T3, Process T4 · LO2 · 20 min in class + 25 min self-study

Produces: RACI matrix, Resource plan

2 Lab 16. Tuckman Diagnosis and Motivation Theory Application

People T3 · LO3 · 20 min in class + 25 min self-study

Produces: Team stage diagnosis, Motivation action plan

3 Lab 17. Conflict Resolution Role-Play across the Five Modes

People T2 · LO3 · 20 min in class + 30 min self-study

Produces: Conflict analysis, Resolution strategy

4 Lab 18. Communication Plan, Channels Formula and Status Report

People T8 · LO4 · 20 min in class + 30 min self-study

Produces: Communications management plan, Status report

HANDS-ON ACTIVITY

Lab 15. RACI Responsibility Assignment Matrix

Case study: Contoso Training Portal Upgrade. People T3, Process T4 · LO2. Produces RACI matrix, Resource plan.

20 min in class

1 List the Lab 11 work packages down the rows and the project roles across the columns.

3 Verify exactly ONE Accountable per row — two accountable parties means nobody is accountable.

5 Reduce excessive Consulted entries — over-consultation is the most common cause of decision latency.

2 Open the RACI tool and assign Responsible, Accountable, Consulted and Informed for every work package.

4 Scan the columns for overload: a role Responsible for most rows is a single point of failure.

6 Review the matrix with the team and have each member confirm their assignments aloud.

DELIVERABLE

RACI matrix, Resource plan

TOOL

RACI Matrix

alfredang.github.io/raci/

Lab 16. Tuckman Diagnosis and Motivation Theory Application

Case study: Contoso Training Portal Upgrade. People T3 · LO3. Produces Team stage diagnosis, Motivation action plan.

20 min in class

1 Assess which Tuckman stage the Contoso team is in, citing the observable behaviours that justify your diagnosis.

2 State the leadership action appropriate to that stage rather than the one you personally prefer.

3 Map each team member's dominant motivator using McClelland: achievement, affiliation or power.

4 Apply Herzberg to separate hygiene complaints from genuine motivator gaps — they need different remedies.

5 Design one specific intervention per team member, with a date and an observable success indicator.

6 Define how you will detect regression to an earlier stage after the next team change.

DELIVERABLE

Team stage diagnosis, Motivation action plan

Lab 17. Conflict Resolution Role-Play across the Five Modes

Case study: Contoso Training Portal Upgrade. People T2 · LO3. Produces Conflict analysis, Resolution strategy.

20 min in class

1 Read the conflict scenario and identify the true source: resources, priorities, technical opinion, or interpersonal style.

2 Assess the conflict's position on Leas' five levels — the level dictates whether the team can still resolve it themselves.

3 Role-play the conflict using each of the five modes in turn: collaborate, compromise, accommodate, force, avoid.

4 Record what each mode produced and the relationship cost it incurred.

5 Select the appropriate mode for this situation and justify it against the urgency and the relationship's importance.

6 Write the agreed resolution, the ground rule that prevents recurrence, and how adherence will be monitored.

DELIVERABLE

Conflict analysis, Resolution strategy

Lab 18. Communication Plan, Channels Formula and Status Report

Case study: Contoso Training Portal Upgrade. People T8 · LO4. Produces Communications management plan, Status report.

20 min in class

1 List every stakeholder group from Lab 07 with the information each actually needs — not everything you have.

2 Calculate the communication channels using $N(N-1)/2$ and state what the result implies for meeting design.

3 Specify format, frequency, method and owner for each communication.

4 Classify each as push, pull or interactive, and confirm anything requiring confirmed understanding is interactive.

5 Draft a one-page status report tailored to the steering committee: exceptions and decisions needed, not raw activity.

6 Draft the same status for the delivery team and note how radically the content differs for the same underlying facts.

DELIVERABLE

Communications management plan, Status report

05

TOPIC

Monitor and Control the Project

Monitoring and controlling is where the plan meets reality. You measure actual performance against the baselines, decide whether the variance matters, and act — through change control, corrective action, quality control, risk response and impediment removal.

- 1 Monitor and control project work and report performance
- 2 Evaluate progress with earned value management
- 3 Control quality with the seven basic statistical tools
- 4 Run integrated change control and the change control board
- 5 Manage risks, issues and impediments to resolution
- 6 Control procurements and scale agile beyond one team

The controlling mindset

You cannot manage what you do not measure — and you must not measure what you will not act on.

Measure against baseline

Variance only has meaning when compared to an approved scope, schedule and cost baseline.

Decide with thresholds

Governance sets the variance thresholds that trigger escalation, a change request or acceptance.

Act, then update artifacts

Every accepted variance ends in an updated plan, log or register so the record stays truthful.

What Monitoring and Controlling Actually Involves

1 Track

Collect work performance data from the work itself — hours booked, defects raised, stories done, invoices received.

2 Review

Convert raw data into work performance information by comparing it to the baseline and asking what the gap means.

3 Regulate

Take corrective or preventive action so future performance moves back toward the approved plan.

4 Report

Package information into work performance reports that a sponsor or steering committee can act on.

5 Change

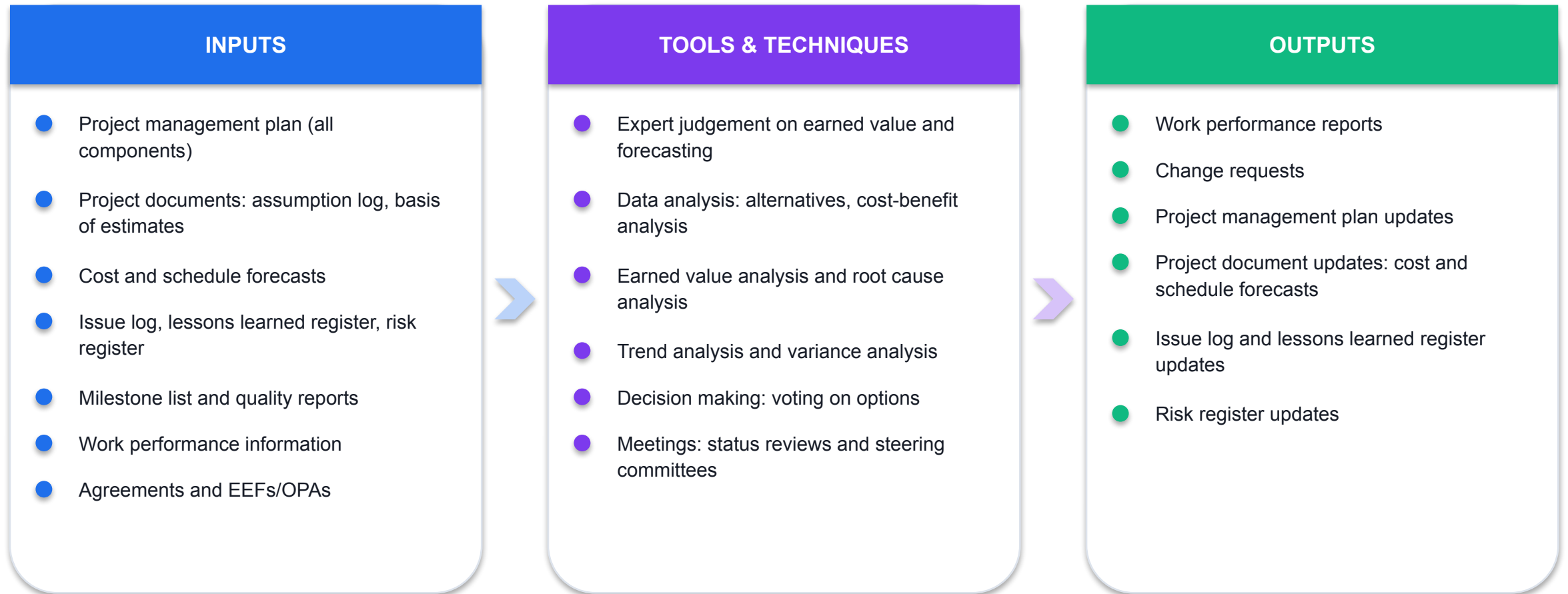
Route anything that alters a baseline through integrated change control — never adjust a baseline informally.

6 Improve

Feed what you learn into process updates and OPAs so the next phase or project starts better than this one did.

Monitor and Control Project Work — ITTO

Purpose: track, review and report overall progress so stakeholders understand true status and the team can forecast where the project will land if nothing changes.



Data to Information to Reports — the Performance Chain

Stage	What it is	Example	Who consumes it
Work performance data	Raw, unanalysed observations and measurements taken during execution	42 hours booked; 3 defects logged; 5 of 8 stories done	Project team, delivery leads
Work performance information	Data analysed in context against the baseline to give it meaning	CPI 0.92, SPI 1.04 — over budget but slightly ahead of schedule	Project manager, functional managers
Work performance reports	Information packaged physically or electronically to drive decisions	Status report, dashboard, milestone schedule, EVM S-curve	Sponsor, steering committee, PMO
Change requests	Formal proposals to alter a baseline, document or deliverable	Add two developers to recover the critical path	Change control board

Exam cue: 'raw' or 'measurements taken' = data. 'Compared to plan' = information. 'Presented to the sponsor' = report.

Report on Performance — the Reporting Toolkit

1 Milestone schedule

High-level visualisation of progress against planned milestone dates — the sponsor's favourite one-pager.

2 Quality reports

Charts and narrative built from collected quality metrics, defect counts and improvement recommendations.

3 EVM reports

Graphs and computed values from the earned value equations, including the PV/EV/AC S-curve and index trends.

4 Variance analysis reports

Graphs and commentary comparing actual results to expected results, with root causes for each gap.

5 Status reports

Period snapshot of scope, schedule, cost, risk and issues, usually with a RAG rating per dimension.

6 Dashboards

Information radiators refreshed automatically so stakeholders self-serve status instead of requesting it.

Developing Project Metrics and Reconciliation

1 Choose metrics that drive action

A metric no one would act on is overhead. Tie each measure to a decision it informs.

2 Baseline and threshold

Define the target and the tolerance band before measuring so variance is judged, not debated.

3 Leading and lagging

Lagging metrics (CPI, defects escaped) confirm; leading metrics (velocity trend, blocker age) warn early.

4 Reconcile the sources

Timesheets, finance ledger and the schedule tool disagree — reconcile them before publishing a number.

5 Cadence and ownership

Every metric needs an owner, a refresh cadence and a documented calculation method.

6 Review effectiveness

ECO 2026 asks you to continually assess whether your metrics and artifacts are still worth maintaining.

Artifact Management Effectiveness (ECO 2026 Emphasis)

1 Identify and tailor artifacts

Select only the registers, logs and plans this project's complexity actually justifies.

2 Create, review, update

An artifact that is not maintained is worse than none — it misleads people who trust it.

3 Ensure accessibility

Stakeholders must be able to find the current version without asking the project manager for it.

4 Single source of truth

One authoritative location per artifact; everything else links to it rather than copying it.

5 Version and configuration

Configuration management controls the deliverable; change control governs the baseline.

6 Assess continually

Retire artifacts that no longer inform a decision and add ones the team keeps recreating informally.

Continuous Improvement — Incremental vs Breakthrough

Small incremental improvement

Kaizen-style, team-owned, low risk

- Many small changes made continuously by the people doing the work
- Low cost, low risk, and reversible if the change does not help
- Driven by retrospectives, suggestion systems and daily problem solving
- Compounds over time — 1% per sprint is transformational over a year
- Typical vehicles: Agile retrospectives, Lean waste reduction

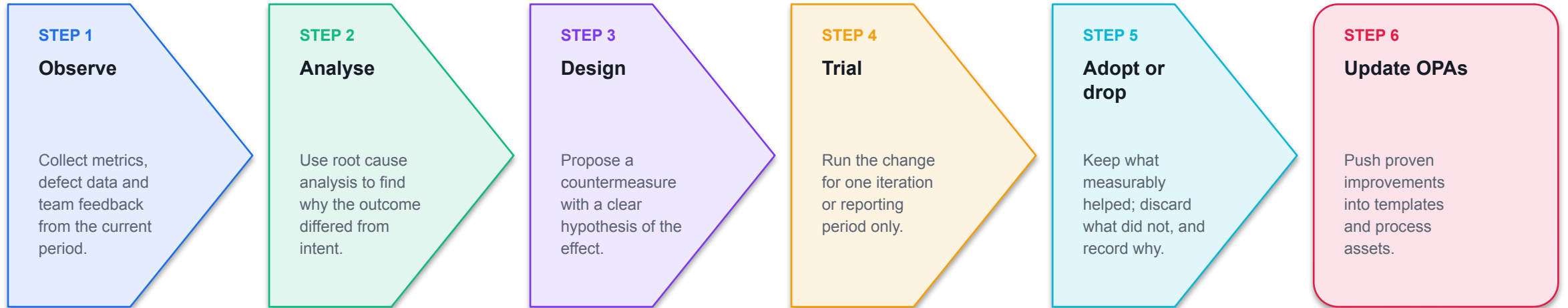
Large breakthrough improvement

Re-engineering, sponsor-owned, high risk

- A step change that redesigns a process, technology or operating model
- High cost, high disruption, needs executive sponsorship and funding
- Usually triggered by a strategic threat or a persistent capability gap
- Delivers a large gain at once but risks organisational rejection
- Typical vehicles: Six Sigma DMAIC projects, business process reengineering

Continuous improvement is a business strategy set at the organisational level and adopted by projects — usually implemented by the PMO through a structured framework such as Agile, Lean or Six Sigma.

The Continuous Improvement Loop on a Live Project



ECO Business Environment T6 asks you to utilise lessons learned, ensure improvement processes are updated, and update organisational process assets — improvement is not real until the OPA changes.

Curate Knowledge as an Asset

1 Document explicit knowledge

Write down what can be written down — decisions, configurations, runbooks — and archive it where others will look.

2 Surface tacit knowledge

Experience and judgement live in people. Pair work, shadowing and communities of practice move it.

3 Treat knowledge as an asset

Knowledge outlives the team. It belongs to the organisation, not to whoever happened to learn it.

4 Build psychological safety

People only share what did not work if admitting failure is safe. Blameless review is the enabler.

5 Make it findable

A lessons learned repository nobody can search has zero value; tag entries by process, phase and risk category.

6 Capture continuously

Record lessons as they occur, not at closure when the detail and the people have already gone.

Realign Team Efforts with Value During Change

1 Prioritise team cohesion

When members join or depart, the team must re-form. Protect cohesion before chasing velocity.

2 Support realignment

Revisit working agreements, roles and the Definition of Done whenever the team's composition changes.

3 Welcome new members

Treat each joiner as a source of new knowledge and motivation, not simply as extra capacity.

4 Restate shared goals

Re-establish shared understanding of project goals and agreements so nobody optimises the wrong thing.

5 Collaborate on value

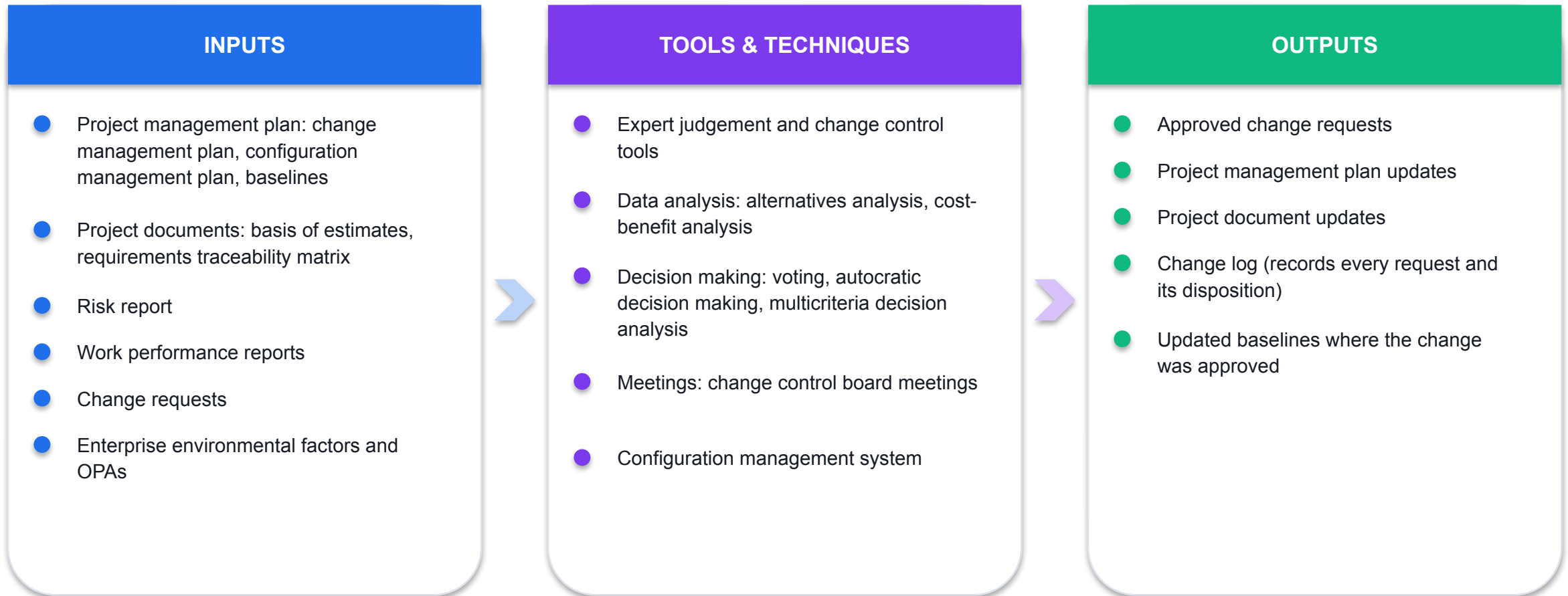
Ask each member explicitly how their work adds value; the answer often exposes hidden waste.

6 Navigate disruption

Expect a dip in performance after disruption; coach through it rather than escalating immediately.

Perform Integrated Change Control — ITTO

Purpose: review, approve or reject every change request, manage changes to deliverables and baselines, and communicate the decision. This process is performed from project start to finish.



Causes of Project Changes

1 Inaccurate initial estimates

Estimates made with the least information ever available prove wrong as detail emerges.

2 New regulations

External legal, tax or safety requirements arrive mid-flight and are non-negotiable.

3 Missed requirements

Elicitation never reached a stakeholder group whose needs surface during review or UAT.

4 Specification changes

The business changes its mind because the market, competitor or strategy changed.

5 Risk response

An identified risk occurs and the agreed response requires a baseline change to implement.

6 Technology and vendor shifts

A supplier discontinues a component or a platform version reaches end of support.

The Four Change Request Types — Know These Cold

Type	Definition	Trigger	Example
Corrective action	Realigns the performance of the project work with the project management plan	Performance has already deviated from plan	Add a second tester because the defect backlog is growing
Preventive action	Ensures the future performance of project work aligns with the plan	A deviation is forecast but has not yet occurred	Book contractor capacity now because a resource gap is predicted
Defect repair	Modifies a nonconforming product, deliverable or component	A deliverable failed inspection or test	Rework a module that failed the acceptance test
Update	Modifies formally controlled project documents or a baseline	Approved content changes require the record to change	Revise the scope baseline after an approved scope addition

Corrective = fix the present. Preventive = protect the future. Defect repair = fix the product. Update = fix the document or baseline.

Corrective vs Preventive Action

Corrective action

Reactive — the variance already happened

- Triggered by measured performance that is already off plan
- Aim is to bring future performance back onto the approved baseline
- Usually discovered through variance analysis or an EVM index below 1.0
- Example: CPI has dropped to 0.85, so rescope low-value work
- Does not by itself change the baseline — it changes how work is done

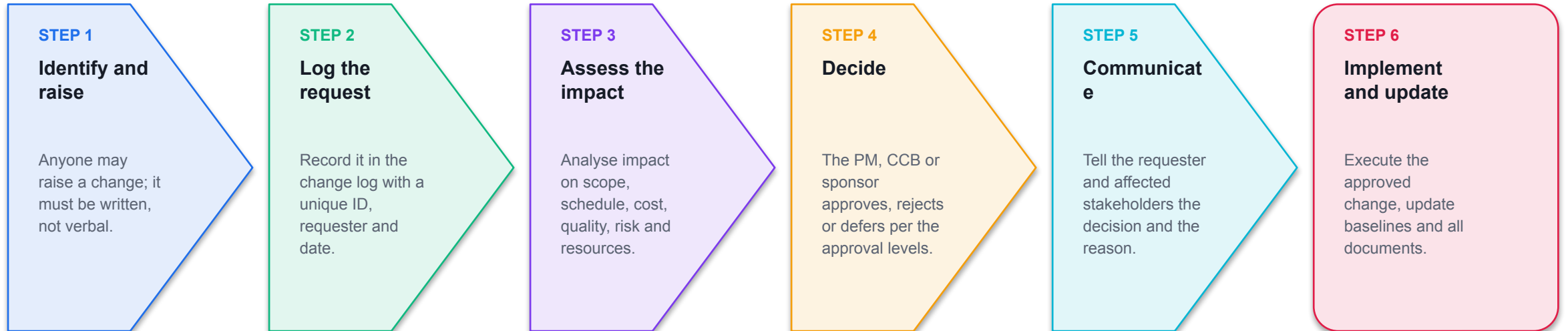
Preventive action

Proactive — the variance is only forecast

- Triggered by a forecast, trend or identified risk, not by actual damage
- Aim is to stop the deviation ever appearing in the actuals
- Usually discovered through trend analysis, risk review or reserve analysis
- Example: cross-train a second engineer before the specialist goes on leave
- Closely related to risk response — prevention is mitigation applied to work

Exam cue: 'has fallen behind' or 'is over budget' points to corrective action. 'May', 'is expected to', or 'trend suggests' points to preventive action.

The Change Request Flow



Rejected changes are still logged. The change log is the audit trail proving that governance was followed — an exam favourite.

Change Control Systems and the Change Control Board

1 Change control system

The forms, tracking methods, processes and approval levels required to authorise or reject a requested change.

2 Change control board (CCB)

A formally chartered group that reviews, evaluates, approves, defers or rejects changes and records decisions.

3 Approval levels

The change management plan defines what the PM may approve alone and what must escalate to the CCB or the sponsor.

4 Configuration management system

Controls the product itself — versions, specifications and physical characteristics.

5 Change log

The register of every request with status, decision, decision date and rationale; shared with stakeholders.

6 Emergency change procedure

A documented fast path for urgent changes, with retrospective CCB ratification.

Exam Traps — Integrated Change Control

IF THE QUESTION SAYS

A stakeholder asks you to add a small feature; there is budget available



THE BEST ANSWER IS

Do not agree on the spot. Assess the impact and submit a change request through integrated change control.

IF THE QUESTION SAYS

A change was already implemented by a team member without approval



THE BEST ANSWER IS

Determine the impact first, then process it through change control and update the affected documents and baselines.

IF THE QUESTION SAYS

The sponsor demands an urgent change during a steering meeting



THE BEST ANSWER IS

Analyse impact on all baselines and present options; the CCB — not the sponsor's verbal instruction — authorises the baseline change.

IF THE QUESTION SAYS

A change request is rejected



THE BEST ANSWER IS

Record the rejection and rationale in the change log and communicate it to the requester — rejected changes are still documented.

Agile Consideration — Handling Change

1 Change is expected

Adaptive approaches assume requirements will change; the process is designed to absorb rather than resist change.

2 Product Owner authority

The Product Owner holds autocratic decision-making power over scope — others must convince them.

3 Backlog is the change log

New or updated features enter the Product Backlog; updated user stories join the next Sprint Backlog.

4 Sprint scope is protected

Change is welcomed between iterations; the current sprint's committed scope stays fixed.

5 Scope evolves in layers

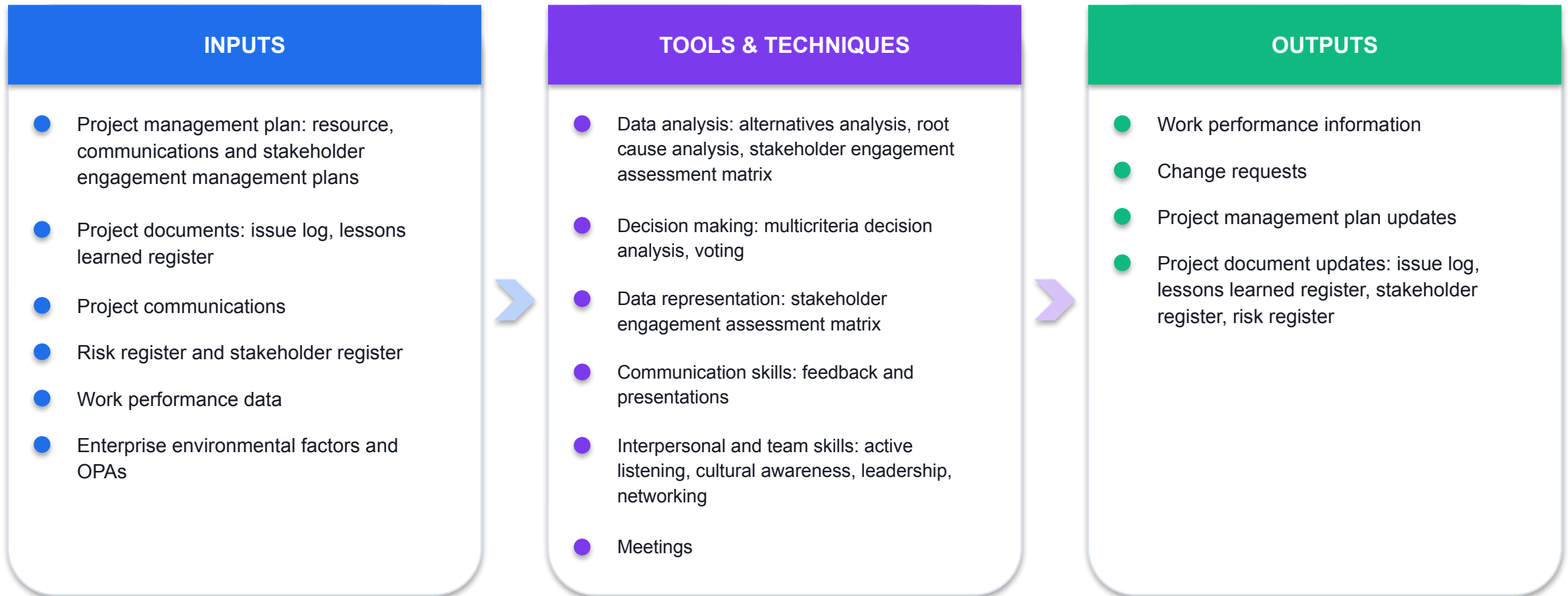
Roadmap to release backlog to iteration backlog — each layer reprioritised as needs are learned.

6 Validate against feedback

Check user stories and the Definition of Done against customer feedback and product requirements.

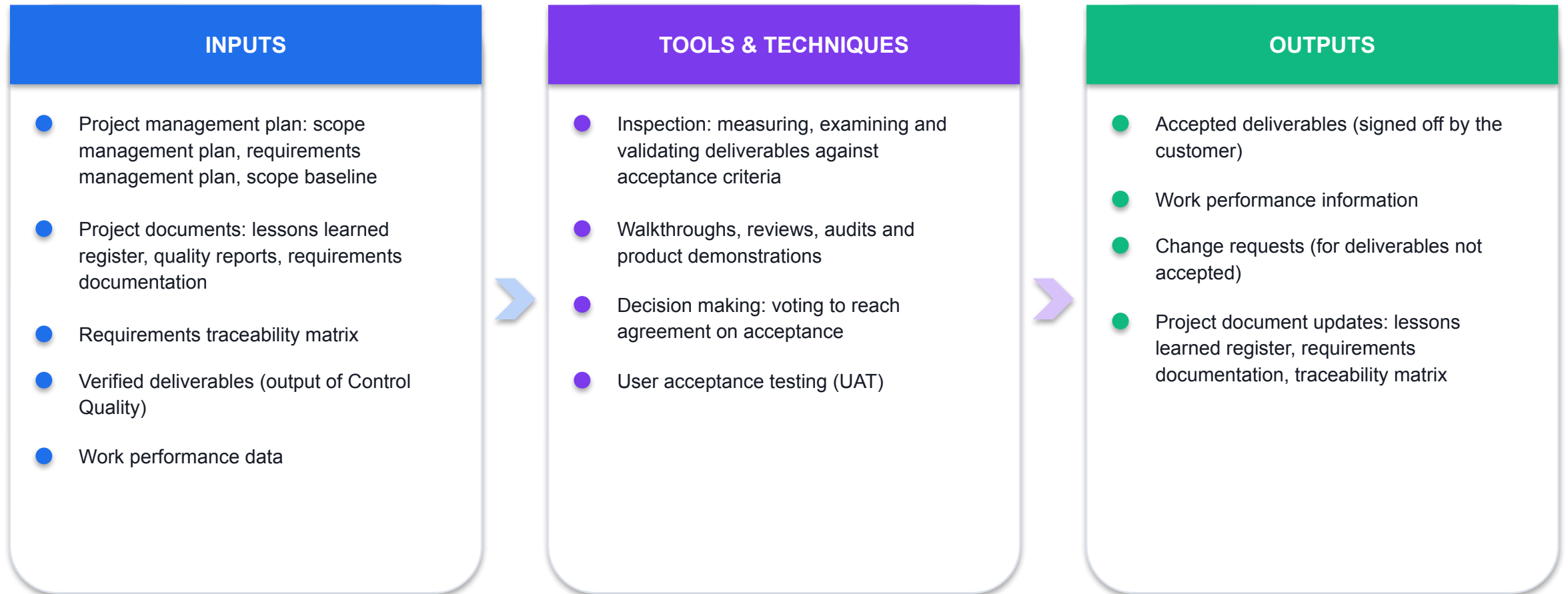
Monitor Stakeholder Engagement — ITTO

Purpose: monitor project stakeholder relationships and tailor strategies to engage stakeholders through modification of engagement strategies and plans.



Validate Scope — ITTO

Purpose: formalise acceptance of completed deliverables. Control Quality verifies correctness internally; Validate Scope obtains the customer's formal acceptance.



Control Quality vs Validate Scope

Control Quality

Internal — is it built right?

- Performed by the project team and quality function, usually before handover
- Checks the deliverable against the quality requirements and specifications
- Uses inspection, testing, measurement and the seven basic quality tools
- Key output: verified deliverables plus quality control measurements
- Nonconformance produces a defect repair change request

Validate Scope

External — is it the right thing?

- Performed with the customer or sponsor, taking verified deliverables as input
- Checks the deliverable against the agreed acceptance criteria
- Uses inspection, demonstration, walkthroughs, UAT and formal sign-off
- Key output: accepted deliverables recorded as formal acceptance
- Non-acceptance produces a change request, not a silent rework

Sequence to remember: Control Quality produces verified deliverables, which become the input to Validate Scope, which produces accepted deliverables, which become the input to Close Project or Phase.

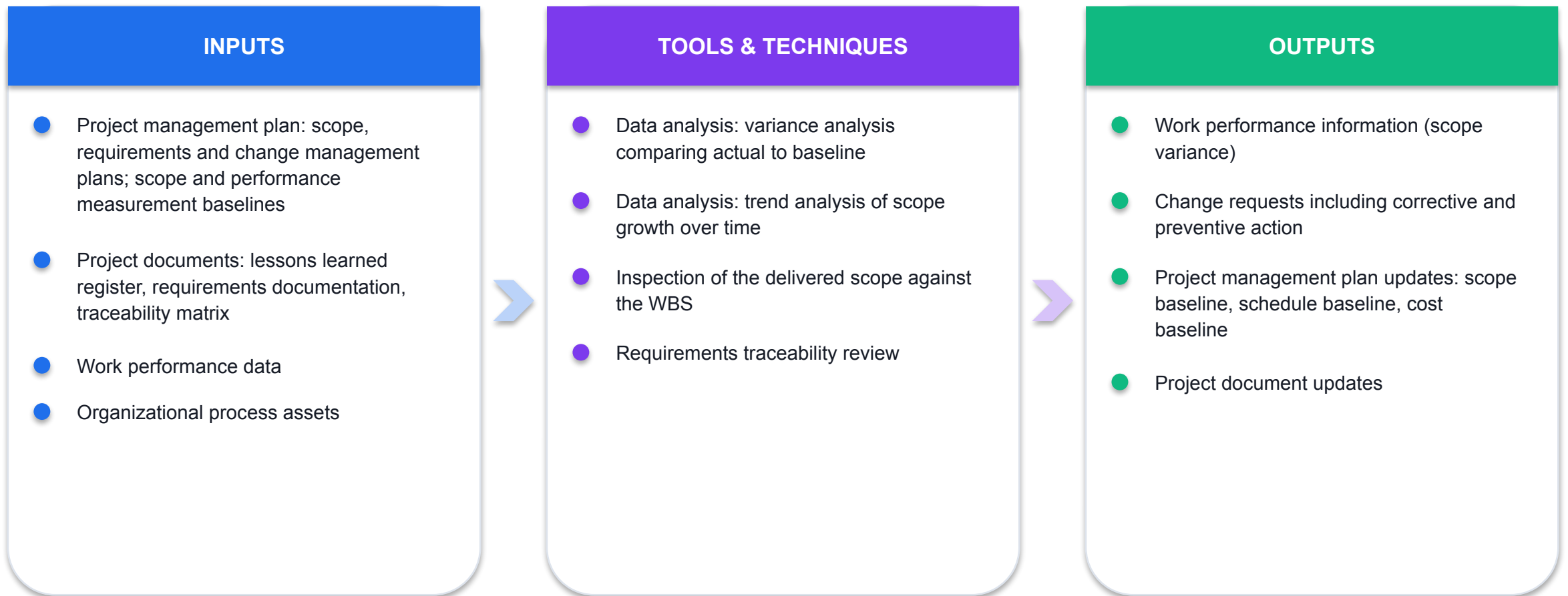
Acceptance Criteria, Definition of Ready, Definition of Done

Concept	Scope of application	Who owns it	What it prevents
Acceptance criteria	Specific to one requirement, user story or deliverable	Product Owner / customer, agreed with the team	Arguing about 'finished' after the work is already built
Definition of Ready (DoR)	Applies to every item before it may enter an iteration	The team, agreed with the Product Owner	Starting work on an item too vague to estimate or complete
Definition of Done (DoD)	Applies to every increment the team produces	The whole delivery team	Hidden work — untested, undocumented, undeployed 'done'
Requirements traceability matrix	Links every requirement to its deliverable, test and acceptance	Project manager / business analyst	Requirements silently dropped between plan and delivery

Acceptance criteria may legitimately be modified during the life cycle — but only through the change control process.

Control Scope — ITTO

Purpose: monitor the status of the project and product scope and manage changes to the scope baseline — this is where scope creep is caught.



Scope Creep vs Gold Plating

Scope creep

Uncontrolled expansion, usually customer-driven

- Requirements added without corresponding schedule, cost or resource change
- Each addition feels small — the cumulative effect destroys the baseline
- Root causes: weak scope baseline, no change control, eager-to-please PM
- Control: a documented scope baseline plus enforced integrated change control
- Detect it with trend analysis on requirement count and story point totals

Gold plating

Unrequested extras, usually team-driven

- The team adds features the customer never asked for and will not pay for
- Consumes budget and introduces risk with zero measured business value
- Root causes: engineer enthusiasm, misread of what 'delighting the customer' means
- Control: Definition of Done tied strictly to acceptance criteria
- PMI's position is unambiguous — gold plating is never acceptable

Both are scope control failures. The response is the same: bring it back to the baseline, and route any genuine addition through a change request with a full impact assessment.

Monitor the External Business Environment

1 Change cuts both ways

External change brings opportunities to add or extend value, not only threats to defend against.

2 Survey the environment

Track regulatory, technology, geopolitical and market changes on a defined cadence, not ad hoc.

3 Stay vigilant for threats

A competitor launch or a regulation can invalidate the business case even while delivery is green.

4 Update the risk register

New external factors become new risks with new probability, impact and threshold values.

5 Reprioritise scope

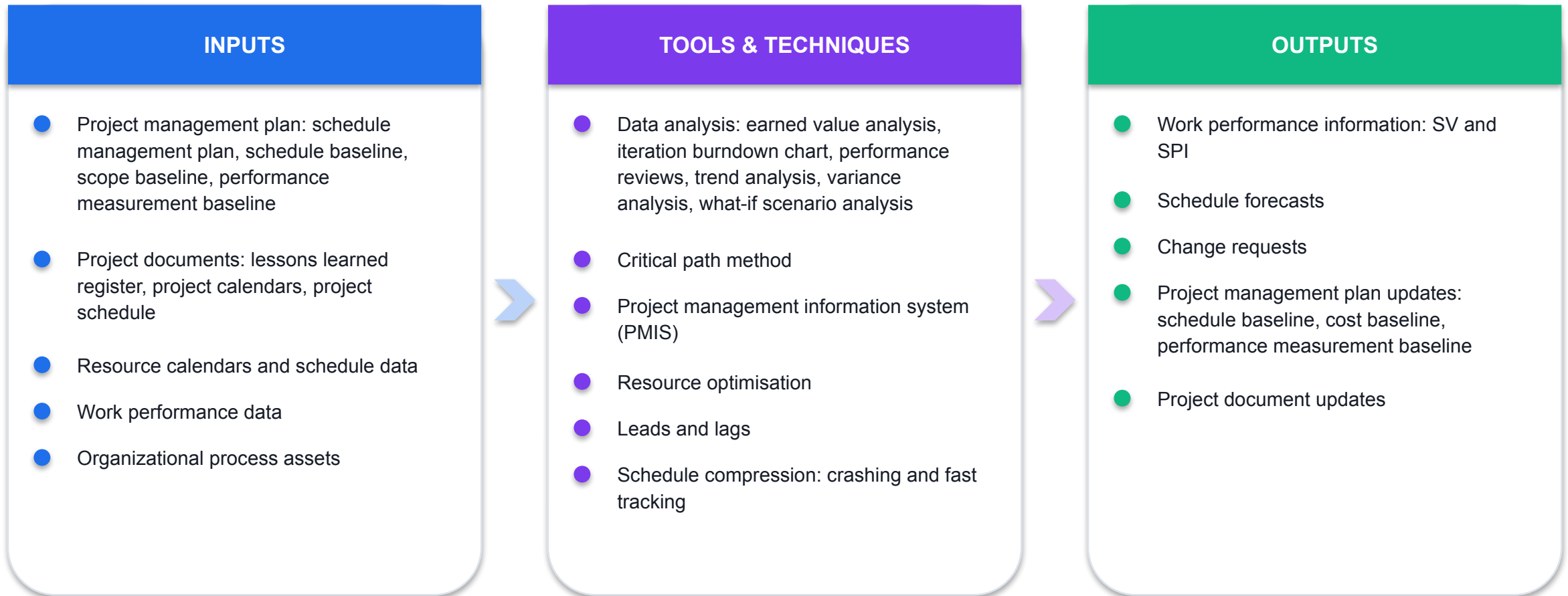
Assess the impact on scope or backlog and reprioritise so the project still delivers the intended value.

6 Use structured tools

PESTLE, SWOT and horizon scanning give the scan discipline rather than relying on the PM's news feed.

Control Schedule — ITTO

Purpose: monitor the status of the project to update the schedule and manage changes to the schedule baseline. Determining the current status and influencing the factors that create schedule change are the core work.



Earned Value Management

EVM answers three questions with one integrated set of numbers: where should we be, where are we, and what did it cost?

Planned Value (PV)

The authorised budget assigned to the work scheduled to be complete by now.

Earned Value (EV)

The measure of work actually performed, expressed in the budget authorised for that work.

Actual Cost (AC)

The realised cost actually incurred for the work performed in the period measured.

EVM Variable Glossary

Symbol	Name	Meaning in plain language	Units
PV	Planned Value	The value of work you planned to have finished by the measurement date	Currency
EV	Earned Value	The budgeted value of the work you actually finished	Currency
AC	Actual Cost	What you actually spent to finish that work	Currency
BAC	Budget at Completion	The total approved budget for the whole project	Currency
CV	Cost Variance	How much more or less you spent than the work was worth	Currency
SV	Schedule Variance	How much work you are ahead of or behind, valued in money	Currency
CPI	Cost Performance Index	Value earned per dollar spent — efficiency of spend	Ratio
SPI	Schedule Performance Index	Rate of progress against the planned rate	Ratio
EAC	Estimate at Completion	Forecast total cost when the project finishes	Currency
ETC	Estimate to Complete	Forecast cost of the remaining work only	Currency
VAC	Variance at Completion	Forecast overrun or underrun against BAC	Currency

EVM Formulas — Variance and Index

Cost Variance

$$CV = EV - AC$$

Negative means you have spent more than the work was worth. Positive is good.

Schedule Variance

$$SV = EV - PV$$

Negative means less work is done than planned. Reaches zero at project end.

Cost Performance Index

$$CPI = EV / AC$$

Above 1.0 is under budget; below 1.0 means you get less than a dollar per dollar.

Schedule Performance Index

$$SPI = EV / PV$$

Above 1.0 is ahead of schedule; below 1.0 means work is arriving slower than planned.

Percent Complete

$$EV / BAC$$

The proportion of total authorised budget that has been earned so far.

Memory aid: every variance is EV minus something and every index is EV divided by something. If the answer is negative or below 1.0, it is bad.

EVM Formulas — Forecasting

EAC — typical variance

$$EAC = BAC / CPI$$

Use when current cost performance is expected to continue for the rest of the project.

EAC — atypical variance

$$EAC = AC + (BAC - EV)$$

Use when the variance was a one-off and remaining work will run at the budgeted rate.

EAC — schedule constrained

$$EAC = AC + (BAC - EV) / (CPI \times SPI)$$

Use when the remaining work must absorb both the cost and the schedule performance.

Estimate to Complete

$$ETC = EAC - AC$$

The forecast cost of the work that still remains after money already spent.

Variance at Completion

$$VAC = BAC - EAC$$

Forecast overrun (negative) or underrun (positive) measured at project completion.

Read the question for the phrase that selects the formula: 'rate will continue' = BAC/CPI ; 'atypical / one-off' = $AC + BAC - EV$; 'must meet the deadline too' = the $CPI \times SPI$ form.

EVM Formulas — To-Complete Performance Index

TCPI against BAC

$$\text{TCPI} = (\text{BAC} - \text{EV}) / (\text{BAC} - \text{AC})$$

The cost efficiency the remaining work must achieve to finish within the original budget.

TCPI against EAC

$$\text{TCPI} = (\text{BAC} - \text{EV}) / (\text{EAC} - \text{AC})$$

Use once the original budget is no longer achievable and a new EAC has been approved.

Interpretation above 1.0

$$\text{TCPI} > 1.0$$

You must perform better than you have been performing — the target is harder than the past.

Interpretation below 1.0

$$\text{TCPI} < 1.0$$

You may perform slightly worse than before and still hit the target — you have slack.

TCPI and CPI move in opposite directions. A CPI of 0.80 with plenty of work left implies a TCPI well above 1.0 — a warning that recovery is unrealistic.

Interpreting the Indices — Above 1.0 vs Below 1.0

1 CPI above 1.0

Cost efficient. You earned more value than you spent. Confirm it is real progress, not under-reported costs.

2 CPI below 1.0

Cost inefficient. Every dollar buys less than a dollar of value. Investigate rate, rework and scope leakage.

3 SPI above 1.0

Ahead of plan. Check whether easy work was pulled forward, leaving the hard work still ahead.

4 SPI below 1.0

Behind plan. Look at the critical path first — non-critical lateness may not delay the project at all.

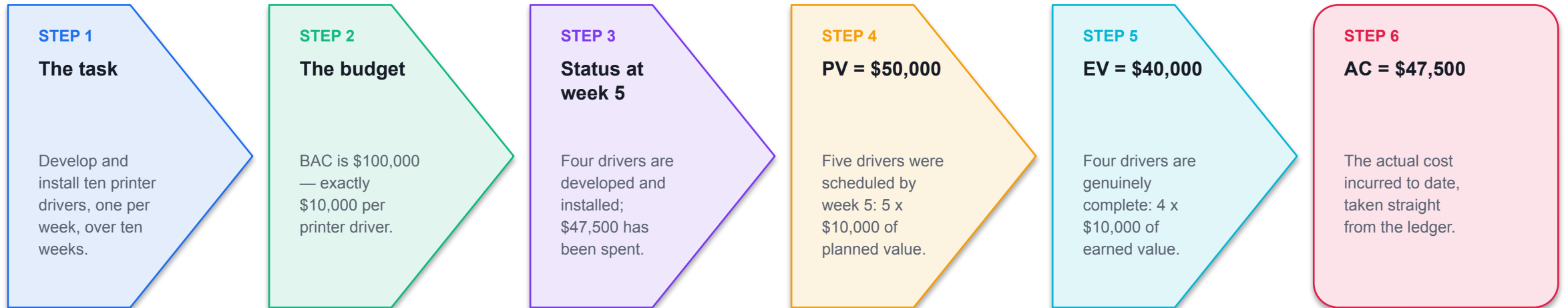
5 CPI low, SPI high

You are buying speed with money — typically overtime or extra resources. Confirm the trade was authorised.

6 CPI high, SPI low

You are underspending because too little work is happening — usually a resource or blocker problem, not thrift.

Worked Example — Ten Printer Drivers



Golden rule: EV is always the budgeted value of completed work — count the deliverables finished, then multiply by their budget. Never use actual cost to compute EV.

Worked Example — Running the Numbers

Cost Variance

$$CV = 40,000 - 47,500 = -7,500$$

Negative: \$7,500 more was spent than the completed work was actually worth.

Schedule Variance

$$SV = 40,000 - 50,000 = -10,000$$

Negative: one full driver's worth of work — \$10,000 — is behind schedule.

Cost Performance Index

$$CPI = 40,000 / 47,500 = 0.84$$

Only 84 cents of value is earned per dollar spent — a serious cost problem.

Schedule Performance Index

$$SPI = 40,000 / 50,000 = 0.80$$

Work is arriving at 80% of the planned rate — four drivers where five were due.

Estimate at Completion

$$EAC = 100,000 / 0.84 = 118,750$$

If this rate continues the project finishes at \$118,750, not \$100,000.

Follow through: $ETC = 118,750 - 47,500 = \$71,250$ still to spend, and $VAC = 100,000 - 118,750 = -\$18,750$, a forecast overrun of \$18,750.

The EVM S-Curve — PV, EV and AC Over Ten Weeks



WHAT IT TELLS YOU

- PV is the straight planned baseline — \$10K of value per week for ten weeks.
- EV sits below PV, so the project is behind schedule (negative SV).
- AC sits above EV, so the project is over budget (negative CV).
- The gap between AC and EV widens each week — the trend, not the point, is the warning.
- Extrapolating AC at the current CPI gives an EAC of roughly \$119K against a \$100K BAC.

CPI and SPI Trend — Is the Project Recovering?



WHAT IT TELLS YOU

- A single index reading is a snapshot; the trend line tells you whether corrective action worked.
- Both indices bottom out around week 5, the point at which corrective action was taken.
- The recovery from week 6 onward suggests the added resources are earning value.
- Neither index has yet reached 1.0, so the project is still both late and over budget.
- Report the trend to the sponsor — 'improving but not recovered' is a defensible status.

Exam Traps — Earned Value

IF THE QUESTION SAYS

You are given AC and percent complete and asked for EV



THE BEST ANSWER IS

$EV = BAC \times \text{percent complete}$. Actual cost never enters the earned value calculation.

IF THE QUESTION SAYS

CPI is 0.9 and the question says the variance was atypical



THE BEST ANSWER IS

Use $EAC = AC + (BAC - EV)$. The atypical wording rules out the BAC/CPI form.

IF THE QUESTION SAYS

SPI is 0.85 but the project manager says the deadline is safe



THE BEST ANSWER IS

Check the critical path. SPI aggregates all work, so non-critical lateness can leave the finish date intact.

IF THE QUESTION SAYS

The question asks what to do about a negative CV of significant size



THE BEST ANSWER IS

Perform variance and root cause analysis first, then raise a change request for corrective action.

Schedule Compression — Crashing vs Fast Tracking

Crashing

Buy time with money

- Shortens duration for the least incremental cost by adding resources
- Typical actions: approve overtime, add staff, pay a supplier to expedite
- Works only on activities that lie on the critical path
- Always increases cost and may increase risk through onboarding and rework
- Does not always produce a viable alternative — some work cannot be parallelised

Fast tracking

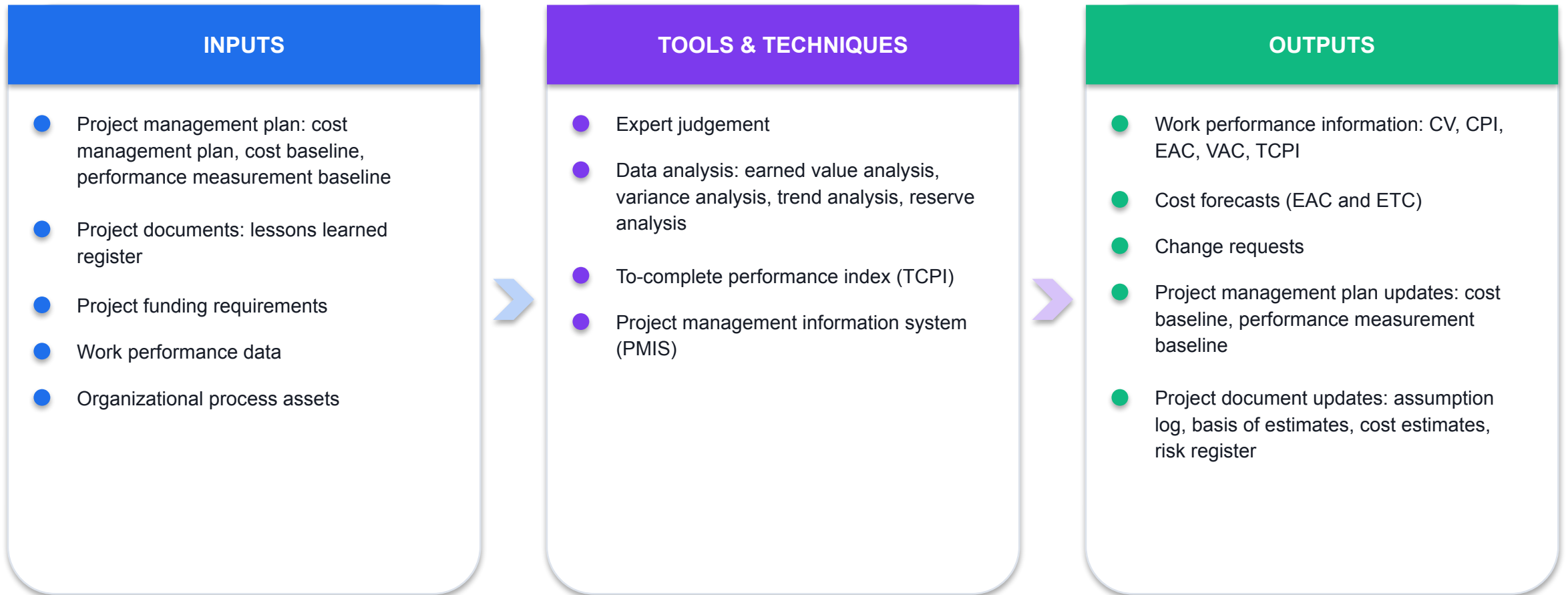
Buy time with risk

- Performs activities or phases in parallel that were planned in sequence
- Typical action: start construction before the design is fully approved
- Costs nothing directly but consumes contingency through rework
- May result in rework, increased risk and ultimately increased cost
- Only applies where the dependency is discretionary, not mandatory

Both techniques apply to the critical path only. Try fast tracking first if the budget is fixed and the dependencies are discretionary; crash first if the risk appetite is low and money is available.

Control Costs — ITTO

Purpose: monitor the status of the project to update project costs and manage changes to the cost baseline. Any increase over authorised funding requires a change request.



Reserve Analysis in Control

1 Contingency reserve

Funds and time set aside for identified risks — the 'known unknowns'. The PM controls and draws on it.

2 Management reserve

Held for unidentified risk — the 'unknown unknowns'. Only management releases it, and it sits outside the baseline.

3 Burn rate check

Compare reserve consumed against risk retired. Burning 60% of reserve at 20% complete is a red flag.

4 Release the surplus

When a risk passes without occurring, release its contingency back — do not silently absorb it as slack.

5 Cost baseline vs budget

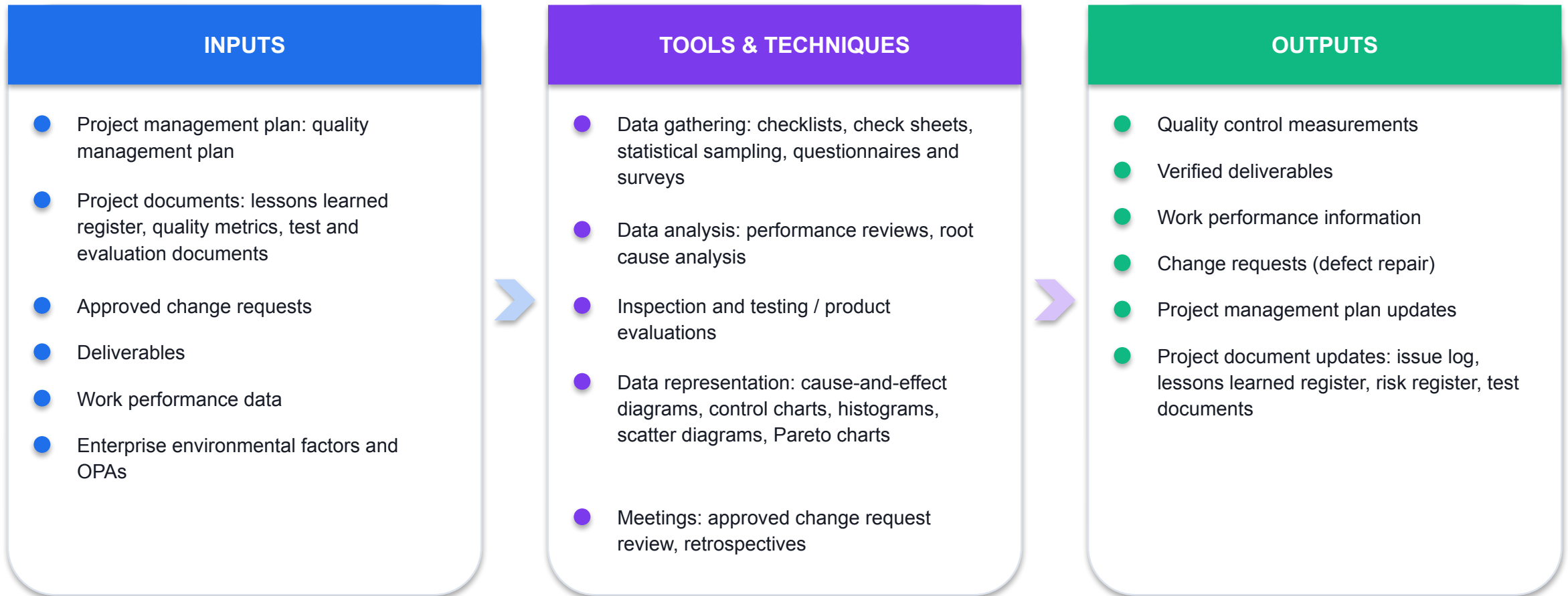
Cost baseline = estimates plus contingency reserve. Project budget = cost baseline plus management reserve.

6 Reserve in adaptive work

Agile teams reserve capacity rather than money — a percentage of each sprint held for emergent work.

Control Quality — ITTO

Purpose: monitor and record the results of executing quality activities to assess performance and ensure the project outputs are complete, correct and meet customer expectations.



What Quality Control Does

1 Assess project approaches

Judge whether the way the work is being done can produce a conforming result at all.

2 Evaluate deliverables

Inspect and test the product against its quality requirements before it reaches the customer.

3 Review processes

Evaluate the quality of project activities and processes through structured reviews and audits.

4 Detect and prevent

Focus on both detecting defects that exist and preventing the causes that keep producing them.

5 Measure, do not guess

Quality control is quantitative — the seven basic tools turn opinion into evidence.

6 Feed improvement

Every defect analysed is a candidate process improvement, not simply a task to rework.

Prevention vs Inspection — and the Cost of Quality

Cost of conformance

Money spent to prevent failure

- Prevention costs: training, documented process, right equipment, time to do it right
- Appraisal costs: testing, destructive test loss, inspection and audit effort
- Spent proactively during the project, inside the approved budget
- PMI's position: prevention is preferred over inspection — quality is planned in
- A dollar spent on prevention typically saves many dollars of failure cost

Cost of nonconformance

Money spent because of failure

- Internal failure costs: rework and scrap found before the customer sees it
- External failure costs: liabilities, warranty work, lost business, reputation
- Spent reactively, usually outside the plan and outside the contingency estimate
- External failure is by far the most expensive category and the hardest to reverse
- Rises exponentially the later in the life cycle the defect is discovered

Quality is planned in, not inspected in. Inspection keeps defects away from the customer but does nothing to stop the process producing them — that requires prevention and root cause analysis.

Quality Audits

1 What it is

A structured, independent review to determine whether project activities comply with organisational and project policies.

2 When it happens

May be scheduled at defined milestones or conducted ad hoc when performance triggers concern.

3 Quality policy

Confirms the quality management policy is understood and genuinely applied, not merely published.

4 Information and methods

Examines how quality information is collected and which analytical methods are used on it.

5 Cost of quality

Reviews whether prevention, appraisal and failure costs are tracked and are trending in the right direction.

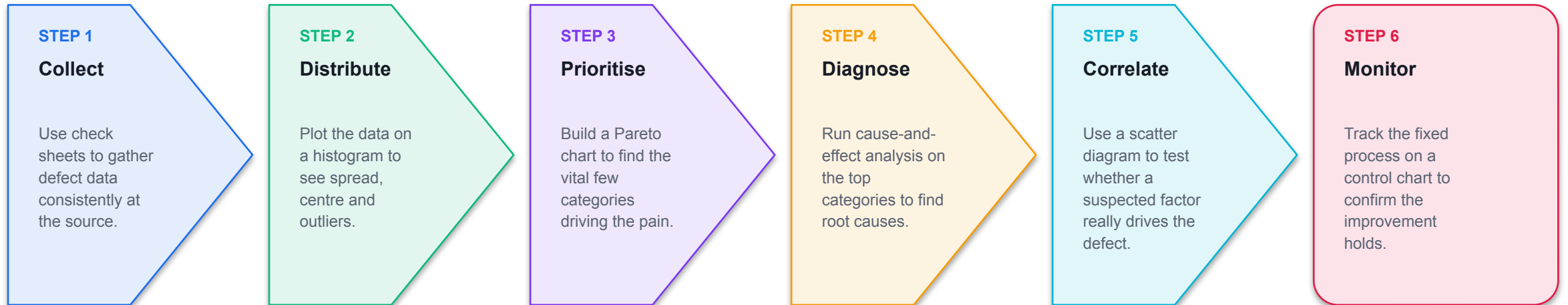
6 Process design

Assesses whether the quality processes themselves are fit for the complexity and risk of this project.

The Seven Basic Quality Tools

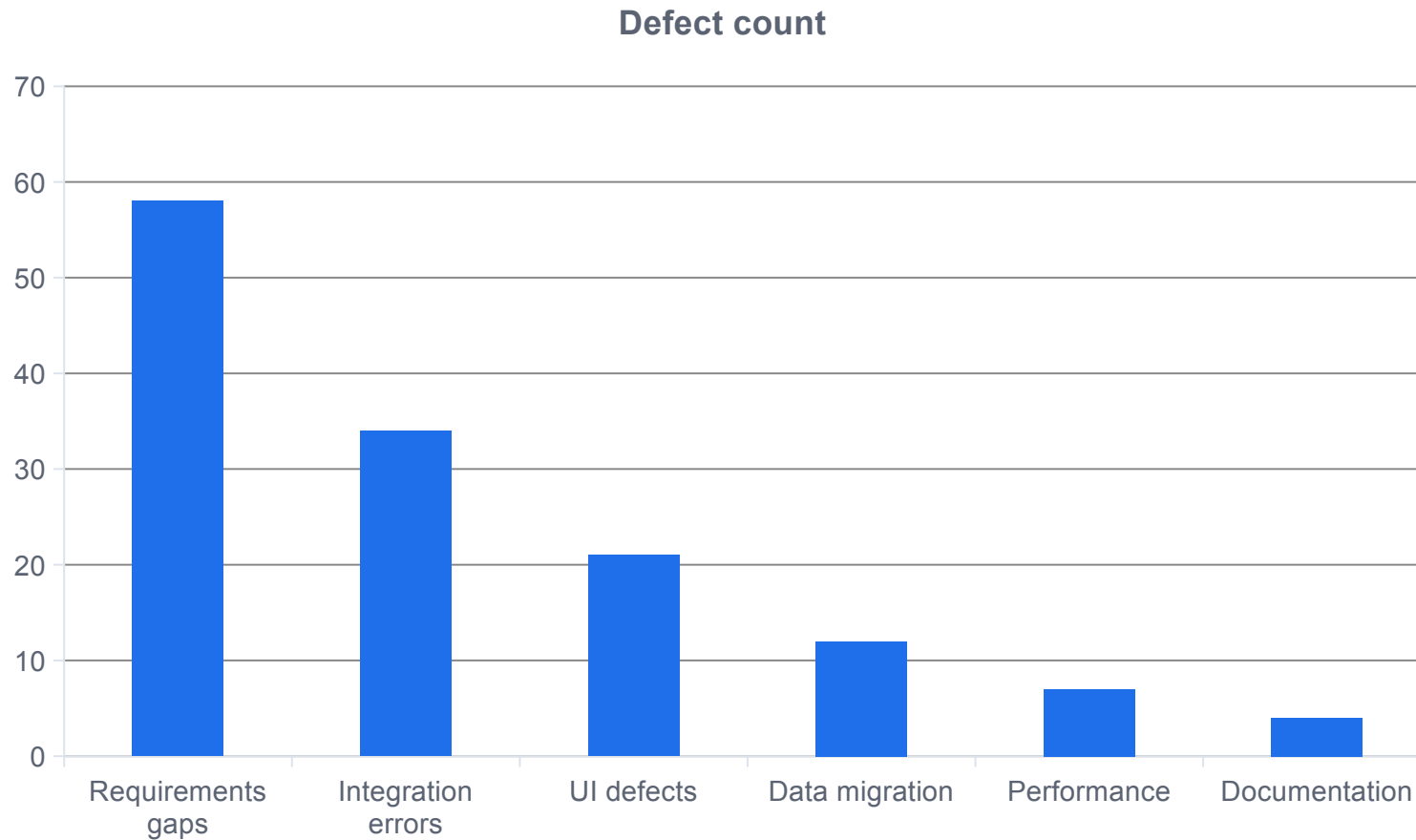
Tool	What it shows	Use it when	Key reading
Check sheet	Tally of occurrences by category, collected at the point of work	You need to gather defect data consistently in the first place	Which category is being tallied most often
Histogram	Distribution and spread of a measured variable across bins	You want to see the shape, centre and spread of the data	Skew, outliers and whether the process is centred on target
Pareto chart	Ranked bar chart of defect categories with a cumulative line	You must decide which few problems to attack first	The 80/20 rule — roughly 80% of effects from 20% of causes
Cause-and-effect (fishbone)	Candidate causes grouped into categories around one problem	You need to explore why a defect keeps recurring	Which branch holds the most plausible root causes
Scatter diagram	Correlation between two variables plotted against each other	You suspect one factor drives a quality defect	Direction and tightness of the relationship, not causation
Control chart	A process plotted over time against mean, UCL and LCL	You must judge whether a process is stable and predictable	Points outside limits or non-random patterns within them
Flowchart	The sequence of steps and decision points in a process	You need to see where handoffs, loops and waste occur	Rework loops, unnecessary approvals and unclear ownership

The Control Quality Analysis Sequence



This sequence turns raw complaints into a defensible improvement case. Skipping straight to a solution without the Pareto and root cause steps is the most common quality mistake — and a common exam distractor.

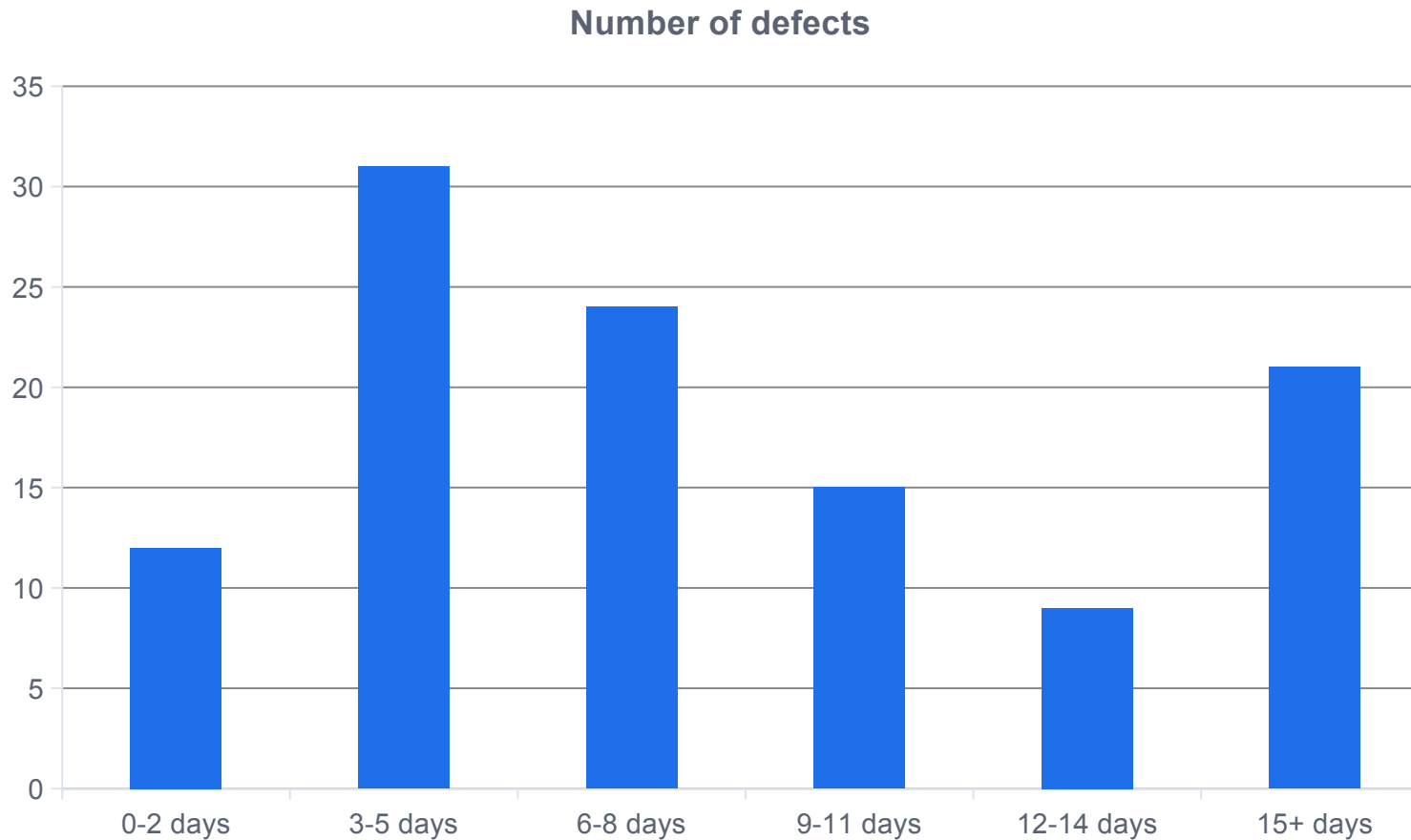
Pareto Chart — Defect Categories on a Software Release



WHAT IT TELLS YOU

- The two leading categories account for roughly 66% of all defects raised.
- The 80/20 rule says fix the vital few, not the trivial many — start left.
- Requirements gaps are an upstream failure; fixing them prevents downstream rework.
- The long tail on the right rarely justifies dedicated corrective action.
- Rebuild the chart after the fix — the ranking should visibly change.

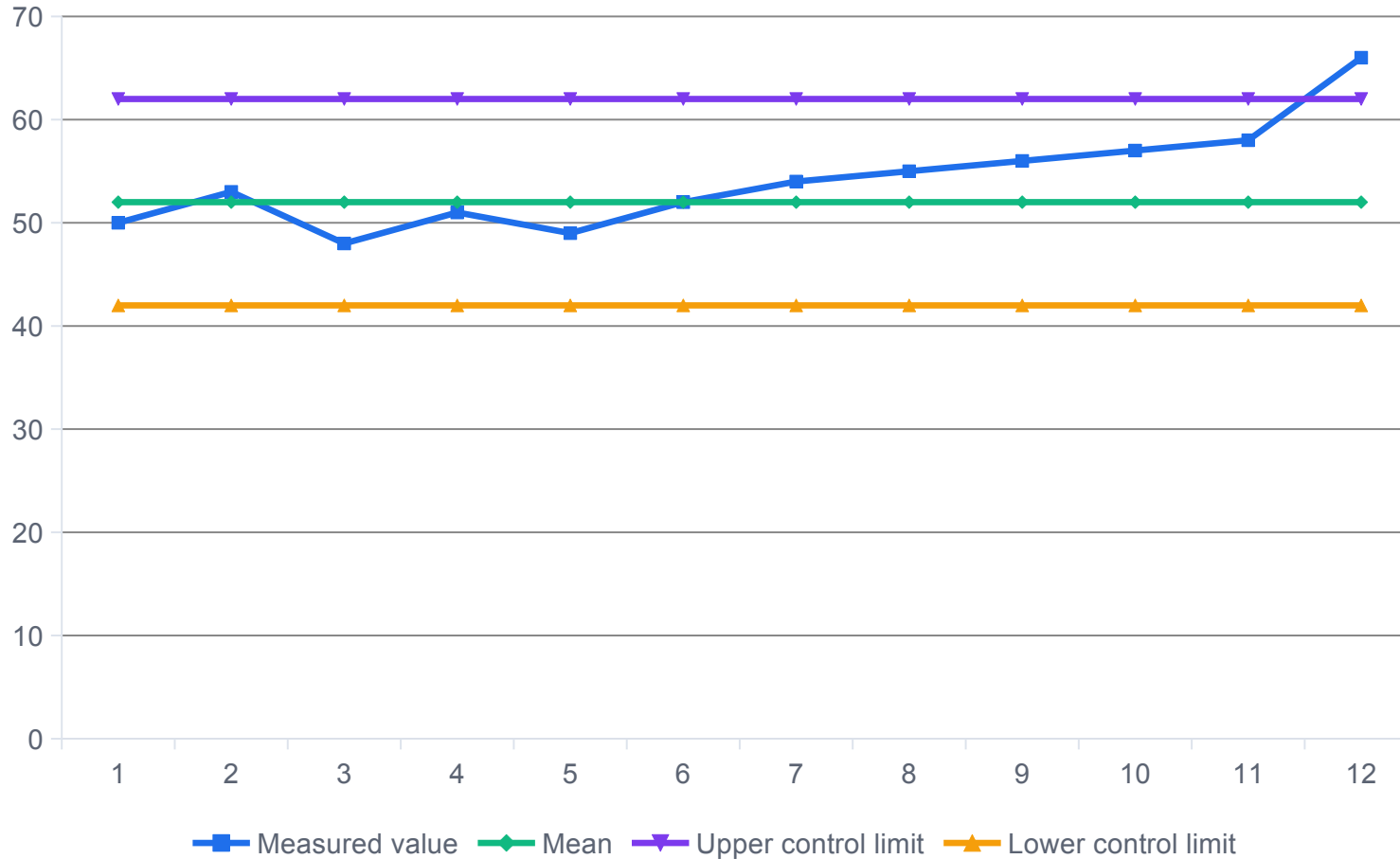
Histogram — Defect Resolution Time Distribution



WHAT IT TELLS YOU

- The histogram shows shape and spread; unlike Pareto it is not ranked by size.
- The peak at 3-5 days is the typical resolution time for routine defects.
- The second cluster at 15+ days is a warning of a distinct second population.
- That long tail is usually a different root cause — escalated or vendor-dependent items.
- Investigate the tail separately; averaging the two populations hides both problems.

Control Chart — Is the Process In Control?



WHAT IT TELLS YOU

- Control limits are set by the process itself (usually plus or minus three sigma).
- Specification limits come from the customer — a process can be in control and still fail spec.
- Point 12 breaches the UCL: an assignable cause that must be investigated.
- Points 5 to 11 rise for seven consecutive readings — the rule of seven flags a trend.
- A process is out of control if a point breaches a limit OR shows a non-random pattern.

Common Cause vs Assignable (Special) Cause

Common cause variation

The noise the process always has

- Natural, random variation inherent in the way the process is designed
- Produces points that fall randomly within the upper and lower control limits
- Present all the time — it is the process behaving exactly as built
- Reducing it requires redesigning the process, which is a management decision
- Reacting to individual common-cause points is 'tampering' and increases variation

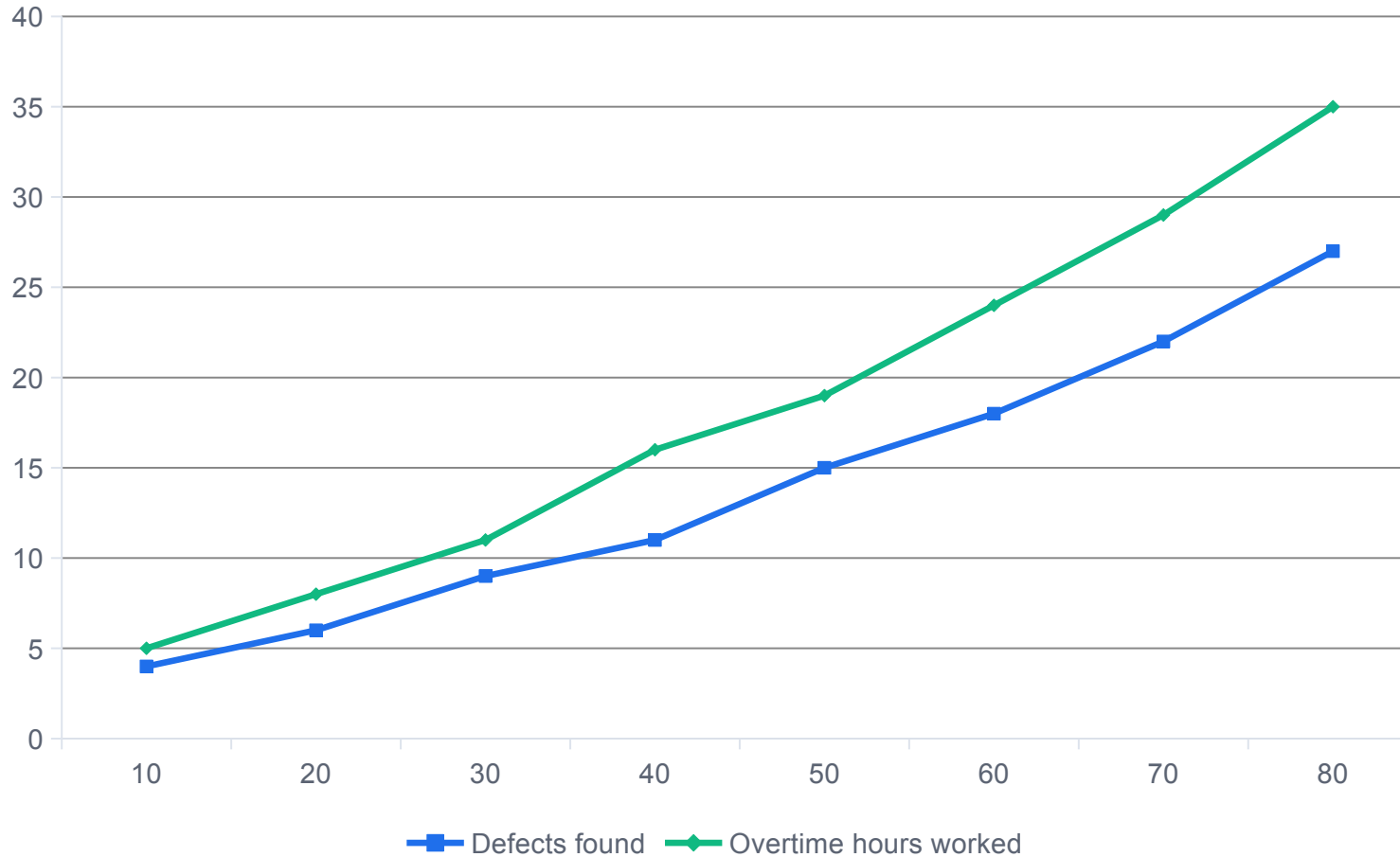
Assignable cause variation

A signal that something changed

- A specific, identifiable event that is not part of normal process behaviour
- Produces a point outside the control limits or a non-random pattern within them
- Examples: a new supplier batch, an untrained operator, a changed tool setting
- Must be investigated and eliminated — this is where corrective action belongs
- The rule of seven catches it even when every point is still inside the limits

Exam cue: 'a point outside the control limits', 'seven consecutive points rising', or 'a sudden shift' all mean assignable cause — investigate. Random scatter within limits means leave it alone.

Scatter Diagram — Correlation Between Two Variables



WHAT IT TELLS YOU

- A scatter diagram shows the relationship between two variables on two axes.
- Here defect count rises steadily with overtime hours — a positive correlation.
- Tight clustering along a line implies a strong relationship worth acting on.
- Correlation is not causation — confirm with root cause analysis before acting.
- Use it to test a hypothesis produced by a fishbone diagram, not to generate one.

Statistical Sampling and Inspection

1 Why sample

Testing every unit is often impossible or destructive; a valid sample infers population quality at a fraction of the cost.

2 Sample size and frequency

Both are determined during Plan Quality Management and documented in the quality management plan.

3 Attribute sampling

The result is binary — the unit conforms or it does not. Used for pass/fail inspection.

4 Variable sampling

The result is measured on a continuous scale, which is what makes control charts possible.

5 Tolerance vs control limit

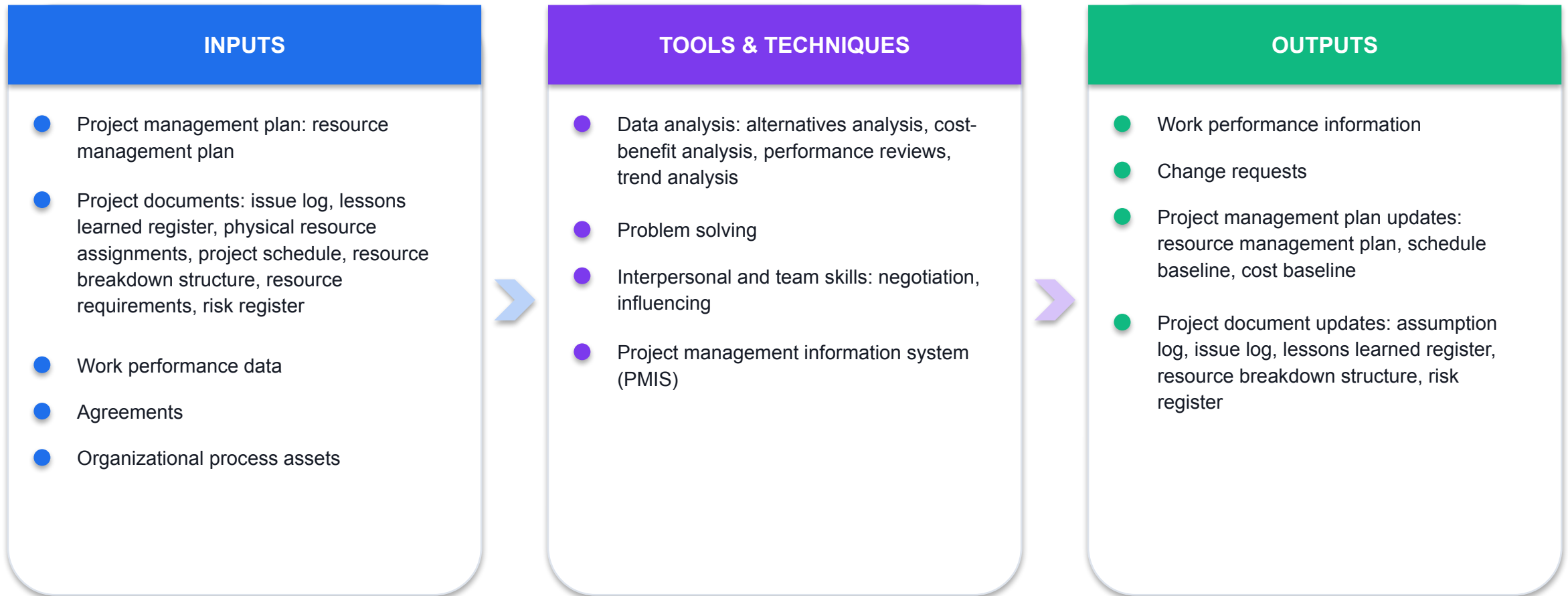
Tolerance is the customer's acceptable range; control limits describe the process's own behaviour.

6 Inspection outcome

Inspection produces verified deliverables and, where the unit fails, a defect repair change request.

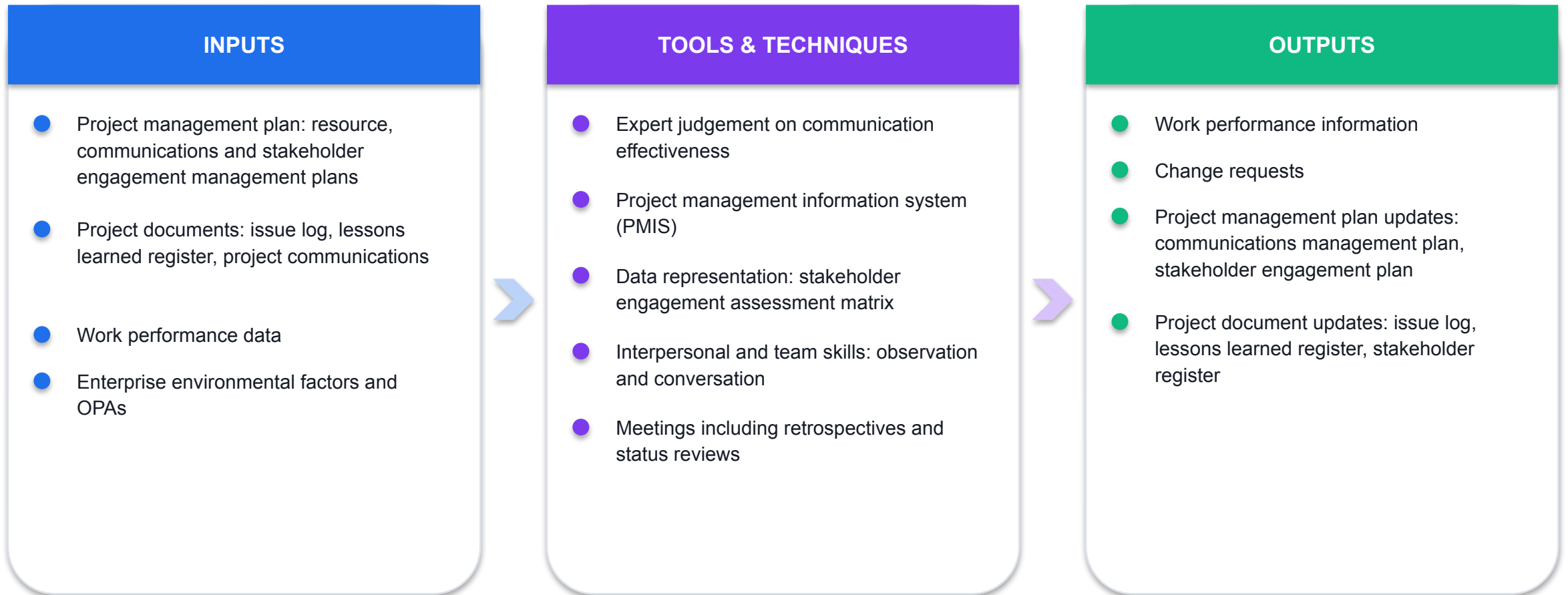
Control Resources — ITTO

Purpose: ensure the physical resources assigned to the project are available as planned, and monitor planned versus actual utilisation, taking corrective action where they diverge.



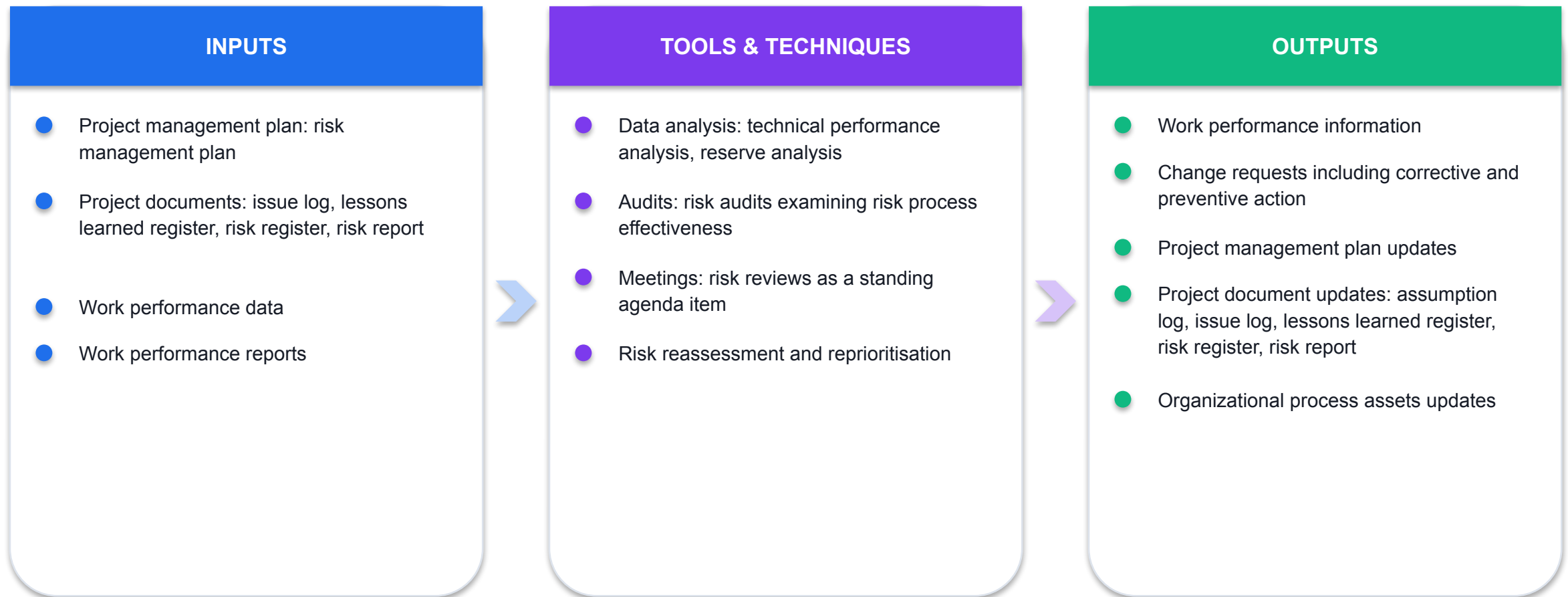
Monitor Communications — ITTO

Purpose: ensure the information needs of the project and its stakeholders are met — the right information, to the right people, in the right format, at the right time.



Monitor Risks — ITTO

Purpose: monitor implementation of agreed risk responses, track identified risks, identify and analyse new risks, and evaluate risk process effectiveness throughout the project.



Monitor Risks — What You Are Actually Doing

1 Enable decisions

Give decision-makers current information on overall risk exposure and on the individual risks that matter.

2 Track status continuously

Monitor each risk's status, probability and impact — all three change as the project moves.

3 Identify new risks

New risks emerge from every change, every phase gate and every shift in the external environment.

4 Reassess current risks

Re-score existing risks; yesterday's low probability risk may now be nearly certain.

5 Close outdated risks

Retire risks whose window has passed and release their contingency reserve back to the project.

6 Improve the process

Risk audits ask whether the risk process itself is working, not just whether individual risks are managed.

When a Risk Becomes an Issue

Risk

A future uncertain event

- Has not happened yet; described with a probability and an impact
- Lives in the risk register with an owner and an agreed response strategy
- Managed proactively — avoid, transfer, mitigate, accept, escalate
- Funded from the contingency reserve if it is an identified risk
- Language cue: 'may', 'could', 'if this occurs'

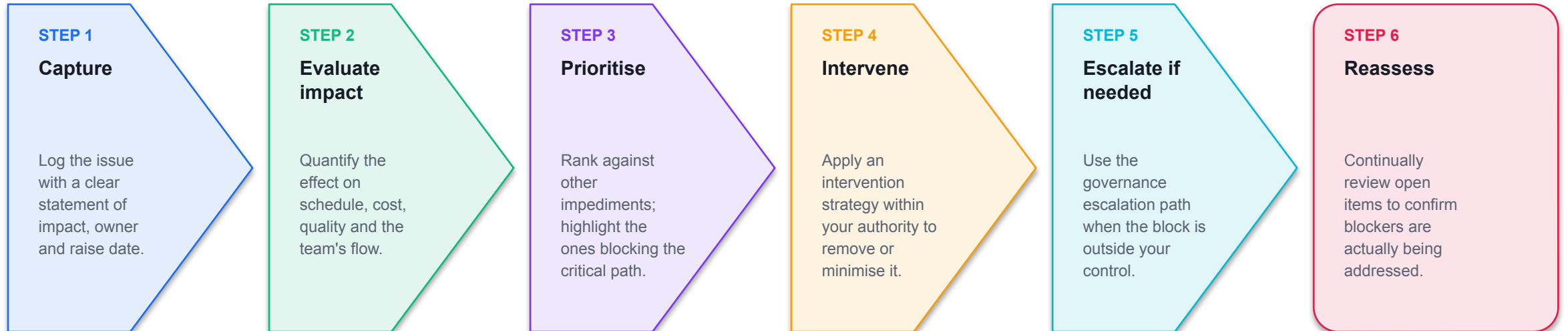
Issue

A present certain condition

- Has already happened and is affecting the project right now
- Lives in the issue log with an owner, an action and a target resolution date
- Managed reactively — resolve, escalate, or accept a workaround
- Consumes real budget and schedule, not contingency probability
- Language cue: 'has occurred', 'is blocking', 'the team cannot proceed'

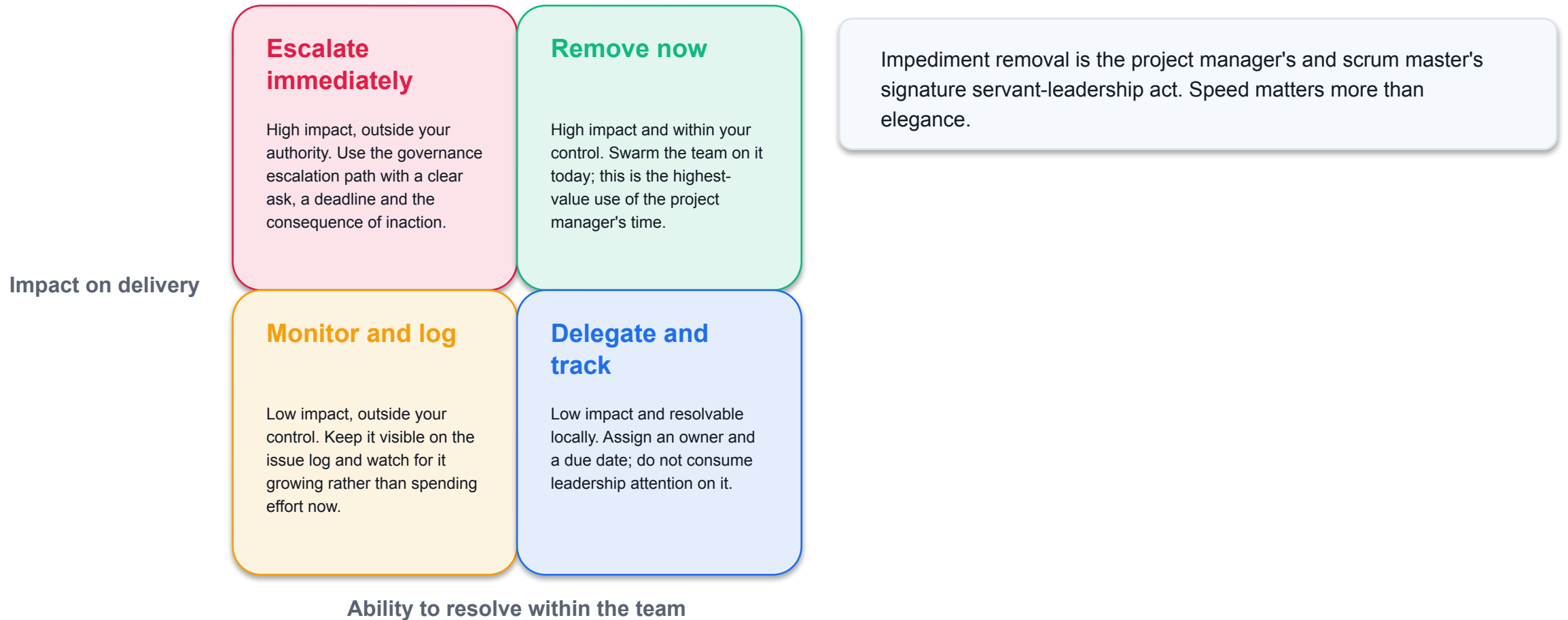
ECO Business Environment T4 explicitly requires you to recognise when a risk becomes an issue. The moment it occurs, move it from the risk register to the issue log — the response strategy becomes an action plan.

Issue and Impediment Resolution



An impediment that stays open past its target date is itself a risk. Ageing reports on the issue log are one of the most useful leading indicators a project manager has.

Prioritising Impediments — Impact vs Ability to Resolve



Sustainability and Security Risk Monitoring (ECO 2026)

1 IT security risk

The ECO explicitly names poor IT security as a risk register entry — monitor vulnerabilities, access and vendor posture.

2 Security response execution

Executing the risk management plan includes delivering security risk responses, not merely logging them.

3 Sustainability risk

Environmental, energy and lifecycle impacts of the deliverable are now monitored risks with real thresholds.

4 Regulatory drift

Compliance obligations change mid-project; a control that was adequate at planning may not be at delivery.

5 Compliance measurement

Measure the extent to which the project is in compliance rather than assuming it is.

6 Communicate risk status

The ECO requires you to communicate the status of a risk's impact — silence is not a risk response.

Manage Compliance During Control

1 Test and validate

Validate deliverables against compliance requirements continuously and at every phase end, not only at closure.

2 Identify approvers

Know which authorised stakeholders can sign a compliance acceptance before you need the signature.

3 Remediate early

Unresolved compliance issues cause timeline damage, cost overruns and increased risk exposure.

4 Early warning

A compliance review programme gives early warning of potential threats before an external auditor finds them.

5 Capture variances

Structured checks let you capture variances and take action while remediation is still cheap.

6 Analyse consequences

Document what noncompliance would actually cost — it is what unlocks the funding to fix it.

Agile Consideration — Risk and Team in Control

1 Risks as user stories

Risks are raised as user stories carrying probability and impact ratings so they compete for capacity.

2 Prioritised in the backlog

Risk stories are prioritised alongside value features — risk work becomes visible, funded work.

3 Retrospective as risk review

Each retrospective doubles as a risk reassessment for the coming iteration.

4 T-shaped teams

Agile teams prefer dedicated, T-shaped members — broad capability with one deep specialism.

5 Skills over roles

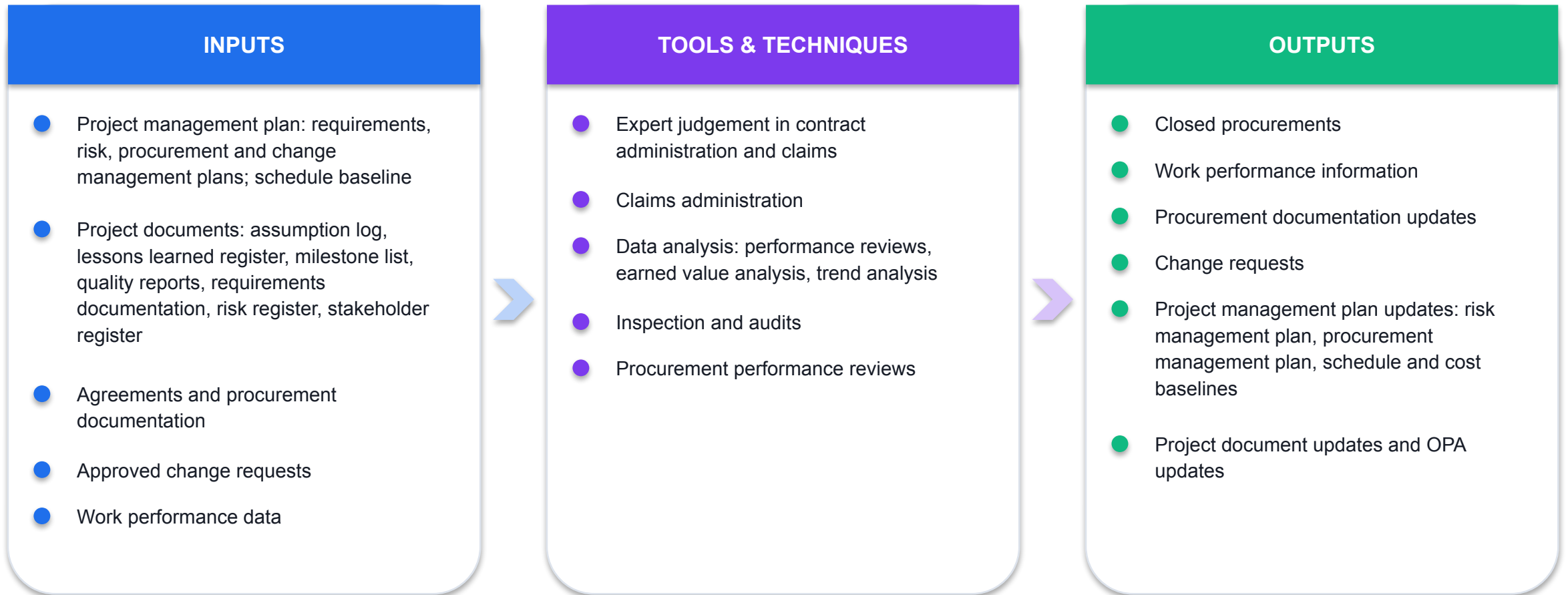
Specific job titles matter less than having every skill needed to complete the feature inside the team.

6 Osmotic communication

Co-located teams learn quickly by osmosis; distributed teams must engineer that flow deliberately.

Control Procurements — ITTO

Purpose: manage procurement relationships, monitor contract performance, make changes and corrections as appropriate, and close out contracts correctly.



Types of Contract Change

Change type	Definition	Typical trigger	Handling
Administrative change	Non-substantive change, usually to the contract administration method	New invoicing address, changed contract officer	Documented by the contracts function; no renegotiation
Contract modification	Substantive change to the contract requirements or product requirements	Approved scope change affecting the vendor's deliverable	Formal, signed by both parties, priced before work starts
Supplemental agreement	An additional agreement related to the contract but negotiated separately	Extra work outside the original statement of work	Negotiated as a separate instrument with its own terms
Constructive change	A change made by the buyer through action or inaction, not by formal order	Buyer directs extra work verbally, or delays an approval	Frequently disputed; the source of most claims
Termination	Ending the contract before completion, for cause or for convenience	Vendor default, or the buyer no longer needs the deliverable	Follows the termination clause; settlement of costs incurred

Constructive change is the exam's favourite: the buyer never issued a change order, but their behaviour effectively directed extra work — and they may still be liable for it.

Managing Disputes and Contract Problems

1 Warranty

A promise, explicit or implied, that goods or services will meet a standard covering reliability, fitness for use and safety.

2 Waiver

A binding provision where one party forfeits a claim without the other becoming liable — waivers can be granted inadvertently.

3 Breach of contract

Failure to meet some or all obligations of a contract, which may entitle the other party to remedy or damages.

4 Work with the vendor first

Most contract problems are solved commercially with the supplier before they become legal ones.

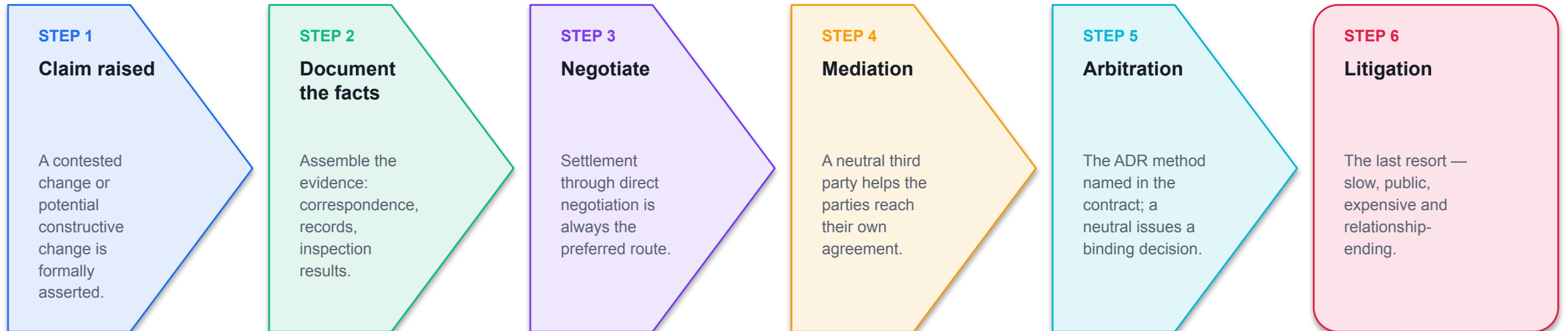
5 Work with your organisation

Engage procurement, finance and functional departments; act within your documented threshold.

6 Get expert help early

Legal problems serious enough to cause project issues need specialist help before positions harden.

Claims Administration and Dispute Escalation



Claims usually arise from a lack of agreement on compensation for a change, or a lack of agreement that a change occurred at all. If unresolved, handle through the alternative dispute resolution method established in the contract.

The Contract Change Control System

1 Purpose

A system specifically dedicated to controlling changes to contracts and the resulting product requirements.

2 Relationship to ICC

May be a component of integrated change control or a separate organisational system, depending on the OPAs.

3 Documentation

Specifies exactly how a contract change is requested, priced, authorised and recorded.

4 Dispute resolution

Includes the dispute-resolution process and the ADR method that applies when the parties disagree.

5 Approval levels

Defines who may sign a contract change at each value band — rarely the project manager alone.

6 Audit trail

Every change instruction must be traceable to an authorised signature; verbal direction creates constructive change risk.

Scaling agile

One Scrum team is a delivery method. Many teams on one product is an organisational design problem.

Coordination overhead

Dependencies between teams grow faster than team count — coordination becomes the constraint.

Shared definition of done

Multiple teams must agree what 'done' and 'integrated' mean or the increment never assembles.

Value stream focus

Scaling frameworks organise around products and value streams rather than functional silos.

eXtreme Programming (XP)

1 Five core values

Communication, Simplicity, Feedback, Respect and Courage — the behavioural foundation the practices rest on.

2 Four core activities

Coding, Testing, Listening and Designing — the only activities XP considers essential to producing software.

3 Whole team

Everyone needed to do the work sits within the team, including the on-site customer representative.

4 Planning game

Planning Poker on user stories: the team estimates, high and low estimators explain, and the team re-estimates to consensus.

5 Pair programming

Two developers on one workstation — continuous review that spreads knowledge and catches defects at the keystroke.

6 Test-driven development

Write the failing test first, then the code that passes it, then refactor — quality is built in.

Feature Driven Development (FDD)

1 Feature teams

Small teams dedicated to delivering one client-valued feature at a time, rather than one technical layer.

2 Develop by feature

Work is decomposed into features small enough to be completed in about two weeks and demonstrated.

3 Individual class ownership

One named owner is responsible for each group of code, giving clear accountability for quality.

4 Domain object modelling

Sequence and context diagrams describe how the code interacts before it is written.

5 Inspections

Formal inspection of both the code and the result — FDD is more document-friendly than most agile methods.

6 Regular builds and CM

Configuration management maintains versions of every change; regular builds merge to the main branch.

Crystal

1 Technologies change techniques

There is no one method — the right practices depend on the technology being used.

2 Cultures change norms

What works in one organisation's culture will not transplant unchanged into another.

3 Distances change communication

Co-location changes what communication mechanisms are necessary and sufficient.

4 Sized by team and criticality

Crystal grades a project by team size and criticality — Clear, Yellow, Orange, Red.

5 More governance when riskier

Larger and riskier projects need more governance; small teams need very little.

6 Agile favours the small

Because agile favours small teams, Crystal Clear and Crystal Yellow are the commonly used variants.

Crystal Core Properties and Techniques

1 Frequent delivery

Ship working software to real users regularly — the single most important property in Crystal.

2 Reflective improvement

Reflection workshops — Crystal's name for the retrospective — adjust the method as you go.

3 Close communication

Osmotic communication in a shared space, plus easy access to expert users when questions arise.

4 Personal safety and focus

People must be able to speak up without fear, and have protected time to concentrate.

5 Methodology shaping

Deliberately tailor the method to this project and team rather than adopting a framework unmodified.

6 Blitz planning and Wideband Delphi

Group planning and group estimation where outliers explain before re-estimating.

Dynamic Systems Development Method (DSDM)

1 The inverted triangle

DSDM is where the alternate constraints triangle comes from: time and cost are fixed, scope is variable.

2 Focus on the business need

Every decision is tested against whether it serves a real, agreed business need.

3 Build on firm foundations

Build incrementally, but only after enough foundation work to know the architecture is sound.

4 Never compromise quality

Quality is fixed alongside time and cost — scope is the only variable DSDM allows you to flex.

5 Timeboxing

Short fixed iterations, plus timeboxed spikes for research or problem solving that would otherwise expand indefinitely.

6 MoSCoW prioritisation

Must have, Should have, Could have, Won't have this time — the mechanism that makes scope flexible.

Scaling Frameworks Compared

Framework	Core idea	Coordination mechanism	Best fit
Scrum of Scrums	A representative from each team attends a shared stand-up	Program and ultimately portfolio Scrums report progress and cross-team blockers	A handful of teams on one product, minimal added structure
LeSS (Large Scale Scrum)	Large Scale Scrum is Scrum — treat many teams as one team	One Product Owner, one backlog, one sprint; transparency via Kanban and radiators	Several teams that can genuinely share a single product backlog
SAFe	Organise around value streams — the products and services themselves	Agile Release Trains: teams of teams pulling in anyone needed to deliver value	Large enterprises needing portfolio-level funding and governance alignment
Disciplined Agile	A decision framework rather than a prescribed method	Guided choices on lifecycle and practices per team context	Organisations with genuinely heterogeneous delivery contexts

LeSS vs SAFe

LeSS — Large Scale Scrum

Descale the organisation

- Applies Scrum ways of working to programs and portfolios only when necessary
- Empirical process control: update and improve the process as you go
- Manages through transparency — Kanban boards and information radiators
- 'More with LeSS': more ownership with fewer roles, more learning with less defined process
- Keeps one Product Owner and one product backlog across all teams

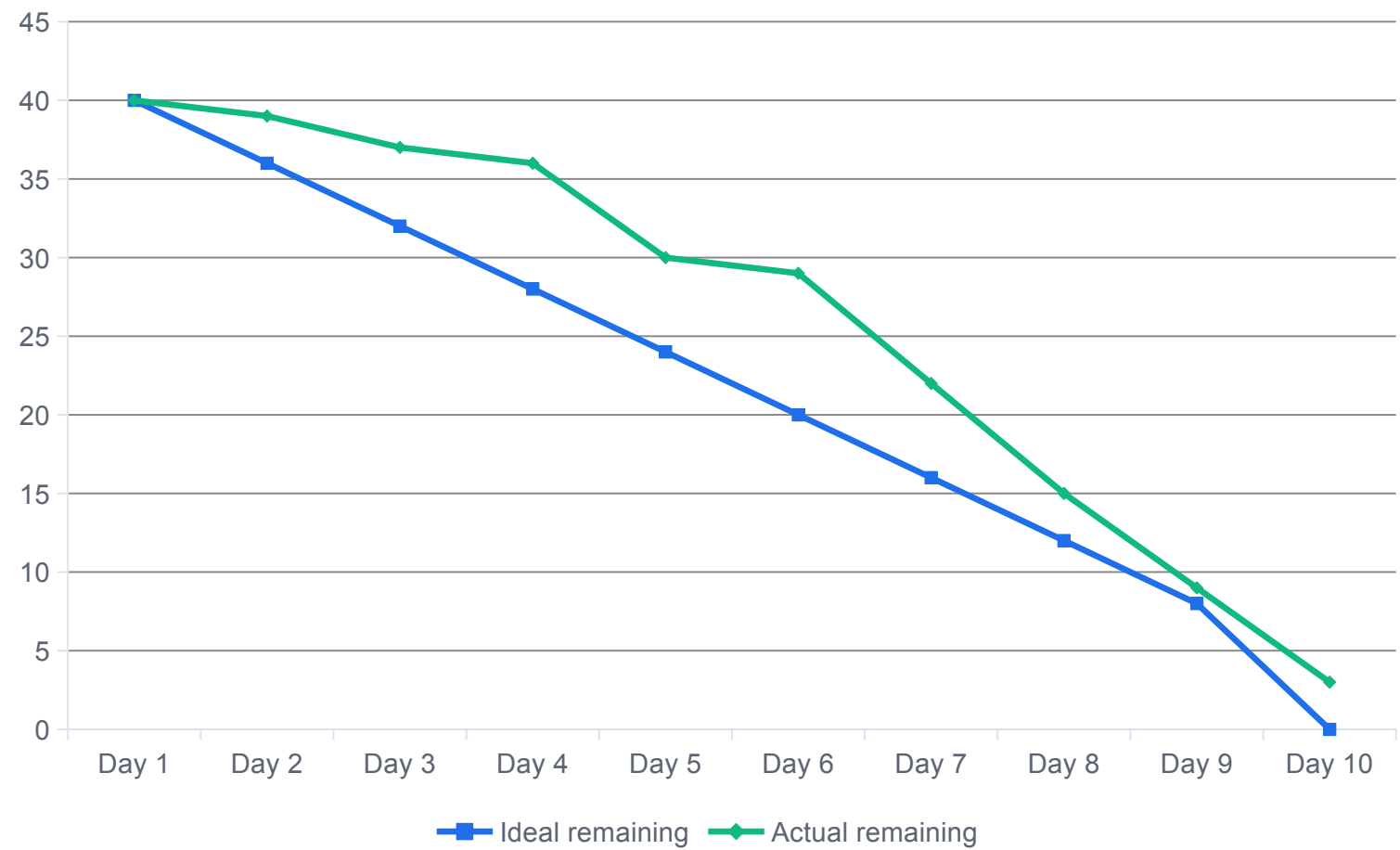
SAFe — Scaled Agile Framework

Add structure to the enterprise

- Organises project teams around value streams — products or services, not functions
- Agile Release Trains are teams of teams, adding anyone needed to deliver the value
- Each value stream carries OKRs: written objectives and measurable key results
- Uses human-centred design and Lean portfolio management above the team level
- Prescriptive: defined roles, cadences and planning events at every level

LeSS reduces organisational structure to let Scrum work at scale; SAFe adds structure to connect agile teams to enterprise funding and governance. Neither is 'more agile' — they solve different constraints.

Iteration Burndown — Reading Adaptive Progress



WHAT IT TELLS YOU

- A burndown is the adaptive equivalent of an EVM S-curve — remaining work over time.
- The flat stretch on days 2-4 signals a blocker, not slow work.
- The steep drop from day 6 is typical of stories finishing in a batch at the end.
- Ending above zero means committed scope was not completed — carry it to the next sprint.
- Trend across several sprints matters more than any one sprint's line.

Exam Traps — Monitoring, Quality and Issues

IF THE QUESTION SAYS

A team member reports a problem that is already affecting the schedule



THE BEST ANSWER IS

It is an issue, not a risk. Record it in the issue log, assign an owner, and determine the impact before escalating.

IF THE QUESTION SAYS

Seven consecutive control chart points fall above the mean, all inside the limits



THE BEST ANSWER IS

The rule of seven signals an assignable cause. Investigate — do not wait for a point to breach the control limit.

IF THE QUESTION SAYS

A deliverable passed the team's testing but the customer refuses to sign



THE BEST ANSWER IS

Control Quality verified it, but Validate Scope has failed. Raise a change request and review the acceptance criteria with the customer.

IF THE QUESTION SAYS

The vendor performed extra work because you delayed an approval



THE BEST ANSWER IS

That is a constructive change. Assess the claim, document the facts, and negotiate before invoking the contract's ADR clause.

Topic 5 Recap — Monitor and Control the Project

1 Data becomes decisions

Work performance data becomes information becomes reports becomes change requests — that chain is the whole domain.

3 Indices below 1.0 are bad

CPI and SPI below 1.0 mean over budget and behind schedule. Trends matter more than single readings.

5 Every change is a request

Nothing changes a baseline without integrated change control — and rejected changes are still logged.

7 Quality is planned in

Prevention beats inspection. The seven basic tools turn defect opinion into a defensible improvement case.

9 Risk becomes issue

The moment a risk occurs, move it to the issue log and switch from response strategy to resolution action.

2 EVM is one integrated system

PV, EV and AC generate every variance, index and forecast. Learn EV first; everything else follows.

4 Choose the right EAC

'Rate continues' uses BAC/CPI. 'Atypical' uses AC+BAC-EV. 'Must hit the date' uses the CPI x SPI form.

6 Corrective vs preventive

Corrective fixes a variance that happened. Preventive stops one that is only forecast.

8 Control chart signals

A breach of the limits or seven consecutive points one side means assignable cause — investigate it.

10 Scaling is organisational

LeSS descales structure; SAFe adds it. Both exist to manage dependencies, not to make teams more agile.

Hands-on Labs — Topic 5

1 Lab 19. Kanban Board with WIP Limits and Lead/Cycle Time

Process T9 · LO5 · 20 min in class + 25 min self-study

Produces: Kanban board, Cycle time measurements

2 Lab 20. Earned Value Management and Schedule Compression

Process T9 · LO5 · 25 min in class + 35 min self-study

Produces: EVM calculation sheet, Forecast (EAC/ETC), Compression decision

3 Lab 21. Root Cause Analysis with 5 Whys, Fishbone and Pareto

BE T4 · LO3 · 25 min in class + 35 min self-study

Produces: 5 Whys chain, Fishbone diagram, Pareto chart

4 Lab 22. SPC Control Chart and Statistical Process Analysis

Process T7 · LO5 · 30 min in class + 30 min self-study

Produces: Control chart, Stability verdict, Capability summary

HANDS-ON ACTIVITY

Lab 19. Kanban Board with WIP Limits and Lead/Cycle Time

Case study: Contoso Training Portal Upgrade. Process T9 · LO5. Produces Kanban board, Cycle time measurements.

20 min in class

1 Open the Kanban tool and create columns matching Contoso's real workflow, including any waiting states.

2 Populate the board with the current sprint's stories from Lab 10.

3 Set explicit WIP limits per column based on team size, then observe what the limits block.

4 Record the start and finish date of each item to measure lead time and cycle time.

5 Identify the bottleneck column — where work queues rather than flows — and propose a specific intervention.

6 Apply Little's Law to forecast delivery from your measured throughput rather than from optimism.

DELIVERABLE

Kanban board, Cycle time measurements

TOOL

Kanban Board

alfredang.github.io/kanban/

Lab 20. Earned Value Management and Schedule Compression

Case study: Contoso Training Portal Upgrade. Process T9 · LO5. Produces EVM calculation sheet, Forecast (EAC/ETC), Compression decision.

25 min in class

1 Record the given Contoso data: BAC, and the PV, EV and AC at the reporting period.

2 Calculate the variances: $CV = EV - AC$ and $SV = EV - PV$, stating whether each is favourable.

3 Calculate the indices: $CPI = EV / AC$ and $SPI = EV / PV$, and interpret each against 1.0.

4 Forecast the outcome: $EAC = BAC / CPI$, then $ETC = EAC - AC$ and $VAC = BAC - EAC$.

5 Calculate $TCPI = (BAC - EV) / (BAC - AC)$ and state whether the required efficiency is realistically achievable.

6 Decide between crashing and fast tracking to recover, quantifying the added cost or the added risk of your choice.

DELIVERABLE

EVM calculation sheet, Forecast (EAC/ETC), Compression decision

TOOL

Statistics

alfredang.github.io/novastats/

Lab 21. Root Cause Analysis with 5 Whys, Fishbone and Pareto

Case study: Contoso Training Portal Upgrade. BE T4 · LO3. Produces 5 Whys chain, Fishbone diagram, Pareto chart.

25 min in class

1 Review the Contoso defect data and write the problem statement as a measurable gap, not an opinion.

2 Open the 5 Whys tool and drill from the presenting symptom to a root cause you can actually act on.

3 Open the Fishbone tool and group candidate causes under people, process, technology, environment, materials and measurement.

4 Open the Pareto tool, enter the defect counts by category and identify the vital few driving roughly 80% of the defects.

5 Compare what each tool surfaced — the three disagree, and the disagreement is the most useful output.

6 Propose a countermeasure for the top two root causes, each with an owner, a date and a verification measure.

DELIVERABLE

5 Whys chain, Fishbone diagram, Pareto chart

TOOL

5 Whys + Fishbone Diagram

alfredang.github.io/5whys/

HANDS-ON ACTIVITY

Lab 22. SPC Control Chart and Statistical Process Analysis

Case study: Contoso Training Portal Upgrade. Process T7 · LO5. Produces Control chart, Stability verdict, Capability summary.

30 min in class

1 Enter the twenty process measurements into the SPC tool and plot them in time order.

2 Calculate the process mean and the upper and lower control limits.

3 Identify any point falling outside the control limits and investigate it as an assignable cause.

4 Apply the rule of seven: seven consecutive points on one side of the mean signals a non-random shift even inside the limits.

5 Use the Statistics tool to compute mean, standard deviation and the distribution shape.

6 State whether the process is in control, and separately whether it is acceptable — the two are different questions.

DELIVERABLE

Control chart, Stability verdict, Capability summary

TOOL

SPC / Control Chart + Statistics

alfredang.github.io/novaspc/

06

TOPIC

Close the Project

Closure is a process, not an event. You obtain formal acceptance, transition the deliverable into operations, settle every contract and account, harvest the knowledge, and confirm the benefits the business case promised — then you turn to the exam itself.

- 1 Close Project or Phase and the closure criteria
- 2 Acceptance, transition and readiness for handover
- 3 Contracts, procurement and financial closure
- 4 Lessons learned, retrospectives and knowledge transfer
- 5 Benefits realisation and confirming delivered value
- 6 PMP exam structure, question types and study strategy

Why closure matters

A project that is not formally closed never stops consuming the organisation's money, attention and risk.

Acceptance ends ambiguity

Formal sign-off is the point at which 'nearly done' becomes contractually and legally done.

Transition protects benefits

Value is realised in operations, not in delivery — a poor handover destroys the business case.

Knowledge outlives the team

Lessons learned move into the repository so the next project starts from your experience.

Three Threads of Closing

1 Project or phase closure

Complete every activity, obtain acceptance, settle contracts and accounts, release resources and archive.

2 Benefits realisation

Confirm the value the business case promised is being measured and delivered, and hand ownership of that measurement over.

3 Knowledge transfer

Move explicit and tacit knowledge from the team into the organisation before the team disbands.

4 Phase gate closure

Closing a phase uses the same process — acceptance, lessons learned, archive — before the next phase is authorised.

5 Administrative closure

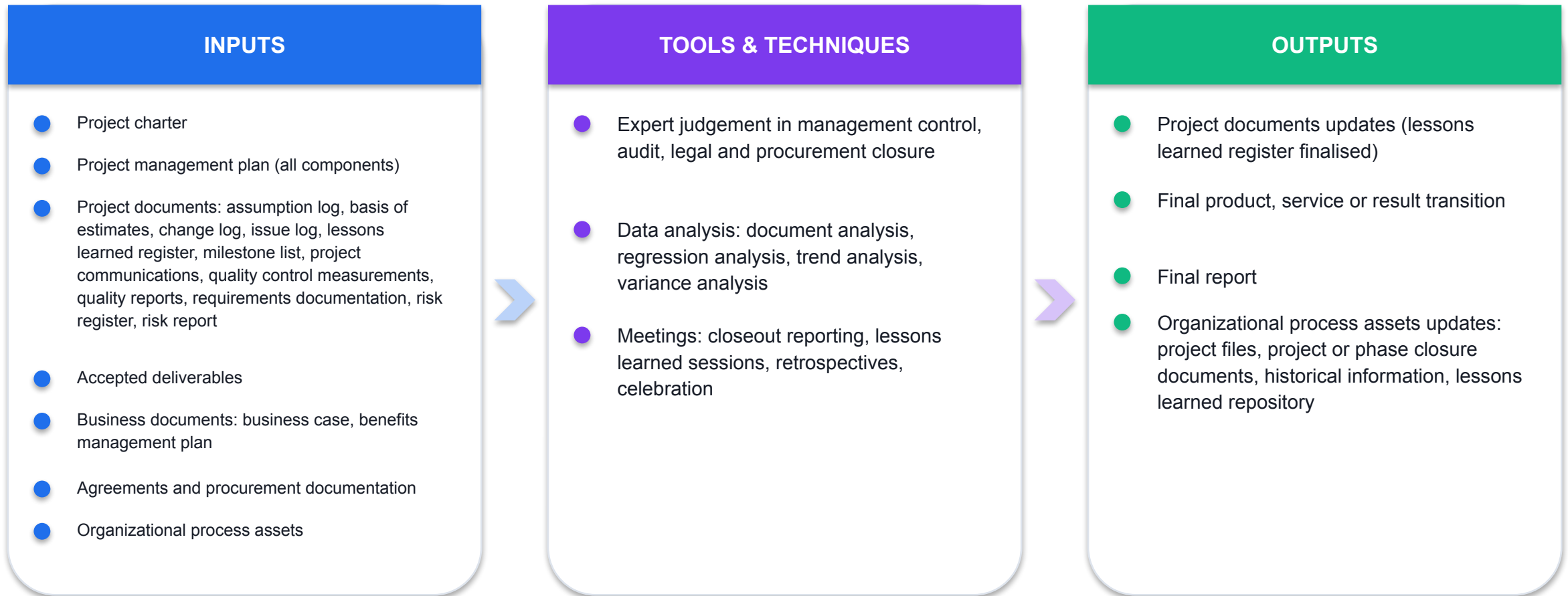
Financial accounts closed, purchase orders settled, systems access revoked, facilities returned.

6 Celebration and recognition

Recognising the team's contribution is a genuine closure activity, not a nicety — it shapes future engagement.

Close Project or Phase — ITTO

Purpose: finalise all activities for the project, phase or contract. This is the only process whose output is the formal, archived record that the project ever happened.



Project Closure vs Phase Closure

Project closure

The whole endeavour ends

- Performed once, when all phases are complete or the project is terminated
- Releases all resources, closes all contracts and all financial accounts
- Transitions the final product, service or result to operations or the customer
- Produces the final report summarising performance against every baseline
- Archives the complete project record into the organisational process assets

Phase closure

A gate between stages

- Performed at the end of each phase in a predictive or hybrid life cycle
- Confirms phase deliverables are accepted before the next phase is authorised
- Resources may be released, retained or reassigned depending on the next phase
- Produces a phase-end report and a go / no-go / hold recommendation
- Captures lessons learned while they are fresh, not at the very end

Both use the same process. A phase gate is the cheapest place to kill a project that has lost its business case — which is precisely why governance defines phase-end criteria in advance.

Why Projects or Phases Close

1 Deliverables accepted

Stakeholders accept deliverables against acceptance criteria established in the project management plan.

2 Criteria may evolve

Acceptance criteria can legitimately be modified during the life cycle — through change control, never informally.

3 Traceability confirms completion

The requirements traceability matrix proves every requirement was completed and approved.

4 Iteration assessment

At the end of an iteration, team and stakeholders assess the product against their mutually agreed Definition of Done.

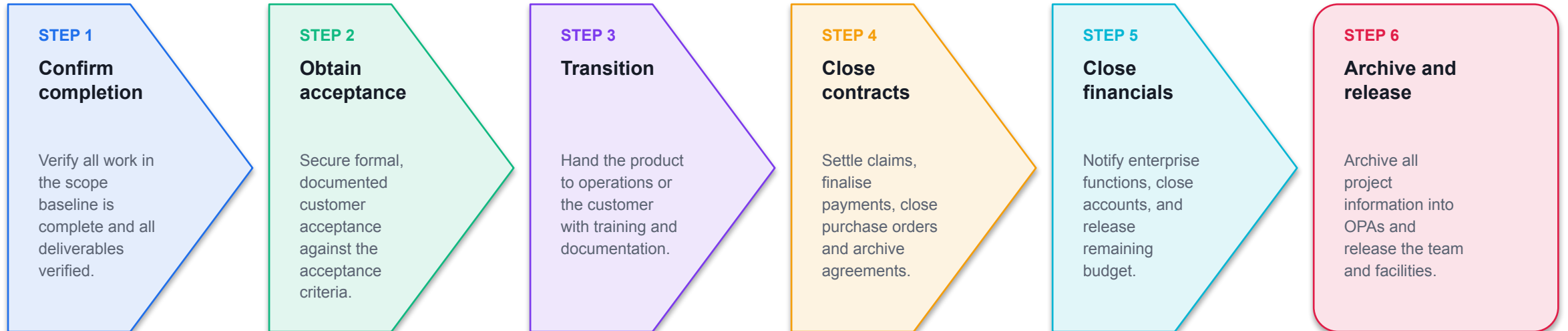
5 Phase gate reached

The phase's exit criteria are met and governance authorises moving to the next phase.

6 Termination

The business case has failed, funding is withdrawn, or the need has disappeared — closure still applies in full.

Close Project or Phase — the Activity Sequence



Conclude external obligations including legal, regulatory and contractual matters — transfer of liability and closure of all accounts in the financial system are frequently forgotten and frequently examined.

Acceptance of Deliverables

1 Acceptance criteria first

Criteria are agreed at the start, in the requirements documentation and the scope baseline.

2 Verified before accepted

Control Quality produces verified deliverables; Validate Scope converts them into accepted deliverables.

3 Formal and documented

Acceptance must be written and signed by an authorised stakeholder — verbal approval is not acceptance.

4 Traceability matrix

Use the requirements traceability matrix to prove every requirement reached completion and approval.

5 Partial acceptance

Where some deliverables fail, accept what passes and raise change requests for the remainder — do not stall everything.

6 Conditional acceptance

Acceptance with a punch list must carry named owners and dates, or the residual work is never done.

Transition and Handover to Operations

1 Adoption drives benefits

Effective transitions enable end-user awareness, increasing adoption and therefore benefits realisation.

2 Training

Users and operators need training on the new product or service before, not after, they depend on it.

3 Documentation

Operating manuals, as-built documentation and troubleshooting guides transfer with the deliverable.

4 Communication

Sustained communication between the project team and the receiving organisation prevents the classic silent handover.

5 Post-implementation support

Agree the hypercare period, the support model and who pays for it, before the team disbands.

6 Warranty and defect window

Define the warranty period and the process for defects discovered after acceptance.

Readiness for Transition



Contents of the Final Report

1 Summary-level description

What the project was, what it delivered, and the period over which it ran.

2 Scope objectives

The scope objectives, the criteria used to evaluate them, and evidence that they were met.

3 Quality objectives

Quality objectives, the criteria applied, and the actual quality achieved against them.

4 Cost and schedule performance

The acceptable variance ranges, the actual variances, and the reason for each.

5 Benefits and business case

Whether the intended benefits were achieved, and if not, when they are expected to be.

6 Risks, issues and recommendations

Summary of how risks and issues were managed and what future projects should do differently.

Finalising Contracts and Procurement Documentation

1 Contract schedule and scope

Collect, index and file the contractual schedule and the final agreed statement of work.

2 Quality and cost performance

File the supplier's quality records and the cost performance data against the agreement.

3 Change documentation

Every contract change, modification and supplemental agreement is archived with its authorisation.

4 Payment and financial records

Payment records, invoices and financial documents complete the commercial audit trail.

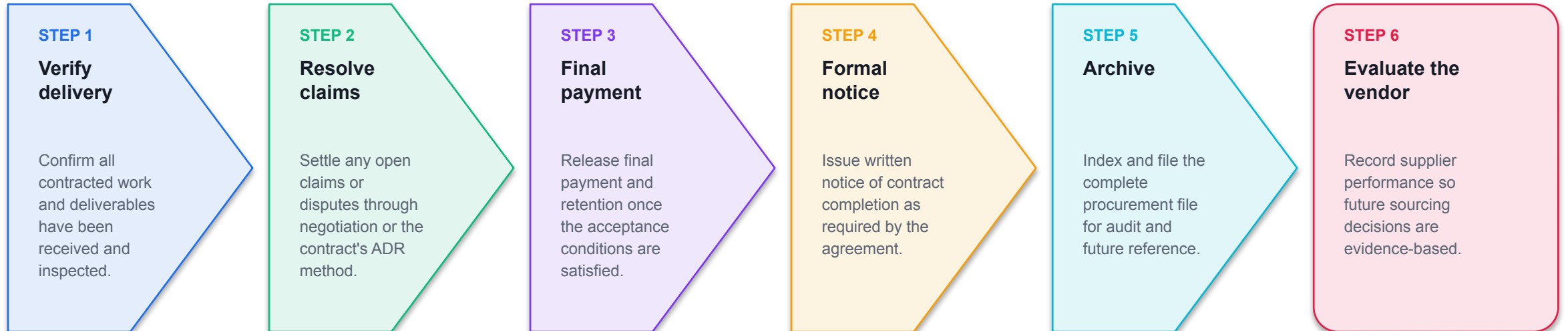
5 Inspection results

Inspection and acceptance test results evidence that the supplier met the agreed standard.

6 As-built documentation

'As-built' or 'as-developed' documents, manuals and troubleshooting and technical documentation.

Procurement Closure Sequence



Early termination is a valid form of procurement closure. The termination clause defines the rights of both parties and the settlement of costs already incurred.

Releasing Resources and Updating OPAs

1 Release people

Return team members to their functional units with a written performance summary and a genuine thank you.

2 Release physical resources

Return equipment, facilities and licences; unclaimed assets keep costing the organisation.

3 Close financial accounts

Close cost centres and purchase orders so no further charges can be booked to a finished project.

4 Revoke access

Remove system and data access — an open security question is a real compliance finding at audit.

5 Update OPA templates

Push improved templates, checklists and estimating data back into the organisational process assets.

6 Update historical databases

Actual durations and costs make the next project's analogous and parametric estimates better.

Lessons Learned Register vs Repository

Lessons learned register

Live, project-level, updated continuously

- A project document created early and updated throughout the whole life cycle
- Captures the situation, the impact, and the recommendation as events happen
- Owned by the project manager and visible to the whole project team
- Feeds every phase gate review, retrospective and status conversation
- Its final state is an input to the closure process and the final report

Lessons learned repository

Organisational, permanent, cross-project

- An organisational process asset holding lessons from all completed projects
- Updated at closure when the register's contents are transferred and indexed
- Owned by the PMO, searchable by process, phase, risk category and domain
- Consulted at the start of new projects to avoid repeating known mistakes
- Its value depends entirely on discipline at closure — the register is not enough

Exam cue: 'during the project' or 'the team recorded' points to the register. 'For future projects' or 'the organisation' points to the repository.

Finalising Lessons Learned for the Final Report

1 Scope changes

What changed, why it changed, and whether the change control process caught it early enough.

2 Schedule impacts

Which estimates proved wrong and what the estimating assumptions should have been.

3 Risks and issues

Which risks materialised, which responses worked, and which risks were never identified at all.

4 Stakeholder relationships

Which engagement strategies built trust and which stakeholders were persistently misjudged.

5 Vendor relationships

Supplier performance, contract type fit, and what to negotiate differently next time.

6 Artifacts and recommendations

Which artifacts earned their keep, and specific recommendations for comparable future projects.

Conducting the Project Retrospective

1 Internalise learning

The goal is that the team internalises learning about both the work product and the process used to build it.

2 Capture successes and challenges

Record what went well as deliberately as what went badly — repeatable success is a finding.

3 Qualitative data

How people felt — morale, safety, frustration — is real data and often explains the numbers.

4 Quantitative data

Measurements: velocity, defect escape rate, cycle time, variance against baseline.

5 Root cause to action plan

Use the data to find root causes, design countermeasures and build an action plan for next time.

6 Praise and motivate

Praise, congratulate and motivate the team — closure is the last chance to reinforce what you want repeated.

Knowledge Transfer Before the Team Disbands

1 Identify critical knowledge

Decide what knowledge is genuinely critical to operating and evolving the deliverable.

2 Gather it deliberately

Interviews, documentation sprints and recorded walkthroughs capture what people carry in their heads.

3 Foster the environment

People share tacit knowledge when they trust the audience and have time set aside to do it.

4 Explicit to the repository

Codified knowledge goes into searchable, indexed assets with clear ownership after the project ends.

5 Tacit through people

Pairing, shadowing and communities of practice are the only reliable transfer mechanisms for judgement.

6 Name the successor

Every critical knowledge area needs a named operational owner before the last team member leaves.

Benefits Realisation and Confirming Value

1 Benefits outlive the project

Most benefits are realised in operations, months after the project team has been released.

2 Benefits management plan

It names each benefit, its metric, its target, its timeframe and its accountable owner.

3 Verify the measurement system

ECO Process T3 requires you to verify a system is actually in place to track the benefits.

4 Hand over the measurement

Transfer benefit tracking to a business owner with the authority and budget to act on the readings.

5 Compare to the business case

Report actual against the promised business case — honestly, including where it fell short.

6 Value in adaptive delivery

Adaptive projects confirm value every increment, so closure confirms cumulative rather than deferred value.

Premature and Terminated Project Closure

1 Termination is legitimate

Stopping a project whose business case has failed protects the organisation — it is not a personal failure.

2 Common triggers

Funding withdrawn, strategy changed, technology superseded, regulator intervened, or the benefit disappeared.

3 Closure still applies fully

Every closure activity is performed: acceptance of partial work, contract settlement, archiving, release.

4 Document the reason

Record precisely why the project was terminated — this is among the most valuable lessons in the repository.

5 Salvage the value

Completed components, research and built assets may be reusable elsewhere; identify them before archiving.

6 Manage the team

Termination hits morale hardest. Communicate early, honestly, and support each member's next assignment.

Exam Traps — Closing

IF THE QUESTION SAYS

The customer has stopped responding and will not sign acceptance



THE BEST ANSWER IS

Follow the escalation path in the communications and governance plans; do not close the project without documented formal acceptance.

IF THE QUESTION SAYS

The project is cancelled halfway through by the sponsor



THE BEST ANSWER IS

Perform Close Project or Phase in full — acceptance of partial work, contract closure, lessons learned and archiving all still apply.

IF THE QUESTION SAYS

A team member asks to move to a new project before closure is complete



THE BEST ANSWER IS

Complete knowledge transfer and lessons learned capture before releasing them; resource release is a closure activity, not a favour.

IF THE QUESTION SAYS

The benefits will not be measurable for another twelve months



THE BEST ANSWER IS

Hand the benefits management plan and its measurement system to an accountable business owner; the project still closes.

Closing Recap

1 Accept before you close

Formal, documented acceptance against agreed criteria is the gateway to every other closure activity.

3 Transition is two-sided

Readiness means the product works and operations can actually run it — check both.

5 Register to repository

The lessons learned register becomes part of the organisational lessons learned repository at closure.

7 Termination closes fully

A cancelled project runs the complete closure process, including documenting why it was stopped.

2 Verified then accepted

Control Quality verifies; Validate Scope accepts; Close Project archives. Learn that sequence.

4 Close every contract

Claims settled, payments made, procurement file archived, vendor performance recorded.

6 Benefits need an owner

Hand measurement to an accountable business owner — the project ends, the benefits do not.

8 Update the OPAs

Templates, estimating data and historical records are the organisation's return on your experience.

PMP exam preparation

The exam does not test what you memorised. It tests what you would do next, in a situation described in four lines.

Situational by design

Most items describe a scenario and ask for the best next action, not a definition.

Approach-agnostic

Predictive, adaptive and hybrid appear throughout all three domains, not in separate sections.

Mindset over method

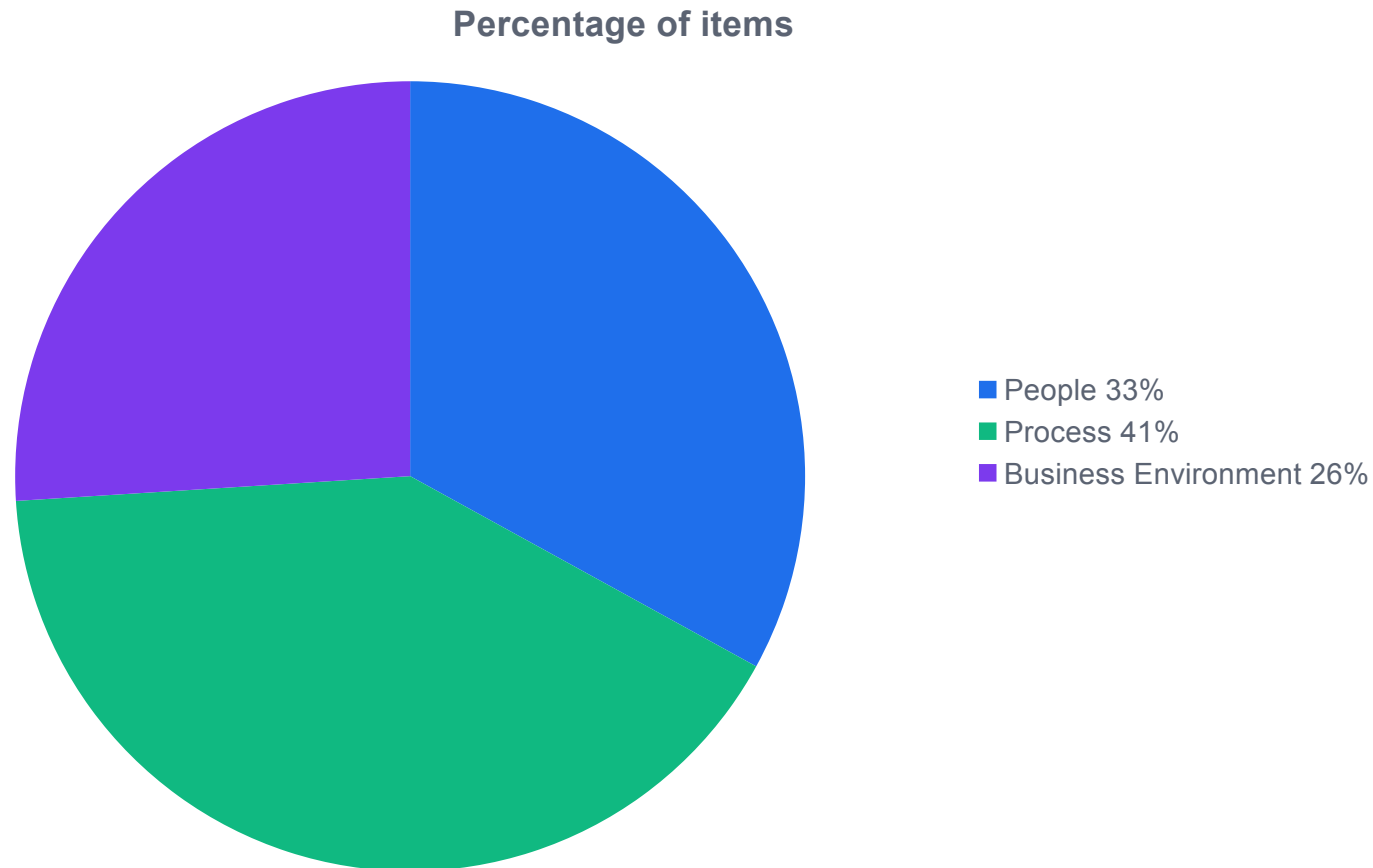
PMI rewards the servant leader who investigates, communicates and follows the process.

PMP Exam Structure — July 2026 Content Outline

Element	Specification	What it means for you
Total questions	180 questions	Every item is presented; you cannot tell which ones count
Scored questions	170 scored items	Your result is calculated from these alone
Pretest questions	10 unscored pretest items	Trialled for future forms — never leave one blank
Time allowed	240 minutes (4 hours)	About 80 seconds per question including reading time
Breaks	Two 10-minute breaks	Two 10-minute breaks — the first after the case-study section, the second midway through the independent questions.
Approach mix	About 40% predictive	About 40% predictive; the remaining 60% split between adaptive/agile and hybrid.
Tutorial and survey	Outside the 240 minutes	The tutorial and the closing survey do not consume exam time
Attempts	Up to three within the one-year eligibility period	After three attempts you must wait before reapplying

Once you review your responses and start a break, you cannot return to the questions in the section you have just completed.

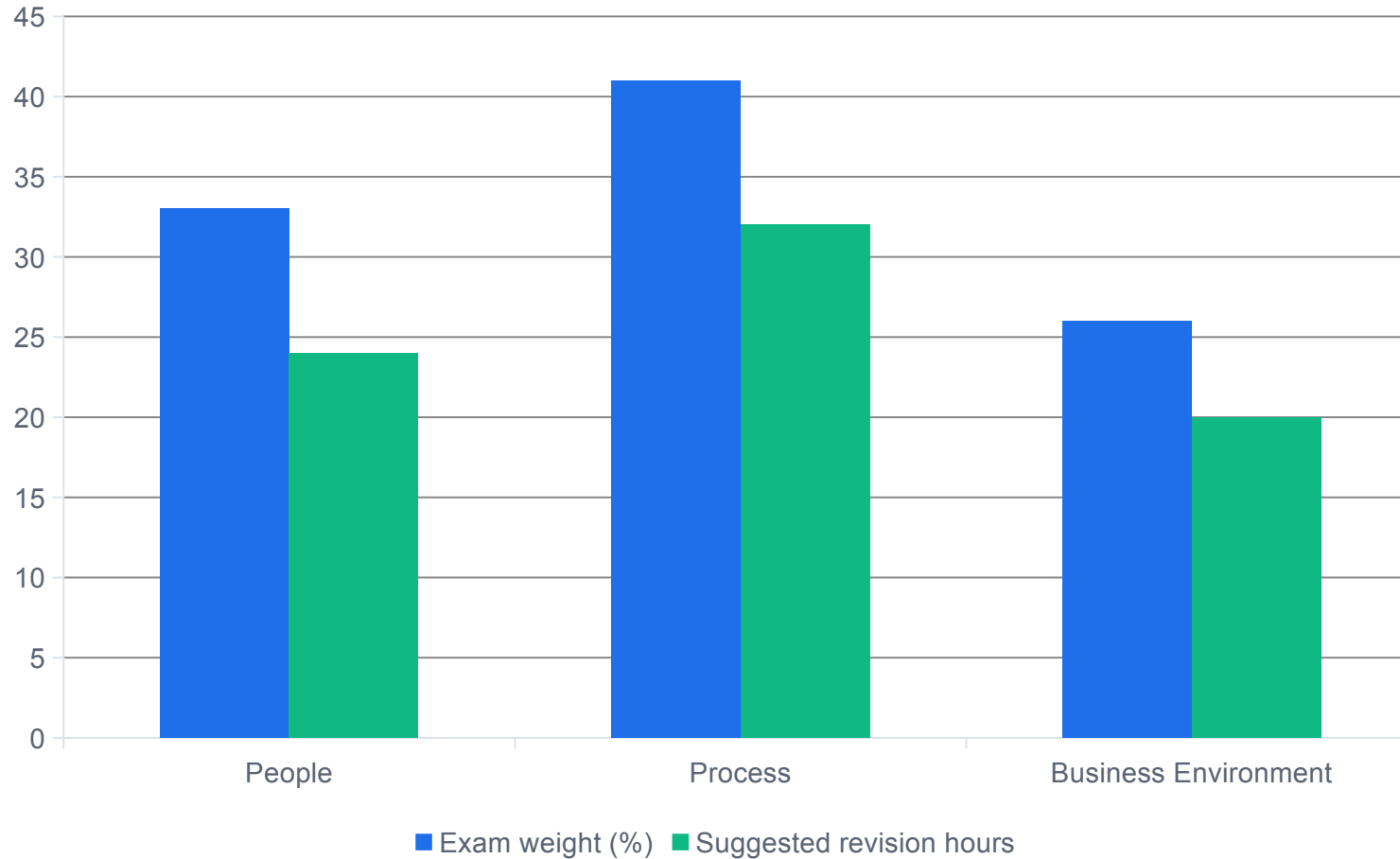
Domain Weighting — Where the Questions Come From



WHAT IT TELLS YOU

- Process at 41% is the largest single domain — roughly 70 of the 170 scored items.
- People at 33% means leadership and stakeholder items outnumber pure Business Environment items.
- Business Environment at 26% covers governance, compliance, change, risk and improvement.
- Weighting applies to items, not to study hours — weak areas deserve disproportionate time.
- Every domain contains predictive, adaptive and hybrid scenarios; none is 'the agile domain'.

Study Effort vs Domain Weight — Plan Your Revision



WHAT IT TELLS YOU

- Allocate revision roughly in proportion to domain weight, then adjust for your mock results.
- Process carries the formulas and the process anatomy — it needs the most contact hours.
- People is high weight and highly situational; practice questions beat re-reading notes.
- Business Environment is compact but dense in compliance and governance vocabulary.
- Re-plan after every mock exam: the domain you scored worst on gets the next block.

The Six Question Types

Type	How it works	Strategy
Multiple choice	One best answer from four options — the bulk of the exam.	Eliminate two options first, then choose between the remaining pair using the PMI mindset.
Multiple response	More than one correct answer; the item tells you how many.	Read the stem for how many answers are required and select exactly that number.
Matching	Drag items from one column to match a second column.	Do the pairs you are certain of first; the remaining items are constrained by elimination.
Enhanced matching	Matching that includes an image or diagram for context.	Study the image before reading the options — the diagram usually resolves the ambiguity.
Graphic-based (NEW)	Read a chart, graph, diagram or image, then answer from it.	Extract the specific data point the question asks for; do not interpret the whole chart.
Case study	A scenario with supporting visuals, then a series of linked questions.	Read the full scenario once, then answer each linked question referring back to the exhibit.

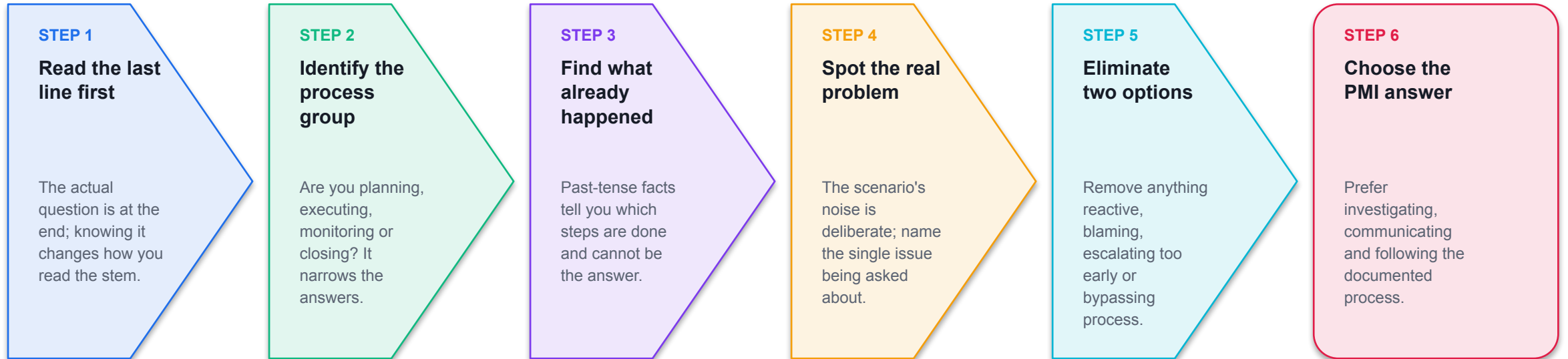
Graphic-based questions are NEW in the July 2026 outline — expect to read control charts, network diagrams, burndowns and EVM S-curves directly.

PMP Certification Eligibility

Educational background	Project management experience	Training
Upper-secondary / secondary school completion (EQF Level 4 / ISCED 3-4)	Minimum 60 months (5 years) of non-overlapping experience leading projects in the past 10 years	35 hours of commercial project management training
Recognised associate-level, higher-certificate or advanced vocational programme (EQF Level 5 / ISCED 5)	Minimum 48 months (4 years) of non-overlapping experience leading projects in the past 10 years	35 hours of commercial project management training
Bachelor's degree or higher, or a qualification mapped to EQF Level 6 / ISCED 6	Minimum 36 months (3 years) of non-overlapping experience leading projects in the past 10 years	35 hours of commercial project management training
Bachelor's or postgraduate degree from a PMI GAC-accredited programme	Minimum 24 months (2 years) of non-overlapping experience leading projects in the past 10 years	35 hours of commercial project management training

Experience must be in leading and directing projects, accrued within the last 10 years. Applications may be audited — keep contact details for verifiers.

How to Read a Situational Question



'Best', 'first', 'next' and 'most likely' are not decoration. 'First' usually means gather information; 'next' means the step after what was described; 'best' means the most complete and process-compliant option.

Common Distractor Patterns

Distractor	Why it looks right	Why it is wrong
Escalate to the sponsor immediately	Escalation is a real governance tool and sounds decisive	PMI expects you to gather facts and act within your authority first
Ask the team to work overtime	It appears to solve a schedule problem directly	It ignores root cause, damages the team, and skips impact analysis
Update the plan / baseline directly	The plan clearly needs to change	Baselines change only through approved change requests
Do nothing and continue monitoring	Sometimes genuinely correct in risk acceptance scenarios	Usually a trap when the scenario describes an active impact
Remove the team member	It removes the visible symptom quickly	Conflict and performance are addressed through coaching first
Add the change because the customer asked	Customer satisfaction sounds like the PMI value	Every scope change requires impact analysis and change control

Formula Sheet Recap — Earned Value

Cost Variance	$CV = EV - AC$	Negative means over budget. Every variance is EV minus something.
Schedule Variance	$SV = EV - PV$	Negative means behind schedule. Always zero at project completion.
Cost Performance Index	$CPI = EV / AC$	Below 1.0 means you earn less than a dollar of value per dollar spent.
Schedule Performance Index	$SPI = EV / PV$	Below 1.0 means work is arriving slower than the plan assumed.
Estimate at Completion	$EAC = BAC / CPI$	The default forecast when current cost performance is expected to continue.

Also carry: $ETC = EAC - AC$, $VAC = BAC - EAC$, and $TCPI = (BAC - EV) / (BAC - AC)$.

Formula Sheet Recap — Estimating, Float and Risk

PERT triangular	$(O + M + P) / 3$	Simple average of optimistic, most likely and pessimistic estimates.
PERT beta	$(O + 4M + P) / 6$	Weighted average giving the most likely estimate four times the weight.
PERT standard deviation	$(P - O) / 6$	The spread of the estimate; a proxy for how uncertain the activity is.
Total float	$LS - ES$ or $LF - EF$	Delay available without delaying project completion. Zero float = critical path.
Expected Monetary Value	$EMV = Probability \times Impact$	Sum the EMV of every branch to value a decision tree; threats are negative.
Communication channels	$n(n-1) / 2$	Channels grow quadratically with people — the case for smaller teams.

Point of confusion to settle now: total float belongs to a path; free float is the delay available without delaying the very next activity.

Exam Day Strategy

1 Pace deliberately

180 questions in 240 minutes is about 80 seconds each. Check your position at every 45-question mark.

2 Answer everything

There is no penalty for a wrong answer, and 10 items do not count anyway — never leave a blank.

3 Mark and move on

If an item takes more than two minutes, choose your best option, flag it and move on.

4 Use the breaks

Take both 10-minute breaks even if you feel fine — fatigue damages accuracy far more than it damages speed.

5 Case study section first

The case-study section comes before the first break; read the exhibit carefully once rather than repeatedly.

6 Trust your first instinct

Change an answer only when you find a specific fact in the stem that you missed the first time.

The PMI Mindset — the Tiebreaker on Every Item

1 Be proactive

Prevent problems rather than react to them; the answer that anticipates usually beats the answer that responds.

2 Gather facts first

Understand the situation before acting, escalating or blaming — 'analyse the impact' is rarely wrong.

3 Serve the team

Remove impediments, empower and coach; do not command, punish or reassign as a first response.

4 Follow the process

Use the documented change, risk, quality and escalation processes — never bypass governance for speed.

5 Communicate directly

Talk to the person concerned before involving their manager or the sponsor.

6 Deliver value

When options conflict, choose the one that protects the business value and the customer's real need.

Your Study Plan After This Course



Submit your PMI application early — it can take up to five business days to review, and audited applications take longer. Your one-year eligibility period starts when the application is approved.

The 35 PDUs and Maintaining Your Certification

1 This course = 35 contact hours

It satisfies PMI's requirement for 35 hours of commercial project management training.

2 Keep your evidence

Retain the certificate of completion and the course outline in case your application is audited.

3 CCR cycle

Once certified you must earn 60 PDUs every three years to maintain the PMP credential.

4 Education PDUs

Minimum 35 of the 60 must be education PDUs across the PMI Talent Triangle.

5 Talent Triangle

Ways of Working, Power Skills, and Business Acumen — at least 8 PDUs in each per cycle.

6 Giving back PDUs

Up to 25 PDUs may come from volunteering, creating content or working as a practitioner.

Final Exam Traps — Whole-Course Review

IF THE QUESTION SAYS

A risk you identified has now occurred and is delaying the team



THE BEST ANSWER IS

It is now an issue. Move it to the issue log, implement the planned response, and reassess the remaining risk exposure.

IF THE QUESTION SAYS

CPI is 0.85 and the sponsor asks whether you will finish within budget



THE BEST ANSWER IS

Calculate EAC. At BAC/CPI the forecast exceeds BAC, so report the forecast overrun and present recovery options.

IF THE QUESTION SAYS

A stakeholder was never identified and now objects to a delivered feature



THE BEST ANSWER IS

Update the stakeholder register, analyse their influence and interest, and revise the engagement plan before negotiating scope.

IF THE QUESTION SAYS

The team wants to add a nice extra feature at no cost to the customer



THE BEST ANSWER IS

That is gold plating. Deliver to the acceptance criteria; route any genuine addition through integrated change control.

Topic 6 Recap — Close the Project and Exam Readiness

1 Closure is a process

Acceptance, transition, contract and financial closure, archiving and release — performed in full, every time.

3 Transition drives benefits

Training, documentation and support determine adoption, and adoption determines value.

5 Retrospectives use both data types

Qualitative feelings and quantitative measures together produce real root causes.

7 Exam: 180 in 240 minutes

170 scored plus 10 pretest items, with two 10-minute breaks and roughly 80 seconds per question.

9 Six question types

Including the new graphic-based items and the case-study section that precedes the first break.

2 Formal acceptance is required

Written sign-off against agreed acceptance criteria is what makes a project complete.

4 Register becomes repository

Lessons learned move from a project document into an organisational process asset at closure.

6 Benefits need an accountable owner

Hand over the measurement system; most benefits arrive after the team has gone.

8 Exam: 33 / 41 / 26

People, Process and Business Environment weighting — with about 40% predictive and 60% adaptive or hybrid.

10 Answer with the PMI mindset

Investigate first, follow the process, serve the team, and protect the business value.

Hands-on Labs — Topic 6

1

Lab 23. Project Closure, Benefits Realisation and Lessons Learned

Process T10, BE T6 · LO5 · 105 min in class
Produces: Closure checklist, Final report, Lessons learned register

2

Lab 24. Capstone: Consolidated Project Plan and Mock Exam

All domains · LO1-LO5 · 175 min in class
Produces: Consolidated project management plan, Mock exam score analysis by domain

3

Everything you have built

These final labs consolidate the artifacts from Labs 01-22 into one integrated plan and an exam-readiness score.

Consolidates: Closure checklist, Consolidated project management plan, Final report, Lessons learned register

Lab 23. Project Closure, Benefits Realisation and Lessons Learned

Case study: Contoso Training Portal Upgrade. Process T10, BE T6 · LO5. Produces Closure checklist, Final report, Lessons learned register.

105 min in class

1 Verify every deliverable against the acceptance criteria agreed in the charter and scope statement.

2 Obtain and record formal stakeholder sign-off — informal agreement is not closure.

3 Confirm transition readiness: documentation, training and support arrangements for the receiving team.

4 Close all procurements, settle final payments and archive the contracts and financial records.

5 Run the retrospective and write the lessons learned into the organisational repository, not just the project folder.

6 Confirm the benefits measurement system is live and the benefits owner accepts accountability past project end.

DELIVERABLE

Closure checklist, Final report, Lessons learned register

Lab 24. Capstone: Consolidated Project Plan and Mock Exam

Case study: Contoso Training Portal Upgrade. All domains · LO1-LO5. Produces Consolidated project management plan, Mock exam score analysis by domain.

175 min in class

1 Consolidate every artifact from Labs 01-23 into one integrated project management plan with a contents page.

2 Verify internal consistency: the WBS, schedule, budget, RACI and risk register must all describe the same project.

3 Present the plan to the class as though to a steering committee, in ten minutes, defending your key trade-offs.

4 Sit the full 180-question mock exam under timed conditions — 240 minutes, closed book, two breaks.

5 Score the mock by ECO domain and compare against the 33/41/26 weighting to expose your weakest area.

6 Rewrite your Lab 01 study plan using the mock results, naming the specific topics to revisit before booking the exam.

DELIVERABLE

Consolidated project management plan, Mock exam score analysis by domain

Your Next 90 Days

1 Submit the PMI application

Use this course for the 35 contact hours; document your project experience against the eligibility rules.

2 Build a study schedule

Plan 8–10 weeks of study, weighting your time by the domain percentages: Process 41%, People 33%, Business Environment 26%.

3 Practise by question type

Drill the new graphic-based and case-study formats, not just multiple choice.

4 Re-run the labs at work

Apply the charter, WBS, EVM and control-chart labs to a live project to consolidate the learning.

5 Take full mock exams

Sit at least three timed 180-question mocks under exam conditions before booking.

6 Maintain your certification

The CCR programme requires 60 PDUs every three years once you are certified.

Recommended Courses

1 Fundamentals of Microsoft Project

Build and control real schedules, baselines and resource plans in MS Project.

2 Effective Project Management for Small Projects

A lighter-weight toolkit for projects that do not warrant full PMP rigour.

3 Certified Lean Six Sigma Green Belt

Go deeper on DMAIC, SPC and the statistical quality tools introduced here.

4 Agile Project Management with Scrum

Extend the adaptive content into full Scrum practice and certification.

5 Business Analysis Fundamentals

Strengthen requirements elicitation, traceability and stakeholder analysis.

6 Data Analytics for Managers

Turn project and operational data into the metrics your governance board wants.

Support

1 Email

enquiry@tertiaryinfotech.com — for course and certificate queries.

2 Telephone

6100 0613 — office hours, Monday to Friday.

3 Course materials

Slides, Learner Guide and the full lab set are shared with you for continued reference.

4 Problem-solving tools

The live 5 Whys, Fishbone, Pareto, SPC and Statistics tools stay available to you after the course.

5 PMI application

Quote this course for the 35 contact hours of project management education PMI requires.

6 Keep practising

Re-run the labs on a live project — applied repetition is what converts these techniques into exam speed.

END OF COURSE

Thank You

Now go and pass the exam.

35 contact hours earned

You have the project management education PMI requires for the application.

24 labs completed

You leave with a full set of working project artifacts.

ECO 2026 covered

Every domain and task on the July 2026 outline has been taught.